# Contents

## Part II The C++ Library 323

# Recently Added Features

# Features Added to C++11

# CHAPTER 1

# GETTING STARTED

## CONTENTS



This chapter introduces most of the basic elements of C++: types, variables, expressions, statements, and functions. Along the way, we'll briefly explain how to compile and execute a program.

After having read this chapter and worked through the exercises, you should be able to write, compile, and execute simple programs. Later chapters will assume that you can use the features introduced in this chapter, and will explain these features in more detail.

*The way to learn* a new programming language is to write programs. In this chapter, we'll write a program to solve a simple problem for a bookstore.

Our store keeps a file of transactions, each of which records the sale of one or more copies of a single book. Each transaction contains three data elements:

```
0-201-70353-X 4 24.99
```

The first element is an ISBN (International Standard Book Number, a unique book identifier), the second is the number of copies sold, and the last is the price at which each of these copies was sold. From time to time, the bookstore owner reads this file and for each book computes the number of copies sold, the total revenue from that book, and the average sales price.

To be able to write this program, we need to cover a few basic C++ features. In addition, we'll need to know how to compile and execute a program.

Although we haven't yet designed our program, it's easy to see that it must

- Define variables
- Do input and output
- Use a data structure to hold the data
- Test whether two records have the same ISBN
- Contain a loop that will process every record in the transaction file

We'll start by reviewing how to solve these subproblems in C++ and then write our bookstore program.

## 1.1 Writing a Simple C++ Program

Every C++ program contains one or more ***functions***, one of which must be named **`main`**. The operating system runs a C++ program by calling `main`. Here is a simple version of `main` that does nothing but return a value to the operating system:

```
int main()
{
    return 0;
}
```

A function definition has four elements: a ***return type***, a ***function name***, a (possibly empty) ***parameter list*** enclosed in parentheses, and a ***function body***. Although `main` is special in some ways, we define `main` the same way we define any other function.

In this example, `main` has an empty parameter list (shown by the `()` with nothing inside). § 6.6.2 (p. 258) will discuss the other parameter types that we can define for `main`.

The `main` function is required to have a return type of `int`, which is a type that represents integers. The `int` type is a **built-in type**, which means that it is one of the types the language defines.

The final part of a function definition, the function body, is a ***block*** of ***statements*** starting with an open **curly brace** and ending with a close curly:

```
{
    return 0;
}
```

The only statement in this block is a `return`, which is a statement that terminates the execution of a function. This particular `return` statement contains the value `0`, which it sends back to the function's caller. In the case of `main`, the caller is the operating system itself.

When a `return` statement includes a value, the value returned must have a type that is compatible with the return type of the function. In this case, the return type of `main` is `int` and the return value is `0`, which is an `int`.

> Note the semicolon at the end of the `return` statement. Semicolons mark the end of most statements in C++. They are easy to overlook but, when forgotten, can lead to mysterious compiler error messages.

On most systems, the value returned from `main` is a status indicator. A return value of `0` indicates success; any other value indicates that an error occurred.

**KEY CONCEPT: TYPES**

**Types are one of the most fundamental concepts in programming. A type defines both the contents of a data element and the operations that are possible on those data. The data that our programs manipulate are stored in variables and every variable has a type. If `T` is the type of a variable named `v`, we often say that "`v` has type `T`" or, interchangeably, that "`v` is a `T`."**

## 1.1.1 Compiling and Executing Our Program

Having written the program, we need to compile it. How you compile a program depends on your operating system and compiler. For details on how your particular compiler works, check the reference manual or ask a knowledgeable colleague.

Many PC-based compilers are run from an integrated development environment (IDE) that bundles the compiler with build and analysis tools. These environments can be a great asset in developing large programs but require a fair bit of time to learn how to use effectively. Learning how to use such environments is well beyond the scope of this book.

Most compilers, including those that come with an IDE, provide a command-line interface. Unless you already know the IDE, you may find it easier to start with the command-line interface. Doing so will let you concentrate on learning C++ first. Moreover, once you understand the language, the IDE is likely to be easier to learn.

### Program Source File Naming Convention

Whether you use a command-line interface or an IDE, most compilers expect program source code to be stored in one or more files. Program files are normally referred to as a ***source files***. On most systems, the name of a source file ends with a suffix, which is a period followed by one or more characters. The suffix tells the system that the file is a C++ program. Different compilers use different suffix conventions; the most common include `.cc`, `.cxx`, `.cpp`, `.cp`, and `.C`.

## Running the Compiler from the Command Line

If we are using a command-line interface, we will typically compile a program in a console window (such as a shell window on a UNIX system or a Command Prompt window on Windows). Assuming that our `main` program is in a file named `prog1.cc`, we might compile it by using a command such as

```
$ CC prog1.cc
```

where `CC` names the compiler and `$` is the system prompt. The compiler generates an executable file. On a Windows system, that executable file is named `prog1.exe`. UNIX compilers tend to put their executables in files named `a.out`.

To run an executable on Windows, we supply the executable file name and can omit the `.exe` file extension:

```
C:\Users\me\Programs> prog1
```

On some systems you must specify the file's location explicitly, even if the file is in the current directory or folder. In such cases, we would write

```
C:\Users\me\Programs> .\prog1
```

The "`.`" followed by a backslash indicates that the file is in the current directory.

To run an executable on UNIX, we use the full file name, including the file extension:

```
$ a.out
```

If we need to specify the file's location, we'd use a "`.`" followed by a forward slash to indicate that our executable is in the current directory:

```
$ ./a.out
```

The value returned from `main` is accessed in a system-dependent manner. On both UNIX and Windows systems, after executing the program, you must issue an appropriate `echo` command.

On UNIX systems, we obtain the status by writing

```
$ echo $?
```

To see the status on a Windows system, we write

```
$ echo %ERRORLEVEL%
```

**RUNNING THE GNU OR MICROSOFT COMPILERS**

**The command used to run the C++ compiler varies across compilers and operating systems. The most common compilers are the GNU compiler and the Microsoft Visual Studio compilers. By default, the command to run the GNU compiler is `g++`:**

```
$ g++ -o prog1 prog1.cc
```

**Here `$` is the system prompt. The `-o prog1` is an argument to the compiler and names the executable file that the compiler will generate. This command generates an executable file named `prog1` or `prog1.exe`, depending on the operating system. On UNIX, executable files have no suffix; on Windows, the suffix is `.exe`. If the `-o prog1` is omitted, the compiler generates an executable named `a.out` on UNIX systems and `a.exe` on Windows. (Note: Depending on the release of the GNU compiler you are using, you may need to specify `-std=c++17` to turn on C++17 support.)**

**The command to run the Microsoft Visual Studio 2010 compiler is `cl`:**

```
C:\Users\me\Programs> cl /EHsc prog1.cpp
```

**Here `C:\Users\me\Programs>` is the system prompt and `\Users\me\Programs` is the name of the current directory (aka the current folder). The `cl` command invokes the compiler, and `/EHsc` is the compiler option that turns on standard exception handling. The Microsoft compiler automatically generates an executable with a name that corresponds to the first source file name. The executable has the suffix `.exe` and the same name as the source file name. In this case, the executable is named `prog1.exe`.**

**Compilers usually include options to generate warnings about problematic constructs. It is usually a good idea to use these options. Our preference is to use `-Wall` with the GNU compiler, and to use `/W4` with the Microsoft compilers.**

**For further information consult your compiler's user's guide.**

**EXERCISES SECTION 1.1.1**

**Exercise 1.1:** Review the documentation for your compiler and determine what file naming convention it uses. Compile and run the `main` program from page 2.

**Exercise 1.2:** Change the program to return `-1`. A return value of `-1` is often treated as an indicator that the program failed. Recompile and rerun your program to see how your system treats a failure indicator from `main`.

## 1.2 A First Look at Input/Output

The C++ language does not define any statements to do input or output (IO). Instead, C++ includes an extensive **standard library** that provides IO (and many other facilities). For many purposes, including the examples in this book, one needs to know only a few basic concepts and operations from the IO library.

Most of the examples in this book use the **`iostream`** library. Fundamental to the `iostream` library are two types named **`istream`** and **`ostream`**, which represent input and output streams, respectively. A stream is a sequence of characters read from or written to an IO device. The term *stream* is intended to suggest that the characters are generated, or consumed, sequentially over time.

### Standard Input and Output Objects

The library defines four IO objects. To handle input, we use an object of type `istream` named **`cin`** (pronounced *see-in*). This object is also referred to as the **standard input**. For output, we use an `ostream` object named **`cout`** (pronounced *see-out*). This object is also known as the **standard output**. The library also defines two other `ostream` objects, named **`cerr`** and **`clog`** (pronounced *see-err* and *see-log*, respectively). We typically use `cerr`, referred to as the **standard error**, for warning and error messages and `clog` for general information about the execution of the program.

Ordinarily, the system associates each of these objects with the window in which the program is executed. So, when we read from `cin`, data are read from the window in which the program is executing, and when we write to `cout`, `cerr`, or `clog`, the output is written to the same window.

### A Program That Uses the IO Library

In our bookstore problem, we'll have several records that we'll want to combine into a single total. As a simpler, related problem, let's look first at how we might compute the total of two numbers. Using the IO library, we can extend our `main` program to prompt the user to give us two numbers and then print their sum:

```
#include <iostream>
int main()
{
    std::cout << "Enter two numbers:" << std::endl;
    int v1 = 0, v2 = 0;
    std::cin >> v1 >> v2;
    std::cout << "The sum of " << v1 << " and " << v2
              << " is " << v1 + v2 << std::endl;
    return 0;
}
```

This program starts by printing

```
Enter two numbers:
```

on the user's screen and then waits for input from the user. If the user enters

```
3 7
```

followed by a newline, then the program produces the following output:

```
The sum of 3 and 7 is 10
```

The first line of our program

```
#include <iostream>
```

tells the compiler that we want to use the `iostream` library. The name inside angle brackets (`iostream` in this case) refers to a **header**. Every program that uses a library facility must include its associated header. The `#include` directive must be written on a single line—the name of the header and the `#include` must appear on the same line. In general, `#include` directives must appear outside any function. To make it easy to see which headers are used, we generally put all the `#include` directives for a program at the beginning of the source file.

## Writing to a Stream

The first statement in the body of `main` executes an **expression**. In C++, an expression yields a result and contains one or more operands and (usually) an operator. The expression in this statement uses the output operator (the **<< operator**) to print a message on the standard output:

```
std::cout << "Enter two numbers:" << std::endl;
```

The `<<` operator takes two operands: The left-hand operand must be an `ostream` object; the right-hand operand is a value to print. The operator writes the given value on the given `ostream`. The result of the output operator is its left-hand operand. That is, the result is the `ostream` on which we wrote the given value.

Our output statement uses the `<<` operator twice. Because the operator returns its left-hand operand, the result of the first operator becomes the left-hand operand of the second. As a result, we can chain together output requests. Thus, our expression is equivalent to

```
(std::cout << "Enter two numbers:") << std::endl;
```

Each operator in the chain has the same object as its left-hand operand, in this case `std::cout`. We could generate the same output by using two statements:

```
std::cout << "Enter two numbers:";
std::cout << std::endl;
```

The first output operator prints a message to the user. That message is a **string literal**, which is a sequence of characters enclosed in double quotation marks. The text between the quotation marks is printed to the standard output.

The second operator prints **`endl`**, which is a special value called a **manipulator**. Writing `endl` has the effect of ending the current line and flushing the ***buffer*** associated with that device. Flushing the buffer ensures that all the output the program has generated so far is actually written to the output stream, rather than sitting in memory waiting to be written.

> **WARNING**
>
> Programmers often add print statements during debugging. Such statements should *always* flush the stream. Otherwise, if the program crashes, output may be left in the buffer, leading to incorrect inferences about where the program crashed.

## Using Names from the Standard Library

Careful readers will note that this program uses `std::cout` and `std::endl` rather than just `cout` and `endl`. The prefix `std::` indicates that the names `cout` and `endl` are defined inside the **namespace** named **`std`**. Namespaces allow us to avoid inadvertent collisions between the names we define and uses of those same names inside a library. All the names defined by the standard library are in the `std` namespace.

One side effect of the library's use of a namespace is that when we use a name from the library, we must say explicitly that we want to use the name from the `std` namespace. Writing `std::cout` uses the scope operator (the **`::` operator**) to say that we want to use the name `cout` that is defined in the namespace `std`. § 3.1 (p. 84) will show a simpler way to access names from the library.

## Reading from a Stream

Having asked the user for input, we next want to read that input. We start by defining two ***variables*** named `v1` and `v2` to hold the input:

```
int v1 = 0, v2 = 0;
```

We define these variables as type `int`, which is a built-in type representing integers. We also ***initialize*** each one to `0`. We initialize a variable by supplying a value for the variable at the same time as the variable is created.

The next statement

```
std::cin >> v1 >> v2;
```

reads the input. The input operator (the **>> operator**) behaves analogously to the output operator: It takes an `istream` as its left-hand operand and an object as its right-hand operand. It reads data from the given `istream` and stores what was read in the given object. Like the output operator, the input operator returns its left-hand operand as its result. Hence, this expression is equivalent to

```
(std::cin >> v1) >> v2;
```

Because the `>>` operator returns its left-hand operand, we can combine a sequence of input requests into a single statement. Our input operation reads two values from `std::cin`, storing the first in `v1` and the second in `v2`. In other words, our input operation executes as

```
std::cin >> v1;
std::cin >> v2;
```

## Completing the Program

What remains is to print our result:

```
std::cout << "The sum of " << v1 << " and " << v2
          << " is " << v1 + v2 << std::endl;
```

This statement, although longer than the one that prompted the user for input, is conceptually similar: It prints each of its operands on the standard output. What is interesting in this example is that the operands are not all the same kinds of values. Some operands are string literals, such as `"The sum of "`. Others are `int` values, such as `v1`, `v2`, and the result of evaluating the expression `v1 + v2`. The library defines versions of the input and output operators that handle operands of each of these differing types.

**EXERCISES SECTION 1.2**

**Exercise 1.3:** Write a program to print `Hello, World` on the standard output.

**Exercise 1.4:** Write a program that uses the multiplication operator, `*`, to multiply two numbers, and prints their product.

**Exercise 1.5:** We wrote the output in one large statement. Rewrite the program to use a separate statement to print each operand.

**Exercise 1.6:** Explain whether the following program fragment is legal.

```
std::cout << "The sum of " << v1;
          << " and " << v2;
          << " is " << v1 + v2 << std::endl;
```

If the program is legal, what does it do? If the program is not legal, why not? How would you fix it?

## 1.3 A Word about Comments

Before our programs get much more complicated, we should see how C++ handles ***comments***. Comments help the human readers of our programs. They are typically used to summarize an algorithm, identify the purpose of a variable, or clarify an otherwise obscure segment of code. The compiler ignores comments, so they have no effect on the program's behavior or performance.

Although the compiler ignores comments, readers of our code do not. Programmers tend to believe comments even when other parts of the system documentation are out of date. An incorrect comment is worse than no comment at all because it may mislead the reader. When you change your code, be sure to update the comments, too!

### Kinds of Comments in C++

There are two kinds of comments in C++: single-line and paired. A single-line comment starts with a double slash (`//`) and ends with the next newline. The compiler ignores everything to the right of the slashes on the current line. A comment of this kind can contain any text, including additional double slashes.

The other kind of comment uses two delimiters (`/*` and `*/`) that are inherited from C. Such comments begin with a `/*` and end with the next `*/`. These comments can include anything that is not a `*/`, including newlines. The compiler treats everything that falls between the `/*` and `*/` as part of the comment.

A comment pair can be placed anywhere a tab, space, or newline is permitted. Comment pairs can span multiple lines of a program but are not required to do so. When a comment pair does span multiple lines, it is often a good idea to indicate visually that the inner lines are part of a multiline comment. Our style is to begin each line in the comment with an asterisk, thus indicating that the entire range is part of a multiline comment.

Programs typically contain a mixture of both comment forms. Comment pairs generally are used for multiline explanations, whereas double-slash comments tend to be used for half-line and single-line remarks:

```
#include <iostream>
/*
 * Simple main function:
 * Read two numbers and write their sum
 */
int main()
{
    // prompt user to enter two numbers
    std::cout << "Enter two numbers:" << std::endl;
    int v1 = 0, v2 = 0;    // variables to hold the input we read
    std::cin >> v1 >> v2; // read input
    std::cout << "The sum of " << v1 << " and " << v2
              << " is " << v1 + v2 << std::endl;
    return 0;
}
```

Note: In this book, we italicize comments to make them stand out from the normal program text. In actual programs, whether comment text is distinguished from the text used for program code depends on the sophistication of the programming environment you are using.

## Comment Pairs Do Not Nest

A comment that begins with `/*` ends with the next `*/`. As a result, one comment pair cannot appear inside another. The compiler error messages that result from this kind of mistake can be mysterious and confusing. As an example, compile the following program on your system:

```
/*
 * comment pairs /*   */ cannot nest.
 * ''cannot nest'' is considered source code,
 * as is the rest of the program
 */
int main()
{
    return 0;
}
```

It is often useful to comment out a block of code during debugging. Because that code might contain nested comment pairs, the best way to comment a block of code is to insert single-line comments at the beginning of each line in the section we want to ignore:

```
// /*
//  * everything inside a single-line comment is ignored
//  * including nested comment pairs
//  */
```

## Exercises Section 1.3

**Exercise 1.7:** Compile a program that has incorrectly nested comments.

**Exercise 1.8:** Indicate which, if any, of the following output statements are legal:

```
std::cout << "/*";
std::cout << "*/";
std::cout << /* "*/" */;
std::cout << /*  "*/" /*  "/*"  */;
```

After you've predicted what will happen, test your answers by compiling a program with each of these statements. Correct any errors you encounter.

# 1.4 Flow of Control

Statements normally execute sequentially: The first statement in a block is executed first, followed by the second, and so on. Of course, few programs—including the one we'll write to solve our bookstore problem—can be written using only sequential execution. Instead, programming languages provide various flow-of-control statements that allow for more complicated execution paths.

## 1.4.1 The `while` Statement

A **`while` statement** repeatedly executes a section of code so long as a given condition is true. We can use a `while` to write a program to sum the numbers from 1 through 10 inclusive as follows:

```
#include <iostream>
int main()
{
    int sum = 0, val = 1;
    // keep executing the while as long as val is less than or equal to 10
    while (val <= 10) {
        sum += val;   // assigns sum + val to sum
        ++val;        // add 1 to val
    }
    std::cout << "Sum of 1 to 10 inclusive is "
              << sum << std::endl;
    return 0;
}
```

When we compile and execute this program, it prints

```
Sum of 1 to 10 inclusive is 55
```

As before, we start by including the `iostream` header and defining `main`. Inside `main` we define two `int` variables: `sum`, which will hold our summation, and `val`, which will represent each of the values from 1 through 10. We give `sum` an initial value of `0` and start `val` off with the value `1`.

The new part of this program is the `while` statement. A `while` has the form

```
while (condition)
      statement
```

A `while` executes by (alternately) testing the *condition* and executing the associated *statement* until the *condition* is false. A **condition** is an expression that yields a result that is either true or false. So long as *condition* is true, *statement* is executed. After executing *statement*, *condition* is tested again. If *condition* is again true, then *statement* is again executed.

In this program, the `while` statement is

```
// keep executing the while as long as val is less than or equal to 10
while (val <= 10) {
    sum += val;   // assigns sum + val to sum
    ++val;        // add 1 to val
}
```

The condition uses the less-than-or-equal operator (the **<= operator**) to compare the current value of `val` and `10`. As long as `val` is less than or equal to 10, the condition is true. If the condition is true, we execute the body of the `while`. In this case, that body is a block with two statements:

```
{
    sum += val;   // assigns sum + val to sum
    ++val;        // add 1 to val
}
```

A block is a sequence of zero or more statements enclosed by curly braces. A block is a statement and may be used wherever a statement is required. The first statement in this block uses the compound assignment operator (the **+= operator**). This operator adds its right-hand operand to its left-hand operand and stores the result in the left-hand operand. It has essentially the same effect as writing an addition and an **assignment**:

```
sum = sum + val;  // assign sum + val to sum
```

Thus, the first statement in the block adds the value of `val` to the current value of `sum` and stores the result back into `sum`.

The next statement

```
++val;        // add 1 to val
```

uses the prefix increment operator (the **++ operator**). The increment operator adds `1` to its operand: Writing `++val` has the same meaning as writing `val = val + 1`.

After executing the `while` body, the loop evaluates the condition again. If the (now incremented) value of `val` is still less than or equal to 10, then the body of the `while` is executed again. The loop continues, testing the condition and executing the body, until `val` is no longer less than or equal to 10.

Once `val` is greater than 10, the program falls out of the `while` loop and continues execution with the statement following the `while`. In this case, that statement prints our output, followed by the `return`, which completes our `main` program.

### EXERCISES SECTION 1.4.1

**Exercise 1.9:** Write a program that uses a `while` to sum the numbers from 50 to 100.

**Exercise 1.10:** In addition to the `++` operator that adds `1` to its operand, there is a decrement operator (`--`) that subtracts `1`. Use the decrement operator to write a `while` that prints the numbers from ten down to zero.

**Exercise 1.11:** Write a program that prompts the user for two integers. Print each number in the range specified by those two integers.

## 1.4.2 The `for` Statement

In our `while` loop, we used the variable `val` to control how many times we executed the loop: We tested the value of `val` in the condition and incremented `val` in the `while` body.

This pattern—using a variable in a condition and modifying that variable in the body—happens so often that the language defines a second statement, the **`for` statement**, to abbreviate code that follows this pattern. We can rewrite this program using a `for` loop to sum the numbers from 1 through 10 as follows:

```
#include <iostream>
int main()
{
    int sum = 0;
    // sum values from 1 through 10 inclusive
    for (int val = 1; val <= 10; ++val)
        sum += val;  // equivalent to sum = sum + val
    std::cout << "Sum of 1 to 10 inclusive is "
              << sum << std::endl;
    return 0;
}
```

As before, we define `sum` and initialize it to zero. In this version, we define `val` as part of the `for` statement itself:

```
for (int val = 1; val <= 10; ++val)
    sum += val;
```

Each `for` statement has two parts: a header and a body. The header controls how often the body is executed. The header itself consists of three parts: an *init-statement*, a *condition*, and an *expression*. Both the *init-statement* and *condition* must be followed by semicolons.

In this loop, the *init-statement*

```
int val = 1;
```

defines an `int` object named `val` and gives it an initial value of `1`. The *init-statement* is executed only once, on entry to the `for`. The variable `val` exists only inside the `for`; it is not possible to use `val` after this loop terminates. The *condition*

## Compilation Revisited

Part of the compiler's job is to look for errors in the program text. A compiler cannot detect whether a program does what its author intends, but it can detect errors in the *form* of the program. The following are the most common kinds of errors a compiler will detect.

*Syntax errors:* The programmer has made a grammatical error in the C++ language. The following program illustrates common syntax errors; each comment describes the error on the following line:

```
// error: missing ) in parameter list for main
int main ( {
    // error: missing semicolon after endl
    std::cout << "Read each file." << std::endl
    // error: missing quotes around string literal
    std::cout << Update master. << std::endl;
    // error: second output operator is missing
    std::cout << "Write new master." std::endl;
    // ok: no syntax errors on this line
    return 0;
}
```

*Type errors:* Each item of data in C++ has an associated type. The value 10, for example, has a type of `int` (or, more colloquially, "is an `int`"). The word `"hello"`, including the double quotation marks, is a string literal, which has a type we'll cover in § 3.5.4 (p. 128) One example of a type error is passing a string literal to a function that expects an `int` argument.

*Declaration errors:* Every name used in a C++ program must be declared before it is used. Failure to declare a name usually results in an error message. The two most common declaration errors are forgetting to use `std::` for a name from the library and misspelling the name of an identifier:

```
#include <iostream>
int main()
{
    int v1 = 0, v2 = 0;
    std::cin >> v >> v2;  // error: uses "v" not "v1"
    // error: cout not defined; should be std::cout
    cout << v1 + v2 << std::endl;
    return 0;
}
```

Error messages usually contain a line number and a brief description of what the compiler believes we have done wrong. It is a good practice to correct errors in the sequence in which they are reported. Often a single error can have a cascading effect and cause a compiler to report more errors than actually are present. It is also a good idea to recompile the code after each fix—or after making at most a small number of obvious fixes. This cycle is known as *edit-compile-debug*.

```
val <= 10
```

compares the current value in `val` to `10`. The *condition* is tested each time through the loop. As long as `val` is less than or equal to `10`, we execute the `for` body.

The *expression* is executed after the `for` body. Here, the *expression*

```
++val
```

uses the prefix increment operator, which adds `1` to the value of `val`. After executing the *expression*, the `for` retests the *condition*. If the new value of `val` is still less than or equal to `10`, then the `for` loop body is executed again. After executing the body, `val` is incremented again. The loop continues until the *condition* fails.

In this loop, the `for` body performs the summation

```
sum += val;   // equivalent to sum = sum + val
```

To recap, the overall execution flow of this `for` is:

1. Create `val` and initialize it to `1`.
2. Test whether `val` is less than or equal to `10`. If the test succeeds, execute the `for` body. If the test fails, exit the loop and continue execution with the first statement following the `for` body.
3. Increment `val`.
4. Repeat the test in step 2, continuing with the remaining steps as long as the condition is true.

**EXERCISES SECTION 1.4.2**

**Exercise 1.12:** What does the following `for` loop do? What is the final value of `sum`?

```
int sum = 0;
for (int i = -100; i <= 100; ++i)
    sum += i;
```

**Exercise 1.13:** Rewrite the first two exercises from § 1.4.1 (p. 13) using `for` loops.

**Exercise 1.14:** Compare and contrast the loops that used a `for` with those using a `while`. Are there advantages or disadvantages to using either form?

**Exercise 1.15:** Write programs that contain the common errors discussed in the box on page 14. Familiarize yourself with the messages the compiler generates.

## 1.4.3 Reading an Unknown Number of Inputs

In the preceding sections, we wrote programs that summed the numbers from 1 through 10. A logical extension of this program would be to ask the user to input a set of numbers to sum. In this case, we won't know how many numbers to add. Instead, we'll keep reading numbers until there are no more numbers to read:

```
#include <iostream>
int main()
{
    int sum = 0;
    // read until end-of-file, calculating a running total of all values read
    for (int value; std::cin >> value; /* no expression */)
        sum += value; // equivalent to sum = sum + value
    std::cout << "Sum is: " << sum << std::endl;
    return 0;
}
```

If we give this program the input

```
3 4 5 6
```

then our output will be

```
Sum is: 18
```

As in our other programs, we start by defining and initializing `sum` to `0`. In the `for` header, we define `value`, which we'll use to hold each number that we read from the input. The condition, `cin >> val`, reads the next input and the `for` body adds `value` to `sum`. No additional work needs to be done, so we omit the *expression* portion of the `for` header.

The interesting part of this program is the condition:

```
for (int value; std::cin >> value; /* no expression */)
```

Evaluating this condition executes the expression

```
std::cin >> value
```

That expression reads the next number from the standard input and stores that number in `value`. The input operator (§ 1.2, p. 8) returns its left operand, which in this case is `std::cin`. This condition, therefore, tests `std::cin`.

When we use an `istream` as a condition, the effect is to test the state of the stream. If the stream is valid—that is, if the stream hasn't encountered an error—then the test succeeds. An `istream` becomes invalid when we hit ***end-of-file*** or encounter an invalid input, such as reading a value that is not an integer. An `istream` that is in an invalid state will cause the condition to yield false.

Thus, our `for` executes until we encounter end-of-file (or an input error). The `for` body uses the compound assignment operator to add the current value to the evolving `sum`. Once the condition fails, the `for` ends. We fall through and execute the next statement, which prints the `sum` followed by `endl`.

One final point about this code. Although we included a comment to indicate that there was no expression in our `for` header, that comment—like any other comment—is strictly optional. We could have written the `for` header as:

```
for (int value; std::cin >> value;)   // another way to omit expression
for (int value; std::cin >> value; )  // equivalent loop
```

Our style is to use the second version, which includes a space where the experssion would ordinarily appear. The extra space is intended to make it clear that we intentionally omitted the expression.

**ENTERING AN END-OF-FILE FROM THE KEYBOARD**

**When we enter input to a program from the keyboard, different operating systems use different conventions to allow us to indicate end-of-file. On Windows systems we enter an end-of-file by typing a control-z—hold down the Ctrl key and press `z`—followed by hitting either the Enter or Return key. On UNIX systems, including on Mac OS X machines, end-of-file is usually control-d at the beginning of an input line.**

**EXERCISES SECTION 1.4.3**

**Exercise 1.16:** Write your own version of a program that prints the sum of a set of integers read from `cin`.

### 1.4.4 The `if` Statement

Like most languages, C++ provides an **`if` statement** that supports conditional execution. We can use an `if` to write a program to count how many consecutive times each distinct value appears in the input:

```
#include <iostream>
int main()
{
    // read first number and ensure that we have data to process
    // currVal is the number we're currently counting
    if (int currVal; std::cin >> currVal) {
        int cnt = 1;  // counter to track how often each value occurs
        // loop to read the remaining values
        for (int val; std::cin >> val; ) { // expression omitted
            if (val == currVal)   // if the values are the same
                ++cnt;            // add 1 to cnt
            else { // otherwise, print the count for the current value
                std::cout << currVal << " occurs "
                          << cnt << " times" << std::endl;
                currVal = val;    // remember the new value
                cnt = 1;          // reset the counter
            }
        }  // for loop ends here
        // remember to print the count for the last value in the file
        std::cout << currVal << " occurs "
                  << cnt << " times" << std::endl;
    } // outermost if statement ends here
    return 0;
}
```

If we give this program the following input:

```
42 42 42 42 42 55 55 62
```

then the output should be

```
42 occurs 5 times
55 occurs 2 times
62 occurs 1 times
```

Much of the code in this program should be familiar. What's new are the two `if` statements. The first `if` ensures that the input is not empty:

```
if (int currVal; std::cin >> currVal) {
    // ...
} // outermost if statement ends here
```

New Like a `while`, an `if` evaluates a condition. Unlike a `while`, an `if` can define a variable as part of the `if` itself. In this case, we define `currVal`, which will hold the number we are currently counting. The condition follows the variable definition (if any). The condition in this `if` reads a value into our newly defined variable. If the read succeeds, then the condition is true and we execute the block that starts with following open curly. That block ends with the close curly just before the `return` statement.

Once we know that there are numbers to count, we define `cnt`, which will count how often each distinct value occurs. We use a `for` loop similar to the one in the previous section to (repeatedly) read values from the standard input.

The body of the `for` is a block that contains the second `if` statement:

```
if (val == currVal)    // if the values are the same
    ++cnt;             // add 1 to cnt
else { // otherwise, print the count for the current value
    std::cout << currVal << " occurs "
              << cnt << " times" << std::endl;
    currVal = val;     // remember the new value
    cnt = 1;           // reset the counter
}
```

The condition in this `if` uses the equality operator (the **== operator**) to test whether `val` is equal to `currVal`. If so, we execute the statement that immediately follows the condition. That statement increments `cnt`, indicating that we have seen `currVal` once more.

If the condition is false—that is, if `val` is not equal to `currVal`—then we execute the statement following the `else`. This statement is a block consisting of an output statement and two assignments. The output statement prints the count for the value we just finished processing. The assignments reset `cnt` to `1` and `currVal` to `val`, which is the number that we just read.

WARNING

> C++ uses = for assignment and == for equality. Either of these operators can appear inside a condition. It is a common mistake to write = when you mean == inside a condition.

## 1.5 Introducing Classes

The only remaining feature we need to understand in order to solve our bookstore problem is how to define a ***data structure*** to represent our transactions. In C++,

### Exercises Section 1.4.4

**Exercise 1.17:** What happens in the program presented in this section if the input values are all equal? What if there are no duplicated values?

**Exercise 1.18:** Compile and run the program from this section giving it only equal values as input. Run it again giving it values in which no number is repeated.

**Exercise 1.19:** Revise the program you wrote for the exercises in § 1.4.1 (p. 13) that printed a range of numbers so that it handles input in which the first number is smaller than the second.

### Key Concept: Indentation and Formatting of C++ Programs

**C++ programs are largely free-format, meaning that where we put spaces, tabs, comments, and newlines usually has no effect on what our programs mean. For example, the curly brace that denotes the beginning of the body of `main` could be on the same line as `main`; positioned as we have done, at the beginning of the next line; or placed anywhere else we'd like. The only requirement is that the open curly must be the first nonblank, noncomment character following `main`'s parameter list.**

**Although we can generally format programs as we wish, the choices we make affect the readability of our programs. We could, for example, write the body of a `main` function on a single long line. Such a definition, although legal, would be hard to read.**

**Endless debates occur as to the right way to format C or C++ programs. Our belief is that there is no single correct style but that there is value in consistency. Most programmers indent subsidiary parts of their programs, as we've done with the statements inside `main` and the bodies of our loops. We tend to put the curly braces that delimit functions on their own lines. We also indent compound IO expressions so that the operators line up. Other indentation conventions will become clear as our programs become more sophisticated.**

**The important thing to keep in mind is that there are many ways to format a program. When you choose a formatting style, think about how it affects readability and comprehension. Once you've chosen a style, use it consistently.**

we define a data structure by defining a **class**.

In this section, we'll describe a simple class that we can use in writing our bookstore program. We'll implement this class in later chapters as we learn more about types, expressions, statements, and functions.

To use a class we need to know three things:

- What is its name?
- Where is it defined?
- What operations does it support?

For our bookstore problem, we'll assume that the class is named `Sales_item` and that it is already defined in a header named `Sales_item.h`.

As we've seen, when we use a library facility, we must include the associated header (§ 1.2, p. 7). Similarly, we use headers to access classes defined for our own applications. Conventionally, a header file name is derived from the name of a class defined in that header. Header files that we write usually have a suffix of `.h`, but some programmers use `.H`, `.hpp`, or `.hxx`. The standard library headers have no suffix at all. Compilers usually don't care about the form of header file names, but IDEs sometimes do.

### 1.5.1 The `Sales_item` Class

The purpose of the `Sales_item` class is to represent the total revenue, number of copies sold, and average sales price for a book. How these data are stored or computed is not our concern. To use a class, we need not care about how it is implemented. Instead, what we need to know is what operations objects of that type can perform.

Every class defines a type. The type name is the same as the name of the class. Hence, our `Sales_item` class defines a type named `Sales_item`. As with the built-in types, we can define a variable of a **class types**. When we write

```
Sales_item item;
```

we are saying that `item` is an object of type `Sales_item`. We often contract the phrase "an object of type `Sales_item`" to "a `Sales_item` object" or even more simply to "a `Sales_item`."

In addition to being able to define variables of type `Sales_item`, we can:

- Call a function named `isbn` to fetch the ISBN from a `Sales_item` object.
- Use the input (`>>`) and output (`<<`) operators to read and write objects of type `Sales_item`.
- Use the assignment operator (`=`) to assign one `Sales_item` object to another.
- Use the addition operator (`+`) to add two `Sales_item` objects that must refer to the same ISBN. The result is a new `Sales_item` object whose ISBN is that of its operands and whose number sold and revenue are the sum of the corresponding values in its operands.
- Use the compound assignment operator (`+=`) to add one `Sales_item` object into another.

**KEY CONCEPT: CLASSES DEFINE BEHAVIOR**

**The important thing to keep in mind when you read these programs is that the author of the `Sales_item` class defines *all* the actions that can be performed by objects of this class. That is, the `Sales_item` class defines what happens when a `Sales_item` object is created and what happens when the assignment, addition, or the input and output operators are applied to `Sales_items`.**

### Reading and Writing `Sales_items`

Now that we know what operations we can use with `Sales_item` objects, we can write programs that use the class. For example, the following program reads data from the standard input into a `Sales_item` object and writes that `Sales_item` back onto the standard output:

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    Sales_item book;
    // read ISBN, number of copies sold, and sales price
    std::cin >> book;
    // write ISBN, number of copies sold, total revenue, and average price
    std::cout << book << std::endl;
    return 0;
}
```

If the input to this program is

```
0-201-70353-X 4 24.99
```

then the output will be

```
0-201-70353-X 4 99.96 24.99
```

Our input says that we sold four copies of the book at $24.99 each, and the output indicates that the total sold was four, the total revenue was $99.96, and the average price per book was $24.99.

This program starts with two `#include` directives, one of which uses a new form. Headers from the standard library are enclosed in angle brackets (`< >`). Those that are not part of the library are enclosed in double quotes (`" "`).

Inside `main` we define an object, named `book`, that we'll use to hold the data that we read from the standard input. The next statement reads into that object, and the third statement prints it to the standard output followed by printing `endl`.

### Adding `Sales_items`

A more interesting example adds two `Sales_item` objects:

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    Sales_item item1, item2;
    std::cin >> item1 >> item2;    // read a pair of transactions
    std::cout << item1 + item2 << std::endl; // print their sum
    return 0;
}
```

If we give this program the following input

```
0-201-78345-X 3 20.00
0-201-78345-X 2 25.00
```

our output is

```
0-201-78345-X 5 110 22
```

This program starts by including the `Sales_item` and `iostream` headers. Next we define two `Sales_item` objects to hold the transactions. We read data into these objects from the standard input. The output expression does the addition and prints the result.

It's worth noting how similar this program looks to the one on page 6: We read two inputs and write their sum. What makes this similarity noteworthy is that instead of reading and printing the sum of two integers, we're reading and printing the sum of two `Sales_item` objects. Moreover, the whole idea of "sum" is different. In the case of `int`s we are adding two numeric values. In the case of `Sales_item` objects we use a conceptually new meaning for sum—the result of adding the components of two `Sales_item` objects.

### Using File Redirection

**It can be tedious to repeatedly type these transactions as input to the programs you are testing. Most operating systems support file redirection, which lets us associate a named file with the standard input and the standard output:**

```
$ addItems <infile >outfile
```

**Assuming that `$` is the system prompt and our addition program has been compiled into an executable file named `addItems.exe` (or `addItems` on UNIX systems), this command will read transactions from a file named `infile` and write its output to a file named `outfile` in the current directory.**

### Exercises Section 1.5.1

**Exercise 1.20:** `http://www.informit.com/title/XXXXXXXXXX` contains a copy of `Sales_item.h` in the Chapter 1 code directory. Copy that file to your working directory. Use it to write a program that reads a set of book sales transactions, writing each transaction to the standard output.

**Exercise 1.21:** Write a program that reads two `Sales_item` objects that have the same ISBN and produces their sum.

**Exercise 1.22:** Write a program that reads several transactions for the same ISBN. Write the sum of all the transactions that were read.

## 1.5.2 A First Look at Member Functions

Our program that adds two `Sales_items` should check whether the objects have the same ISBN. We'll do so as follows:

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    Sales_item item1, item2;
    std::cin >> item1 >> item2;
    // first check that item1 and item2 represent the same book
    if (item1.isbn() == item2.isbn()) {
        std::cout << item1 + item2 << std::endl;
        return 0;   // indicate success
    } else {
        std::cerr << "Data must refer to same ISBN"
                  << std::endl;
        return -1;  // indicate failure
    }
}
```

The difference between this program and the previous version is the `if` and its associated `else` branch. Even without understanding the `if` condition, we know what this program does. If the condition succeeds, then we write the same output as before and return `0`, indicating success. If the condition fails, we execute the block following the `else`, which prints a message and returns an error indicator.

### What Is a Member Function?

The `if` condition

```
item1.isbn() == item2.isbn()
```

calls a **member function** named `isbn` (twice). A member function is a function that is defined as part of a class. Member functions are sometimes referred to as **methods**. Ordinarily, we call a member function on behalf of an object. For example, the first part of the left-hand operand of the equality expression

```
item1.isbn
```

uses the dot operator (the **"." operator**) to say that we want "the `isbn` member of the object named `item1`." The left-hand operand of the dot operator must be an object of class type, and the right-hand operand must be the name of a member of that class. The result of the dot operator is the member of left-hand object named by the right-hand operand.

When we use the dot operator to access a member function, we usually do so to call that function. We call a function using the call operator (the **() operator**). The call operator is a pair of parentheses that enclose a (possibly empty) list of ***arguments***. The `isbn` member function does not take an argument. Thus,

```
item1.isbn()
```

calls the `isbn` function that is a member of the object named `item1`. This function returns the ISBN stored in `item1`. The right-hand operand of the equality operator executes in the same way—it returns the ISBN stored in `item2`. If the ISBNs are the same, the condition is true; otherwise it is false.

**EXERCISES SECTION 1.5.2**

**Exercise 1.23:** Write a program that reads several transactions and counts how many transactions occur for each ISBN.

**Exercise 1.24:** Test the previous program by giving multiple transactions representing multiple ISBNs. The records for each ISBN should be grouped together.

## 1.6 The Bookstore Program

We are now ready to solve our original bookstore problem. We need to read a file of sales transactions and produce a report that shows, for each book, the total number of copies sold, the total revenue, and the average sales price. We'll assume that all the transactions for each ISBN are grouped together in the input.

Our program will combine the data for each ISBN in a variable named `total`. We'll use a second variable named `trans` to hold each transaction we read. If `trans` and `total` refer to the same ISBN, we'll update `total`. Otherwise we'll print `total` and reset it using the transaction we just read:

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    // read the first transaction into total to ensure that there are data to process
    if (Sales_item total; std::cin >> total) {
        // if we are here there was at least one transaction in the input
        // read and process the remaining transactions
        for (Sales_item trans; std::cin >> trans; ) {
            // if we're still processing the same book
            if (total.isbn() == trans.isbn())
                total += trans; // update the running total
            else {
                // print results for the previous book
                std::cout << total << std::endl;
                total = trans;  // total now refers to the next book
            }
        }
        std::cout << total << std::endl; // print the last transaction
    }
    return 0;
}
```

This program is the most complicated one we've seen so far, but it uses only facilities that we have already seen.

As usual, we begin by including the headers that we use: `iostream` from the library and our own `Sales_item.h`. Inside `main` we use an `if` statement to read the first transation. That `if` defines the variable `total`, which we'll use to sum the data for a given ISBN. The condition reads the first transaction into `total` and tests whether the read was successful. If the read fails, then there are no records to process, so we fall through to the outermost close curly.

Assuming that we have successfully read a record, we execute the block following the outermost `if`. That block starts with a `for` statement, which will read all the remaining records. As in our earlier programs, the `for` header defines a variable to hold the data as we read them and its condition reads a value from the standard input. In this case, we read a `Sales_item` object into `trans`. As long as the read succeeds, we execute the body of the `while`.

The body of the `for` is a block that contains a single `if` statement. The `if` checks whether the ISBNs are equal. If so, we use the compound assignment operator to add `trans` to `total`. If the ISBNs are not equal, we print the value stored in `total` and reset `total` by assigning `trans` to it. After executing the `if`, we return to the condition in the `for`, reading the next transaction, and so on until we run out of records.

When the `for` terminates, `total` contains the data for the last ISBN in the file. We write the data for the last ISBN in the last statement of the block that concludes the outermost `if` statement.

## EXERCISES SECTION 1.6

**Exercise 1.25:** Using the `Sales_item.h` header from the Web site, compile and execute the bookstore program presented in this section.

# Chapter Summary

This chapter introduced enough of C++ to let you compile and execute simple C++ programs. We saw how to define a `main` function, which is the function that the operating system calls to execute our program. We also saw how to define variables, how to do input and output, and how to write `if`, `for`, and `while` statements. The chapter closed by introducing the most fundamental facility in C++: the class. In this chapter, we saw how to create and use objects of a class that someone else has defined. Later chapters will show how to define our own classes.

# Defined Terms

**argument** Value passed to a function.

**assignment** Obliterates an object's current value and replaces that value by a new one.

**block** Sequence of zero or more statements enclosed in curly braces.

**buffer** A region of storage used to hold data. IO facilities often store input (or output) in a buffer and read or write the buffer independently from actions in the program.

**built-in type** Type, such as `int`, defined by the language.

**cerr** `ostream` object tied to the standard error, which often writes to the same device as the standard output. By default, writes to `cerr` are not buffered. Usually used for error messages or other output that is not part of the normal logic of the program.

**character string literal** Another term for string literal.

**cin** `istream` object used to read from the standard input.

**class** Facility for defining our own data structures together with associated operations. The class is one of the most fundamental features in C++. Library types, such as `istream` and `ostream`, are classes.

**class type** A type defined by a class. The name of the type is the class name.

**clog** `ostream` object tied to the standard error. By default, writes to `clog` are buffered. Usually used to report information about program execution to a log file.

**comments** Program text that is ignored by the compiler. C++ has two kinds of comments: single-line and paired. Single-line comments start with a `//`. Everything from the `//` to the end of the line is a comment. Paired comments begin with a `/*` and include all text up to the next `*/`.

**condition** An expression that is evaluated as true or false. A value of zero is false; any other value is considered true.

**cout** `ostream` object used to write to the standard output. Ordinarily used to write the output of a program.

**curly brace** Curly braces delimit blocks. An open curly (`{`) starts a block; a close curly (`}`) ends one.

**data structure** A logical grouping of data and operations on that data.

**edit-compile-debug** The process of getting a program to execute properly.

**end-of-file** System-specific marker that indicates that there is no more input in a file.

**endl** Manipulator that ends the current line of output and flushes the output buffer.

**expression** The smallest unit of computation. An expression consists of one or more operands and usually one or more operators. An expression is evaluated to produce a result. For example, assuming that `i` and j are ints, then i + j is an expression and yields the sum of the two int values.

**for statement** Iteration statement that executes an initialization statement before the first iteration. On each iteration the loop tests a condition, executes its body, and then executes an expression.

**function** Named unit of computation.

**function body** Block that defines the actions performed by a function.

**function name** Name by which a function is known and can be called.

**header** Mechanism whereby the definitions of a class or other names are made available to multiple programs. A program uses a header through a #include directive.

**if statement** Conditional execution based on the value of a specified condition. If the condition is true, the if body is executed. If not, the else body is executed if there is one.

**initialize** Give an object a value at the same time that it is created.

**iostream** Header that provides the library types for stream-oriented input and output.

**istream** Library type providing stream-oriented input.

**library type** Type, such as istream, defined by the standard library.

**main** Function called by the operating system to execute a C++ program. Each program must have one and only one function named main.

**manipulator** Object, such as std::endl, that when read or written "manipulates" the stream itself.

**member function** Operation defined by a class. A member function (typically) is called to operate on a specific object.

**method** Synonym for member function.

**namespace** Mechanism for putting names defined by a library into a single place. Namespaces help avoid inadvertent name clashes. The names defined by the C++ library are in the namespace std.

**ostream** Library type providing stream-oriented output.

**parameter list** Part of the definition of a function. Possibly empty list that specifies what arguments can be used to call the function.

**return type** Type of the value returned by a function.

**source file** Term used to describe a file that contains all or part of a C++ program.

**standard error** Output stream used for error reporting. Ordinarily, the standard output and the standard error are tied to the window in which the program is executed.

**standard input** Input stream usually associated with the window in which the program executes.

**standard library** Collection of types and functions that every C++ compiler must support. The library provides the types that support IO. C++ programmers tend to talk about "the library," meaning the entire standard library. They also tend to refer to particular parts of the library by referring to a library type, such as the "iostream library," meaning the part of the standard library that defines the IO classes.

**standard output** Output stream usually associated with the window in which the program executes.

**statement** A part of a program that specifies an action to take place when the program is executed. An expression followed by a semicolon is a statement; other kinds of statements include blocks and if, for, and while statements, all of which contain other statements within themselves.

**std** Name of the namespace used by the standard library. `std::cout` indicates that we're using the name `cout` defined in the `std` namespace.

**string literal** Sequence of zero or more characters enclosed in double quotes (`"a string literal"`).

**uninitialized variable** Variable that is not given an initial value. Variables of class type for which no initial value is specified are initialized as specified by the class definition. Variables of built-in type defined inside a function are uninitialized unless explicitly initialized. It is an error to try to use the value of an uninitialized variable. *Uninitialized variables are a rich source of bugs*.

**variable** A named object.

**while statement** Iteration statement that provides iterative execution so long as a specified condition is true.

**() operator** Call operator. A pair of parentheses "`()`" following a function name. The operator causes a function to be invoked. Arguments to the function may be passed inside the parentheses.

**++ operator** Increment operator. Adds `1` to an operand; `++i` is equivalent to `i = i + 1`.

**+= operator** Compound assignment operator that adds the right-hand operand to the left and stores the result in the left-hand operand; `a += b` is equivalent to `a = a + b`.

**. operator** Dot operator. Left-hand operand must be an object of class type and the right-hand operand must be the name of a member of that object. The operator yields the named member of the given object.

**:: operator** Scope operator. Among other uses, the scope operator is used to access names in a namespace. For example, `std::cout` denotes the name `cout` from the namespace `std`.

**= operator** Assignment operator. Assigns the value of the right-hand operand to the object denoted by the left-hand operand.

**-- operator** Decrement operator. Subtracts `1` from the operand; `--i` is equivalent to `i = i - 1`.

**<< operator** Output operator. Writes the right-hand operand to the output stream indicated by the left-hand operand: `cout << "hi"` writes `hi` to the standard output. Output operations can be chained together: `cout << "hi" << "bye"` writes `hibye`.

**>> operator** Input operator. Reads from the input stream specified by the left-hand operand into the right-hand operand: `cin >> i` reads the next value on the standard input into `i`. Input operations can be chained together: `cin >> i >> j` reads first into `i` and then into `j`.

**#include** Directive that makes code in a header available to a program.

**== operator** The equality operator. Tests whether the left-hand operand is equal to the right-hand operand.

**!= operator** The inequality operator. Tests whether the left-hand operand is not equal to the right-hand operand.

**<= operator** The less-than-or-equal operator. Tests whether the left-hand operand is less than or equal to the right-hand operand.

**< operator** The less-than operator. Tests whether the left-hand operand is less than the right-hand operand.

**>= operator** Greater-than-or-equal operator. Tests whether the left-hand operand is greater than or equal to the right-hand operand.

**> operator** Greater-than operator. Tests whether the left-hand operand is greater than the right-hand operand.

# PART I

## THE BASICS

### CONTENTS



Every widely used programming language provides a common set of features, which differ in detail from one language to another. Understanding the details of how a language provides these features is the first step toward understanding the language. Among the most fundamental of these common features are

- Built-in types such as integers, characters, and so forth
- Variables, which let us give names to the objects we use
- Expressions and statements to manipulate values of these types
- Control structures, such as `if` or `while`, that allow us to conditionally or repeatedly execute a set of actions
- Functions that let us define callable units of computation

Most programming languages supplement these basic features in two ways: They let programmers extend the language by defining their own types, and they provide library routines that define useful functions and types not otherwise built into the language.

In C++, as in most programming languages, the type of an object determines what operations can be performed on it. Whether a particular expression is legal depends on the type of the objects in that expression. Some languages, such as Smalltalk and Python, check types at run time. In contrast, C++ is a statically typed language; type checking is done at compile time. As a consequence, the compiler must know the type of every name used in the program.

C++ provides a set of built-in types, operators to manipulate those types, and a small set of statements for program flow control. These elements form an alphabet from which we can write large, complicated, real-world systems. At this basic level, C++ is a simple language. Its expressive power arises from its support for mechanisms that allow the programmer to define new data structures. Using these facilities, programmers can shape the language to their own purposes without the language designers having to anticipate the programmers' needs.

Perhaps the most important feature in C++ is the class, which lets programmers define their own types. In C++ such types are sometimes called "class types" to distinguish them from the types that are built into the language. Some languages let programmers define types that specify only what data make up the type. Others, like C++, allow programmers to define types that include operations as well as data. A major design goal of C++ is to let programmers define their own types that are as easy to use as the built-in types. The Standard C++ library uses these features to implement a rich library of class types and associated functions.

The first step in mastering C++—learning the basics of the language and library—is the topic of Part I. Chapter 2 covers the built-in types and looks briefly at the mechanisms for defining our own new types. Chapter 3 introduces two of the most fundamental library types: `string` and `vector`. That chapter also covers arrays, which are a lower-level data structure built into C++ and many other languages. Chapters 4 through 6 cover expressions, statements, and functions. This part concludes in Chapter 7, which describes the basics of building our own class types. As we'll see, defining our own types brings together all that we've learned before, because writing a class entails using the facilities covered in Part I.

# CHAPTER 2

## VARIABLES AND BASIC TYPES

### CONTENTS



Types are fundamental to any program: They tell us what our data mean and what operations we can perform on those data.

C++ has extensive support for types. The language defines several primitive types (characters, integers, floating-point numbers, etc.) and provides mechanisms that let us define our own data types. The library uses these mechanisms to define more complicated types such as variable-length character strings, vectors, and so on. This chapter covers the built-in types and begins our coverage of how C++ supports more complicated types.

*Types determine* the meaning of the data and operations in our programs. The meaning of even as simple an expression as

```
i + j;
```

depends on the types of `i` and `j`. If `i` and `j` are integers, this statement has the ordinary, arithmetic meaning of `+`. However, if `i` and `j` are `Sales_item` objects (§ 1.5.1, p. 20), this statement adds the components of these two objects.

## 2.1 Primitive Built-in Types

C++ defines a set of primitive types that include the **arithmetic types** and a special type named **`void`**. The arithmetic types represent characters, integers, boolean values, and floating-point numbers. The `void` type has no associated values and can be used in only a few circumstances, most commonly as the return type for functions that do not return a value.

### 2.1.1 Arithmetic Types

The arithmetic types are divided into two categories: **integral types** (which include character and boolean types) and floating-point types.

The size of—that is, the number of bits in—the arithmetic types varies across machines. The standard guarantees minimum sizes as listed in Table 2.1. However, compilers are allowed to use larger sizes for these types. Because the number of bits varies, the largest (or smallest) value that a type can represent also varies.

**Table 2.1: C++: Arithmetic Types**

| Type | Meaning | Minimum Size |
|---|---|---|
| `bool` | boolean | NA |
| `char` | character | 8 bits |
| `wchar_t` | wide character | 16 bits |
| `char16_t` | Unicode character | 16 bits |
| `char32_t` | Unicode character | 32 bits |
| `short` | short integer | 16 bits |
| `int` | integer | 16 bits |
| `long` | long integer | 32 bits |
| `long long` | long integer | 64 bits |
| `float` | single-precision floating-point | 6 significant digits |
| `double` | double-precision floating-point | 10 significant digits |
| `long double` | extended-precision floating-point | 10 significant digits |

The `bool` type represents the truth values `true` and `false`.

There are several character types, most of which exist to support internationalization. The basic character type is `char`. A `char` is guaranteed to be big enough to hold numeric values corresponding to the characters in the machine's basic character set. That is, a `char` is no smaller than a single machine byte.

The remaining character types—`wchar_t`, `char16_t`, and `char32_t`—are used for extended character sets. The `wchar_t` type is guaranteed to be large enough to hold any character in the machine's largest extended character set. The types `char16_t` and `char32_t` are intended for Unicode characters. (Unicode is a standard for representing characters used in essentially any natural language.)

The remaining integral types represent integer values of (potentially) different sizes. The language guarantees that an `int` will be at least as large as `short`, a `long` at least as large as an `int`, and `long long` at least as large as `long`.

### Machine-Level Representation of the Built-in Types

**Computers store data as a sequence of bits, each holding a 0 or 1, such as**

```
00011011011100010110010000111011 ...
```

**Most computers deal with memory as chunks of bits, with the size of each chunk being a power of 2. The smallest chunk of addressable memory is referred to as a "byte." The basic unit of storage, usually a small number of bytes, is referred to as a "word." In C++ a byte has at least as many bits as are needed to hold a character in the machine's basic character set. On most machines a byte contains 8 bits and a word is either 32 or 64 bits, that is, 4 or 8 bytes.**

**Most computers associate a number (called an "address") with each byte. On a machine with 8-bit bytes and 32-bit words, we might view a word of memory as**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 736424 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 |
| 736425 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 736426 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| 736427 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |

**Here, the byte's address is on the left, with the 8 bits of the byte following the address.**

**We can use an address to refer to any of several variously sized collections of bits starting at that address. It is possible to speak of the word at address 736424 or the byte at address 736427. To give meaning to memory at a given address, we must know the type of the value stored there. The type determines how many bits are used and how to interpret those bits.**

**If the object at location 736424 is an `unsigned char` on a machine using the ISO-Latin-1 character set, then the byte at that address represents a semicolon. Alternatively, if the object at location 736424 has type `float` and if `floats` on this machine are stored in 32 bits, then we know that the object at that address spans the entire word. The value of that `float` depends on the details of how the machine stores floating-point numbers.**

The floating-point types represent single-, double-, and extended-precision values. The standard specifies a minimum number of significant digits for each of these types. Most compilers provide more precision than the specified minimum. Typically, `floats` are represented in one word (32 bits), `doubles` in two words (64 bits), and `long doubles` in either three or four words (96 or 128 bits). The `float` and `double` types typically yield about 7 and 16 significant digits, respectively. The type `long double` is often used as a way to accommodate special-purpose floating-point hardware; its precision is more likely to vary across machines.

### Signed and Unsigned Types

Except for `bool` and the extended character types, each integral type may be **signed** or **unsigned**. A signed type represents negative or positive numbers (including zero); an unsigned type represents only values greater than or equal to zero. The types `short`, `int`, `long`, and `long long` are all signed. We obtain the corresponding unsigned type by adding `unsigned` to the type, such as `unsigned long`. The type `unsigned int` may be abbreviated as `unsigned`.

Unlike the other integer types, there are three distinct basic character types: `char`, `signed char`, and `unsigned char`. In particular, `char` is not the same type as `signed char`. Nevertheless, although there are three character types, there are only two representations: signed and unsigned. The (plain) `char` type uses one of these representations. Which of the other two character representations is equivalent to `char` depends on the compiler.

In an unsigned type, all the bits represent the value. For example, an 8-bit `unsigned char` can hold the values from 0 through 255 inclusive.

The standard does not define how signed types are represented, but does specify that the range should be as evenly divided as possible between positive and negative values. Hence, an 8-bit `signed char` is guaranteed to be able to hold values from –127 through 127; most modern machines use representations that allow values from –128 through 127.

**EXERCISES SECTION 2.1.1**

**Exercise 2.1:** What are the differences between `int`, `long`, `long long`, and `short`? Between an unsigned and a signed type? Between a `float` and a `double`?

## 2.1.2 Type Conversions

The type of an object defines the data that an object might contain and what operations that object can perform. Among the operations that many types support is the ability to **convert** objects of the given type to other, related types.

Type conversions happen automatically when we use an object of one type where an object of another type is expected. We'll have more to say about conversions in § 4.11 (p. 169), but for now it is useful to understand what happens when we assign a value of one arithmetic type to an object of another arithmetic type:

```
bool b = 42;            // b is true
int i = b;              // i has value 1
i = 3.14;               // i has value 3
double pi = i;          // pi has value 3.0
unsigned char c = -1;   // assuming 8-bit chars, c has value 255
signed char c2 = 256;   // assuming 8-bit chars, the value of c2 is undefined
```

**ADVICE: DECIDING WHICH TYPE TO USE**

**C++, like C, is designed to let programs get close to the hardware when necessary. The arithmetic types are defined to cater to the peculiarities of various kinds of hardware. Accordingly, the number of arithmetic types in C++ can be bewildering. Most programmers can (and should) ignore these complexities by restricting the types they use. A few rules of thumb can be useful in deciding which type to use:**

- **Use an unsigned type when you know that the values cannot be negative.**
- **Use `int` for integer arithmetic. `short` is usually too small and, in practice, `long` often has the same size as `int`. If your data values are larger than the minimum guaranteed size of an `int`, then use `long long`.**
- **Do not use plain `char` or `bool` in arithmetic expressions. Use them *only* to hold characters or truth values. Computations using `char` are especially problematic because `char` is `signed` on some machines and `unsigned` on others. If you need a tiny integer, explicitly specify either `signed char` or `unsigned char`.**
- **Use `double` for floating-point computations; `float` usually does not have enough precision, and the cost of double-precision calculations versus single-precision is negligible. In fact, on some machines, double-precision operations are faster than single. The precision offered by `long double` usually is unnecessary and often entails considerable run-time cost.**

What happens depends on the range of the values that the types permit:

- When we assign one of the non`bool` arithmetic types to a `bool` object, the result is `false` if the value is `0` and `true` otherwise.
- When we assign a `bool` to one of the other arithmetic types, the resulting value is `1` if the `bool` is `true` and `0` if the `bool` is `false`.
- When we assign a floating-point value to an object of integral type, the value is truncated. The value that is stored is the part before the decimal point.
- When we assign an integral value to an object of floating-point type, the fractional part is zero. Precision may be lost if the integer has more bits than the floating-point object can accommodate.
- If we assign an out-of-range value to an object of unsigned type, the value "wraps around:" The result is the remainder of the value modulo the number of values the target type can hold. For example, an 8-bit `unsigned char` can hold values from 0 through 255, inclusive. If we assign a value outside this range, the compiler assigns the remainder of that value modulo 256. Therefore, assigning –1 to an 8-bit `unsigned char` gives that object the value 255.
- If we assign an out-of-range value to an object of signed type, the result is **undefined**. The program might appear to work, it might crash, or it might produce garbage values.

The compiler applies these same type conversions when we use a value of one arithmetic type where a value of another arithmetic type is expected. For example,

> **ADVICE: AVOID UNDEFINED AND IMPLEMENTATION-DEFINED BEHAVIOR**
>
> **Undefined behavior results from errors that the compiler is not required (and sometimes is not able) to detect. Even if the code compiles, a program that executes an undefined expression is in error.**
>
> **Unfortunately, programs that contain undefined behavior can appear to execute correctly in some circumstances and/or on some compilers. There is no guarantee that the same program, compiled under a different compiler or even a subsequent release of the same compiler, will continue to run correctly. Nor is there any guarantee that what works with one set of inputs will work with another.**
>
> **Similarly, programs usually should avoid implementation-defined behavior, such as assuming that the size of an `int` is a fixed and known value. Such programs are said to be *nonportable*. When the program is moved to another machine, code that relied on implementation-defined behavior may fail. Tracking down these sorts of problems in previously working programs is, mildly put, unpleasant.**

when we use a non`bool` value as a condition (§ 1.4.1, p. 12), the arithmetic value is converted to `bool` in the same way that it would be converted if we had assigned that arithmetic value to a `bool` variable:

```
int i = 42;
if (i) //  condition will evaluate as true
    i = 0;
```

If the value to be converted is `0`, then the condition is `false`; all other (nonzero) values yield `true`. If we use a `bool` in an arithmetic expression, the `bool` value `false` converts to `0`; `true` converts to `1`.

## Expressions Involving Unsigned Types

Although we are unlikely to assign a negative value to an object of unsigned type directly, we can (all too easily) write code that does so implicitly. For example, if we use both `unsigned` and `int` values in an arithmetic expression, the `int` value ordinarily is converted to `unsigned`. Converting an `int` to `unsigned` executes the same way as if we assigned the `int` to an `unsigned`:

```
unsigned u = 10;
int i = -42;
std::cout << i + i << std::endl;  //  prints -84
std::cout << u + i << std::endl;  //  if 32-bit ints, prints 4294967264
```

In the first expression, we add two (negative) `int` values and obtain the expected result. In the second expression, the `int` value `-42` is converted to `unsigned` before the addition is done. Converting a negative value to `unsigned` "wraps around" as described above. On our machine `-42` converted to `unsigned` yields `4294967254`.

Any time we subtract a value from an unsigned, we must be sure that the result cannot be negative:

```
unsigned u1 = 42, u2 = 10;
```

```
std::cout << u1 - u2 << std::endl; // ok: result is 32
std::cout << u2 - u1 << std::endl; // ok: but the result will wrap around
```

The fact that an unsigned cannot be less than zero also affects how we write loops. For example, in the exercises to § 1.4.1 (p. 13), you were to write a loop that used the decrement operator to print the numbers from `10` down to `0`. The loop you wrote probably looked something like

```
for (int i = 10; i >= 0; --i)
    std::cout << i << std::endl;
```

We might think we could rewrite this loop using an `unsigned`. After all, we don't plan to print negative numbers. However, this simple change in type means that our loop will never terminate:

```
// WRONG: u can never be less than 0; the condition will always succeed
for (unsigned u = 10; u >= 0; --u)
    std::cout << u << std::endl;
```

Consider what happens when `u` is `0`. On that iteration, we'll print `0` and then execute the expression in the `for` loop. That expression, `--u`, subtracts `1` from `u`. That result, `-1`, doesn't fit in an `unsigned` value. Like any other out-of-range value, `-1` will be transformed to an `unsigned` value. Assuming 32-bit `int`s, the result of `--u`, when `u` is `0`, is `4294967295`.

One way to write this loop is to use a `while` instead of a `for`. Using a `while` lets us decrement before (rather than after) printing our value:

```
unsigned u = 11; // start the loop one past the first element we want to print
while (u > 0) {
    --u;        // decrement first, so that the last iteration will print 0
    std::cout << u << std::endl;
}
```

This loop starts by decrementing the value of the loop control variable. On the last iteration, `u` will be `1` on entry to the loop. We'll decrement that value, meaning that we'll print `0` on this iteration. When we next test `u` in the `while` condition, its value will be `0` and the loop will exit. Because we start by decrementing `u`, we have to initialize `u` to a value one greater than the first value we want to print. Hence, we initialize `u` to `11`, so that the first value printed is `10`.

### CAUTION: DON'T MIX SIGNED AND UNSIGNED TYPES

**Expressions that mix signed and unsigned values can yield surprising results when the signed value is negative. It is essential to remember that signed values are automatically converted to unsigned. For example, in an expression like `a * b`, if `a` is `-1` and `b` is `1`, then if both `a` and `b` are `int`s, the value is, as expected `-1`. However, if `a` is `int` and `b` is an `unsigned`, then the value of this expression depends on how many bits an `int` has on the particular machine. On our machine, this expression yields `4294967295`.**

**EXERCISES SECTION 2.1.2**

**Exercise 2.2:** What output will the following code produce?

```
unsigned u = 10, u2 = 42;
std::cout << u2 - u << std::endl;
std::cout << u - u2 << std::endl;

int i = 10, i2 = 42;
std::cout << i2 - i << std::endl;
std::cout << i - i2 << std::endl;

std::cout << i - u << std::endl;
std::cout << u - i << std::endl;
```

**Exercise 2.3:** Write a program to check whether your predictions were correct. If not, study this section until you understand what the problem is.

### 2.1.3 Literals

A value, such as `42`, is known as a **literal** because its value is self-evident. As with variables, every literal has a type. The form and value of a literal determine its type.

#### Integer and Floating-Point Literals

We can write an integer literal using decimal, octal, or hexadecimal notation and New in up-to-date compilers we can also use binary notation. Integer literals that begin with `0` (zero) are interpreted as octal. Literals that begin with `0b` or `0B` are interpreted as binary; those that begin with `0x` or `0X` are interpreted as hexadecimal. For example, we can write the value `42` in any of the following four ways:

```
 42 /* decimal */   0b101010 /* binary */
052 /* octal */     0x2a     /* hexadecimal */
```

The type of an integer literal depends on its value and notation. By default, decimal literals are signed whereas binary, octal and hexadecimal literals can be either signed or unsigned types. A decimal literal has the smallest type of `int`, `long`, or `long long` (i.e., the first type in this list) in which the literal's value fits. Binary, octal and hexadecimal literals have the first type in the following list that can contain the literal's value: `int`, `unsigned int`, `long`, `unsigned long`, `long long`, or `unsigned long long`. It is an error to use a literal that is too large to fit in the largest related type. There are no literals of type `short`. We'll see in Table 2.2 (p. 40) that we can override these defaults by using a suffix.

Although integer literals may be stored in signed types, the value of a decimal literal is never a negative number. If we write what appears to be a negative decimal literal, for example, `-42`, the minus sign is *not* part of the literal. The minus sign is an operator that negates the value of its (literal) operand.

Floating-point literals may be specified in decimal or (in new compilers) hex- New adecimal. Decimal floating-point literals may either use a decimal point or be written in scientific notation in which the exponent is preceded by either `E` or `e`. Hexadecimal floating-point literals are specified using base-2 scientific notation: The base part is in hexadecimal and the exponent is scaled as a power of two. Hexadecimal floating-point literals start with `0x` or `0X`—to indicate that the base part is in hexadecimal—and the (power of 2) exponent follows a `p` or `P`. The decimal point is optional in hexadecimal floating-point literals and in decimal literals that use scientific notation:

```
0.  .000    0e0          0x0p+0
3.14159     3.14159E0    0xc.90fcf80dc337p-2
```

By default, floating-point literals have type `double`. We can override the default using a suffix from Table 2.2 (overleaf).

In the past, numeric literals could contain only digits (including the hexadecimal digits represented by characters). Under current compilers we can use a single quotation mark (`'`) between any two digits (including in an exponent) to make large literals easier to read: New

```
1000000000 /* one trillion */     1'000'000'000 /* same value */
0xc.90fcf80dc337p-2     /* value of pi */
0xc.90f'cf8'0dc'337p-2 /* same value */
```

## Character and Character String Literals

A character enclosed within single quotes is a literal of type `char`. A string literal is zero or more characters enclosed in double quotation marks:

```
'a'   // character literal
"Hello World!"  // string literal
```

The type of a string literal is ***array*** of constant `char`s, a type we'll discuss in § 3.5.4 (p. 128). The compiler appends a null character to every string literal. Thus, the actual size of a string literal is one more than its apparent size. For example, the literal `'A'` represents the single character `A`, whereas the string literal `"A"` represents an array of two characters, the letter `A` and the null character.

A string literal may not contain a newline. However, two string literals that appear adjacent to one another and that are separated only by spaces, tabs, or newlines are concatenated into a single literal. We use this form of literal when we need to write a literal that does not fit comfortably on a single line:

```
// multiline string literal
std::cout << "a really, really long string literal "
             "that spans two lines" << std::endl;
```

As we'll see in § 2.2.1 (p. 42) the C++ library defines its own `string` type. It is important to note that what we refer to as "string literals" are not the same type as the library `string`s. This distinction will make more sense once we cover `string`s more fully in Chapter 3. In particular, in § 3.2.2 (p. 92) we'll see how to define literals that are of `string` type.

### Specifying the Type of a Literal

We can override the default type of an integer, floating-point, character, or character string literal by supplying a suffix or prefix as listed in Table 2.2.

```
L'a'      // wide character literal, type is wchar_t
u8"!"     // utf-8 character literal (new in C++17)
u8"hi!"   // utf-8 character string literal (each character is in utf-8)
42ULL     // unsigned integer literal, type is unsigned long long
1E-3F     // single-precision floating-point literal, type is float
3.14159L  // extended-precision floating-point literal, type is long double
```

Best Practices

When you write a `long` literal, use the uppercase `L`; the lowercase letter `l` is too easily mistaken for the digit 1.

**Table 2.2: Specifying the Type of a Literal**

**Character and Character String Literals**

| Prefix | Meaning | Type |
|---|---|---|
| `u` | Unicode 16 character | `char16_t` |
| `U` | Unicode 32 character | `char32_t` |
| `L` | wide character | `wchar_t` |
| `u8` | utf-8 character | `char` |

**Library `string` literals see § 3.2.2 (p. 92)**

| Integer Literals | | Floating-Point Literals | |
|---|---|---|---|
| **Suffix** | **Minimum Type** | **Suffix** | **Type** |
| `u` or `U` | `unsigned` | `f` or `F` | `float` |
| `l` or `L` | `long` | `l` or `L` | `long double` |
| `ll` or `LL` | `long long` | | |

We can independently specify the signedness and size of an integral literal. If the suffix contains a `U`, then the literal has an unsigned type, so a decimal, octal, or hexadecimal literal with a `U` suffix has the first type of `unsigned int`, `unsigned long`, or `unsigned long long` in which the literal's value fits. If the suffix contains an `L`, then the literal's type will be at least `long`; if the suffix contains `LL`, then the literal's type will be either `long long` or `unsigned long long`. We can combine `U` with either `L` or `LL`. For example, a literal with a suffix of `UL` will be either `unsigned long` or `unsigned long long`, depending on whether its value fits in `unsigned long`.

### Escape Sequences

Some characters, such as backspace or control characters, have no visible image. Such characters are **nonprintable**. Other characters (single and double quotation marks, question mark, and backslash) have restrictions on the contexts in which we can use them. The language defines several **escape sequences** to use when we need one of these special characters. An escape sequence begins with a backslash:

| | | | | | |
|---|---|---|---|---|---|
| newline | `\n` | horizontal tab | `\t` | alert (bell) | `\a` |
| vertical tab | `\v` | backspace | `\b` | double quote | `\"` |
| backslash | `\\` | question mark | `\?` | single quote | `\'` |
| carriage return | `\r` | formfeed | `\f` | | |

We use an escape sequence as if it were a single character:

```
std::cout << '\n';        // prints a newline
std::cout << "\tHi!\n";   // prints a tab followd by "Hi!" and a newline
```

We can also write a generalized escape sequence, which is `\x` followed by one or more hexadecimal digits or a `\` followed by one, two, or three octal digits. The value represents the numerical value of the character. Some examples (assuming the Latin-1 character set):

```
\7 (bell)    \12 (newline)     \40 (blank)
\0 (null)    \115 ('M')        \x4d ('M')
```

As with an escape sequence defined by the language, we use these escape sequences as we would any other character:

```
std::cout << "Hi \x4dO\115!\n"; // prints Hi MOM! followed by a newline
std::cout << '\115' << '\n';    // prints M followed by a newline
```

Note that if a `\` is followed by more than three octal digits, only the first three are associated with the `\`. For example, `"\1234"` represents two characters: the character represented by the octal value 123 and the character 4. In contrast, `\x` uses up all the hex digits following it; `"\x1234"` represents a single, 16-bit character composed from the bits corresponding to these four hexadecimal digits. Because most machines have 8-bit `char`s, such values are unlikely to be useful. Ordinarily, hexadecimal characters with more than 8 bits are used with extended characters sets using one of the prefixes from Table 2.2.

### Other Literals

There are three more literals: The words `true` and `false` are literals of type `bool` and `nullptr` is a literal that we'll describe in § 2.3.2 (p. 54):

## 2.2 Variables

A ***variable*** provides us with named storage that our programs can manipulate. Each variable in C++ has a type. The type determines the size and layout of the variable's memory, the range of values that can be stored within that memory, and the set of operations that can be applied to the variable. C++ programmers tend to refer to variables as "variables" or "objects" interchangeably.

### 2.2.1 Variable Definitions

A simple variable definition consists of a **type specifier**, followed by a list of one or more variable names separated by commas, and ends with a semicolon. Each name

### Exercises Section 2.1.3

**Exercise 2.4:** Determine the type of each of the following literals. Explain the differences among the literals in each of the four examples:

(a) `'a', L'a', "a", L"a"` (c) `3.14, 3.14f, 3.14L`
(b) `10, 10u, 10L, 10uL, 012, 0xC` (d) `10, 10u, 10., 10e-2`

**Exercise 2.5:** What, if any, are the differences between the following definitions:

```
int month = 9, day = 7;
int month = 09, day = 07;
```

**Exercise 2.6:** What values do these literals represent? What type does each have?

(a) `"Who goes with F\145rgus?\012"`
(b) `3.14e1L` (c) `1024f` (d) `3.14L`

**Exercise 2.7:** Using escape sequences, write a program to print `2M` followed by a newline. Modify the program to print `2`, then a tab, then an `M`, followed by a newline.

in the list has the type defined by the type specifier. A definition may (optionally) provide an initial value for one or more of the names it defines:

```
// sum, value, and units_sold have type int
// sum and units_sold have initial value 42
int sum = 42, value, units_sold = 42;
Sales_item item;      // item has type Sales_item (see § 1.5.1 (p. 20))
// string is a library type, representing a variable-length sequence of characters
std::string book("0-201-78345-X"); // book initialized from string literal
```

The definition of `book` uses the `std::string` library type. Like `iostream` (§ 1.2, p. 7), `string` is defined in namespace `std`. We'll have more to say about the `string` type in Chapter 3. For now, what's useful to know is that a `string` is a type that represents a variable-length sequence of characters. The `string` library gives us several ways to initialize `string` objects. One of these ways is as a copy of a string literal (§ 2.1.3, p. 39). Thus, `book` is initialized to hold the characters `0-201-78345-X`.

## Initializers

An object that is **initialized** gets the specified value at the moment it is created. The values used to initialize a variable can be arbitrarily complicated expressions. When a definition defines two or more variables, the name of each object becomes visible immediately. Thus, it is possible to initialize a variable in a way that depends on the value of one defined earlier in the same definition.

```
// ok: price is defined and initialized before it is used to initialize discount
double price = 109.99, discount = price * 0.16;
// ok: call applyDiscount and use the return value to initialize salePrice
double salePrice = applyDiscount(price, discount);
```

**TERMINOLOGY: WHAT IS AN OBJECT?**

**C++ programmers tend to be cavalier in their use of the term *object*. Most generally, an object is a region of memory that can contain data and has a type.**

**Some use the term *object* only to refer to variables or values of class types. Others distinguish between named and unnamed objects, using the term *variable* to refer to named objects. Still others distinguish between objects and values, using the term *object* for data that can be changed by the program and the term *value* for data that are read-only.**

**In this book, we'll follow the more general usage that an object is a region of memory that has a type. We will freely use the term *object* regardless of whether the object has built-in or class type, is named or unnamed, or can be read or written.**

Initialization in C++ is a surprisingly complicated topic to which we will return again and again. Many programmers are confused by the use of the = symbol to initialize a variable. It is tempting to think of initialization as a form of assignment, but initialization and assignment are different operations in C++. This concept is particularly confusing because in many languages the distinction is irrelevant and can be ignored. Moreover, even in C++ the distinction often doesn't matter. Nonetheless, it is a crucial concept and one we will reiterate throughout the text.

WARNING

Initialization is not assignment. Initialization happens when a variable is given a value when it is created. Assignment obliterates an object's current value and replaces that value with a new one.

## Initialization Forms

One reason that initialization is complicated is that the language defines several different forms of initialization. For example, we can use any of the following four different ways to define an `int` variable named `units_sold` and initialize it to `0`:

```
int units_sold = 0;
int units_sold(0);
int units_sold = {0};
int units_sold{0};
```

For reasons we'll cover in § 3.2.1 (p. 86), initializations that use an = are referred to as **copy initialization**; those without are **direct initialization**. As we'll see in § 3.3.1 (p. 102), the last two initializations are referred to as **list initialization**.

These forms of initialization differ from each other in ways that are often subtle. One such subtlety is that when we list-initialize a variable of built-in type the compiler will not let us use an initializer that might lead to the loss of information:

```
double ld = 3.1415926536;
int a{ld}, b = {ld}; // error: narrowing conversion required
int c(ld), d = ld;   // ok: but value will be truncated
```

The compiler rejects the initializations of `a` and `b` because using a `long double` to initialize an `int` is likely to lose data. At a minimum, the fractional part of `ld` will be truncated. In addition, the integer part in `ld` might be too large to fit in an `int`.

As presented here, the distinction might seem trivial—after all, we'd be unlikely to directly initialize an `int` from a `long double`. However, as we'll see in Chapter 16, such initializations might happen unintentionally. We'll say more about these forms of initialization in § 3.2.1 (p. 86) and § 3.3.1 (p. 102).

### Default Initialization

When we define a variable without an initializer, the variable is **default initialized**. Such variables are given a "default" value that depends on the type of the variable and may also depend on where the variable is defined.

For example, the default value of an object of built-in type that is defined outside any function body is zero. With one exception, which we cover in § 6.1.1 (p. 217), variables of built-in type defined inside a function are **uninitialized**. The value of such a variable is undefined (§ 2.1.2, p. 36). It is an error to copy or otherwise try to access the value of a variable whose value is undefined.

Each class controls how we initialize objects of that class type. In particular, it is up to the class whether we can define objects of that type without supplying an initializer. If we can omit the initializer, the class determines what value the resulting object will have.

For example, the library `string` class generates an empty string if we do not supply our own intializer:

```
std::string s;          // s implicitly initialized to the empty string
```

Some classes require that every object be explicitly initialized. The compiler will complain if we try to create an object of such a class with no initializer.

> Uninitialized objects of built-in type defined inside a function body have undefined value. Objects of class type that we do not explicitly initialize have a value that is defined by the class.

## 2.2.2 Variable Declarations and Definitions

To allow programs to be written in logical parts, C++ supports what is commonly known as ***separate compilation***. Separate compilation lets us split our programs into several files, each of which can be compiled independently.

When we separate a program into multiple files, we need a way to share code across those files. For example, code that appears in one file may need to use a variable defined in another file. As a concrete example, consider `std::cout` and `std::cin`. These are objects defined somewhere in the standard library, yet our programs can use these objects.

To support separate compilation, C++ distinguishes between declarations and definitions. A **declaration** makes a name known to the program. A **definition** is a declaration that also creates the associated entity. Names must be declared before

**CAUTION: UNINITIALIZED VARIABLES CAUSE RUN-TIME PROBLEMS**

**An uninitialized variable of built-in type has an indeterminate value. Trying to use value of such a variable is an error that is often hard to debug. Moreover, the compiler is not required to detect such errors, although most will warn about at least some uses of uninitialized variables.**

**What happens when we use an uninitialized variable is undefined. Sometimes, we're lucky and our program crashes as soon as we access the object. Once we track down the location of the crash, it is usually easy to see that the variable was not properly initialized. Other times, the program completes but produces erroneous results. Even worse, the results may appear correct on one run of our program but fail on a subsequent run, or fail when we run it on a different machine or use a different compiler. Moreover, adding code to the program in an unrelated location can cause what we thought was a correct program to start producing incorrect results.**

Tip

**We recommend initializing every object of built-in type. It is not always necessary, but it is easier and safer to provide an initializer until you can be certain it is safe to omit the initializer.**

## EXERCISES SECTION 2.2.1

**Exercise 2.8:** Explain the following definitions. For those that are illegal, explain what's wrong and how to correct it.

(a) `std::cin >> int input_value;` (b) `int i = { 3.14 };`
(c) `double salary = wage = 9999.99;` (d) `int i = 3.14;`

**Exercise 2.9:** What are the initial values, if any, of each of the following variables?

```
std::string global_str;
int global_int;
int main()
{
    int local_int;
    std::string local_str;
}
```

we can use them. With some exceptions, names may be declared many times but must be defined exactly once.

As we've seen (§ 2.2.1, p. 41) a variable definition specifies a type and the variable's name along with an optional initializer. A definition is also a declaration.

In order to use a variable that is defined elsewhere, we need to declare but not define that variable. We use the keyword `extern` to indicate that we are declaring a variable that is defined elsewhere. Like a variable definition, a variable declaration specifies the type and name of a variable:

```
extern int i;    // declares but does not define i
int j;           // declares and defines j
```

We can provide an initializer on a variable defined as `extern`, but doing so overrides the `extern`. Every declaration that has an initializer is a definition:

```
extern double pi = 3.1416; // definition
```

It is an error to provide an initializer on an `extern` inside a function.

The distinction between a declaration and a definition may seem obscure at this point but is important. To use a variable in more than one file requires declarations that are separate from the variable's definition. To use the same variable in multiple files, we must define that variable in one—and only one—file. Other files that use that variable must declare—but not define—that variable.

> **Note:** Variables must be defined exactly once but can be declared many times.

### `inline` Variables

We said that there are some exceptions to the rule that names may be declared multiple times but must be defined exactly once. One of those exceptions was introduced by the C++17 standard: **inline variables**:

```
inline unsigned counter = 0;  // declaration and definition
```

Here the keyword `inline` says that we are defining the `int counter` and that this same definition may appear in multiple files. No matter how many times `counter` is defined, only one object is created and all references to `counter` will refer to the same storage.

Each file may contain only one defition for a given `inline` variable. The definition in each file must match exactly—for example, it would be an error to initialize `counter` to different values in different files. The `inline` specifier may not appear on names defined inside a function.

#### Exercises Section 2.2.2

**Exercise 2.10:** Explain whether each of the following is a declaration or a definition:

(a) `extern int ix = 1024;`
(b) `int iy;`
(c) `extern int iz;`

### 2.2.3 Identifiers

***Identifiers*** in C++ can contain letters, digits, and the underscore character. The language imposes no limit on name length. Identifiers must begin with either a letter or an underscore. Identifiers are case-sensitive; upper- and lowercase letters are distinct:

```
// defines four different int variables
int somename, someName, SomeName, SOMENAME;
```

**KEY CONCEPT: STATIC TYPING**

**C++ is a *statically typed* language, which means that types are checked at compile time. The process by which types are checked is referred to as *type checking*.**

**As we've seen, the type of an object constrains the operations that the object can perform. In C++, the compiler checks whether the operations we write are supported by the types we use. If we try to do things that the type does not support, the compiler usually generates an error message and does not produce an executable file.**

**As our programs get more complicated, we'll see that static type checking can help find bugs. However, a consequence of static checking is that the type of every entity we use must be known to the compiler. As one example, we must declare the type of a variable before we can use that variable.**

The language reserves a set of names, listed in Tables 2.3 (overleaf) and Table 2.4 (overleaf), for its own use. These names may not be used as identifiers.

The standard also reserves a set of names for use in the standard library: The identifiers that we define may not contain two consecutive underscores, nor can an identifier begin with an underscore followed immediately by an uppercase letter. Moreover, identifiers defined outside a function may not begin with an underscore.

### Conventions for Variable Names

There are some generally accepted conventions for naming variables that help to make our programs more readable:

- An identifier should give some indication of its meaning.
- Variable names normally are lowercase—`index`, not `Index` or `INDEX`.
- Classes that we define, such as `Sales_item`, usually begin with an uppercase letter.
- Identifiers with multiple words should visually distinguish each word, for example, `student_loan` or `studentLoan`, not `studentloan`.

Best Practices: Naming conventions are most useful when followed consistently.

**EXERCISES SECTION 2.2.3**

**Exercise 2.11:** Which, if any, of the following names are invalid?

(a) `int double = 3.14;`
(b) `int _;`
(c) `int catch-22;`
(d) `int 1_or_2 = 1;`
(e) `double Double = 3.14;`

**Table 2.3: C++ Keywords**

| | | | | |
|---|---|---|---|---|
| alignas | continue | friend | register | true |
| alignof | decltype | goto | reinterpret_cast | try |
| asm | default | if | return | typedef |
| auto | delete | inline | short | typeid |
| bool | do | int | signed | typename |
| break | double | long | sizeof | union |
| case | dynamic_cast | mutable | static | unsigned |
| catch | else | namespace | static_assert | using |
| char | enum | new | static_cast | virtual |
| char16_t | explicit | noexcept | struct | void |
| char32_t | export | nullptr | switch | volatile |
| class | extern | operator | template | wchar_t |
| const | false | private | this | while |
| constexpr | float | protected | thread_local | |
| const_cast | for | public | throw | |

**Table 2.4: C++ Alternative Operator Names**

| | | | | | |
|---|---|---|---|---|---|
| and | bitand | compl | not_eq | or_eq | xor_eq |
| and_eq | bitor | not | or | xor | |

## 2.2.4 Scope of a Name

At any particular point in a program, each name that is in use refers to a specific entity—a variable, function, type, and so on. However, a given name can be reused to refer to different entities at different points in the program.

A **scope** is a part of the program in which a name has a particular meaning. Most scopes in C++ are delimited by curly braces.

The same name can refer to different entities in different scopes. Names are visible from the point where they are declared until the end of the scope in which the declaration appears. As an example, consider the program from § 1.4.2 (p. 13):

```
#include <iostream>
int main()
{
    int sum = 0;
    // sum values from 1 through 10 inclusive
    for (int val = 1; val <= 10; ++val)
        sum += val;  // equivalent to sum = sum + val
    std::cout << "Sum of 1 to 10 inclusive is "
              << sum << std::endl;
    return 0;
}
```

This program defines three names—`main`, `sum`, and `val`—and uses the namespace name `std`, along with two names from that namespace—`cout` and `endl`.

The name `main` is defined outside any curly braces. Like most names defined outside a function, the name `main` has **global scope**. Once declared, names at the global scope are accessible throughout the program. The name `sum` is defined within the scope of the block that is the body of the `main` function. It is accessible from its point of declaration throughout the rest of the `main` function but not outside of it. The variable `sum` has **block scope**. The name `val` is defined in the scope of the `for` statement. It can be used in that statement but not elsewhere in `main`.

**ADVICE: DEFINE VARIABLES WHERE YOU FIRST USE THEM**

**It is usually a good idea to define a variable where it is first used. Doing so improves readability by making it easy to find the definition of the variable. More importantly, it is often easier to give the variable a useful initial value when the variable is defined close to where it is first used.**

## Nested Scopes

Scopes can contain other scopes. The contained (or nested) scope is referred to as an **inner scope**, the containing scope is the **outer scope**.

Once a name has been declared in a scope, that name can be used by scopes nested inside that scope. Names declared in the outer scope can also be redefined in an inner scope:

```
#include <iostream>
// Program for illustration purposes only: It is bad style for a function
// to use a global variable and also define a local variable with the same name
int reused = 42;  // reused has global scope
int main()
{
    int unique = 0; // unique has block scope
    // output #1: uses global reused; prints 42 0
    std::cout << reused << " " << unique << std::endl;
    int reused = 0; // new, local object named reused hides global reused
    // output #2: uses local reused; prints 0 0
    std::cout << reused << " " <<  unique << std::endl;
    // output #3: explicitly requests the global reused; prints 42 0
    std::cout << ::reused << " " <<  unique << std::endl;
    return 0;
}
```

Output #1 appears before the local definition of `reused`. Therefore, this output statement uses the name `reused` that is defined in the global scope. This statement prints `42 0`. Output #2 occurs after the local definition of `reused`. The local `reused` is now **in scope**. Thus, this second output statement uses the local object named `reused` rather than the global one and prints `0 0`. Output #3 uses the scope operator (§ 1.2, p. 8) to override the default scoping rules. The global scope has no name. Hence, when the scope operator has an empty left-hand side, it is a request to fetch the name on the right-hand side from the global scope. Thus, this expression uses the global `reused` and prints `42 0`.

WARNING

It is almost always a bad idea to define a local variable with the same name as a global variable that the function uses or might use.

**EXERCISES SECTION 2.2.4**

**Exercise 2.12:** What is the value of `j` in the following program?

```
int i = 42;
int main()
{
    int i = 100;
    int j = i;
}
```

**Exercise 2.13:** Is the following code legal? If so, what values are printed?

```
int i = 100, sum = 0;
for (int i = 0; i != 10; ++i)
     sum += i;
std::cout << i << " " << sum << std::endl;
```

## 2.3 Compound Types

A **compound type** is a type that is defined in terms of another type. C++ has several compound types, two of which—references and pointers—we'll cover in this chapter.

Defining variables of compound type is more complicated than the declarations we've seen so far. In § 2.2 (p. 41) we said that a simple declaration consists of a type followed by a list of variable names. More generally, a declaration is a **base type** followed by a list of **declarators**. Each declarator names a variable and gives the variable a type that is related to the base type.

The declarations that we have seen so far have declarators that are nothing more than variable names. The type of such variables is the base type of the declaration. More complicated declarators specify variables with compound types that are built from the base type of the declaration.

### 2.3.1 References

A **reference** defines an alternative name for an object. A reference type "refers to" another type. C++ has two kinds of references: "rvalue references" and "lvalue references." Rvalue references, which we'll cover in § 13.5.1 (p. 585), are somewhat special purpose and are primarily intended for use inside classes. Technically speaking, when we (and most C++ programmers) use the term *reference*, we mean "lvalue reference."

We define a reference type by writing a declarator of the form `&d`, where `d` is the name being declared:

```
int ival = 1024;
int &refVal = ival;  // refVal refers to (is another name for) ival
int &refVal2;        // error: a reference must be initialized
```

Ordinarily, when we initialize a variable, the value of the initializer is copied into the object we are creating. When we define a reference, instead of copying the initializer's value, we **bind** the reference to its initializer. Once it is defined, a reference remains bound to the object with which it was initialized. There is no way to rebind a reference to refer to a different object. Because there is no way to rebind a reference, references *must* be initialized.

## A Reference Is an Alias

Note

A reference is not an object. Instead, a reference is *just another name for an already existing object*.

After a reference has been defined, *all* operations on that reference are actually operations on the object to which the reference is bound:

```
refVal = 2;         // assigns 2 to the object to which refVal refers, i.e., to ival
int ii = refVal;    // same as ii = ival
```

When we assign to a reference, we are assigning to the object to which the reference is bound. When we fetch the value of a reference, we are fetching the value of the object to which the reference is bound. Similarly, when we use a reference as an initializer, we are using the object to which the reference is bound:

```
// ok: refVal3 is bound to the object to which refVal is bound, i.e., to ival
int &refVal3 = refVal;
// initializes i from the value in the object to which refVal is bound
int i = refVal;    // ok: initializes i to the same value as ival
```

## Reference Definitions

We can define multiple references in a single definition. The `&` must be repeated for each reference:

```
int i = 1024, i2 = 2048;   // i and i2 are both ints
int &r = i, r2 = i2;       // r is a reference bound to i; r2 is an int
int i3 = 1024, &ri = i3;   // i3 is an int; ri is a reference bound to i3
int &r3 = i3, &r4 = i2;    // both r3 and r4 are references
```

With two exceptions that we'll cover in § 2.4.1 (p. 62) and § 15.2.3 (p. 656), the type of a reference and the object to which the reference refers must match exactly. Moreover, for reasons we'll explore in § 2.4.1, a reference may be bound only to an object, not to a literal or to the result of a more general expression:

```
int &refVal4 = 10;    // error: initializer must be an object
double dval = 3.14;
int &refVal5 = dval; // error: initializer must be an int object
```

### Exercises Section 2.3.1

**Exercise 2.14:** Which of the following definitions, if any, are invalid? Why?

(a) `int ival = 1.01;`     (b) `int &rval1 = 1.01;`
(c) `int &rval2 = ival;`   (d) `int &rval3;`

**Exercise 2.15:** Which, if any, of the following assignments are invalid? If they are valid, explain what they do.

```
int i = 0, &r1 = i; double d = 0, &r2 = d;
```

(a) `r2 = 3.14159;`   (b) `r2 = r1;`
(c) `i = r2;`         (d) `r1 = d;`

**Exercise 2.16:** What does the following code print?

```
int i, &ri = i;
i = 5; ri = 10;
std::cout << i << " " << ri << std::endl;
```

## 2.3.2 Pointers

A **pointer** is a compound type that "points to" another type. Like references, pointers are used for indirect access to other objects. Unlike a reference, a pointer is an object in its own right. Pointers can be assigned and copied; a single pointer can point to several different objects over its lifetime. Unlike a reference, a pointer need not be initialized at the time it is defined. Like other built-in types, pointers defined at block scope that are not initialized have undefined value.

WARNING

Pointers are often hard to understand. Debugging problems due to pointer errors bedevil even experienced programmers.

We define a pointer type by writing a declarator of the form `*d`, where `d` is the name being defined. The `*` must be repeated for each pointer variable:

```
int *ip1, *ip2;  // both ip1 and ip2 are pointers to int
double dp, *dp2; // dp2 is a pointer to double; dp is a double
```

### Taking the Address of an Object

A pointer holds the address of an object. We get the address of an object by using the address-of operator (the **& operator**):

```
int ival = 42;
int *p = &ival; // p holds the address of ival; p is a pointer to ival
```

The second statement defines p as a pointer to int and initializes p to point to the int object named ival. Because references are not objects, they don't have addresses. Hence, we may not define a pointer to a reference type.

With two exceptions, which we cover in § 2.4.2 (p. 63) and § 15.2.3 (p. 656), the types of a pointer and the object to which it points must match:

```
double dval;
double *pd = &dval;   // ok: initializer is the address of a double
double *pd2 = pd;     // ok: initializer is a pointer to double
int *pi = pd;   // error: types of pi and pd differ
pi = &dval;     // error: assigning the address of a double to a pointer to int
```

The types must match because the type of the pointer is used to infer the type of the object to which the pointer points. If a pointer addressed an object of another type, operations performed on the underlying object would fail.

## Pointer Value

The value (i.e., the address) stored in a pointer can be in one of four states:

1. It can point to an object.
2. It can point to the location immediately past the end of an object.
3. It can be a null pointer, indicating that it is not bound to any object.
4. It can be invalid; values other than the preceding three are invalid.

It is an error to copy or otherwise try to access the value of an invalid pointer. As when we use an uninitialized variable, this error is one that the compiler is unlikely to detect. The result of accessing an invalid pointer is undefined. Therefore, we must always know whether a given pointer is valid.

Although pointers in cases 2 and 3 are valid, there are limits on what we can do with such pointers. Because these pointers do not point to any object, we may not use them to access the (supposed) object to which the pointer points. If we do attempt to access an object through such pointers, the behavior is undefined.

## Using a Pointer to Access an Object

When a pointer points to an object, we can use the dereference operator (the *** operator**) to access that object:

```
int ival = 42;
int *p = &ival; // p holds the address of ival; p is a pointer to ival
cout << *p;     // *p yields the object to which p points; prints 42
```

Dereferencing a pointer yields the object to which the pointer points. We can assign to that object by assigning to the result of the dereference:

```
*p = 0;       // * yields the object; we assign a new value to ival through p
cout << *p; // prints 0
```

When we assign to `*p`, we are assigning to the object to which `p` points.

Note: We may dereference a pointer only if that pointer points to an object.

**KEY CONCEPT: SOME SYMBOLS HAVE MULTIPLE MEANINGS**

**Some symbols, such as `&` and `*`, are used as both an operator in an expression and as part of a declaration. The context in which a symbol is used determines what the symbol means:**

```
int i = 42;
int &r = i;    // & is part of a declarator; r is a reference
int *p;        // * is part of a declarator; p is a pointer
p = &i;        // & is used in an expression as the address-of operator
*p = i;        // * is used in an expression as the dereference operator
int &r2 = *p;  // & is part of the declarator; * is the dereference operator
               // r2 refers to the object to which p points
```

**In declarators, `&` and `*` are used to form compound types. In expressions, these same symbols denote operators. Because the same symbol is used with very different meanings, it can be helpful to think of them as if they were different symbols.**

### Null Pointers

A **null pointer** is a valid pointer that does not point to an object. We can check whether a pointer is null before attempting to use it. There are several ways to obtain a null pointer:

```
int *p1 = nullptr; // equivalent to int *p1 = 0;
int *p2 = 0;       // directly initializes p2 from the literal constant 0
// must #include cstdlib
int *p3 = NULL;    // equivalent to int *p3 = 0;
```

The most direct approach is to initialize the pointer using the literal **`nullptr`**. `nullptr` is a literal that has a special type that can be converted (§ 2.1.2, p. 34) to any other pointer type. Alternatively, we can initialize a pointer to the literal `0`, as we do in the definition of `p2`.

Old programs sometimes use a **preprocessor variable** named `NULL`, which the `cstdlib` header defines as `0`. When we use a preprocessor variable, the preprocessor automatically replaces the variable by its value. Hence, initializing a pointer to `NULL` is equivalent to initializing it to `0`.

We'll describe the preprocessor in a bit more detail in § 2.6.3 (p. 79). What's useful to know now is that the preprocessor is a program that runs before the compiler. Preprocessor variables are not part of the `std` namespace. As a result, we refer to them directly without the `std::` prefix.

Best Practices: C++ programs should avoid using `NULL` and use `nullptr` instead.

It is illegal to assign an `int` variable to a pointer, even if the variable's value happens to be `0`.

```
int zero = 0;
int *p4 = zero;    // error: cannot assign an int to a pointer
```

**ADVICE: INITIALIZE ALL POINTERS**

**Uninitialized pointers are a common source of run-time errors.**

**As with any other uninitialized variable, what happens when we use an uninitialized pointer is undefined. Using an uninitialized pointer almost always results in a run-time crash. However, debugging the resulting crashes can be surprisingly hard.**

**Under most compilers, when we use an uninitialized pointer, the bits in the memory in which the pointer resides are used as an address. Using an uninitialized pointer is a request to access a supposed object at that supposed location. There is no way to distinguish a valid address from an invalid one formed from the bits that happen to be in the memory in which the pointer was allocated.**

**Our recommendation to initialize all variables is particularly important for pointers. If possible, define a pointer only after the object to which it should point has been defined. If there is no object to bind to a pointer, then initialize the pointer to `nullptr` or zero. That way, the program can detect that the pointer does not point to an object.**

## Assignment and Pointers

Both pointers and references give indirect access to other objects. However, there are important differences in how they do so. The most important is that a reference is not an object. Once we have defined a reference, there is no way to make that reference refer to a different object. When we use a reference, we always get the object to which the reference was initially bound.

There is no such identity between a pointer and the address that it holds. As with any other (nonreference) variable, when we assign to a pointer, we give the pointer itself a new value. Assignment makes the pointer point to a different object:

```
int i = 42;
int *pi = nullptr; // pi is initialized but addresses no object
int *pi2 = &i;     // pi2 initialized to hold the address of i
int *pi3;          // if pi3 is defined inside a block, pi3 is uninitialized
pi3 = pi2;         // pi3 and pi2 address the same object, e.g., i
pi2 = nullptr;     // pi2 now addresses no object
```

It can be hard to keep straight whether an assignment changes the pointer or the object to which the pointer points. The important thing to keep in mind is that assignment changes its left-hand operand. When we write

```
pi = &ival; // value in pi is changed; pi now points to ival
```

we assign a new value to `pi`, which changes the address that `pi` holds. On the other hand, when we write

```
*pi = 0;      // value in ival is changed; pi is unchanged
```

it is `*pi` (i.e., the value to which `pi` points) that is changed.

## Other Pointer Operations

So long as a pointer has a valid value, we can use that pointer as a condition. Just as when we use an arithmetic value in a condition (§ 2.1.2, p. 35), if the pointer is 0, then the condition is `false`:

```
int ival = 1024;
int *pi = nullptr; // pi is a valid, null pointer
int *pi2 = &ival;  // pi2 is a valid pointer that holds the address of ival
if (pi)  // pi has value 0, so condition evaluates as false
    // ...
if (pi2) // pi2 points to ival, so it is not 0; the condition evaluates as true
    // ...
```

Any nonzero pointer evaluates as `true`

Given two valid pointers of the same type, we can compare them using the equality (`==`) or inequality (`!=`) operators. The result of these operators has type `bool`. Two pointers are equal if they hold the same address and unequal otherwise. That is, two pointers are equal if they are both null, if they address the same object, or if they are both pointers one past the same object. Note that it is possible for a pointer to an object and a pointer one past the end of a different object to hold the same address. Such pointers will compare equal.

Because these operations use the value of the pointer, a pointer used in a condition or in a comparsion must be a valid pointer. Using an invalid pointer as a condition or in a comparison is undefined.

§ 3.5.3 (p. 123) will cover additional pointer operations.

## `void*` Pointers

The type **`void*`** is a special pointer type that can hold the address of any object. Like any other pointer, a `void*` pointer holds an address, but the type of the object at that address is unknown:

```
double obj = 3.14, *pd = &obj;
// ok: void* can hold the address value of any data pointer type
void *pv = &obj;  // obj can be an object of any type
pv = pd;          // pv can hold a pointer to any type
```

We can use a `void*` pointer in only a limited number of ways: We can compare it to another pointer, we can pass it to or return it from a function, and we can assign it to another `void*` pointer. We cannot use a `void*` to operate on the object it addresses—we don't know that object's type, and the type determines what operations we can perform on the object.

Generally, we use a `void*` pointer to deal with memory as memory, rather than using the pointer to access the object stored in that memory. We'll cover using `void*` pointers in this way in § 19.1.1 (p. 873). § 4.11.3 (p. 174) will show how we can retrieve the address stored in a `void*` pointer.

**EXERCISES SECTION 2.3.2**

**Exercise 2.17:** Write code to change the value of a pointer. Write code to change the value to which the pointer points.

**Exercise 2.18:** Explain the key differences between pointers and references.

**Exercise 2.19:** What does the following program do?

```
int i = 42;   int *p1 = &i;
*p1 = *p1 * *p1;
```

**Exercise 2.20:** Assuming `p` is a pointer to `int`, explain the following code:

```
if (p) // ...
if (*p) // ...
```

**Exercise 2.21:** Given a pointer `p`, can you determine whether `p` points to a valid object? If so, how? If not, why not?

**Exercise 2.22:** Why is the initialization of `p` legal but that of `lp` illegal?

```
int i = 42;    void *p = &i;     long *lp = &i;
```

## 2.3.3 Understanding Compound Type Declarations

As we've seen, a variable definition consists of a base type and a list of declarators. Each declarator can relate its variable to the base type differently from the other declarators in the same definition. Thus, a single definition might define variables of different types:

```
// i is an int; p is a pointer to int; r is a reference to int
int i = 1024, *p = &i, &r = i;
```

**WARNING** Many programmers are confused by the interaction between the base type and the type modification that may be part of a declarator.

### Defining Multiple Variables

It is a common misconception to think that the type modifier (`*` or `&`) applies to all the variables defined in a single statement. Part of the problem arises because we can put whitespace between the type modifier and the name being declared:

```
int* p;   // legal but might be misleading
```

We say that this definition might be misleading because it suggests that `int*` is the type of each variable declared in that statement. Despite appearances, the base type of this declaration is `int`, not `int*`. The `*` modifies the type of `p`. It says nothing about any other objects that might be declared in the same statement:

```
int* p1, p2; // p1 is a pointer to int; p2 is an int
```

There are two common styles used to define multiple variables with pointer or reference type. The first places the type modifier adjacent to the identifier:

```
int *p1, *p2; // both p1 and p2 are pointers to int
```

This style emphasizes that the variable has the indicated compound type.

The second places the type modifier with the type but defines only one variable per statement:

```
int* p1;       // p1 is a pointer to int
int* p2;       // p2 is a pointer to int
```

This style emphasizes that the declaration defines a compound type.

Tip: There is no single right way to define pointers or references. The important thing is to choose a style and use it consistently.

In this book we use the first style and place the `*` (or the `&`) with the variable name.

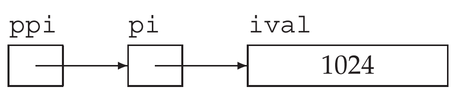
## Pointers to Pointers

In general, there are no limits to how many type modifiers can be applied to a declarator. When there is more than one modifier, they combine in ways that are logical but not always obvious. As one example, consider a pointer. A pointer is an object in memory, so like any object it has an address. Therefore, we can store the address of a pointer in another pointer.

We indicate each pointer level by its own `*`. That is, we write `**` for a pointer to a pointer, `***` for a pointer to a pointer to a pointer, and so on:

```
int ival = 1024;
int *pi = &ival;    // pi points to an int
int **ppi = &pi;    // ppi points to a pointer to an int
```

Here `pi` is a pointer to an `int` and `ppi` is a pointer to a pointer to an `int`. We might represent these objects as




Just as dereferencing a pointer to an `int` yields an `int`, dereferencing a pointer to a pointer yields a pointer. To access the underlying object, we must dereference the original pointer twice:

```
cout << "The value of ival\n"
     << "direct value: " << ival << "\n"
     << "indirect value: " << *pi << "\n"
     << "doubly indirect value: " << **ppi
     << endl;
```

This program prints the value of `ival` three different ways: first, directly; then, through the pointer to `int` in `pi`; and finally, by dereferencing `ppi` twice to get to the underlying value in `ival`.

### References to Pointers

A reference is not an object. Hence, we may not have a pointer to a reference. However, because a pointer is an object, we can define a reference to a pointer:

```
int i = 42;
int *p;        // p is a pointer to int
int *&r = p;   // r is a reference to the pointer p
r = &i; //  r refers to a pointer; assigning &i to r makes p point to i
*r = 0; //  dereferencing r yields i, the object to which p points; changes i to 0
```

The easiest way to understand the type of `r` is to read the definition right to left. The symbol closest to the name of the variable (in this case the `&` in `&r`) is the one that has the most immediate effect on the variable's type. Thus, we know that `r` is a reference. The rest of the declarator determines the type to which `r` refers. The next symbol, `*` in this case, says that the type `r` refers to is a pointer type. Finally, the base type of the declaration says that `r` is a reference to a pointer to an `int`.

Tip

It can be easier to understand complicated pointer or reference declarations if you read them from right to left.

#### EXERCISES SECTION 2.3.3

**Exercise 2.23:** Determine the types and values of each of the following variables.

(a) `int* ip, i, &r = i;` (b) `int i, *ip = 0;` (c) `int* ip, ip2;`

## 2.4 `const` Qualifier

Sometimes we want to define a variable whose value, once initialized, cannot subsequently be changed. For example, we might want to use a variable to refer to the size of a buffer. Using a variable makes it easy for us to rewrite our program to give the buffer a different size. On the other hand, we'd also like to prevent code from inadvertently giving a new value to the variable that we use to represent the buffer size. We can make a variable unchangeable by including the keyword **`const`** in the base type of the variable's declaration:

```
const int bufSize = 512;    // input buffer size
```

defines `bufSize` as a constant. Any attempt to assign to `bufSize` is an error:

```
bufSize = 512; // error: attempt to write to const object
```

Because we can't change the value of a `const` object after we create it, it must be initialized. As usual, the initializer may be an arbitrarily complicated expression:

```
const int i = get_size();  // ok: initialized at run time
const int j = 42;          // ok: initialized at compile time
const int k;               // error: k is uninitialized const
```

## Initialization and `const`

As we have observed many times, the type of an object defines the operations that can be performed by that object. A `const` type can use most but not all of the same operations as its non`const` version. The one restriction is that we may use only those operations that cannot change an object. So, for example, we can use a `const int` in arithmetic expressions in exactly the same way as a plain, non`const` `int`. A `const int` converts to `bool` the same way as a plain `int`, and so on.

Among the operations that don't change the value of an object is initialization—when we use an object to initialize another object, it doesn't matter whether either or both of the objects are `const`s:

```
int i = 42;
const int ci = i;     // ok: the value in i is copied into ci
int j = ci;           // ok: the value in ci is copied into j
```

Although `ci` is a `const int`, the value in `ci` is an `int`. The `const`ness of `ci` matters only for operations that might change `ci`. When we copy `ci` to initialize `j`, we don't care that `ci` is a `const`. Copying an object doesn't change that object. Once the copy is made, the new object has no further access to the original object.

## By Default, `const` Objects Are Local to a File

When a `const` object is initialized from a compile-time constant, such as in our definition of `bufSize`:

```
const int bufSize = 512;     // input buffer size
```

the compiler will usually replace uses of the variable with its corresponding value during compilation. That is, the compiler will generate code using the value `512` in the places that our code uses `bufSize`.

To substitute the value for the variable, the compiler has to see the variable's initializer. When we split a program into multiple files, every file that uses the `const` must have access to its initializer. In order to see the initializer, the variable must be defined in every file that wants to use the variable's value (§ 2.2.2, p. 45). To support this usage, yet avoid multiple definitions of the same variable, `const` variables are defined as local to the file. When we define a `const` with the same name in multiple files, it is as if we had written definitions for separate variables in each file.

Sometimes we have a `const` variable that we want to share across multiple files but whose initializer is not a constant expression. In this case, we don't want the compiler to generate a separate variable in each file. Instead, we want the `const` object to behave like other (non`const`) variables: We want to define the `const` in one file, and declare it in the other files that use that object.

To define a single instance of a `const` variable, we use the keyword `extern` on both its definition and declaration(s):

```
// file_1.cc defines and initializes a const that is accessible to other files
extern const int bufSize = fcn();
// file_1.h
extern const int bufSize; // same bufSize as defined in file_1.cc
```

In this program, `file_1.cc` defines and initializes `bufSize`. Because this declaration includes an initializer, it is (as usual) a definition. However, because `bufSize` is `const`, we must specify `extern` in order for `bufSize` to be used in other files.

The declaration in `file_1.h` is also `extern`. In this case, the `extern` signifies that `bufSize` is not local to this file and that its definition will occur elsewhere.

Note: To share a `const` object among multiple files, you must define the variable as `extern` or `inline` (§ 2.2.2, p. 46).

**EXERCISES SECTION 2.4**

**Exercise 2.24:** Which of the following are legal? For those that are illegal, explain why.

(a) `const int buf;` (b) `int cnt = 0;`
(c) `const int sz = cnt;` (d) `++cnt; ++sz;`

## 2.4.1 References to `const`

As with any other object, we can bind a reference to an object of a `const` type. To do so we use a **reference to `const`**, which is a reference that refers to a `const` type. Unlike an ordinary reference, a reference to `const` cannot be used to change the object to which the reference is bound:

```
const int ci = 1024;
const int &r1 =  ci;  // ok: both reference and underlying object are const
r1 = 42;              // error: r1 is a reference to const
int &r2 = ci;         // error: nonconst reference to a const object
```

Because we cannot assign directly to `ci`, we also should not be able to use a reference to change `ci`. Therefore, the initialization of `r2` is an error. If this initialization were legal, we could use `r2` to change the value of its underlying object.

**TERMINOLOGY: CONST REFERENCE IS A REFERENCE TO CONST**

**C++ programmers tend to abbreviate the phrase "reference to `const`" as "`const` reference." This abbreviation makes sense—if you remember that it is an abbreviation.**

**Technically speaking, there are no `const` references. A reference is not an object, so we cannot make a reference that is itself `const`. Indeed, once we've defined a reference there is no way make that reference refer to a different object. Therefore, in some sense all references are `const`. Whether a reference refers to a `const` or nonconst type affects what we can do with the object to which that reference refers, not whether we can alter the binding of the reference itself.**

## Initialization and References to `const`

Ordinarily, the type of a reference must match the type of the object to which the reference refers (§ 2.3.1, p. 51). However, as an exception to this rule, we can initialize a reference to `const` from any expression that can be converted (§ 2.1.2, p. 34) to the type of the reference. In particular, we can bind a reference to `const` to a non`const` object, a literal, or a more general expression:

```
int i = 42;
const int &r1 = i;      // we can bind a const int& to a plain int object
const int &r2 = 42;     // ok: r2 is a reference to const
const int &r3 = r1 * 2; // ok: r3 is a reference to const
int &r4 = r1 * 2;       // error: r4 is a plain, nonconst reference
```

The easiest way to understand this difference in initialization rules is to consider what happens when we bind a reference to an object of a different type:

```
double dval = 3.14;
const int &ri = dval;
```

Here `ri` refers to a `const int`. Operations on `ri` will be integer operations, but `dval` is a floating-point number, not an integer. To ensure that the object to which `ri` is bound is a `const int`, the compiler transforms this code into something like

```
const int temp = dval;   // create a temporary const int from the double
const int &ri = temp;    // bind ri to that temporary
```

In this case, `ri` is bound to a **temporary** object. A temporary object is an unnamed object created by the compiler when it needs a place to store the result from evaluating an expression. C++ programmers often use the word temporary as an abbreviation for temporary object. As the name "temporary" implies, temporaries generally exist only for the duration of the expression in which they are created.

Now consider what could happen if this initialization were allowed but `ri` was not `const`. If `ri` weren't `const`, we could assign to `ri`. Doing so would change the object to which `ri` is bound. That object is a temporary, not `dval`. The programmer who made `ri` refer to `dval` would probably expect that assigning to `ri` would change `dval`. After all, why assign to `ri` unless the intent is to change the object to which `ri` is bound? Because binding a reference to a temporary is almost surely *not* what the programmer intended, the language makes it illegal.

## A Reference to `const` May Refer to an Object That Is Not `const`

It is important to realize that a reference to `const` restricts only what we can do through that reference. Binding a reference to `const` to an object says nothing about whether the underlying object itself is `const`. Because the underlying object might be non`const`, it might be changed by other means:

```
int i = 42;
int &r1 = i;          // r1 bound to i
const int &r2 = i;    // r2 also bound to i; but cannot be used to change i
r1 = 0;               // r1 is not const; i (and r2) now 0
r2 = 0;               // error: r2 is a reference to const
```

Binding `r2` to the (non`const`) `int i` is legal. However, we cannot use `r2` to change `i`. Even so, the value in `i` still might change. We can change `i` by assigning to it directly, or by assigning to another reference bound to `i`, such as `r1`.

## 2.4.2 Pointers and `const`

As with references, we can define pointers that point to either `const` or non`const` types. Like a reference to `const`, a **pointer to `const`** may not be used to change the object to which the pointer points. We may store the address of a `const` object only in a pointer to `const`:

```
const double pd = 3.14;    // pd is const; its value may not be changed
double *ptr = &pd;         // error: ptr is a plain pointer
const double *cptr = &pd; // ok: cptr may point to a double that is const
*cptr = 42;                // error: can't assign to *cptr; it points to const
```

Ordinarily, the type of a pointer must match the type of the object to which the pointer points (§ 2.3.2, p. 53). However, as an exception to this rule, we can use a pointer to `const` to point to a non`const` object:

```
double dval = 3.14;        // dval is a double; its value can be changed
cptr = &dval;              // ok: but can't change dval through cptr
```

Like a reference to `const`, a pointer to `const` says nothing about whether the object to which the pointer points is `const`. Defining a pointer as a pointer to `const` affects only what we can do with the pointer. It is important to remember that there is no guarantee that the object pointed to by a pointer to `const` won't change.

Tip: It may be helpful to think of pointers and references to `const` as pointers or references "that *think* they point or refer to `const`."

### `const` Pointers

Unlike references, pointers are objects. Hence, as with any other object type, we can have a pointer that is itself `const`. **`const` pointer** We indicate that the pointer is `const` by putting the `const` after the `*`. This placement indicates that it is the pointer, not the pointed-to type, that is `const`. Like any other `const` object, a `const` pointer must be initialized, and once initialized, its value (i.e., the address that it holds) may not be changed:

```
int errNumb = 0;
int *const curErr = &errNumb;   // curErr will always point to errNumb
const double pd = 3.14159;
const double *const pdp = &pd; // pdp is a const pointer to a const object
```

As we saw in § 2.3.3 (p. 59), the easiest way to understand these declarations is to read them from right to left. In this case, the symbol closest to `curErr` is `const`, which means that `curErr` itself will be a `const` object. The type of that object is formed from the rest of the declarator. The next symbol in the declarator is `*`, which means that `curErr` is a `const` pointer. Finally, the base type of the declaration completes the type of `curErr`, which is a `const` pointer to an object of type `int`. Similarly, `pdp` is a `const` pointer to an object of type `const double`.

The fact that a pointer is itself `const` says nothing about whether we can use the pointer to change the underlying object. Whether we can change that object depends entirely on the type to which the pointer points. For example, `pdp` is a `const` pointer to `const`. Neither the value of the object addressed by `pdp` nor the address stored in `pdp` can be changed. On the other hand, `curErr` addresses a plain, non`const` `int`. We can use `curErr` to change the value of `errNumb`:

```
*pdp = 2.72;      // error: pdp is a pointer to const
// if the object to which curErr points (i.e., errNumb) is nonzero
if (*curErr) {
    errorHandler();
    *curErr = 0; // ok: reset the value of the object to which curErr is bound
}
```

**EXERCISES SECTION 2.4.2**

**Exercise 2.25:** Which of the following initializations are legal? Explain why.

(a) `int i = -1, &r = 0;` (b) `int *const p2 = &i2;`
(c) `const int i = -1, &r = 0;` (d) `const int *const p3 = &i2;`
(e) `const int *p1 = &i2;` (f) `const int &const r2;`
(g) `const int i2 = i, &r = i;`

**Exercise 2.26:** Explain the following definitions. Identify any that are illegal.

(a) `int i, *const cp;` (b) `int *p1, *const p2;`
(c) `const int ic, &r = ic;` (d) `const int *const p3;`
(e) `const int *p;`

**Exercise 2.27:** Uing the variables in the previous exercise, which of the following assignments are legal? Explain why.

(a) `i = ic;` (b) `p1 = p3;`
(c) `p1 = ⁣` (d) `p3 = ⁣`
(e) `p2 = p1;` (f) `ic = *p3;`

### 2.4.3 Top-Level `const`

As we've seen, a pointer is an object that can point to a different object. As a result, we can talk independently about whether a pointer itself is `const` and whether the objects to which a pointer can point are `const`:

- **Top-level `const`** indicates that an object is `const`. Top-level `const` can appear in any object type (i.e., one of the built-in arithmetic types, a class type, or a pointer type) but not on a reference type.

- **Low-level `const`**. appears only in the base type of compound types such as pointers or references.

Unlike most other types, pointers can have both top-level and low-level `const`:

```
int i = 0;
int *const p1 = &i;  // we can't change the value of p1; const is top-level
const int ci = 42;   // we cannot change the value of ci; const is top-level
const int *p2 = &ci; // we can change p2; const is low-level
const int *const p3 = p2; // right-most const is top-level, left-most is not
const int &r = ci;  // const in reference types is always low-level
```

The distinction between top-level and low-level matters when we copy an object. When we copy an object, top-level `const`s are ignored: Copying an object doesn't change the copied object. As a result, it is immaterial whether the object copied from or copied into is `const`:

```
i = ci;  // ok: copying the value of ci; top-level const in ci is ignored
p2 = p3; // ok: pointed-to type matches; top-level const in p3 is ignored
```

On the other hand, low-level `const` is never ignored. When we copy an object, both objects must have the same low-level `const` qualification or there must be a conversion between the types of the two objects. In general, we can convert a low-level non`const` to `const` but not the other way round:

```
int *p = p3; // error: p3 has a low-level const but p doesn't
p2 = p3;     // ok: p2 has the same low-level const qualification as p3
p2 = &i;     // ok: we can convert int* to const int*
int &r = ci; // error: can't bind an ordinary int& to a const int object
const int &r2 = i; // ok: can bind const int& to plain int
```

`p3` has both a top-level and low-level `const`. When we copy `p3`, we can ignore its top-level `const` but not the fact that it points to a `const` type. Hence, we cannot use `p3` to initialize `p`, which points to a plain (non`const`) `int`. On the other hand, we can assign `p3` to `p2`. Both pointers have the same (low-level `const`) type. The fact that `p3` is a `const` pointer (i.e., that it has a top-level `const`) doesn't matter.

### EXERCISES SECTION 2.4.3

**Exercise 2.28:** For each of the following declarations, indicate whether the object being declared has top-level or low-level `const`.

```
const int v2 = 0;    int v1 = v2;
int *p1 = &v1, &r1 = v1;
const int *p2 = &v2, *const p3 = &i, &r2 = v2;
```

**Exercise 2.29:** Given the declarations in the previous exercise say whether the following assignments are legal and explain how top-level or low-level `const` applies:

```
r1 = v2;   p1 = p2;   p2 = p1;   p1 = p3;   p2 = p3;
```

### 2.4.4 `constexpr` and Constant Expressions

A **constant expression** is an expression whose value cannot change and that can be evaluated at compile time. A literal is a constant expression. A `const` object that is initialized from a constant expression is also a constant expression. As we'll see, several contexts in the language require constant expressions.

Whether a given object (or expression) is a constant expression depends on the types and the initializers. For example:

```
const int max_files = 20;     // max_files is a constant expression
const int limit = max_files + 1; // limit is a constant expression
int staff_size = 27;        // staff_size is not a constant expression
// if len is an ordinary function, sz is not a constant expression
const int sz = len();
```

Although `staff_size` is initialized from a literal, it is not a constant expression because `staff_size` is a plain `int`, not a `const int`. Whether `sz` is a constant expression depends how `len` is defined. Ordinary functions cannot be evaluated until run time, so if `len` is an ordinary function, `sz` is not a constant expression.

#### `constexpr` Variables

In a large system, it can be difficult to determine (for certain) that an initializer is a constant expression. We might define a `const` variable with an initializer that we think is a constant expression. Sometime later, when we use that variable in a context that requires a constant expression, we may discover that the initializer was not, in fact, a constant expression. In general, the definition of an object and its use in such a context can be widely separated.

We can ask the compiler to verify that a variable is a constant expression by declaring it in a **`constexpr`** declaration. Specifying `constexpr` on a variable definition does two things: it implicitly defines each variable in that definition as (top-level) `const` and `inline` (§ 2.2.2, p. 46), and the compiler checks that the initializers are constant expressions:

```
constexpr int mf = 20;         // 20 is a constant expression
constexpr int limit = mf + 1; // mf + 1 is a constant expression
// ok only if len is constexpr and returns a constant expression
constexpr int sz = len();
```

Although we cannot initialize a `constexpr` variable from a call to an ordinary function, in § 6.5.1 (p. 247) we'll see that we can define `constexpr` functions. Such functions must be simple enough that the compiler can evaluate them at compile time. We can call a `constexpr` function to initialize a `constexpr` variable.

Best Practices

Use `constexpr` for variables that you use as constant expressions.

#### Pointers and `constexpr`

We can define both pointers and references as `constexprs`. To do so, it is important to know that when we said that `constexpr` variables are implicitly `const`, that `const` is top-level—the `const` applies to the variable itself. Thus, when we define a pointer in a `constexpr` we're saying that the pointer itself is `const`:

```
const int *p1 = nullptr;      // p1 is a pointer to a const int
constexpr int *p2 = nullptr;  // p2 is a const pointer to int
constexpr const int *p3 = p2; // p3 is a const pointer to const int
```

Despite appearances, the types of `p1` and `p2` are quite different; `p1` is a pointer to `const`, whereas `p2` is a constant pointer.

The objects we can use to initialize a `constexpr` pointer or reference are strictly limited. We can initialize a `constexpr` pointer from the `nullptr` literal or the literal (i.e., constant expression) `0`. We can also point to (or refer to) an object that remains at a fixed address. For reasons we'll cover in § 6.1.1 (p. 216), variables defined inside a function ordinarily are not stored at a fixed address. Hence, we cannot use a `constexpr` pointer (or reference) to point to (or refer to) such variables. On the other hand, the address of an object defined outside of any function is a constant expression, and so may be used to initialize a `constexpr` pointer or reference. We'll see in § 6.1.1 (p. 217), that functions may define variables that persist across calls to that function. Such objects do have fixed addresses. Therefore, we can use a `constexpr` pointer to point to (or reference to refer to) such variables.

### Literal Types

Because a constant expression is one that can be evaluated at compile time, there are limits on the types that we can use in a `constexpr` declaration. The types we can use in a `constexpr` are known as "literal types" because they are simple enough to have literal values.

Of the types we have used so far, the arithmetic, reference, and pointer types are literal types. Our `Sales_item` class and the library IO and `string` types are not literal types. Hence, we cannot define variables of these types as `constexpr`s. We'll see other kinds of literal types in § 7.6.2 (p. 314) and § 19.3 (p. 884).

## 2.5 Dealing with Types

As our programs get more complicated, we'll see that dealing with types also becomes more complicated. Sometimes we want to use a our own name in place of the name of a particular type. Other times it may be hard to determine the exact type we need. The language gives us ways to deal with both these problems.

### 2.5.1 Type Aliases

Sometimes we want to use a different name for a particular type. For example, some types are hard to "spell." That is, they have a form that is tedious and error-prone to write. Other times, we might want to use an alternative name to emphasize the purpose or meaning of the type. In both cases we can use a **type alias** to define a name that is a synonym for another type.

One way to define a type alias is to use an **alias declaration**:

```
using wages = double;   // wages is a synonym for double
using bonus = wages*;   // bonus is a synonym for pointer to double
```

An alias declaration starts with the keyword `using` followed by the alias name and an `=`. The right-hand side of the `=` must specify a type. The alias declaration defines the name on the left-hand side of the `=` as an alias for the type that appears on the right-hand side.

Alternatively, we can use a **typedef**:

```
typedef double wages;        // wages is a synonym for double
// base is a synonym for double; bonus is a synonum for double*
typedef wages base, *bonus;
```

The keyword `typedef` may appear as part of the base type of a declaration (§ 2.3, p. 50). Such declarations define type aliases rather than variables. Unlike an alias declaration, a single `typedef` can define multiple names. As with any other declaration, the names declared in a `typedef` are built up from the base type of the declaration.

A type alias is a type name and can appear wherever a type name can appear:

```
wages hourly, weekly;    // same as double hourly, weekly;
```

### Pointers, `const`, and Type Aliases

Consider the following declarations that use the type `pstring`, which is an alias for the type `char*`:

```
using pstring = char*;
const pstring cstr = 0; // cstr is a constant pointer to char
const pstring *ps;      // ps is a pointer to a constant pointer to char
```

The base type in these declarations is `const pstring`. As usual, a `const` that appears in the base type modifies the given type. The type of `pstring` is "pointer to `char`." So, `const pstring` is a constant pointer to `char`—not a pointer to `const char`.

It can be tempting, albeit incorrect, to interpret a declaration that uses a type alias by conceptually replacing the alias with its corresponding type:

```
const char *cstr = 0;   // wrong interpretation of const pstring cstr
```

Here we've replaced `pstring` by `char*`, which we might think would yield the equivalent type. However, this interpretation is wrong. The base type in this declaration is `const char`, whereas when we use `pstring`, the base type is a pointer type. This rewrite declares `cstr` as a pointer to `const char` rather than as a `const` pointer to `char`.

## 2.5.2 The `auto` Type Specifier

It is not uncommon to want to store the value of an expression in a variable. To declare the variable, we have to know the type of that expression. Unfortunately, it can be surprisingly difficult—and sometimes even impossible—to determine the type of an expression. Fortunately, we can ask the compiler figure out the type by using the **`auto`** type specifier. Unlike type specifiers, such as `int`, that name a specific type, `auto` tells the compiler to deduce the type from the initializer. By implication, a variable that uses `auto` as its type specifier must have an initializer:

```
// the type of item is deduced from the type of the result of adding val1 and val2
auto item = val1 + val2; // item initialized to the result of val1 + val2
```

Here the compiler will deduce the type of `item` from the type returned by applying `+` to `val1` and `val2`. If `val1` and `val2` are `Sales_item` objects (§ 1.5, p. 19), `item` will have type `Sales_item`. If those variables are type `double`, then `item` has type `double`, and so on.

As with any other type specifier, we can define multiple variables using `auto`. Because a declaration can involve only a single base type, the initializers for all the variables in the declaration must have types that are consistent with each other:

```
auto i = 0, *p = &i;      // ok: i is int and p is a pointer to int
auto sz = 0, pi = 3.14;   // error: inconsistent types for sz and pi
```

## Compound Types, `const`, and `auto`

When we use `auto` to define a variable, the variable's type is not necessairly exactly the same as the type of the initializer. To deduce a variable's type, the compiler adjusts the type of the initializer to conform to normal initialization rules.

For example, as we've seen, when we use a reference, we are really using the object to which the reference refers. Thus, when we use a reference as an initializer, the initializer is the object to which that reference refers. The compiler uses that object's type for `auto`'s type deduction:

```
int i = 0, &r = i; // i is an int, r is int&
auto a = r;        // a is an int (r is an alias for i, which has type int)
```

Similarly, as is usual in initializations, `auto`'s type deduction ignores top-level `const`s and retains low-level `const`s in the initializer: (§ 2.4.3, p. 64):

```
const int ci = i, &cr = ci;
auto b = ci;  // b is an int (top-level const in ci is dropped)
auto c = cr;  // c is an int (cr is an alias for ci whose const is top-level)
auto d = &i;  // d is an int* (& of an int object is int*)
auto e = &ci; // e is const int* (& of a const object is low-level const)
```

If we want to use `auto` to declare a reference variable or a variable with a top-level `const` in its type, we must do so explicitly:

```
const auto f = ci;  // auto deduced as int; f has type const int
auto &g = ci;       // g is a const int& that is bound to ci
auto &h = 42; // error: auto deduced as int, h is int& can't bind to a literal
const auto &j = 42; // ok: j is const int&
```

Moreover, when we define several variables, all of the initializers must have the same `auto`-deduced type:

```
auto k = ci, &l = i; // k is int (top-level const in ci is ignored); l is int&
auto &m = ci, *p = &ci; // m is a const int&; p is a pointer to const int
// error: type deduced from i is int; type deduced from &ci is const int
auto &n = i, *p2 = &ci;
```

### List Initialization and `auto`

New

As with other types, we can list initialize (§ 2.2.1, p. 43) a variable defined with `auto`. When we do so, the deduced type depends on which form of initialization we use. If we use direct-initialization (without an =), we can supply only one initializer and the variable's type will be deduced from the initializer:

```
auto a{1};      // type is int, equivalent to auto a = 1
auto a2{1,2};   // error: auto with direct initialization takes only one initializer
```

If we use copy-initialization (with an =), we can supply a comma-separated list of initializers, which must all have the same type. The variable's type will be deduced as a library type named `initializer_list`, which we'll cover in § 6.5.2 (p. 249):

```
auto a3 = {1}; // deduced as initializer_list see § 6.5.2 (p. 249)
auto a4 = {0,1,2,3}; // deduced as initializer_list
auto a5 = {1, 2.0};  // error: element types must match exactly
```

Note

When `auto` is used with list initialization, the type of the variable depends on the form of initialization:

```
auto a{1};      // type is int, equivalent to auto a = 1
auto a3 = {1}; // deduced as initializer_list see § 6.5.2 (p. 249)
```

It is worth noting that the rules for `auto` deduction from a braced initializer list changed in the C++17 release of the standard. Earlier versions of the language, *always* deduced `initializer_list` and the direct-form of initialization could be used with lists of more than one element:

```
auto a{1};     // C++17 type is int, prior releases deduced initializer_list
auto a2{1,2}; // C++17 error, prior releases deduced initializer_list
```

WARNING

The behavior of `auto` with a braced initializer list changed between C++14 and C++17.

## 2.5.3 `decltype`

Sometimes we want to define a variable with a type deduced from an expression but we do not want to use that expression to initialize the variable. For such cases, we can use **`decltype`**, which returns the type of its operand. The compiler analyzes the expression used with `decltype` to determine its type but does not evaluate that expression:

```
decltype(f()) sum = x; // sum has whatever type f returns
```

## EXERCISES SECTION 2.5.2

**Exercise 2.30:** Using the variable definitions from this section, determine which of these assignments are legal and what the result is in those that are:

```
a = 42;   b = 42;   c = 42; d = 42;   e = 42;   g = 42;
```

**Exercise 2.31:** Write a program to check your answers to the previous exercise. If your predictions were wrong, study the examples until you can convince yourself you know what led you to the wrong conclusion.

**Exercise 2.32:** Determine the types deduced in each of the following definitions. Once you've figured out the types, write a program to see whether you were correct.

```
const int i = 42;
auto j = i; const auto &k = i; auto *p = &i;
const auto j2 = i, &k2 = i;
```

Here, the compiler does not call `f`. Instead, the compiler determines what type a call to `f` would return and uses that type as the type for `sum`.

Unlike `auto`, `decltype` does not use the initialization rules to guide its type deduction. Instead, `decltype` returns the "declared type" of the expression to which it is applied. In particular, unlike `auto`, `decltype` preserves top-level `const` and references:

```
const int ci = 0, &cj = ci;
decltype(ci) x = 0; // x has type const int
decltype(cj) y = x; // y has type const int& and is bound to x
decltype(cj) z;     // error: z is a reference and must be initialized
```

Because `cj` was declared as a reference, `decltype(cj)` is a reference type. Like any other reference, `z` must be initialized.

Note: `decltype` is the *only* context in which a variable defined as a reference is not treated as a synonym for the object to which it refers.

### `decltype` and References

When we apply `decltype` to an expression that is not a variable, we get the type that that expression yields. Moreover, as we'll see in § 4.1.1 (p. 143), some expressions will cause `decltype` to yield a reference type. Generally speaking, `decltype` returns a reference type for expressions that yield objects that can stand on the left-hand side of the assignment:

```
// decltype of an expression can be a reference type
int i = 42, *p = &i, &r = i;
decltype(r) a = i;   // ok: a is a reference to the int i
decltype(r + 0) b;   // ok: addition yields an int; b is an (uninitialized) int
decltype(*p) c;      // error: c is int& and must be initialized
```

Here `r` is a reference, so `decltype(r)` is a reference type. If we want the type to which `r` refers, we can use `r` in an expression, such as `r + 0`, which is an expression that yields a value that has a nonreference type.

Unlike addition, the dereference operator yields a result that can appear in the left-hand side of an assignment (§ 2.3.2, p. 55): When we dereference a pointer, we get the object to which the pointer points and we can assign to that resulting object. Thus, the type deduced by `decltype(*p)` is `int&`, not plain `int`.

### Two Ways of Using `decltype` on a Variable

Assuming `v` is a variable, we can write either `decltype(v)` or `decltype((v))` (note the extra parentheses). Whether the variable is parenthesized or not affects the type returned by `decltype`:

- When we apply `decltype` to a variable without any parentheses, we get the declared type of that variable.
- When we wrap the variable's name in one or more sets of parentheses, the compiler evaluates the variable as an expression.

A variable expression can be the left-hand side of an assignment. Thus, `decltype` of a parenthesized variable yields a reference to the type of that variable:

```
int i = 0;
// decltype of a parenthesized variable is always a reference
decltype((i)) d;     // error: d is int& and must be initialized
decltype(i) e;       // ok: e is an (uninitialized) int
```

WARNING

`decltype((`*variable*`))` (note, double parentheses) is always a reference type; `decltype(`*variable*`)` is a reference only if *variable* is a reference.

We'll see in § 16.2.3 (p. 740) that C++14 introduced another form of type deduction: `decltype(auto)`.

## 2.6 Defining Our Own Data Structures

At the most basic level, a data structure is a way to group together related data elements and a strategy for using those data. As one example, our `Sales_item` class groups an ISBN, a count of how many copies of that book had been sold, and the revenue associated with those sales. It also provides a set of operations such as the `isbn` function and the `>>`, `<<`, `+`, and `+=` operators.

In C++, we define our own data types by defining a class. The library types `string`, `istream`, and `ostream` are all defined as classes, as is the `Sales_item` type we used in Chapter 1. C++ support for classes is extensive—in fact, Parts III and IV are largely devoted to describing class-related features. Even though the `Sales_item` class is pretty simple, we won't be able to fully define that class until we learn how to write our own operators in Chapter 14.

### Exercises Section 2.5.3

**Exercise 2.33:** In the following code, determine the type of each variable and the value each variable has when the code finishes:

```
int a = 3, b = 4;
decltype(a) c = a;
decltype((b)) d = a;
++c;
++d;
```

**Exercise 2.34:** Assignment is an example of an expression that yields a reference type. The type is a reference to the type of the left-hand operand. That is, if `i` is an `int`, then the type of the expression `i = x` is `int&`. Using that knowledge, determine the type and value of each variable in this code:

```
int a = 3, b = 4;
decltype(a) c = a;
decltype(a = b) d = a;
```

**Exercise 2.35:** Describe the differences in type deduction between `decltype` and `auto`. Give an example of an expression where `auto` and `decltype` will deduce the same type and an example where they will deduce differing types.

## 2.6.1 Defining the `Sales_data` Type

Although we can't yet write our `Sales_item` class, we can write a more concrete class that groups the same data elements. Our strategy for using this class is that users will be able to access the data elements directly and must implement needed operations for themselves.

Because our data structure does not support any operations, we'll name our version `Sales_data` to distinguish it from `Sales_item`. We'll define our class as follows:

```
struct Sales_data {
    std::string bookNo;
    unsigned units_sold = 0;
    double revenue = 0.0;
};
```

Our class begins with the keyword **`struct`**, followed by the name of the class and a (possibly empty) class body. The class body is surrounded by curly braces and forms a new scope (§ 2.2.4, p. 48). The names defined inside the class must be unique within the class but can reuse names defined outside the class.

The close curly that ends the class body must be followed by a semicolon. The semicolon is needed because we can define variables after the class body:

```
struct Sales_data { /* ... */ } accum, trans, *salesptr;
struct Sales_data { /* ... */ };
Sales_data accum, trans, *salesptr; // better way to define these objects
```

The semicolon marks the end of the (usually empty) list of declarators. Ordinarily, it is a bad idea to define an object as part of a class definition. Doing so obscures the code by combining the definitions of two different entities—the class and a variable—in a single statement.

> **WARNING** It is a common mistake among new programmers to forget the semicolon at the end of a class definition.

## Class Data Members

The class body defines the **members** of the class. Our class has only **data members**. The data members of a class define the contents of the objects of that class type. Each object has its own copy of the class data members. Modifying the data members of one object does not change the data in any other `Sales_data` object.

We define data members the same way that we define normal variables: We specify a base type followed by a list of one or more declarators. Our class has three data members: a member of type `string` named `bookNo`, an `unsigned` member named `units_sold`, and a member of type `double` named `revenue`. Each `Sales_data` object will have these three data members.

Most classes use a feature that we'll cover in § 7.1.4 (p. 276) to initialize data members, but for simple classes such as the current version of `Sales_data`, it can be simpler to supply a **member initializer**. Whenever a `Sales_data` object is created, the member initializers will be used to initialize the data members of that object. Members without an initializer are default initialized (§ 2.2.1, p. 44). Thus, when we define `Sales_data` objects, `units_sold` and `revenue` will be initialized to 0, and `bookNo` will be initialized to the empty string.

In-class initializers are restricted as to the form (§ 2.2.1, p. 43) we can use: They must either be enclosed inside curly braces or follow an = sign. We may not specify an in-class initializer inside parentheses.

In § 7.2 (p. 282), we'll see that C++ has a second keyword, `class`, that we can use to define our own data structures. We'll explain in that section why we use `struct` here. Until we cover additional class-related features in Chapter 7, you should use `struct` to define your own data structures.

## No `inline` or `constexpr` Data Members

Recall that `inline` is used to define an object that can be shared across multiple files (§ 2.2.2, p. 46). As we've just seen, the data members of a class define the data contained in each object of the class' type. That is, each object contains its own data; the data members of one object are independent from and not shared with any other object.

As such, it should not be surprising that we can have no (ordinary) data members that are `inline`. Because `constexpr`s are implicitly `inline` (§ 2.4.4, p. 66), (ordinary) data members also cannot be `constexpr`. We'll see in § 7.7 (p. 315) that classes can define data members that are independent of any object of the class type. As we'll see, those members can be `inline` and `constexpr`.

**EXERCISES SECTION 2.6.1**

**Exercise 2.36:** Compile the following program to see what happens when you forget the semicolon after a class definition. Remember the message for future reference.

```
struct Foo { /* empty */ } // Note: no semicolon
int main()
{
    return 0;
}
```

**Exercise 2.37:** Write your own version of the `Sales_data` class.

## 2.6.2 Using the `Sales_data` Class

Unlike the `Sales_item` class, our `Sales_data` class does not provide any operations. Users of `Sales_data` have to write whatever operations they need. As an example, we'll write a version of the program from § 1.5.2 (p. 23) that printed the sum of two transactions. The input to our program will be transactions such as

```
0-201-78345-X 3 20.00
0-201-78345-X 2 25.00
```

Each transaction holds an ISBN, the count of how many books were sold, and the price at which each book was sold.

### Adding Two `Sales_data` Objects

Because `Sales_data` provides no operations, we will have to write our own code to do the input, output, and addition operations. We'll assume that our `Sales_data` class is defined inside `Sales_data.h`. We'll see how to define this header in § 2.6.3 (p. 78).

Because this program will be longer than any we've written so far, we'll explain it in separate parts. Overall, our program will have the following structure:

```
#include <iostream>
#include <string>
#include "Sales_data.h"
int main()
{
    Sales_data data1, data2;
    // code to read into data1 and data2
    // code to check whether data1 and data2 have the same ISBN
    //      and if so print the sum of data1 and data2
}
```

As in our original program, we begin by including the headers we'll need and defining variables to hold the input. Note that unlike the `Sales_item` version, our new program includes the `string` header. We need that header because our code will have to manage the `bookNo` member, which has type `string`.

### Reading Data into a `Sales_data` Object

Although we won't describe the library `string` type in detail until Chapters 3 and 10, we need to know only a little bit about `string`s in order to define and use our ISBN member. The `string` type holds a sequence of characters. Its operations include the `>>`, `<<`, and `==` operators to read, write, and compare `string`s, respectively. With this knowledge we can write the code to read our two transactions:

```
// code to read into data1 and data2
double price = 0;   // price per book, used to calculate total revenue
// read the first transactions: ISBN, number of books sold, price per book
std::cin >> data1.bookNo >> data1.units_sold >> price;
// calculate total revenue from price and units_sold
data1.revenue = data1.units_sold * price;
// read the second transaction
std::cin >> data2.bookNo >> data2.units_sold >> price;
data2.revenue = data2.units_sold * price;
```

We start by defining a local variable named `price` to hold the price element of our transactions. When we read the transactions, we can store the ISBN and number of books sold into the `Sales_data` object directly. We'll use `price` and `unit_sold` to calculate `revenue`. The input statement uses the dot operator (§ 1.5.2, p. 23) to read into the `bookNo` and `units_sold` members of the object named `data1`. The next statement assigns the product of `data1.units_sold` and `price` into the `revenue` member of `data1`. Our program then repeats the same code to read a transaction into `data2`.

### Printing the Sum of Two `Sales_data` Objects

Our other task is to check that the transactions are for the same ISBN. If so, we'll print their sum; otherwise, we'll print an error message:

```
// code to check whether data1 and data2 have the same ISBN
//     and if so print the sum of data1 and data2
if (data1.bookNo == data2.bookNo) {
    unsigned totalCnt = data1.units_sold + data2.units_sold;
    double totalRevenue = data1.revenue + data2.revenue;
    // print: ISBN, total sold, total revenue, average price per book
    std::cout << data1.bookNo << " " << totalCnt
              << " " << totalRevenue << " ";
    if (totalCnt != 0)
        std::cout << totalRevenue/totalCnt << std::endl;
    else
        std::cout  << "(no sales)" << std::endl;
    return 0;   // indicate success
} else {   // transactions weren't for the same ISBN
    std::cerr << "Data must refer to the same ISBN"
              << std::endl;
    return -1; // indicate failure
}
```

In the first if, we compare the bookNo members of data1 and data2. If those members are the same ISBN, we execute the code inside the curly braces. That code adds the components of our two variables. Because we'll need to print the average price, we start by computing the total of units_sold and revenue and store those in totalCnt and totalRevenue, respectively. We print those values. Next we check that there were books sold and, if so, print the computed average price per book. If there were no sales, we print a message noting that fact.

## Structured Bindings to Access the Members in a Class-Type Object

Starting with C++17, we have another way to access the data members of simple classes such as Sales_data. We can use a **structured binding** declaration. A structured binding has the form: New

```
auto [identifier-list] = expression;
auto [identifier-list] { expression };
```

where *identifier-list* is a comma separated list of identifer names and *expression* is an initializer. A structured binding declaration decomposes the given *expression* into its constituent parts and stores those parts into the variables named in *identifier-list*. The *expression* must be a type that meets certain requirements that we'll cover in § 7.6 (p. 311), § 3.5.2 (p. 122), and § 17.1 (p. 776). Our Sales_data class meets these requirements, so we can use a structured binding to access the data members of an object of type Sales_data.

When used with a class type object, the number of names in *identifier-list* must be exactly equal to the number of data members in the class. The data members are used to initialize the variables in the same order as they appear in the class definition. Thus, a structured binding applied to a Sales_data object must supply three variable names. The first variable will have type string and be initialized by the bookNo member. The second and third variables will be an unsigned and a double, respectively. These variables will be initialized from the units_sold and revenue members:

```
// copies the data members from Sales_data object named data1
auto [book, units, revenue] = data1;
```

This statement is equivalent to:

```
// equivalent to auto [book, units, revenue] = data1
std::string book = data1.bookNo;
unsigned units= data1.units_sold;
double revenue = data1.revenue;
```

We can also use a structure binding to bind a reference to each data member. For example, we might use a structured binding to simplify code to read into a Sales_data object:

```
// these variables are aliases for the data members of data1; nothing is copied
auto &[book, units, revenue] = data1; // note variables are references
std::cin >> book >> units >> revenue; // read into the members of data1
// remember to calculate the total revenue;
revenue *= units; // equivalent to revenue = revenue * units
```

Note the our declaration now says `auto &`. The reference symbol applies to each of the variables in the list. In this case, we are defining three reference variables, a `string&`, an `unsigned&`, and a `double&`. As with any other reference, we can write to the contents of the object to which the reference is bound (§ 2.3.1, p. 51), which we do in the input statements. That is, `std::cin >> book` changes the value of `data1.bookNo`.

If we only need to read, but not write, to the data members we can define the variables as references to `const` (§ 2.4.1, p. 61):

```
// read-only access to data1
const auto &[book, units, revenue] = data1;
std::cout << book << " " << units << " " << revenue;
if (units != 0)
    std::cout << " " << revenue/units << std::endl;
else
    std::cout << 0 << std::endl;
```

**EXERCISES SECTION 2.6.2**

**Exercise 2.38:** Use your `Sales_data` class to rewrite the exercises in § 1.5.1 (p. 22), § 1.5.2 (p. 24), and § 1.6 (p. 25). For now, you should define your `Sales_data` class in the same file as your `main` function.

### 2.6.3 Writing Our Own Header Files

Ordinarily, a given class is used in more than one file. Like other entities that are shared across files, the class definition must be the same in each file. To ensure that the files share the same definition, classes are stored in headers. Typically, the header name is derived from the name of the class. For example, as we've already seen, we will define `Sales_data` in a file named `Sales_data.h`.

Headers (usually) contain entities `inline`, `const`, and `constexpr` variables that can be defined only once in any given file. However, headers often need to use facilities from other headers. For example, because our `Sales_data` class has a `string` member, `Sales_data.h` must `#include` the `string` header. Programs that use `Sales_data` also need to include the `string` header in order to use the `bookNo` member. As a result, programs that use `Sales_data` will include the `string` header twice: once directly and once as a side effect of including `Sales_data.h`. Because a header might be included more than once, we need to write our headers in a way that is safe even if the header is included multiple times.

Note

Whenever a header is updated, the source files that use that header must be recompiled to get the new or changed declarations.

## A Brief Introduction to the Preprocessor

We can use facilities defined by the **preprocessor** to avoid multiple inclusion. The preprocessor—which C++ inherits from C—is a program that runs before the compiler and changes the source text of our programs. Our programs already use one preprocessor facility, `#include`. When the preprocessor sees a `#include`, it replaces the `#include` with the contents of the specified header.

C++ programs also use the preprocessor to define **header guards**. Header guards rely on preprocessor variables (§ 2.3.2, p. 54). Preprocessor variables have one of two possible states: defined or not defined. The **`#define`** directive takes a name and defines that name as a preprocessor variable. Two other directives **`#ifdef`** and **`#ifndef`** test whether a given preprocessor variable has or has not been defined. If the variable has been defined, then everything following `#ifdef` is processed up to the matching **`#endif`**. Similarly, if the variable has not been defined, code following `#ifndef` is processed. We can use these facilities to guard against multiple inclusion as follows:

```
#ifndef SALES_DATA_H
#define SALES_DATA_H
#include <string>
struct Sales_data {
    std::string bookNo;
    unsigned units_sold = 0;
    double revenue = 0.0;
};
#endif
```

The first time `Sales_data.h` is included, the `#ifndef` test will succeed. The preprocessor will process the lines following `#ifndef` up to the `#endif`. As a result, the preprocessor variable `SALES_DATA_H` will be defined and the contents of `Sales_data.h` will be copied into our program. If we include `Sales_data.h` later on in the same file, the `#ifndef` directive will be false. The lines between it and the `#endif` directive will be ignored.

**WARNING** Preprocessor names do not respect C++ scoping rules. Therefore, these names must not duplicate any other name used in the program.

Preprocessor variables, including names of header guards, must be unique throughout the program. Typically we ensure uniqueness by basing the guard's name on the name of a class in the header. To avoid name clashes with other entities in our programs, preprocessor variables usually are written in all uppercase.

### Exercises Section 2.6.3

**Exercise 2.39:** Write your own `Sales_data.h` and use it to compile the exercise from § 2.6.2 (p. 78).

# Chapter Summary

Types are fundamental to all programming in C++. Each type defines the storage requirements and the operations that may be performed on objects of that type. The language provides a set of fundamental built-in types such as `int` and `char`, which are closely tied to their representation on the machine's hardware. Types can optionally be `const`; a `const` object must be initialized and, once initialized, its value may not be changed. In addition, we can define compound types, such as pointers or references, which are defined in terms of other types. The language lets us define our own types by defining classes. The library uses the class facility to provide a set of higher-level abstractions such as the IO and `string` types.

# Defined Terms

**address** Number by which a byte in memory can be found.

**alias declaration** Defines a synonym for another type: `using` *name* = *type* declares *name* as a synonym for the type *type*.

**arithmetic types** Built-in types representing boolean values, characters, integers, and floating-point numbers.

**array** Data structure that holds a collection of objects. See § 3.5 (p. 119).

**auto** Type specifier that deduces the type of a variable from its initializer.

**base type** type specifier, possibly qualified by `const`, that precedes the declarators in a declaration. The base type provides the common type on which the declarators in a declaration can build.

**bind** Associating a name with a given entity so that uses of the name are uses of the underlying entity. For example, a reference is a name that is bound to an object.

**byte** Smallest addressable unit of memory. On most machines a byte is 8 bits.

**class member** Part of a class.

**compound type** A type that is defined in terms of another type.

**const** Type qualifier used to define objects that may not be changed. `const` objects must be initialized, because there is no way to give them a value after they are defined.

**const pointer** Pointer that is `const`.

**const reference** Colloquial synonym for reference to `const`.

**constant expression** Expression that can be evaluated at compile time.

**constexpr** Variable that represents a constant expression. § 6.5.1 (p. 247) covers `constexpr` functions.

**conversion** Process whereby a value of one type is transformed into a value of another type. The language defines conversions among the built-in types.

**copy initialization** Form of initialization that uses an `=`.

**data member** Data elements that constitute an object. Every object of a given class has its own copies of the class' data members. Data members may be initialized when declared inside the class.

**declaration** Asserts the existence of a variable, function, or type defined elsewhere.

**declarator** Part of a declaration that includes a name and optional type modifiers.

**decltype** Type specifier that deduces the type of a variable or an expression.

**default initialization** How objects are initialized if no initializer is given. Objects of built-in type defined at global scope are initialized to 0; those defined at local scope have undefined values. Each class defines how objects of its type are initialized.

**definition** A declaration that allocates storage for a variable of a specified type and optionally initializes the variable.

**direct initialization** Form of initialization that does not include an =.

**escape sequence** Alternative mechanism for representing characters, particularly for those without printable representations. An escape sequence is a backslash followed by a character, three or fewer octal digits, or an `x` followed by a hexadecimal number.

**global scope** The scope that is outside all other scopes.

**header guard** Preprocessor variable used to prevent a header from being included more than once in a single file.

**identifier** Sequence of characters that make up a name. Identifiers are case-sensitive.

**in scope** Name that is visible from the current scope.

**initialized** A variable given an initial value when it is defined. Variables usually should be initialized.

**inline variable** Variable that may be defined in more than one file. Every file that uses the variable must include an `inline` definition for it and the definitions must match in all files.

**inner scope** Scope that is nested inside another scope.

**integral type** See arithmetic type.

**list initialization** Form of initialization that uses curly braces.

**literal** A value such as a number, character, or string of characters Literal characters are enclosed in single quotes, literal strings in double quotes.

**local scope** Synonym for block scope.

**low-level const** A `const` that is integral to the type and is never ignored.

**member** Part of a class.

**member initializer** Initializer provided as part of the declaration of a class data member. Member initializers must follow an = symbol or be enclosed inside curly braces.

**nonprintable character** A character with no visible representation, such as a control character, a backspace, newline, and so on.

**null pointer** Pointer whose value is 0. A null pointer is valid but does not point to any object.

**nullptr** Literal constant that denotes the null pointer.

**object** A region of memory that has a type. A variable is an object that has a name.

**outer scope** Scope that encloses another scope.

**pointer** An object that can hold the address of an object, the address one past the end of an object, or zero.

**pointer to const** Pointer that can hold the address of a `const` object. A pointer to `const` may not be used to change the value of the object to which it points.

**preprocessor** Program that runs as part of compilation of a C++ program.

**preprocessor variable** Variable managed by the preprocessor. The preprocessor replaces each preprocessor variable by its value before our program is compiled.

**reference** An alias for another object.

**reference to `const`** A reference that may not change the value of the object to which it refers. A reference to `const` may be bound to a `const` object, a non`const` object, or the result of an expression.

**scope** The portion of a program in which names have meaning. C++ has several levels of scope:

- **global**—names defined outside any other scope
- **class**—names defined inside a class
- **namespace**—names defined inside a namespace
- **block**—names defined inside a block

Scopes nest. Once a name is declared, it is accessible until the end of the scope in which it was declared.

**separate compilation** Ability to split a program into multiple separate source files.

**signed** Integer type that holds negative or positive values, including zero.

**string** Library type representing variable-length sequences of characters.

**struct** Keyword used to define a class.

**structured binding** A declaration that decomposes an aggregate object into a list of named variables. In the case of an array, the elements of the array are stored in order into a specified list of variables.

**temporary** Unnamed object created by the compiler while evaluating an expression. A temporary exists for the duration of the full expression in which it was created.

**top-level const** The `const` that specifies that an object may not be changed.

**type alias** A name that is a synonym for another type. Defined through either a `typedef` or an alias declaration.

**type checking** Term used to describe the process by which the compiler verifies that the way objects of a given type are used is consistent with the definition of that type.

**type specifier** The name of a type.

**typedef** Defines an alias for another type. When `typedef` appears in the base type of a declaration, the names defined in the declaration are type names.

**undefined** Usage for which the language does not specify a meaning. Knowingly or unknowingly relying on undefined behavior is a great source of hard-to-track runtime errors, security problems, and portability problems.

**uninitialized** Variable defined without an initial value. In general, trying to access the value of an uninitialized variable results in undefined behavior.

**unsigned** Integer type that holds only values greater than or equal to zero.

**variable** A named object or reference.

**void*** Pointer type that can point to any non`const` type. Such pointers may not be dereferenced.

**void type** Special-purpose type that has no operations and no value. It is not possible to define a variable of type `void`.

**word** The natural unit of integer computation on a given machine. Usually a word is large enough to hold an address. On a 32-bit machine a word is typically 4 bytes.

**& operator** Address-of operator. Yields the address of the object to which it is applied.

*** operator** Dereference operator. Dereferencing a pointer returns the object to which the pointer points. Assigning to the result of a dereference assigns a new value to the underlying object.

**#define** Preprocessor directive that defines a preprocessor variable.

**#endif** Preprocessor directive that ends an `#ifdef` or `#ifndef` region.

**#ifdef** Preprocessor directive that determines whether a given variable is defined.

**#ifndef** Preprocessor directive that determines whether a given variable is not defined.

# CHAPTER 3
# STRINGS, VECTORS, AND ARRAYS

## CONTENTS



In addition to the built-in types covered in Chapter 2, C++ defines a rich library of abstract data types. Among the most important library types are `string`, which supports variable-length character strings, and `vector`, which defines variable-size collections. Associated with `string` and `vector` are companion types known as iterators, which are used to access the characters in a `string` or the elements in a `vector`.

The `string` and `vector` types defined by the library are abstractions of the more primitive built-in array type. This chapter covers arrays and introduces the library `vector` and `string` types.

*The built-in types* that we covered in Chapter 2 are defined directly by the C++ language. These types represent facilities present in most computer hardware, such as numbers or characters. The standard library defines a number of additional types of a higher-level nature that computer hardware usually does not implement directly.

In this chapter, we'll introduce two of the most important library types: `string` and `vector`. A `string` is a variable-length sequence of characters. A `vector` holds a variable-length sequence of objects of a given type. We'll also cover the built-in array type. Like the other built-in types, arrays represent facilities of the hardware. As a result, arrays are less convenient to use than the library `string` and `vector` types.

Before beginning our exploration of the library types, we'll look at a mechanism for simplifying access to the names defined in the library.

## 3.1 Namespace `using` Declarations

Up to now, our programs have explicitly indicated that each library name we use is in the `std` namespace. For example, to read from the standard input, we write `std::cin`. These names use the scope operator (`::`) (§ 1.2, p. 8), which says that the compiler should look in the scope of the left-hand operand for the name of the right-hand operand. Thus, `std::cin` says that we want to use the name `cin` from the namespace `std`.

Referring to library names with this notation can be cumbersome. Fortunately, there are easier ways to use namespace members. The safest way is a **`using` declaration**. § 18.2.2 (p. 845) covers another way to use names from a namespace.

A `using` declaration lets us use a name from a namespace without qualifying the name with a `namespace_name::` prefix. A `using` declaration has the form

```
using namespace::name1, namespace::name2, ...;
```

which includes a comma-separated list of one or more names each of which is qualified by the name of the `namespace` in which that name can be found. Once a `using` declaration has been made, we can access whichever names were declared directly:

```
#include <iostream>
// using declaration; subsequent uses of cin or cout
// are synonyms for std::cin and std::cout, respectively
using std::cin, std::cout;
int main()
{
    int i;
    cin >> i;           // ok: cin is a synonym for std::cin
    cout << i << endl; // error: no using declaration for endl
    cout << i << std::endl; // ok: explicitly use endl from namepsace std
    return 0;
}
```

Prior to C++17, a using declaration could declare only one name. For older compilers, we'd have to write these declarations as: New

```
using std::cin; using std::cout;
using std::endl;
```

Because C++ programs are largely free-format, we can write more than one using declaration on each line. However, each using declaration forms a statement and so must end with a semicolon.

### Headers Should Not Include using Declarations

Like any other name, a name introduced by a using declaration has a scope. (§ 2.2.4, p. 48). Like a variable declaration, a using declaration may appear at global scope (outside any curlies) or in a local scope (inside a block) and the name is visible from the point of the using declaration until the end of the scope in which the using appeared.

Code inside headers (§ 2.6.3, p. 78) ordinarily should only have using declarations that appear inside a block. The reason is that the contents of a header are copied into the including program's text. If a header has a using declaration in the global scope, then every program that includes that header gets that same using declaration. As a result, a program that didn't intend to use the specified library name might encounter unexpected name conflicts.

### A Note to the Reader

From this point on, our examples will assume that using declarations have been made for the names we use from the standard library. Thus, we will refer to cin, not std::cin, in the text and in code examples.

Moreover, to keep the code examples short, we won't show the using declarations, nor will we show the necessary #include directives. Table A.1 (p. 922) in Appendix A lists the names and corresponding headers for standard library names we use in this Primer.

WARNING

Readers should be aware that they must add appropriate #include and using declarations to our examples before compiling them.

**EXERCISES SECTION 3.1**

**Exercise 3.1:** Rewrite the exercises from § 1.4.1 (p. 13) and § 2.6.2 (p. 78) with appropriate using declarations.

## 3.2 Library string Type

A **string** is a variable-length sequence of characters. To use the string type, we must include the string header. Because it is part of the library, string is defined in the `std` namespace. Our examples assume the following code:

```
#include <string>
using std::string;
```

This section describes the most common `string` operations; § 9.5 (p. 396) will cover additional operations.

### 3.2.1 Defining and Initializing `strings`

Each class controls how objects of its type can be defined and initialized. Because class objects tend to be more complex than the simple types defined by the language, a class often requires that more than one value be supplied in order to initialize objects of its type.

In C++, when we define and initialize an object of class type, we say that we **construct** that object. A class may define many different ways to construct objects of its type. Each way must be distinguished from the others either by the number or type(s) of the values that must be supplied. Table 3.1 lists the most common ways to initialize `strings`. Some examples:

```
string s1;            // default initialization; s1 is the empty string
string s2 = s1;       // s2 is a copy of s1
string s3 = "hiya";   // s3 is hiya
string s4(10, 'c');   // s4 is cccccccccc
```

We can default initialize a `string` (§ 2.2.1, p. 44), which creates an empty `string`; that is, a `string` with no characters. When we supply a string literal (§ 2.1.3, p. 39), the characters from that literal—up to but not including the null character at the end of the literal—are copied into the newly created `string`. When we supply a count and a character, the `string` contains that many copies of the given character.

**Table 3.1: Ways to Initialize a `string`**

| | |
|---|---|
| `string s1` | Default initialization; `s1` is the empty string. |
| `string s2(s1)` | Initialize `s2` as a copy of `s1`. |
| `string s2 = s1` | |
| `string s3("value")` | Initialize `s3` as a copy of the characters in the string literal, not including the null. |
| `string s3 = "value"` | |
| `string s4(n, 'c')` | Initialize `s4` with `n` copies of the character `'c'`. |

#### Direct and Copy Forms of Initialization

In § 2.2.1 (p. 43) we saw that C++ has several different forms of initialization. Using `strings`, we can start to understand how these forms differ from one another. The `=` form takes a value and copies that value into the object being constructed. Hence, this form is known as **copy initialization**. When we omit the `=`, we're using **direct**

**initialization**. This form can take any number of values. The class uses those values in a class-dependent way to construct the object.

When we want to construct an object from a single value, we can use either the direct or copy form of initialization. When we need to supply more than one value, we must use the direct form of initialization:

```
string s5 = "hiya";   // copy initialization
string s6("hiya");    // direct initialization
string s7(10, 'c');   // direct initialization; s7 is cccccccccc
```

If we want to use the copy form of initialization and we need to supply more than one value, we can do so by explicitly constructing a `string` to copy:

```
string s8 = string(10, 'c'); // copy initialization; s8 is cccccccccc
```

**KEY CONCEPT: COPY INITIALIZATION**

**Conceptually, when we define a variable and use `=` to initialize it we are saying that the new variable should be a copy of the value of the initializer. For example, when we say `int i = j`, we are saying that `i` is a copy of whatever value is stored in `j`. Simiarly, when we write**

```
string s8 = string(10, 'c'); // copy initialization; s8 is cccccccccc
```

**we're asking the compiler to copy the value generated by `string(10, c)` into the variable `s8`.**

**Although conceptually we have asked to copy the the value `string(10, c)` into `s8`, the compiler knows that the the object we are copying is a temporary (§ 2.4.1, p. 62). That temporary will go away as soon as it is copied into `s8`. Rather than creating the temporary in order to copy it, the compiler directly initializes `s8` as a string of ten `c`s. That is, the declaration of `s8` executes exactly the same way as the declaration of `s7`.**

**In the early days of C++, compilers generally would create and copy temporaries in contexts such as the initialization of `s8`. Over time, compilers got smarter about avoiding such copies and typically an initialization like this one would not involve a copy. However, prior to the C++17 standard, compliers were allowed to generate and copy a temporary in these cases. Therefore, even though a copy is not made, for historical reasons, we still refer to this form of initialization as "copy" initialization**

**For many types, the guarantee that a copy isn't made has at most a performance impact. We'll see in § 13.5 (p. 584) that for some classes the fact that there is no copy is crucial. We'll have more to say about copy initialization in § 13.1.1 (p. 553)**

### 3.2.2 Operations on `strings`

Along with defining how objects are created and initialized, a class also defines the operations that objects of the class type can perform. A class can define operations with names, such as the `isbn` function of our `Sales_item` class (§ 1.5.2, p. 23). A class also can define what various operator symbols, such as `<<` or `+`, mean when applied to objects of the class' type. Table 3.2 (overleaf) lists the most common `string` operations.

**Table 3.2: `string` Operations**

| | |
|---|---|
| `os << s` | Writes `s` onto output stream `os`. Returns `os`. |
| `is >> s` | Reads whitespace-separated string from `is` into `s`. Returns `is`. |
| `getline(is, s)` | Reads a line of input from `is` into `s`. Returns `is`. |
| `s.empty()` | Returns `true` if `s` is empty; otherwise returns `false`. |
| `s.size()` | Returns the number of characters in `s`. |
| `s[n]` | Returns a reference to the `char` at position `n` in `s`; positions start at 0. |
| `s1 + s2` | Returns a `string` that is the concatenation of `s1` and `s2`. |
| `s1 = s2` | Replaces characters in `s1` with a copy of `s2`. |
| `s += c` | Appends a copy of the character `c` to `s`. |
| `s1 += s2` | Appends a copy of the characters in `s2` to `s1`. |
| `s1 == s2`<br>`s1 != s2` | The `strings` `s1` and `s2` are equal if and only if they contain the same characters. Equality is case-sensitive. |
| `<, <=, >, >=` | Comparisons are case-sensitive and use dictionary ordering. |

## Reading and Writing `strings`

As we saw in Chapter 1, we use the `iostream` library to read and write values of built-in types such as `int`, `double`, and so on. We use the same IO operators to read and write `strings`:

```
// Note: #include and using declarations must be added to compile this code
int main()
{
    string s;          // empty string
    cin >> s;          // read a whitespace-separated string into s
    cout << s << endl; // write s to the output
    return 0;
}
```

This program begins by defining a default-initialized `string` named `s`. The next line reads the standard input, storing what is read in `s`. The `string` input operator reads and discards any leading whitespace (e.g., spaces, newlines, tabs). It then reads characters until the next whitespace character is encountered.

So, if the input to this program is **`   Hello World!   `** (note leading and trailing spaces), then the output will be **`Hello`** with no extra spaces.

Like the input and output operators on the built-in types, the `string` operators return their left-hand operand as their result. Thus, we can chain together multiple reads or writes:

```
string s1, s2;
cin >> s1 >> s2; // read first input into s1, second into s2
cout << s1 << s2 << endl; // write both strings
```

If we give this code the same input, our output would be "**`HelloWorld!`**"

### Reading an Unknown Number of `strings`

In § 1.4.3 (p. 15) we wrote a program that read an unknown number of `int` values. We can write a similar program that reads `string`s instead:

```
int main()
{
    for (string word; cin >> word; )   //  read until end-of-file
        cout << word << endl; //  write each word followed by a new line
    return 0;
}
```

In this program, we read into a `string`, not an `int`. Otherwise, the `for` condition executes similarly to the one in our previous program. The condition tests the stream after the read completes. If the stream is valid—it hasn't hit end-of-file or encountered an invalid input—then the body of the `for` is executed. In this case, the `for` body prints the value we read on the standard output. Once we hit end-of-file (or invalid input), we fall out of the `for`.

### Using `getline` to Read an Entire Line

Sometimes we do not want to ignore the whitespace in our input. In such cases, we can use the **`getline`** function instead of the `>>` operator. The `getline` function takes an input stream and a `string`. This function reads the given stream up to and including the first newline and stores what it read—*not including* the newline—in its `string` argument. After `getline` sees a newline, even if it is the first character in the input, it stops reading and returns. If the first character in the input is a newline, then the resulting `string` is the empty `string`.

Like the input operator, `getline` returns its `istream` argument. As a result, we can use `getline` as a condition just as we can use the input operator as a condition (§ 1.4.3, p. 15). For example, we can rewrite the previous program that wrote one word per line to write a line at a time instead:

```
int main()
{
    //  read input a line at a time until end-of-file
    for (string line; getline(cin, line); )
        cout << line << endl;   //  print the line we just read
    return 0;
}
```

Because `line` does not contain a newline, we must write our own. As usual, we use `endl` to end the current line and flush the buffer.

> **Note:** The newline that causes `getline` to return is discarded; the newline is *not* stored in the `string`.

### The `string` `empty` and `size` Operations

The **`empty`** function does what one would expect: It returns a `bool` (§ 2.1, p. 32) indicating whether the `string` is empty. Like the `isbn` member of `Sales_item`

(§ 1.5.2, p. 23), `empty` is a member function of `string`. To call this function, we use the dot operator (§ 1.5.2, p. 23) to specify the object on which we want to run the `empty` function.

We can revise the previous program to print only lines that are not empty:

```
// read input a line at a time and discard blank lines
for (string line; getline(cin, line); )
    if (!line.empty())
        cout << line << endl;
```

The condition uses the logical NOT operator (the **`!` operator**). This operator returns the inverse of the `bool` value of its operand. In this case, the condition is `true` if `str` is not empty.

The **`size`** member returns the length of a `string` (i.e., the number of characters in it). We can use `size` to print only lines longer than 80 characters:

```
// read input a line at a time and print lines that are longer than 80 characters
for (string line; getline(cin, line); )
    if (line.size() > 80)
        cout << line << endl;
```

### The `string::size_type` Type

It might be logical to expect that `size` returns an `int` or, thinking back to § 2.1.1 (p. 35), an `unsigned`. Instead, `size` returns a `string::size_type` value. This type requires a bit of explanation.

The `string` class—and most other library types—defines several companion types. These companion types make it possible to use the library types in a machine-independent manner. The type **`size_type`** is one of these companion types. To use the `size_type` defined by `string`, we use the scope operator to say that the name `size_type` is defined in the `string` class.

Although we don't know the precise type of `string::size_type`, we do know that it is a type alias (§ 2.5.1, p. 67) for an unsigned type (§ 2.1.1, p. 32) big enough to hold the size of any `string`. Any variable used to store the result from the `string` `size` operation should be of type `string::size_type`.

Admittedly, it can be tedious to type `string::size_type`. Fortunately, we can use `auto` or `decltype` (§ 2.5.2, p. 68) and let the compiler do the work:

```
auto len = line.size(); // len has type string::size_type
```

Because `size` returns an unsigned type, it is essential to remember that expressions that mix signed and unsigned data can have surprising results (§ 2.1.2, p. 36). For example, if `n` is an `int` that holds a negative value, then `s.size() < n` will almost surely evaluate as `true`. It yields `true` because the negative value in `n` will convert to a large unsigned value.

Tip

> You can avoid problems due to conversion between `unsigned` and `int` by not using `int`s in expressions that use `size()`.

### Comparing strings

The string class defines several operators that compare strings. These operators work by comparing the characters of the strings. The comparisons are case-sensitive—upper- and lowercase versions of a letter are different characters.

The equality operators (== and !=) test whether two strings are equal or unequal, respectively. Two strings are equal if they are the same length and contain the same characters. The relational operators <, <=, >, >= test whether one string is less than, less than or equal to, greater than, or greater than or equal to another. These operators use the same strategy as a (case-sensitive) dictionary:

1. If two strings have different lengths and if every character in the shorter string is equal to the corresponding character of the longer string, then the shorter string is less than the longer one.
2. If any characters at corresponding positions in the two strings differ, then the result of the string comparison is the result of comparing the first character at which the strings differ.

As an example, consider the following strings:

```
string str = "Hello";
string phrase = "Hello World";
string slang  = "Hiya";
```

Using rule 1, we see that str is less than phrase. By applying rule 2, we see that slang is greater than both str and phrase.

### Assignment for strings

In general, the library types strive to make it as easy to use a library type as it is to use a built-in type. To this end, most of the library types support assignment. In the case of strings, we can assign one string object to another:

```
string st1(10, 'c'), st2; // st1 is cccccccccc; st2 is an empty string
st1 = st2; // assignment: replace contents of st1 with a copy of st2
           // both st1 and st2 are now the empty string
```

### Adding Two strings

Adding two strings yields a new string that is the concatenation of the left-hand followed by the right-hand operand. That is, when we use the plus operator (+) on strings, the result is a new string whose characters are a copy of those in the left-hand operand followed by those from the right-hand operand. The compound assignment operator (+=) (§ 1.4.1, p. 12) appends the right-hand operand to the left-hand string:

```
string s1  = "hello, ", s2 = "world\n";
string s3 = s1 + s2;   // s3 is hello, world\n
s1 += s2;   // equivalent to s1 = s1 + s2
```

### Adding Character String Literals and `strings`

As we saw in § 2.1.2 (p. 34), we can use one type where another type is expected if there is a conversion from the given type to the expected type. The `string` library lets us convert both character literals and character string literals (§ 2.1.3, p. 39) to `strings`. Because we can use these literals where a `string` is expected, we can rewrite the previous program as follows:

```
string s1 = "hello", s2 = "world";  // no punctuation in s1 or s2
string s3 = s1 + ", " + s2 + '\n';
```

When we mix `strings` and string or character literals, at least one operand to each + operator must be of `string` type:

```
string s4 = s1 + ", ";           // ok: adding a string and a literal
string s5 = "hello" + ", ";      // error: no string operand
string s6 = s1 + ", " + "world"; // ok: each + has a string operand
string s7 = "hello" + ", " + s2; // error: can't add string literals
```

The initializations of `s4` and `s5` involve only a single operation each, so it is easy to see whether the initialization is legal. The initialization of `s6` may appear surprising, but it works in much the same way as when we chain together input or output expressions (§ 1.2, p. 7). This initialization groups as

```
string s6 = (s1 + ", ") + "world";
```

The subexpression `s1 + ", "` returns a `string`, which forms the left-hand operand of the second + operator. It is as if we had written

```
string tmp = s1 + ", ";  // ok: + has a string operand
s6 = tmp + "world";      // ok: + has a string operand
```

On the other hand, the initialization of `s7` is illegal, which we can see if we parenthesize the expression:

```
string s7 = ("hello" + ", ") + s2;  // error: can't add string literals
```

Now it should be easy to see that the first subexpression adds two string literals. There is no way to do so, and so the statement is in error.

> **WARNING** For historical reasons, and for compatibility with C, character string literals do *not* have type `string`. It is essential to remember that these types differ when mixing character string literals and library `strings`.

### Literals of Type `string`

When we use character-string literals with library `strings` we must remember that the type of a character-string literal is array of `const char`, not `string` (§ 2.1.3, p. 39). As we've just seen, in some contexts, the difference in type between character-string literals and `strings` can be surprising.

New The `string` library now allows us to write literals that are of type `string`. As with built-in literals (§ 2.1.3, p. 38), we use a suffix to distinguish among different types of literals. In the case of `string` literals, the suffix is `s`. For example:

```
"hello"s
```

is a literal of type `string`. However, unlike built-in literals, literals of type `string` are generated at runtime by calling a library function. When we write `"hello"s`, the compiler transforms this code into something like:

```
special_function("hello", 5)
```

where *special_function* is a name that we'll explain in § 14.7 (p. 623) and `5` is the number of characters in the given literal.

Now let's look again at the initialization of `s5` above:

```
string s5 = "hello" + ", ";       // error: no string operand
```

If we rewrite this expression using a `string` literal, it compiles as expected:

```
string s5 = "hello"s + ", "; // ok: first operand has type string
```

Alternatively, we could write:

```
string s5 = "hello" + ", "s; // ok: second operand has type string
```

with the same effect.

The fact that literals of type `string` are really function calls has two implications. First, even though we don't write a function call explicitly, we still have to provide a `using` declaration so that the compiler knows we want to use the function that generates literals of type `string`. To access this function, we need a new form of `using` declaration, which we'll explain in § 18.2.2 (p. 845):

```
#include <string>
using std::string;                      // as usual
using namespace std::string_literals;   // needed to use string literals
```

The second implication is that as with any other function call, there is some overhead involved in calling this function.

**WARNING** Unlike most literals, literals of type `string` are allocated at runtime.

## 3.2.3 Dealing with the Characters in a `string`

Often we need to deal with the individual characters in a `string`. We might want to check to see whether a `string` contains any whitespace, or to change the characters to lowercase, or to see whether a given character is present, and so on. Programs that deal with character data often use one or more the functions described in Table 3.3 (p. 95). These functions are defined in the `cctype` header.

### Processing Every Character? Use Range-Based `for`

If we want to sequentially process each character in a `string`, by far the best approach is to use a **range `for`** statement. This statement iterates through the elements in a given sequence and performs some operation on each value in that sequence. The syntactic form is

**EXERCISES SECTION 3.2.2**

**Exercise 3.2:** Write a program to read the standard input a line at a time. Modify your program to read a word at a time.

**Exercise 3.3:** Explain how whitespace characters are handled in the `string` input operator and in the `getline` function.

**Exercise 3.4:** Write a program to read two `string`s and report whether the `string`s are equal. If not, report which of the two is larger. Now, change the program to report whether the `string`s have the same length, and if not, report which is longer.

**Exercise 3.5:** Write a program to read `string`s from the standard input, concatenating what is read into one large `string`. Print the concatenated `string`. Next, change the program to separate adjacent input `string`s by a space.

```
for (declaration : expression)
    statement
```

where *expression* is an object of a type that represents a sequence, and *declaration* defines the variable that we'll use to access the underlying elements in the sequence. On each iteration, the variable in *declaration* is initialized from the value of the next element in *expression*.

A `string` represents a sequence of characters, so we can use a `string` as the *expression* in a range `for`. As a simple example, we can use a range `for` to print each character from a `string` on its own line of output:

```
string str("some string");
// print the characters in str one character to a line
for (auto c : str)       // for every char in str
    cout << c << endl;   // print the current character followed by a newline
```

The `for` loop associates the variable `c` with `str`. We define the loop control variable the same way we do any other variable. Here, we use `auto` (§ 2.5.2, p. 68) to let the compiler determine the type of `c`, which in this case will be `char`. On each iteration, the next character in `str` will be copied into `c`. Thus, we can read this loop as saying, "For every character `c` in the `string str`," do something. The "something" in this case is to print the character followed by a newline.

As a somewhat more complicated example, we'll use a range `for` and the `ispunct` function to count the number of punctuation characters in a `string`:

```
string s("Hello World!!!");
// punct_cnt has the same type that s.size returns; see § 2.5.3 (p. 70)
decltype(s.size()) punct_cnt = 0;
// count the number of punctuation characters in s
for (auto c : s)          // for every char in s
    if (ispunct(c))       // if the character is punctuation
        ++punct_cnt;      // increment the punctuation counter
cout << punct_cnt
     << " punctuation characters in " << s << endl;
```

The output of this program is

```
3 punctuation characters in Hello World!!!
```

Here we use `decltype` (§ 2.5.3, p. 70) to declare our counter, `punct_cnt`. Its type is the type returned by calling `s.size`, which is `string::size_type`. We use a range `for` to process each character in the `string`. This time we check whether each character is punctuation. If so, we use the increment operator (§ 1.4.1, p. 12) to add `1` to the counter. When the range `for` completes, we print the result.

**Table 3.3: `cctype` Functions**

| | |
|---|---|
| `isalnum(c)` | `true` if `c` is a letter or a digit. |
| `isalpha(c)` | `true` if `c` is a letter. |
| `iscntrl(c)` | `true` if `c` is a control character. |
| `isdigit(c)` | `true` if `c` is a digit. |
| `isgraph(c)` | `true` if `c` is not a space but is printable. |
| `islower(c)` | `true` if `c` is a lowercase letter. |
| `isprint(c)` | `true` if `c` is a printable character (i.e., a space or a character that has a visible representation). |
| `ispunct(c)` | `true` if `c` is a punctuation character (i.e., a character that is not a control character, a digit, a letter, or a printable whitespace). |
| `isspace(c)` | `true` if `c` is whitespace (i.e., a space, tab, vertical tab, return, newline, or formfeed). |
| `isupper(c)` | `true` if `c` is an uppercase letter. |
| `isxdigit(c)` | `true` if `c` is a hexadecimal digit. |
| `tolower(c)` | If `c` is an uppercase letter, returns its lowercase equivalent; otherwise returns `c` unchanged. |
| `toupper(c)` | If `c` is a lowercase letter, returns its uppercase equivalent; otherwise returns `c` unchanged. |

### USING C LIBRARY HEADERS FROM C++ PROGRAMS

**In addition to facilities defined specifically for C++, the C++ library incorporates the C library. Headers in C have names of the form *name*`.h`. The C++ versions of these headers are named `c`*name*—they remove the `.h` suffix and precede the *name* with the letter `c`. The `c` indicates that the header is part of the C library.**

**Hence, `cctype` has the same contents as `ctype.h`, but in a form that is appropriate for C++ programs. In particular, the names defined in the `c`*name* headers are defined inside the `std` namespace, whereas those defined in the `.h` versions are not.**

**Ordinarily, C++ programs should use the `c`*name* versions of headers and not the *name*`.h` versions. That way names from the standard library are consistently found in the `std` namespace. Using the `.h` headers puts the burden on the programmer to remember which library names are inherited from C and which are unique to C++.**

## Using a Range `for` to Change the Characters in a `string`

If we want to change the value of the characters in a `string`, we must define the loop variable as a reference type (§ 2.3.1, p. 50). Remember that a reference is just another name for a given object. When we use a reference as our control variable, that variable is bound to each element in the sequence in turn. Using the reference, we can change the character to which the reference is bound.

Suppose that instead of counting punctuation, we wanted to convert a `string` to all uppercase letters. To do so, we can use the library `toupper` function, which takes a character and returns the uppercase version of that character. To convert the whole `string`, we need to call `toupper` on each character and put the result back in that character:

```
string s("Hello World!!!");
// convert s to uppercase
for (auto &c : s)    // for every char in s (note: c is a reference)
    c = toupper(c); // c is a reference, so the assignment changes the char in s
cout << s << endl;
```

The output of this code is

```
HELLO WORLD!!!
```

On each iteration, `c` refers to the next character in `s`. When we assign to `c`, we are changing the underlying character in `s`. So, when we execute

```
c = toupper(c); // c is a reference, so the assignment changes the char in s
```

we're changing the value of the character to which `c` is bound. When this loop completes, all the characters in `s` will be uppercase.

## Don't Forget the `&`

It is, unfortunately, very easy to forget the `&` when writing loops such as the previous one that capitalizes the letters in a `string`. If we forget to make the loop variable a reference:

```
for (auto c : s)      // we forgot the &, c is a copy of a character in s
    c = toupper(c); // c isn't a reference, s is unchanged!
cout << s << endl;  // prints our original, unchanged s, Hello World!!!
```

our program will still compile but its output will not be as expected:

```
Hello World!!!
```

Although we intended `c` to be a reference to a character in `s`, we declared `c` as a nonreference type. On each trip through the loop the next character in `s` is copied into `c`.

Finding a bug like this one is particularly hard because the `&` is visually so small and our expectation that we have declared the variable correctly can overpower even a careful reading of the code.

## Processing Only Some Characters?

A range `for` works well when we need to process every character. However, sometimes we need to access only a single character or to access characters until some condition is reached. For example, we might want to capitalize only the first character or only the first word in a `string`.

There are two ways to access individual characters in a `string`: In this section, we'll show how to use a subscript; § 3.4 (p. 111) will cover the second way.

To use the `string` subscript operator (the **`[]` operator**) we supply a value of type `string::size_type` (§ 3.2.2, p. 90) that denotes the position of the character that we want to access. The operator returns a reference to the character at the given position.

Subscripts for `string`s start at zero; if `s` is a `string` with at least two characters, then `s[0]` is the first character, `s[1]` is the second, and the last character is in `s[s.size() - 1]`.

> Note
>
> The values we use to subscript a `string` must be `>= 0` and `< size()`.
> **The result of using an index outside this range is undefined.**
> **By implication, subscripting an empty `string` is undefined.**

The value in the `[ ]` is referred to as "a subscript" or "an **index**." The index we supply can be any expression that yields an integral value. However, if our index has a signed type, its value will be converted to the unsigned type that `string::size_type` represents (§ 2.1.2, p. 36).

The following example uses the subscript operator to print the first character in a `string`:

```
if (!s.empty())            // make sure there's a character to print
    cout << s[0] << endl;  // print the first character in s
```

Before accessing the character, we check that `s` is not empty. Any time we use a subscript, we must ensure that there is a value at the given location. If `s` is empty, then `s[0]` is undefined.

So long as the `string` is not `const` (§ 2.4, p. 59), we can assign a new value to the character that the subscript operator returns. For example, we can capitalize the first letter as follows:

```
string s("some string");
if (!s.empty())             // make sure there's a character in s[0]
    s[0] = toupper(s[0]);   // assign a new value to the first character in s
```

The output of this program is

```
Some string
```

## Using a Subscript for Iteration

As a another example, we'll change the first word in `s` to all uppercase:

```
// process characters in s until we run out of characters or we hit a whitespace
for (decltype(s.size()) index = 0;
     index != s.size() && !isspace(s[index]); ++index)
        s[index] = toupper(s[index]); // capitalize the current character
```

This program generates

```
SOME string
```

Our `for` loop (§ 1.4.2, p. 13) uses `index` to subscript `s`. We use `decltype` to give `index` the appropriate type. We initialize `index` to `0` so that the first iteration will start on the first character in `s`. On each iteration we increment `index` to look at the next character in `s`. In the body of the loop we capitalize the current letter.

The new part in this loop is the condition in the `for`. That condition uses the logical AND operator (the **`&&` operator**). This operator yields `true` if both operands are `true` and `false` otherwise. The important part about this operator is that we are guaranteed that it evaluates its right-hand operand *only* if the left-hand operand is `true`. In this case, we are guaranteed that we will not subscript `s` unless we know that `index` is in range. That is, `s[index]` is executed only if `index` is not equal to `s.size()`. Because `index` is never incremented beyond the value of `s.size()`, we know that `index` will always be less than `s.size()`.

### CAUTION: SUBSCRIPTS ARE UNCHECKED

**When we use a subscript, we must ensure that the subscript is in range. That is, the subscript must be `>= 0` and `<` the `size()` of the `string`. One way to simplify code that uses subscripts is *always* to use a variable of type `string::size_type` as the subscript. Because that type is `unsigned`, we ensure that the subscript cannot be less than zero. When we use a `size_type` value as the subscript, we need to check only that our subscript is less than value returned by `size()`.**

WARNING

**The library is not required to check the value of a subscript. The result of using an out-of-range subscript is undefined.**

## Using a Subscript for Random Access

In the previous example we advanced our subscript one position at a time to capitalize each character in sequence. We can also calculate a subscript and directly fetch the indicated character. There is no need to access characters in sequence.

As an example, let's assume we have a number between 0 and 15 and we want to generate the hexadecimal representation of that number. We can do so using a `string` that is initialized to hold the 16 hexadecimal "digits":

```
const string hexdigits = "0123456789ABCDEF";  // possible hex digits
cout << "Enter a series of numbers between 0 and 15"
     << " separated by spaces.  Hit ENTER when finished: "
     << endl;
string result; // will hold the hexify'd string
for (string::size_type n; cin >> n; ) // read each number from the input
    if (n < hexdigits.size())         // process only valid input
        result += hexdigits[n];       // fetch the indicated hex digit
cout << "Your hex number is: " << result << endl;
```

If we give this program the input

```
12 0 5 15 8 15
```

the output will be

```
Your hex number is: C05F8F
```

We start by initializing `hexdigits` to hold the hexadecimal digits `0` through `F`. We make that `string const` (§ 2.4, p. 59) because we do not want these values to change. Inside the loop we use the input value `n` to subscript `hexdigits`. The value of `hexdigits[n]` is the `char` that appears at position `n` in `hexdigits`. For example, if `n` is `15`, then the result is `F`; if it's `12`, the result is `C`; and so on. We append that digit to `result`, which we print once we have read all the input.

Whenever we use a subscript, we must ensure that our indicies are in range. In this program, our subscript, `n`, is a `string::size_type`, which as we know is an unsigned type. As a result, we know that `n` is guaranteed to be greater than or equal to `0`. Before we use `n` to subscript `hexdigits`, we verify that it is less than the `size` of `hexdigits`.

**EXERCISES SECTION 3.2.3**

**Exercise 3.6:** Use a range `for` to change all the characters in a `string` to `X`.

**Exercise 3.7:** What would happen if you define the loop control variable in the previous exercise as type `char`? Predict the results and then change your program to use a `char` to see if you were right.

**Exercise 3.8:** Rewrite the program in the first exercise, first using a `while` and again using a traditional `for` loop. Which of the three approaches do you prefer and why?

**Exercise 3.9:** What does the following program do? Is it valid? If not, why not?

```
string s;
cout << s[0] << endl;
```

**Exercise 3.10:** Write a program that reads a string of characters including punctuation and writes what was read but with the punctuation removed.

**Exercise 3.11:** Is the following range `for` legal? If so, what is the type of `c`?

```
const string s = "Keep out!";
for (auto &c : s) { /* ...  */ }
```

## 3.3 Library `vector` Type

A **`vector`** is a collection of objects, all of which have the same type. Every object in the collection has an associated index, which gives access to that object. A `vector` is often referred to as a **container** because it "contains" other objects. We'll have much more to say about containers in Part II.

To use a `vector`, we must include the appropriate header. In our examples, we also assume that an appropriate `using` declaration is made:

```
#include <vector>
using std::vector;
```

A `vector` is a **class template**. C++ has both class and function templates. Writing a template requires a fairly deep understanding of C++. Indeed, we won't see how to create our own templates until Chapter 16! Fortunately, we can use templates without knowing how to write them.

Templates are not themselves functions or classes. Instead, they can be thought of as instructions to the compiler for generating classes or functions. The process that the compiler uses to create classes or functions from templates is called **instantiation**. When we use a template, we specify—explicitly or implicitly—what kind of class or function we want the compiler to instantiate.

For a class template, we usually explicitly specify which class to instantiate by supplying additional information, the nature of which depends on the template. How we specify the information is always the same: We supply it inside a pair of angle brackets following the template's name.

In the case of `vector`, the additional information we need to supply is the type of the objects that the `vector` will hold:

```
vector<int> ivec;             // ivec holds objects of type int
vector<Sales_item> Sales_vec; // holds Sales_items
vector<vector<string>> file;  // vector whose elements are vectors
```

In this example, the compiler generates three distinct types from the `vector` template: `vector<int>`, `vector<Sales_item>`, and `vector<vector<string>>`.

We can define `vector`s to hold objects of most any type. Because references are not objects (§ 2.3.1, p. 51), we cannot have a `vector` of references. However, we can have `vector`s of most other (nonreference) built-in types and most class types. In particular, we can have `vector`s whose elements are themselves `vector`s.

**WARNING**

Old compilers may require a space between the closing angle bracket of the outer `vector` and its element type, e.g., `vector<vector<int> >` rather than `vector<vector<int>>`.

### 3.3.1 Defining and Initializing `vector`s

As with any class type, the `vector` template controls how we define and initialize `vector`s. Table 3.4 (p. 103) lists the most common ways to define `vector`s.

We can default initialize a `vector` (§ 2.2.1, p. 44), which creates an empty `vector` of the specified type:

```
vector<string> svec; // default initialization; svec has no elements
```

It might seem that an empty `vector` would be of little use. However, as we'll see shortly, we can (efficiently) add elements to a `vector` at run time. Indeed, the most common way of using `vector`s is to define an initially empty `vector` to which elements are added as their values become known at run time.

Alternatively, we can supply initial value(s) for the element(s) when we define a `vector`. For example, we can copy elements from another `vector`. When we copy a vector, each element in the new vector is a copy of the corresponding element in the original vector. The two vectors must have the same type:

```
vector<int> ivec;              // initially empty
// give ivec some values
vector<int> ivec2(ivec);       // copy elements of ivec into ivec2
vector<string> svec(ivec2);    // error: svec holds strings, not ints
// error: element types must match exactly; elements are not converted (see § 2.1.2 (p. 34))
vector<double> dvec = ivec;    // error: element types must match exactly
```

We can also initialize a vector from a count and an element value. The count determines how many elements the vector will have; the value provides the initial value for each of those elements:

```
vector<int> ivec(10, -1);         // ten int elements, each initialized to -1
vector<string> svec(10, "hi!");   // ten strings; each element is "hi!"
```

## Element Conversion and vector Initialization

When using class template types, it can be confusing to understand when conversions can happen and when not. As we've just seen, when we copy one vector to another, the types of the elements must match exactly. Yet, when we defined svec, we supplied an element initializer that is not of type string. The difference is subtle and can be hard to grasp.

When we supply an element initializer, the compiler expects a value of the vector's element type. If the intializer that we supply doesn't match that type, the compiler will convert that initializer to the element type if it is possible to do so. When we passed "hi" as the element initializer for svec, the compiler knows that it needs a string. It also knows that there is a conversion from character-string literal to type string (§ 3.2.2, p. 92). It performs that conversion and uses the resulting string value to initialize the elements of svec.

When we copy one vector to another, the initializer is itself a vector. That vector has a type, such as vector<int> or vector<double>, etc. There are no conversions from one vector type to another. It doesn't matter if the two vectors hold elements that can be converted to one another (e.g., if one vector holds ints and the other holds doubles). In order to to copy a vector<int> into a vector<double>, there would have to be a way to convert the vector<int> to vector<double>, but there is no such conversion. As a result, if we want to initialize a vector as a copy of another vector, the types must match exactly.

## Value Initialization

For most types of vectors, we can define a vector by just supplying a size and omitting the element initializer. In this case the library creates a **value-initialized** element initializer for us. This library-generated value is used to initialize each element in the container. The value of the element initializer depends on the type of the elements stored in the vector.

If the vector holds elements of a built-in type, such as int, then the element is 0. If the elements are of a class type, such as string, then the element initializer is itself default initialized:

```
vector<int> ivec(10);       // ten elements, each initialized to 0
vector<string> svec(10);    // ten elements, each an empty string
```

To use this form of initialization, the `vector` must hold elements of a type that has a default value (§ 2.2.1, p. 44). Some classes have no default. If our `vector` holds objects of a type has no default, then we must supply an initial element value as well as the size.

When we supply an element count without also supplying an initial value, we must use the direct form of initialization:

```
vector<int> vi = 10;        // error: must use direct initialization to supply a size
```

Here we are using 10 to instruct `vector` how to create the `vector`—we want a `vector` with ten value-initialized elements. We are not "copying" 10 into the `vector`. Hence, we cannot use the copy form of initialization. We'll see more about how this restriction works in § 7.5.4 (p. 310).

### List Initializing a `vector`

Using `vector`, we can now better understand why we refer to an initialization that uses curly braces as list initialization (§ 2.2.1, p. 43). We can use the braces to enclose a comma-separated list of zero or more values:

```
vector<int> digits = {0,1,2,3,4,5,6,7,8,9};// digits has 10 elements
vector<int> digits2{0,1,2,3,4,5,6,7,8,9};  // equivalent definition
// element initailizers will be converted to type string; resulting vector has 3 elements
vector<string> articles = {"a", "an", "the"};
vector<string> empty = {};  // empty vector<string>
```

In each case, the resulting `vector` has as many elements as there are values in the list. If the list is empty, then the `vector` has no elements. If the list is not empty, the first element in the resulting `vector` is initialized by the first value in the list, the second element by the next value, and so on. Thus, `digits` has ten elements; the first has the value `0`, the second is `1`, and so on.

If we want to supply a list of element values, we *must* use curly braces. We cannot supply a list of initializers using parentheses:

```
vector<int> v1{0,1,2,3,4,5,6,7,8,9};        // list initialization
vector<int> v2(0,1,2,3,4,5,6,7,8,9);        // error
vector<string> vs1{"hi"};   // ok: vs1 has one element
vector<string> vs2("hi");   // error
```

> **Note** We *must* use curly braces to supply a list of element initializers.

### List Initializer or Element Count?

In a few cases, what an initialization means depends upon whether we use curly braces or parentheses. For example, when we initialize a `vector<int>` from a

**Table 3.4: Ways to Initialize a vector**

| | |
|---|---|
| vector<T> v1 | vector that holds objects of type T. Default initialization; v1 is empty. |
| vector<T> v2(v1) | v2 has a copy of each element in v1. |
| vector<T> v2 = v1 | Equivalent to v2(v1), v2 is a copy of the elements in v1. |
| vector<T> v3(n, val) | v3 has n elements with value val. |
| vector<T> v4(n) | v4 has n copies of a value-initialized object. |
| vector<T> v5{a,b,c ...} | v5 has as many elements as there are values in the list; elements are initialized by corresponding initializers. |
| vector<T> v5 = {a,b,c ...} | Equivalent to v5{a,b,c ...}. |

single int value, that value might represent the vector's size or it might be an element value. Similarly, if we supply exactly two ints, those ints could be a size and an initial value, or they could be values for a two-element vector. We specify which meaning we want by whether we use curly braces or parentheses:

```
vector<int> v1(10);      // v1 has ten elements with value 0
vector<int> v2(10, 1);   // v2 has ten elements with value 1
vector<int> v3{10};      // v3 has one element with value 10
vector<int> v4{10, 1};   // v4 has two elements with values 10 and 1
```

When we use parentheses, we are saying that the values we supply are to be used to construct the object. Thus, the value used to define v1 determines the vector's size, and the values with v2 specify its size and element values, respectively.

When we use curly braces, {...}, we're saying that, if it is possible to do so, the compiler should use the values inside the braces as a list of element initializers. The values we supplied for v3 and v4 can be used as element values. The resulting vectors have one and two elements, respectively.

If it is not possible to use the values inside curly braces as element initializers, then the compiler will consider other ways to initialize the object:

```
vector<string> vs1(5); // vs1 has 5 elements, each of which is an empty string
vector<string> vs2{5}; // equivalent definition; 5 isn't a string
```

We cannot create a string from the value 5: The only possible meaning for either of these initializations is to use 5 as specifying the vector's size.

## 3.3.2 Type Deduction for Class Templates

It is important to understand that templates themselves are not types; instead they tell the compiler how to instantiate (§ 3.3, p. 99) a range of classes. Each template specifies what information must be supplied in order to instantiate (i.e., generate) an actual type from that template. In the case of vector, that information is the element type. Until C++17, we always had to *explicitly* supply the necessary information inside angle brackets following the name of the template:

## Exercises Section 3.3.1

**Exercise 3.12:** Which, if any, of the following `vector` definitions are in error? For those that are legal, explain what the definition does. For those that are not legal, explain why they are illegal.

(a) `vector<vector<int>> ivec;`
(b) `vector<string> sv = ivec;`
(c) `vector<string> svec(10, "null");`
(d) `vector<int> dvec{ 1,2.0,3,4.0 };`

**Exercise 3.13:** How many elements are there in each of the following `vector`s? What are the values of the elements?

(a) `vector<int> v1;` (b) `vector<int> v2(10);`
(c) `vector<int> v3(10, 42);` (d) `vector<int> v4{10};`
(e) `vector<int> v5{10, 42};` (f) `vector<string> v6{10};`
(g) `vector<string> v7{10, "hi"};`

```
// ok: <int> tells the complier to instantiate a class
//     that is a vector that can hold int elements
vector<int> v1{1,2}; // ok: explicitly says what kind of vector to instantiate
// error: prior to c++17 must always explicitly specify the type of vector we want
vector v2{1,2}; // ok: in c++17 the compiler deduces vector<int>
```

With new compilers, we can (sometimes) *implicitly* specify the needed information. In particular, when we define an object of a class template type, we can omit the angle brackets if the template type we want can be inferred from the type(s) of the values used to construct the object:

```
vector<int> v1;
vector v2 = v1;          // v2 same type as v1, i.e., vector<int>
vector v3{1,2,3,4,5};    // v3 is vector<int>
vector v4(10, 0);        // v4 is vector<int>
```

In each of these cases, it can be inferred that we're defining a `vector<int>`: we do not have to (explicitly) specify `<int>`. As we saw above, when we copy a `vector`, the types of the `vector`s must match, therefore, `v2` must have the same type as `v1`. Similarly, when we provide a list of element intializers, the type of those elements determines the type of the `vector`. When we pass a count and an element value, the `vector`'s type is determined by the type of the element value.

We can omit the element type only when there is enough information available to infer the type of the `vector`. For example, it is an error to omit the element type when we create a `vector` from a count:

```
vector v6(10);          // error: 10 elements, but of what type?
```

Here there is no way to infer what type of elements we want `v6` to hold.

One implication of implicitly specifying a `vector`'s type is that element initializer(s) won't be converted. For example, if we initialize a `vector<string>`

from a brace-enclosed list of character string literals, the initializers are converted to `string`:

```
// ok: each literal is converted to string
vector<string> articles = {"a", "an", "the"};
```

This seemingly equivalent definition:

```
vector a2 = {"a", "an", "the"}; // type is vector<const char*>
```

yields a `vector` that holds `const char*`! If we want the `vector` to hold `string`s then we need to specify that explicitly or use element initializers that are of type `string`, which we could do by using `string` literals (§ 3.2.2, p. 92):

```
vector a3 = {"a"s, "an"s, "the"s}; // type is vector<string>
```

If we list initialize a `vector`, we can implicitly specify the `vector`'s type only if the element initializers all have the exact same type:

```
vector<double> v1{1,2.0}; // ok: complier converts 1 to double
vector<int>    v2{1,2.0}; // ok: complier converts 2.0 to int
// error: the compiler won't choose between vector<int> or vector<double>
vector v3{1, 2.0};         // error: element types don't match
```

## Exercises Section 3.3.2

**Exercise 3.14:** Identify which, if any of these definitions is illegal. For the legal ones, explain the type of the resulting `vector`:

(a) `vector<vector<int>> ivec;`
(b) `vector v1 = ivec;`
(c) `vector v2(10, 42);`
(d) `vector v3{ 1,2.0,3,4.0 };`

### 3.3.3 Adding Elements to a `vector`

Directly initializing the elements of a `vector` is feasible only if we have a small number of known initial values, if we want to make a copy of another `vector`, or if we want to initialize all the elements to the same value. More commonly, when we create a `vector`, we don't know how many elements we'll need, or we don't know the value of those elements. Even if we do know all the values, if we have a large number of different initial element values, it can be cumbersome to specify them when we create the `vector`.

As one example, if we need a `vector` with values from 0 to 9, we can easily use list initialization. What if we wanted elements from 0 to 99 or 0 to 999? List initialization would be too unwieldy. In such cases, it is better to create an empty `vector` and use a `vector` member named **`push_back`** to add elements at run time. The `push_back` operation takes a value and "pushes" that value as a new last element onto the "back" of the `vector`. For example:

```
vector<int> v2;          // empty vector
for (int i = 0; i != 100; ++i)
    v2.push_back(i);     // append sequential integers to v2
// at end of loop v2 has 100 elements, values 0 . . . 99
```

Even though we know that we will have 100 elements, we define `v2` as empty. Each iteration appends the next sequential integer as a new element in `v2`.

We use the same approach when we want to create a `vector` where we don't know until run time how many elements the `vector` should have. For example, we might read the input, storing the values we read in the `vector`:

```
// read words from the standard input and store them as elements in a vector
vector<string> text;       // empty vector
for (string word; cin >> word; ) {
    text.push_back(word);  // append word to text
}
```

Again, we start with an initially empty `vector`. This time, we read and store an unknown number of values in `text`.

**KEY CONCEPT: `vector`S GROW EFFICIENTLY**

**The standard requires that `vector` implementations can efficiently add elements at run time. Because `vector`s grow efficiently, it is often unnecessary—and can result in poorer performance—to define a `vector` of a specific size. The exception to this rule is if *all* the elements actually need the same value. If differing element values are needed, it is usually more efficient to define an empty `vector` and add elements as the values we need become known at run time. Moreover, as we'll see in § 9.4 (p. 391), `vector` offers capabilities to allow us to further enhance run-time performance when we add elements.**

**Starting with an empty `vector` and adding elements at run time is distinctly different from how we use built-in arrays in C and in most other languages. In particular, if you are accustomed to using C, Java, or Python, you might expect that it would be best to define the `vector` at its expected size. In fact, the contrary is usually the case.**

## Programming Implications of Adding Elements to a `vector`

The fact that we can easily and efficiently add elements to a `vector` greatly simplifies many programming tasks. However, this simplicity imposes a new obligation on our programs: We must ensure that any loops we write are correct even if the loop changes the size of the `vector`.

Other implications that follow from the dynamic nature of `vector`s will become clearer as we learn more about using them. However, there is one implication that is worth noting already: For reasons we'll explore in § 5.4.3 (p. 201), we cannot use a range `for` if the body of the loop adds elements to the `vector`.

WARNING

The body of a range `for` must not change the size of the sequence over which it is iterating.

**EXERCISES SECTION 3.3.3**

**Exercise 3.15:** Write a program to read a sequence of ints from cin and store those values in a vector.

**Exercise 3.16:** Repeat the previous program but read strings this time.

### 3.3.4 Other vector Operations

In addition to push_back, vector provides only a few other operations, most of which are similar to the corresponding operations on strings. Table 3.5 (overleaf) lists the most important ones.

We access the elements of a vector the same way that we access the characters in a string: through their position in the vector. For example, we can use a range for (§ 3.2.3, p. 93) to process all the elements in a vector:

```
vector v{1,2,3,4,5,6,7,8,9}; // implicitly vector<int> see § 3.3.2 (p. 103)
for (auto &i : v)        // for each element in v (note: i is a reference)
    i *= i;              // square the element value
for (auto i : v)         // for each element in v
    cout << i << " ";    // print the element
cout << endl;
```

In the first loop, we define our control variable, i, as a reference so that we can use i to assign new values to the elements in v. We let auto deduce the type of i. This loop uses a new form of the compound assignment operator (§ 1.4.1, p. 12). As we've seen, += adds the right-hand operand to the left and stores the result in the left-hand operand. The *= operator behaves similarly, except that it multiplies the left- and right-hand operands, storing the result in the left-hand one. The second range for prints each element.

The empty and size members behave as do the corresponding string members (§ 3.2.2, p. 89): empty returns a bool indicating whether the vector has any elements, and size returns a value of the size_type defined by the corresponding vector type indicating the number of elements in the vector.

> *Note*
>
> To use size_type, we must name the type in which it is defined. A vector type *always* includes its element type (§ 3.3, p. 100):
>
> ```
> vector<int>::size_type // ok
> vector::size_type      // error
> ```

The equality and relational operators have the same behavior as the corresponding string operations (§ 3.2.2, p. 91). Two vectors are equal if they have the same number of elements and if the corresponding elements all have the same value. The relational operators apply a dictionary ordering: If the vectors have differing sizes, but the elements that are in common are equal, then the vector with fewer elements is less than the one with more elements. If the elements have differing values, then the relationship between the vectors is determined by the relationship between the first elements that differ.

We can compare two `vector`s only if we can compare the elements in those `vector`s. Some class types, such as `string`, define the meaning of the equality and relational operators. Others, such as our `Sales_item` class, do not. The only operations `Sales_item` supports are those listed in § 1.5.1 (p. 20). Those operations did not include the equality or relational operators. As a result, we cannot compare two `vector<Sales_item>` objects.

**Table 3.5: `vector` Operations**

| | |
|---|---|
| `v.empty()` | Returns `true` if `v` is empty; otherwise returns `false`. |
| `v.size()` | Returns the number of elements in `v`. |
| `v.push_back(t)` | Adds an element with value `t` to end of `v`. |
| `v[n]` | Returns a reference to the element at position `n` in `v`. |
| `v1 = v2` | Replaces the elements in `v1` with a copy of the elements in `v2`. |
| `v1 = {a,b,c...}` | Replaces the elements in `v1` with a copy of the elements in the comma-separated list. |
| `v1 == v2`<br>`v1 != v2` | `v1` and `v2` are equal if they have the same number of elements and each element in `v1` is equal to the corresponding element in `v2`. |
| `<, <=, >, >=` | Have their normal meanings using dictionary ordering. |

## Computing a `vector` Index

Just as we can use a subscript to fetch a given character from a `string` (§ 3.2.3, p. 97), we can use a subscript to fetch a given element from a `vector`. As with `string`s, subscripts for `vector` start at 0; the type of the index is the corresponding `size_type`; we can compute an index and fetch the element at the computed position (§ 3.2.3, p. 98); and—unless the `vector` is `const`—we can write to the element returned by the subscript operator.

As an example, let's assume that we have a collection of grades that range from 0 through 100. We'd like to count how many grades fall into various clusters of 10. Between zero and 100 there are 101 possible grades. These grades can be represented by 11 clusters: 10 clusters of 10 grades each plus one cluster for the perfect score of 100. The first cluster will count grades of 0 through 9, the second will count grades from 10 through 19, and so on. The final cluster counts how many scores of 100 were achieved.

Clustering the grades this way, if our input is

```
42 65 95 100 39 67 95 76 88 76 83 92 76 93
```

then the output should be

```
0 0 0 1 1 0 2 3 2 4 1
```

which indicates that there were no grades below 30, one grade in the 30s, one in the 40s, none in the 50s, two in the 60s, three in the 70s, two in the 80s, four in the 90s, and one grade of 100.

### CAUTION: SUBSCRIPT ONLY ELEMENTS THAT ARE KNOWN TO EXIST!

**It is crucially important to understand that we may use the subscript operator (the `[]` operator) to fetch only elements that actually exist. For example,**

```
vector<int> ivec;       // empty vector
cout << ivec[0];        // error: ivec has no elements!

vector<int> ivec2(10);  // vector with ten elements
cout << ivec2[10];      // error: ivec2 has elements 0 . . . 9
```

**It is an error to subscript an element that doesn't exist, but it is an error that the compiler is unlikely to detect. Instead, the value we get at run time is undefined.**

**Attempting to subscript elements that do not exist is, unfortunately, an extremely common and pernicious programming error. So-called *buffer overflow* errors are the result of subscripting elements that don't exist. Such bugs are the most common cause of security problems in PC and other applications.**

Tip

**A good way to ensure that subscripts are in range is to avoid subscripting altogether by using a range `for` whenever possible.**

We'll use a `vector` with 11 elements to hold the counters for each cluster. We can determine the cluster index for a given grade by dividing that grade by 10. When we divide two integers, we get an integer in which the fractional part is truncated. For example, 42/10 is 4, 65/10 is 6 and 100/10 is 10. Once we've computed an index, we'll use it fetch the counter we want to increment:

```
// count the number of grades by clusters of ten: 0--9, 10--19, . .. 90--99, 100
vector<unsigned> scores(11, 0); // 11 buckets, all initially 0
for (unsigned grade; cin >> grade; )     // read the grades
    if (grade <= 100)               // handle only valid grades
        ++scores[grade/10]; // increment the counter for the current cluster
```

We start by defining a `vector` to hold the cluster counts. In this case, we do want each element to have the same value, so we allocate all 11 elements, each of which is initialized to `0`. The `for` condition reads the grades. Inside the loop, we check that the grade we read has a valid value (i.e., that it is less than or equal to 100). Assuming the grade is valid, we increment the appropriate counter for `grade`.

The statement that does the increment is a good example of the kind of terse code characteristic of C++ programs. This expression

```
++scores[grade/10]; // increment the counter for the current cluster
```

is equivalent to

```
auto ind = grade/10; // get the bucket index
scores[ind] = scores[ind] + 1; // increment the count
```

We compute the bucket index by dividing `grade` by 10 and use the result of the division to index `scores`. Subscripting `scores` fetches the appropriate counter for this grade. We increment the value of that element to indicate the occurrence of a score in the given range.

As we've seen, when we use a subscript, we should think about how we know that the indices are in range (§ 3.2.3, p. 99). In this program, because `grade` is `unsigned`, we know `grade` is `>= 0` and we check that `grade` is `<= 100`. Thus, we know that the indices we can compute are between `0` and `10`. These indices are between `0` and `scores.size() - 1`.

## Subscripting Does Not Add Elements

Programmers new to C++ sometimes think that subscripting a `vector` adds elements; it does not. The following code intends to add ten elements to `ivec`:

```
vector<int> ivec;   // empty vector
for (decltype(ivec.size()) ix = 0; ix != 10; ++ix)
    ivec[ix] = ix;  // disaster: ivec has no elements
```

However, it is in error: `ivec` is an empty `vector`; there are no elements to subscript! As we've seen, the right way to write this loop is to use `push_back`:

```
for (decltype(ivec.size()) ix = 0; ix != 10; ++ix)
    ivec.push_back(ix);  // ok: adds a new element with value ix
```

WARNING

The subscript operator on `vector` (and `string`) fetches an existing element; it does *not* add an element.

### Exercises Section 3.3.4

**Exercise 3.17:** Write a program to print the size and contents of the `vectors` from exercise 3.13. Check whether your answers to that exercise were correct.

**Exercise 3.18:** Identify any errors in the following and fix them:

```
vector<unsigned> scores;
for (unsigned grade; cin >> grade; )
    ++scores;
```

**Exercise 3.19:** Read a sequence of words from `cin` and store the values a `vector`. After you've read all the words, process the `vector` and change each word to uppercase. Print the transformed elements, eight words to a line.

**Exercise 3.20:** List three ways to define a `vector` and give it ten elements, each with the value 42. Indicate whether there is a preferred way to do so and why.

**Exercise 3.21:** Read a set of integers into a `vector`. Print the sum of each pair of adjacent elements. Change your program so that it prints the sum of the first and last elements, followed by the sum of the second and second-to-last, and so on.

## 3.4 Introducing Iterators

In addition to using subscripts to access the characters of a `string` or the elements in a `vector`, we can use a more general mechanism—known as **iterators**—for the same purpose. As we'll see in Part II, in addition to `vector`, the library defines several other kinds of containers. All of the library containers have iterators, but only a few of them support the subscript operator. Technically speaking, a `string` is not a container type, but `string` supports many of the container operations. As we've seen, `string` like `vector`, has a subscript operator. Like `vector`s, `string`s also have iterators.

Like pointers (§ 2.3.2, p. 52), iterators give us indirect access to an object. In the case of an iterator, that object is an element in a container or a character in a `string`. We can use an iterator to fetch an element and can move an iterator from one element to another. As with pointers, an iterator may be valid or invalid. A valid iterator either denotes an element or denotes a position one past the last element in a container. All other iterator values are invalid.

### 3.4.1 Using Iterators

Unlike how we obtain a pointer, we do not use the address-of operator to obtain an iterator. Instead, types that have iterators have members that return iterators. In particular, these types have members named **`begin`** and **`end`**. The `begin` member returns an iterator that denotes the first element (or first character), if there is one:

```
// the compiler determines the type of b and e; see § 2.5.2 (p. 68)
// b denotes the first element and e denotes one past the last element in v
auto b = v.begin(), e = v.end(); // b and e have the same type
```

The iterator returned by `end` is an iterator positioned "one past the end" of the associated container (or `string`). This iterator denotes a nonexistent element "off the end" of the container. It is used as a marker indicating when we have processed all the elements. The iterator returned by `end` is often referred to as the **off-the-end iterator** or abbreviated as "the `end` iterator." If the container is empty, `begin` returns the same iterator as the one returned by `end`.

Note

If the container is empty, the iterators returned by `begin` and `end` are equal—they are both off-the-end iterators.

In general, we do not know (or care about) the precise type that an iterator has. In this example, we used `auto` to define `b` and `e` (§ 2.5.2, p. 68). As a result, these variables have whatever type is returned by the `begin` and `end` members, respectively. We'll have more to say about those types on page 115.

#### Iterator Operations

Iterators support only a few operations, which are listed in Table 3.6 (overleaf). We can compare two valid iterators using `==` or `!=`. Iterators are equal if they denote the same element or if they are both off-the-end iterators for the same container. Otherwise, they are unequal.

As with pointers, we can dereference an iterator to obtain the element that the iterator denotes. Also, as with pointers, we may dereference only a valid iterator that denotes an element (§ 2.3.2, p. 53). Dereferencing an invalid iterator or an off-the-end iterator has undefined behavior.

As an example, we'll rewrite the program from § 3.2.3 (p. 97) that capitalized the first character of a `string` using an iterator instead of a subscript:

```
string s("some string");
if (auto it = s.begin(); it != s.end()) // make sure s is not empty
    *it = toupper(*it);                 // make that character uppercase
```

As in our original program, we need to check that `s` isn't empty. In this case, we do so by first defining an iterator named `it` inside the `if` and initializing `it` to the value returned by `s.begin()`. The condition in the `if` compares that iterator to the one returned by `end`. Those iterators are equal if the `string` is empty. If they are unequal, there is at least one character in `s`.

In the `if` body, we dereference `it` to pass the underlying character to `toupper`. We also dereference `it` on the left-hand side of the assignment in order to assign the character returned from `toupper` to the first character in `s`. As in our original program, the output of this code will be:

```
Some string
```

**Table 3.6: Standard Container Iterator Operations**

| | |
|---|---|
| `*iter` | Returns a reference to the element denoted by the iterator `iter`. |
| `iter->mem` | Dereferences `iter` and fetches the member named `mem` from the underlying element. Equivalent to `(*iter).mem`. |
| `++iter` | Increments `iter` to refer to the next element in the container. |
| `--iter` | Decrements `iter` to refer to the previous element in the container. |
| `iter1 == iter2`<br>`iter1 != iter2` | Compares two iterators for equality (inequality). Two iterators are equal if they denote the same element or if they are the off-the-end iterator for the same container. |

## Moving Iterators from One Element to Another

Iterators use the increment (`++`) operator (§ 1.4.1, p. 12) to move from one element to the next. Incrementing an iterator is a logically similar operation to incrementing an integer. In the case of integers, the effect is to "add `1`" to the integer's value. In the case of iterators, the effect is to "advance the iterator by one position."

Note: Because the iterator returned from `end` does not denote an element, it may not be incremented or dereferenced.

Using the increment operator, we can rewrite our program that changed the case of the first word in a `string` to use iterators instead:

```
// process characters in s until we run out of characters or we hit a whitespace
for (auto it = s.begin(); it != s.end() && !isspace(*it); ++it)
    *it = toupper(*it); // capitalize the current character
```

This loop, like the one in § 3.2.3 (p. 97), iterates through the characters in `s`, stopping when we encounter a whitespace character. However, this loop accesses these characters using an iterator, not a subscript.

The loop starts by initializing `it` from `s.begin`, meaning that `it` denotes the first character (if any) in `s`. The condition checks whether `it` has reached the `end` of `s`. If not, the condition next dereferences `it` to pass the current character to `isspace` to see whether we're done. At the end of each iteration, we execute `++it` to advance the iterator to access the next character in `s`.

The body of this loop, is the same as the last statement in the previous `if`. We dereference `it` to pass the current character to `toupper` and assign the resulting uppercase letter back into the character denoted by `it`.

**KEY CONCEPT: GENERIC PROGRAMMING**

**Programmers coming to C++ from C or Java might be surprised that we used `!=` rather than < in our `for` loops such as the one above and in the one on page 97. C++ programmers use `!=` as a matter of habit. They do so for the same reason that they use iterators rather than subscripts: This coding style applies equally well to various kinds of containers provided by the library.**

**We use iterators because only a few library types, `vector` and `string` among them, have the subscript operator. Similarly, most of the library containers have iterators that define `==` and `!=` operators but do not have the < operator. By routinely using iterators and `!=`, we don't have to worry about the precise type of container we're processing.**

## Combining Dereference and Member Access

When we dereference an iterator, we get the object that the iterator denotes. If that object has a class type, we may want to access a member of that object. For example, we might have a `vector` of `string`s and we might need to know whether a given element is empty. Assuming `it` is an iterator into this `vector`, we can check whether the `string` that `it` denotes is empty as follows:

```
(*it).empty()
```

For reasons we'll cover in § 4.1.2 (p. 144), the parentheses in `(*it).empty()` are necessary. The parentheses say to apply the dereference operator to `it` and to apply the dot operator (§ 1.5.2, p. 23) to the result of dereferencing `it`. Without parentheses, the dot operator would apply to `it`, not to the resulting object:

```
(*it).empty() // dereferences it and calls the member empty on the resulting object
*it.empty()   // error: attempts to fetch the member named empty from it
              //        but it is an iterator and has no member named empty
```

`*it.empty()` is interpreted as if we had written `*(it.empty())`, which says to fetch the `empty` member from the object named `it`. However, `it` is an iterator and has no member named `empty`. Hence, the second expression is in error.

To simplify expressions such as this one, the language defines the arrow operator (the **`->` operator**). The arrow operator combines dereference and member access into a single operation. That is, `it->mem` is a synonym for `(*it).mem`.

For example, assume we have a `vector<string>` named `text` that holds the data from a text file. Each element in the `vector` is either a sentence or an empty `string` representing a paragraph break. If we want to print the contents of the first paragraph from `text`, we'd write a loop that iterates through `text` until we encounter an element that is empty:

```
// print each line in text up to the first blank line
for (auto it = text.begin();
      it != text.end() && !it->empty(); ++it)
     cout << *it << endl;
```

We start by initializing `it` to denote the first element in `text`. The loop continues until either we process every element in `text` or we find an element that is empty. So long as there are elements and we haven't seen an empty element, we print the current element.

## Using Iterators for Initialization

In addition to the ways to construct `strings` and `vectors` listed in Tables 3.1 (p. 86) and 3.4 (p. 103), respectively, we can also initialize objects of these types from a pair of iterators. The iterators mark the range of values to use to initialize the object we are constructing. The first iterator denotes the first element to copy and the second denotees one past the last element to be copied.

As one example, when we want to copy one `vector` into another `vector`, the two `vectors` must hold exactly the same element type (§ 3.3.1, p. 100). Using a pair of iterators, we can copy elements from one `vector` into another so long as it is possible to convert the element type we are copying to the type of the `vector` we are initializing:

```
vector<int> digits = {0,1,2,3,4,5,6,7,8,9};
vector<double> fpDigits = digits; // error: element types don't match
// ok: copy each int element from digits into fpDigits
vector<double> fpDigits(digits.begin(), digits.end());
```

Moreover, using iterators, we can even use a `string` to initialize a `vector` or vice versa:

```
string letters = "abcdefghijklmnopqrstuvwxyz";
vector<char> alphabet(letters.begin(), letters.end());
```

Our `vector` will contain 26 characters, each initialized as a copy of the corresponding character from `letters`. That is, `alphabet[0]` will be `'a'`, the next element will be `'b'`, and so on up to `alphabet[25]`, which will be `'z'`.

**Table 3.7: Using Iterators to Initialize `string` and `vector` Objects**

| `b` and `e` are iterators that must refer to the same sequence | |
|---|---|
| `string s(b, e)` | `s` is a copy of the characters in the range denoted by `b` and `e`. The elements in the sequence must be convertible to `char`. |
| `vector<T> v(b, e)` | `v` is a copy of the elements in the range denoted by `b` and `e`. The type of the elements in the sequence must be convertible to `T` (i.e., the element type of `v`). |

## Iterator Types

Just as we do not know the precise type of a `vector`'s or `string`'s `size_type` member (§ 3.2.2, p. 90), so too, we generally do not know—and do not need to know—the precise type of an iterator. Instead, as with `size_type`, the library types that have iterators define type aliases (§ 2.5.1, p. 67) named `iterator` and `const_iterator` that represent actual iterator types:

```
vector<int>::iterator it; //  it can read and write vector<int> elements
string::iterator it2;     //  it2 can read and write characters in a string

vector<int>::const_iterator it3; //  it3 can read but not write elements
string::const_iterator it4;      //  it4 can read but not write characters
```

A `const_iterator` behaves like a pointer to `const` (§ 2.4.2, p. 63). Like a pointer to `const`, a `const_iterator` may read but not write the element it denotes; an object of type `iterator` can both read and write. If a `vector` or `string` is `const`, we may use only its `const_iterator` type. With a nonconst `vector` or `string`, we can use either `iterator` or `const_iterator`.

## The `begin` and `end` Operations

The type returned by `begin` and `end` depends on whether the object on which they operate is `const`. If the object is `const`, then `begin` and `end` return a `const_iterator`; if the object is not `const`, they return `iterator`:

```
vector<int> v;
const vector<int> cv;
auto it1 = v.begin();  //  it1 has type vector<int>::iterator
auto it2 = cv.begin(); //  it2 has type vector<int>::const_iterator
```

Often this default behavior is not what we want. For reasons we'll explain in § 6.2.2 (p. 223), it is usually best to use a `const` type (such as `const_iterator`) when we need to read but do not need to write to an object. To get the `const_iterator` type we call `cbegin` and `cend`:

```
auto it3 = v.cbegin(); //  it3 has type vector<int>::const_iterator
```

Like the `begin` and `end` members, these members return iterators to the first and one past the last element in the container. However, regardless of whether the `vector` (or `string`) is `const`, they return a `const_iterator`.

**TERMINOLOGY: ITERATORS AND ITERATOR TYPES**

**The term iterator is used to refer to three different entities. We might mean the *concept* of an iterator, or we might refer to the `iterator` *type* defined by a container, or we might refer to an *object* as an iterator.**

**What's important to understand is that there is a collection of types that are related conceptually. A type is an iterator if it supports a common set of actions. Those actions let us access an element in a container and let us move from one element to another.**

**Each container class defines a type named `iterator`; that `iterator` type supports the actions of an (conceptual) iterator.**

### Some `vector` Operations Invalidate Iterators

In § 3.3.3 (p. 106) we noted that there are implications of the fact that `vector`s can grow dynamically. We also noted that one such implication is that we cannot add elements to a `vector` inside a range `for` loop. Another implication is that any operation, such as `push_back`, that changes the size of a `vector` potentially invalidates all iterators into that `vector`. We'll explore how iterators become invalid in more detail in § 9.3.6 (p. 389).

WARNING

For now, it is important to realize that loops that use iterators should not add elements to the container to which the iterators refer.

**EXERCISES SECTION 3.4.1**

**Exercise 3.22:** Redo the first exercise from § 3.3.4 (p. 110) using iterators.

**Exercise 3.23:** Revise the loop from page 114 that printed the first paragraph in `text` to instead change the elements in `text` that correspond to the first paragraph to all uppercase. After you've updated `text`, print its contents.

**Exercise 3.24:** Write a program to create a `vector` with ten `int` elements. Using an iterator, assign each element a value that is twice its current value. Test your program by printing the `vector`.

## 3.4.2 Iterator Arithmetic

Incrementing an iterator moves the iterator one element at a time. All the library containers have iterators that support increment. Similarly, we can use `==` and `!=` to compare two valid iterators (§ 3.4, p. 111) into any of the library container types.

Iterators for `string` and `vector` support additional operations that can move an iterator multiple elements at a time. They also support all the relational operators. These operations, which are often referred to as **iterator arithmetic**, are described in Table 3.8.

**Table 3.8: Operations Supported by `vector` and `string` Iterators**

| | |
|---|---|
| **Only valid on iterators that denote elements in, or one past the end, of the same container** | |
| `iter + n`<br>`iter - n` | Adding (subtracting) an integral value `n` to (from) an iterator yields an iterator that many elements forward (backward) within the container. We can write `iter + n` or `n + iter`. |
| `iter1 += n`<br>`iter1 -= n` | Compound-assignment for iterator addition and subtraction. Assigns to `iter1` the value of adding `n` to, or subtracting `n` from, `iter1`. |
| `iter1 - iter2` | Subtracting two iterators yields the number that when added to the right-hand iterator yields the left-hand iterator. |
| `>`, `>=`, `<`, `<=` | Relational operators on iterators. One iterator is less than another if it refers to an element that appears in the container before the one referred to by the other iterator. |

## Arithmetic Operations on Iterators

We can add (or subtract) an integral value (which can be negative) and an iterator. Doing so returns an iterator positioned forward (or backward) that many elements. When we add or subtract an integral value and an iterator, the result must denote an element in the same `vector` (or `string`) or denote one past the end of the associated `vector` (or `string`). As an example, we can compute an iterator to the element nearest the middle of a `vector`:

```
// compute an iterator to the element closest to the midpoint of vi
auto mid = vi.begin() + vi.size() / 2;
```

If `vi` has 20 elements, then `vi.size()/2` is `10`. In this case, we'd set `mid` equal to `vi.begin() + 10`. Remembering that subscripts start at 0, this element is the same as `vi[10]`, the element ten past the first.

In addition to comparing two iterators for equality, we can compare `vector` and `string` iterators using the relational operators (`<`, `<=`, `>`, `>=`). The iterators must be valid and must denote elements in (or one past the end of) the same `vector` or `string`. For example, assuming `it` is an iterator into the same `vector` as `mid`, we can check whether `it` denotes an element before or after `mid` as follows:

```
if (it < mid)
    // process elements in the first half of vi
```

We can also subtract two iterators so long as they refer to elements in, or one off the end of, the same `vector` or `string`. The result is the distance between the iterators. By distance we mean the amount by which we'd have to change one iterator to get the other. The result type is a signed integral type named **`difference_type`**. Both `vector` and `string` define `difference_type`. This type is signed, because subtraction might have a negative result.

## Using Iterator Arithmetic

A classic algorithm that uses iterator arithmetic is binary search. A binary search looks for a particular value in a sorted sequence. The search starts by looking at the element closest to the middle of the sequence. If that element is the one we want, we're done. Otherwise, if that element is smaller than the one we want, we continue our search by looking only at elements after the rejected one. If the middle element is larger than the one we want, we continue by looking only in the first half. We compute a new middle element in the reduced range and continue looking until we either find the element or run out of elements.

We can do a binary search using iterators as follows:

```
// text must be sorted; beg and end will denote the range we're searching
auto beg = text.begin(), end = text.end();
auto mid = beg + (end - beg)/2; // original midpoint
// while there are still elements to look at and we haven't yet found sought
while (mid != end && *mid != sought) {
    if (sought < *mid)      // is the element we want in the first half?
        end = mid;          // if so, adjust the range to ignore the second half
    else                    // the element we want is in the second half
        beg = mid + 1;      // start looking with the element just after mid
    mid = beg + (end - beg)/2;  // new midpoint
}
```

We start by defining three iterators: `beg` will be the first element in the range, `end` one past the last element, and `mid` the element closest to the middle.

Our loop first checks that the range is not empty. If `mid` is equal to the current value of `end`, then we've run out of elements to search. In this case, the condition fails and we exit the `while`. Otherwise, `mid` refers to an element and we check whether `mid` denotes the one we want. If so, we're done and we exit the loop.

If we still have elements to process, the code inside the `while` adjusts the range by moving `end` or `beg`. If `sought` is less than the element denoted by `mid`, we know that if `sought` is in `text`, it will appear before the element denoted by `mid`. Therefore, we can ignore elements after `mid`, which we do by assigning `mid` to `end`. If `*mid` is smaller than `sought`, the element that we are seeking must be in the range of elements after the one denoted by `mid`. In this case, we adjust the range by making `beg` denote the element just after `mid`. We already know that `mid` is not the one we want, so we can eliminate it from the range.

At the end of the `while`, `mid` will be equal to `end` or it will denote the element for which we are looking. If `mid` equals `end`, then the element was not in `text`.

### Exercises Section 3.4.2

**Exercise 3.25:** Redo the last exercise from § 3.3.4 (p. 110) using iterators.

**Exercise 3.26:** Rewrite the grade clustering program from § 3.3.4 (p. 109) using iterators instead of subscripts.

**Exercise 3.27:** In the binary search program on page 118, why did we write `mid = beg + (end - beg) / 2;` instead of `mid = (beg + end) /2;`?

## 3.5 Arrays

An array is a data structure that is similar to the library `vector` type (§ 3.3, p. 99) but offers a different trade-off between performance and flexibility. Like a `vector`, an array is a container of unnamed objects of a single type that we access by position. Unlike a `vector`, arrays have fixed size. Because arrays have fixed size, they sometimes offer better run-time performance for specialized applications. However, that run-time advantage comes at the cost of lost flexibility.

> **Best Practices** Modern C++ programs rarely need to use arrays. Ordinarily, it's best to use a library container such as `vector` and use an array only when you are certain that you must.

### 3.5.1 Defining and Initializing Built-in Arrays

Arrays are a compound type (§ 2.3, p. 50). An array declarator has the form `a[d]`, where `a` is the name being defined and `d` is the dimension of the array. The dimension specifies the number of elements and must be greater than zero. The number of elements in an array is part of the array's type. As a result, the dimension must be known at compile time, which means that the dimension must be a constant expression (§ 2.4.4, p. 66):

```
unsigned cnt = 42;            // not a constant expression
constexpr unsigned sz = 42; // constexpr see § 2.4.4 (p. 66)
int arr[10];               // array of ten ints
int *parr[sz];             // array of 42 pointers to int
string bad[cnt];           // error: cnt is not a constant expression
string strs[get_size()]; // ok if get_size is constexpr, error otherwise
```

By default, the elements in an array are default initialized (§ 2.2.1, p. 44).

> **WARNING** As with variables of built-in type, a default-initialized array of built-in type that is defined inside a function will have undefined values.

#### Explicitly Initializing Array Elements

We can list initialize (§ 3.3.1, p. 102) the elements in an array. When we do so, we can omit the dimension; the compiler infers it from the number of initializers. If we specify a dimension, the number of initializers must not exceed the specified size. If the dimension is greater than the number of initializers, the initializers are used for the first elements and any remaining elements are value initialized (§ 3.3.1, p. 101):

```
const unsigned sz = 3;
int ia1[sz] = {0,1,2};          // array of three ints with values 0, 1, 2
int a2[] = {0, 1, 2};           // an array of dimension 3
int a3[5] = {0, 1, 2};          // equivalent to a3[] = {0, 1, 2, 0, 0}
string a4[3] = {"hi", "bye"}; // same as a4[] = {"hi", "bye", ""}
int a5[2] = {0,1,2};            // error: too many initializers
```

### Character Arrays Are Special

Character arrays have an additional form of initialization: We can initialize such arrays from a string literal (§ 2.1.3, p. 39). When we use this form of initialization, it is important to remember that string literals end with a null character. That null character is copied into the array along with the characters in the literal:

```
char a1[] = {'C', '+', '+'};         // list initialization, no null
char a2[] = {'C', '+', '+', '\0'}; // list initialization, explicit null
char a3[] = "C++";                   // null terminator added automatically
const char a4[6] = "Daniel";         // error: no space for the null!
const char a5[30] = "Daniel"; // characters after a5[5] are value-initialized
```

The dimension of `a1` is 3; the dimensions of `a2` and `a3` are both 4. The definition of `a4` is in error. Although the literal contains only six characters, the array size must be at least seven—six to hold the literal and one for the null. As with any other array initialization, we can supply an initializer that initializes only the first portion of the array, as we do for `a5`. In this case the remaining elements are value-initialized; that is, the characters from `a5[6]` through `s5[29]` are nulls.

### No Copy or Assignment

We cannot initialize an array as a copy of another array, nor can we assign one array to another:

```
int a[] = {0, 1, 2}; // array of three ints
int a2[] = a;         // error: cannot initialize one array with another
a2 = a;               // error: cannot assign one array to another
```

WARNING

Some compilers allow array assignment as a **compiler extension**. It is usually a good idea to avoid using nonstandard features. Programs that use such features, will not work with a different compiler.

### Understanding Complicated Array Declarations

Like `vector`s, arrays can hold objects of most any type. As with `vector`, there are no arrays of references. Because an array is an object, we can define both pointers and references to arrays. However, it's a bit complicated to define pointers or references to an array. Fortunately, modern C++ programs can use `auto` or `decltype` and let the compiler figure out the type:

```
int arr[10] = {0};   // arr is an array of ten ints all zero
auto Parray = arr;   // Parray points to an array of ten ints
auto &arrRef = arr; // arrRef refers to an array of ten ints

int *ptrs[10];       // ptrs is an array of ten pointers to int
auto &arry = ptrs;   // arry is a reference to an array of ten pointers to int
```

Nonetheless, C++ programmers still need to be able to read and write complicated array declarations. Because the array dimension follows the name being declared, it can be useful to understand array declarations from the inside out:

Start by finding the name of the variable and then look to the right to find the dimension and then to the left to find the element type:

```
int (*Parray)[10] = &arr; //  Parray points to an array of ten ints
int (&arrRef)[10] = arr;  //  arrRef refers to an array of ten ints
int *(&arry)[10] = ptrs;  //  arry is a reference to an array of ten pointers
```

Reading from the inside out, we start by observing that the parentheses around `*Parray` mean that `Parray` is a pointer. Looking right, we see that `Parray` points to an array of size 10. Looking left, we see that the elements in that array are `int`s. Thus, `Parray` is a pointer to an array of ten `int`s. Similarly, `(&arrRef)` says that `arrRef` is a reference. The type to which it refers is an array of size 10. That array holds elements of type `int`. As shown in the definition of `arry`, there are no limits on how many type modifiers can be used. Reading the declaration of `arry` from the inside out, we see that `arry` is a reference. Looking right, we see that the object to which `arry` refers is an array of size 10. Looking left, we see that the element type is pointer to `int`. Thus, `arry` is a reference to an array of ten pointers.

Tip

It can be easier to understand array declarations by starting with the array's name and reading them from the inside out.

### Exercises Section 3.5.1

**Exercise 3.28:** Assuming `txt_size` is a function that takes no arguments and returns an `int` value, which of the following definitions are illegal? Explain why.

```
unsigned buf_size = 1024;
(a) int ia[buf_size];        (b) int ia[4 * 7 - 14];
(c) int ia[txt_size()];      (d) char st[11] = "fundamental";
```

**Exercise 3.29:** What are the values in the following arrays?

```
string sa[10];
int ia[10];
int main() {
    string sa2[10];
    int    ia2[10];
}
```

## 3.5.2 Accessing the Elements of an Array

With the exception that arrays are fixed size, we use arrays in ways that are similar to how we use `vector`s: In particular, we can use a range `for` or a subscript to access elements of an array. As usual, the indices start at 0. For an array of ten elements, the indices are 0 through 9, not 1 through 10.

When we use a variable to subscript an array, we normally should define that variable to have type **`size_t`**. `size_t` is a machine-specific unsigned type that is guaranteed to be large enough to hold the size of any object in memory. The `size_t` type is defined in the `cstddef` header.

As an example, we'll reimplement our grading program from § 3.3.4 (p. 109) to use an array to hold the cluster counters:

```
// count the number of grades by clusters of ten: 0--9, 10--19, . . . 90--99, 100
unsigned scores[11] = {};   // 11 buckets, all value initialized to 0
for (unsigned grade; cin >> grade; )
    if (grade <= 100)
        ++scores[grade/10]; // increment the counter for the current cluster
```

The only obvious difference between this program and the one on page 109 is the declaration of `scores`. In this program, `scores` is an array of 11 `unsigned` elements. The not so obvious difference between these programs is that the subscript operator in this program is the one that is defined as part of the language. The subscript operator used in the program on page 109 was defined by the library `vector` template and applies to operands of type `vector`.

As in the case of `string` or `vector`, it is best to use a range `for` when we want to traverse the entire array. For example, we can print the resulting `scores`:

```
for (auto i : scores)        // for each counter in scores
    cout << i << " ";        // print the value of that counter
cout << endl;
```

## Checking Subscript Values

As with `string` and `vector`, it is up to the programmer to ensure that the subscript value is in range—that is, that the index value is equal to or greater than zero and less than the size of the array. Nothing stops a program from stepping across an array boundary except careful attention to detail and thorough testing of the code. It is possible for programs to compile and execute yet still be fatally wrong.

WARNING

The most common source of security problems are buffer overflow bugs. Such bugs occur when a program mistakenly uses memory outside the range of an array or similar data structure.

## Structured Bindings to Access All the Elements in an Array

New

We can use a structured binding declaration (§ 2.6.2, p. 77) to map each element in an array to a named variable. When applied to an array, a structured binding must declare exactly as many identifiers as there are elements in the array. Each variable will have the same type as the elements in the array. The variables are initialized by the values of the elements, in order. That is, the first variable is initialized by the first element, the second element initializes the second variable, and so on.

For example, we might have a three-element array of `doubles` that we use to represent the three dimensions of a solid. A structured binding gives us a convenient way to define variables corresponding to each dimension:

```
// three doubles initialized from the elements in solid
auto [ht,wd,len] = solid;
```

which is equivalent to:

```
// equivalent to auto [ht,wd,len] = solid
double ht = solid[0], wd = solid[1], len = solid[2];
```

### EXERCISES SECTION 3.5.2

**Exercise 3.30:** Identify the indexing errors in the following code:

```
constexpr size_t array_size = 10;
int ia[array_size];
for (size_t ix = 1; ix <= array_size; ++ix)
      ia[ix] = ix;
```

**Exercise 3.31:** Write a program to define an array of ten `int`s. Give each element the same value as its position in the array.

**Exercise 3.32:** Copy the array you defined in the previous exercise into another array. Rewrite your program to use `vector`s.

**Exercise 3.33:** What would happen if we did not initialize the `scores` array in the program on page 122?

## 3.5.3 Pointers and Arrays

Normally, we obtain a pointer to an object by using the address-of operator (§ 2.3.2, p. 52). Generally speaking, the address-of operator may be applied to any object. The elements in an array are objects. When we subscript an array, the result is the object at that location in the array. As with any other object, we can obtain a pointer to an array element by taking the address of that element:

```
string nums[] = {"one", "two", "three"};  // array of strings
string *p = &nums[0];    // p points to the first element in nums
```

In addition, arrays have a special property—in most places when we use an array, the compiler automatically substitutes a pointer to the first element:

```
string *p2 = nums;       // equivalent to p2 = &nums[0]
```

Note: In most expressions, when we use an object of array type, we are really using a pointer to the first element in that array.

There are various implications of the fact that operations on arrays are often really operations on pointers. One such implication is that when we use an array as an initializer for a variable defined using `auto` (§ 2.5.2, p. 68), the deduced type is a pointer, not an array:

```
auto p(nums); // p is a string * that points to the first element in nums
*p = "zero";  // ok: p is a pointer, changes the value of nums[0]
```

Although `nums` is an array of three `string`s, when we use `nums` as an initializer, the compiler treats that initialization as if we had written

```
auto p(&nums[0]); // now it's clear that p has type string *
```

It is worth noting that this conversion does not happen when we use `decltype` (§ 2.5.3, p. 70). The type returned by `decltype(nums)` is array of three `string`s:

```
// articles is an array of 3 strings
decltype (nums) articles = {"a", "an", "the"};
```

## Pointers Are Iterators

Pointers that address elements in an array have additional operations beyond those we described in § 2.3.2 (p. 52). In particular, pointers to array elements support the same operations as iterators on `vector`s or `string`s (§ 3.4, p. 111). For example, we can use the increment operator to move from one element in an array to the next:

```
int arr[] = {0,1,2,3,4,5,6,7,8,9};
int *p = arr; // p points to the first element in arr; equivalent to *p = &arr[0]
++p;          // p points to arr[1]
```

Just as we can use iterators to traverse the elements in a `vector`, we can use pointers to traverse the elements in an array. Of course, to do so, we need to obtain pointers to the first and one past the last element. As we've just seen, we can obtain a pointer to the first element by using the array itself or by taking the address-of the first element. We can obtain an off-the-end pointer by using another special property of arrays. We can take the address of the nonexistent element one past the last element of an array:

```
int *e = &arr[10]; // pointer just past the last element in arr
```

Here we used the subscript operator to index a nonexisting element; `arr` has ten elements, so the last element in `arr` is at index position 9. The only thing we can do with this element is take its address, which we do to initialize `e`. Like an off-the-end iterator (§ 3.4.1, p. 111), an off-the-end pointer does not point to an element. As a result, we may not dereference or increment an off-the-end pointer.

Using these pointers we can write a loop to print the elements in `arr` as follows:

```
for (int *b = arr; b != e; ++b)
    cout << *b << endl; // print the elements in arr
```

## Library Range and Size Functions

Although we can compute an off-the-end pointer, doing so is error-prone. Instead of managing the array pointers ourselves, we can use library functions, named

New `begin` and `end` (added in C++11), or `cbegin` and `cend` (added in C++14). These functions act like the similarly named container members (§ 3.4.1, p. 111) and (§ 3.4.1, p. 115). They are defined in the `iterator` header.

When applied to an array, `begin` and `cbegin` return a pointer to the first, and `end` and `cend` return a pointer one past the last element in a given array. Like the container operations, the `c` versions let us read, but not write to, the elements in the array. The `c` versions return pointers to `const` (§ 2.4.2, p. 63).

Because arrays are not class types, an array does not have member functions. Instead, these functions are ordinary functions that can be called with an array:

```
int ia[] = {0,1,2,3,4,5,6,7,8,9}; // ia is an array of ten ints
int *beg = begin(ia); // pointer to the first element in ia
int *last = end(ia);  // pointer one past the last element in ia
// cbegin and cend return pointers to const
const int *b2 = cbegin(ia), *e2 = cend(ia);
```

Using `begin` and `end`, it is easy to write a loop to process the elements in an array. For example, assuming `arr` is an array that holds `int` values, we might find the first negative value in `arr` as follows:

```
// pbeg points to the first
int *pbeg = begin(arr);
// find the first negative element, stopping if we've seen all the elements
while (pbeg != end(arr) && *pbeg >= 0)
    ++pbeg;
```

We start by defining `pbeg` to denote the first element. The `while` condition checks that `pbeg` hasn't gotten to the end of `arr`. That check ensures that it is safe to dereference `pbeg`. If there are still elements, we dereference `pbeg` and check whether the underlying element is negative. If so, the condition fails and we exit the loop. If not, we increment the pointer to look at the next element.

**WARNING** A pointer "one past" the end of a built-in array behaves the same way as the iterator returned by the `end` operation of a `vector`. In particular, we may not dereference or increment an off-the-end pointer.

The C++17 library also provides `size` and `empty` functions, which operate like the corresponding container operations: `size` returns the number of elements in a given array. Because there are no arrays of size zero (§ 3.5.1, p. 119), `empty` applied to an array always returns `false`. We can use `size` to rewrite the previous loop that finds the first negative value to use subscripts instead of pointers: New

```
size_t idx = 0;    // we'll use idx to fetch the elements in arr
// this loop ends when we've seen every element or we find a negative value
while (idx < size(arr) && arr[idx] >= 0)
    ++idx;
```

In this version, we define a `size_t` named `idx` that we'll use to iterate through the elements of `arr`. The `while` condition first checks that `idx` is a valid subscript. So long as `idx` is less than the number of elements in `arr`, we can use it to fetch an element from that array. If `idx` is valid, we use it to fetch the next element and check whether that element is a negative value. If so, we fall out of the loop, otherwise we increment our index and look at the next element.

The `size` function is `constexpr` (§ 2.4.4, p. 66) so we can use it to declare an array with the same size as some other array:

```
int a2[size(arr)];  // a2 has as many elements as in the array arr
```

### Using an Array to Initialize a `vector`

Because pointers act like iterators, we can use an array to initialize a `vector` (§ 3.4.1, p. 114):

```
int int_arr[] = {0, 1, 2, 3, 4, 5};
// ivec has six elements; each is a copy of the corresponding element in int_arr
vector<int> ivec(begin(int_arr), end(int_arr));
```

Here we use the library `begin` and `end` functions to pass pointers to the first and one past the last elements in `int_arr`. As a result, `ivec` will have six elements, each of which will have the same value as the corresponding element in `int_arr`.

The specified range can be a subset of the array:

```
// copies three elements: int_arr[1], int_arr[2], int_arr[3]
vector<int> subVec(int_arr + 1, int_arr + 4);
```

This initialization creates `subVec` with three elements. The values of these elements are copies of the values in `int_arr[1]` through `int_arr[3]`.

### Pointer Arithmetic

Pointers that address array elements can use all the iterator operations listed in Table 3.6 (p. 112) and Table 3.8 (p. 117). These operations—dereference, increment, comparisons, addition (or subtraction) of an integral value, subtraction of two pointers—have the same meaning when applied to pointers that point at elements in a built-in array as they do when applied to iterators.

When we add (or subtract) an integral value to (or from) a pointer, the result is a new pointer. That new pointer points to the element the given number ahead of (or behind) the original pointer:

```
constexpr size_t sz = 5;
int arr[sz] = {1,2,3,4,5};
int *ip = arr;      // equivalent to int *ip = &arr[0]
int *ip2 = ip + 4; // ip2 points to arr[4], the last element in arr
```

The result of adding `4` to `ip` is a pointer that points to the element four elements further on in the array from the one to which `ip` currently points.

The result of adding an integral value to a pointer must be a pointer to an element in the same array, or a pointer just past the end of the array:

```
// ok: arr is converted to a pointer to its first element; p points one past the end of arr
int *p = arr + sz; // use caution -- do not dereference!
int *p2 = arr + 10; // error: arr has only 5 elements; p2 has undefined value
```

When we add `sz` to `arr`, the compiler converts `arr` to a pointer to the first element in `arr`. When we add `sz` to that pointer, we get a pointer that points `sz` positions (i.e., `5` positions) past the first one. That is, it points one past the last element in `arr`. Computing a pointer more than one past the last element is an error, although the compiler is unlikely to detect such errors.

As with iterators, subtracting two pointers gives us the distance between those pointers. The pointers must point to elements in (or one past the end of) the same array:

```
auto n = end(arr) - begin(arr); // n is 5, the number of elements in arr
```

The result of subtracting two pointers is a library type named **ptrdiff_t**. Like size_t, the ptrdiff_t type is a machine-specific type and is defined in the cstddef header. Because subtraction might yield a negative distance, ptrdiff_t is a signed integral type.

We can use the relational operators to compare pointers that point to elements of an array, or one past the last element in that array. For example, we can traverse the elements in arr as follows:

```
int *b = arr, *e = arr + size(arr);
while (b < e) {
    // use *b
    ++b;
}
```

Although we can use equality operators (== and !=) to compare any two pointers, we cannot use the relational operators on pointers to two unrelated objects:

```
int i = 0, sz = 42;
int *p = &i, *e = &sz;
while (p < e) // undefined: p and e are unrelated; comparison is meaningless!
```

Although the utility may be obscure at this point, it is worth noting that pointer arithmetic is also valid for null pointers (§ 2.3.2, p. 54) and for pointers that point to an object that is not an array. In the latter case, the pointers must point to the same object, or one past that object. If p is a null pointer, we can add or subtract an integral constant expression (§ 2.4.4, p. 66) whose value is 0 to p. We can also subtract two null pointers from one another, in which case the result is 0.

## Interaction between Dereference and Pointer Arithmetic

The result of adding an integral value to a pointer is itself a pointer. Assuming the resulting pointer points to an element, we can dereference the resulting pointer:

```
int ia[] = {0,2,4,6,8};  // array with 5 elements of type int
int last = *(ia + 4);    // ok: initializes last to 8, the value of ia[4]
```

The expression *(ia + 4) calculates the address four elements past ia and dereferences the resulting pointer. This expression is equivalent to writing ia[4].

Note that, the parentheses around this pointer addition are essential. Writing

```
last = *ia + 4;   // ok: last = 4, equivalent to ia[0] + 4
```

means dereference ia and add 4 to the dereferenced value. We'll cover the reasons for this behavior in § 4.1.2 (p. 144).

## Subscripts and Pointers

As we've seen, in most places when we use the name of an array, we are really using a pointer to the first element in that array. One place where the compiler does this transformation is when we subscript an array. Given

```
int ia[] = {0,2,4,6,8};  // array with 5 elements of type int
```

if we write `ia[0]`, that is an expression that uses the name of an array. When we subscript an array, we are really subscripting a pointer to an element in that array:

```
int i = ia[2];  // ia is converted to a pointer to the first element in ia
                // ia[2] fetches the element to which (ia + 2) points
int *p = ia;    // p points to the first element in ia
i = *(p + 2);   // equivalent to i = ia[2]
```

We can use the subscript operator on any pointer, as long as that pointer points to an element (or one past the last element) in an array:

```
int *p = &ia[2];  // p points to the element indexed by 2
int j = p[1];     // p[1] is equivalent to *(p + 1),
                  // p[1] is the same element as ia[3]
int k = p[-2];    // p[-2] is the same element as ia[0]
```

This last example points out an important difference between arrays and library types such as `vector` and `string` that have subscript operators. The library types force the index used with a subscript to be an unsigned value. The built-in subscript operator does not. The index used with the built-in subscript operator can be a negative value. Of course, the resulting address must point to an element in (or one past the end of) the array to which the original pointer points.

WARNING

Unlike subscripts for `vector` and `string`, the index of the built-in subscript operator has a signed type.

**EXERCISES SECTION 3.5.3**

**Exercise 3.34:** Given that `p1` and `p2` point to elements in the same array, what does the following code do? Are there values of `p1` or `p2` that make this code illegal?

```
p1 += (p2 - p1);
```

**Exercise 3.35:** Using pointers, write a program to set the elements in an array to zero.

**Exercise 3.36:** Write a program to compare two arrays for equality. Write a similar program to compare two `vector`s.

## 3.5.4 C-Style Character Strings

WARNING

Although C++ supports C-style strings, they should be used only to initialize library `string` objects. C-style strings are a rich source of bugs and are the root cause of many security problems.

Character string literals are an instance of a more general construct that C++ inherits from C: **C-style character strings**. C-style strings are not a type. Instead, they are a convention for how to represent and use character strings. Strings that follow this convention are stored in character arrays and are **null terminated**; the last character in the string is followed by a null character (`'\0'`).

Ordinarily we use pointers to manipulate c-style strings. For example, assuming `cp` is a character pointer, the following loop:

```
// check that cp itself isn't null and if not
// loop through the characters in the array until we hit the null-terminator
while (cp && *cp) {
    // do something with *cp
    ++cp;
}
```

first checks that `cp` itself is not a null pointer. (§ 2.3.2, p. 54). If not, `cp` points to a character, which we dereference. If the resulting character is the null-terminator, `*cp` is `0` and the condition will fail. Otherwise, `*cp` is a character that is processed in the body of the `while`.

## C Library String Functions

The Standard C library provides a set of functions, listed in Table 3.9 (overleaf), that operate on C-style strings. These functions are defined in the `cstring` header.

WARNING

The functions in Table 3.9 do not verify their string parameters.

The pointer(s) passed to these routines must point to null-terminated array(s):

```
char ca[] = {'C', '+', '+'};   // not null terminated
cout << strlen(ca) << endl;    // disaster: ca isn't null terminated
```

In this case, `ca` is an array of `char` but is not null terminated. The result is undefined. The most likely effect of this call is that `strlen` will keep looking through the memory that follows `ca` until it encounters a null character.

Although `strlen` and `size` are similar operations there are important differences between the two. In particular, the value returned by these two functions differ from one another: `size` returns the actual size of the array, including any terminating null, `strlen` returns the size up to the first null character. Another difference is that we can call `strlen` with either a pointer or an array object but can call `size` only on an array object, not a pointer. When we call `strlen` with an array, the compiler automatically converts the array to a pointer to the first element (§ 3.5.3, p. 123). The argument to `size` must be an array object (not a pointer) because `size` uses the dimension, which is part of the array type. A pointer does not remember the size of the array to which it points. For example,

```
const char arr[] = "a c-style string";
cout << size(arr) << endl;    // includes the null-terminator (prints 17)
cout << strlen(arr) << endl;  // excludes the null-terminator (prints 16)
auto cp = arr; // cp is a pointer type, not an array type
cout << size(cp) << endl;     // error: cp doesn't have array type
cout << strlen(cp) << endl;   // ok: prints 16
```

**Table 3.9: C-Style Character String Functions**

| | |
|---|---|
| `strlen(p)` | Returns the length of `p`, *not counting the null*. |
| `strcmp(p1, p2)` | Compares `p1` and `p2` for equality. Returns `0` if `p1 == p2`, a positive value if `p1 > p2`, a negative value if `p1 < p2`. |
| `strcat(p1, p2)` | Appends `p2` to `p1`. Returns `p1`. |
| `strcpy(p1, p2)` | Copies `p2` into `p1`. Returns `p1`. |

## Comparing Strings

Comparing two C-style strings is done quite differently from how we compare library `strings`. When we compare two library `strings`, we use the normal relational or equality operators:

```
string s1 = "A string example";
string s2 = "A different string";
if (s1 < s2)  // false: s2 is less than s1
```

Using these operators on similarly defined C-style strings compares the pointer values, not the strings themselves:

```
const char ca1[] = "A string example";
const char ca2[] = "A different string";
if (ca1 < ca2)  // undefined: compares two unrelated addresses
```

Remember that when we use an array, we are really using a pointer to the first element in the array (§ 3.5.3, p. 123). Hence, this condition actually compares two `const char*` values. Those pointers do not address the same object, so the comparison is undefined.

To compare the strings, rather than the pointer values, we can call `strcmp`. That function returns 0 if the strings are equal, or a positive or negative value, depending on whether the first string is larger or smaller than the second:

```
if (strcmp(ca1, ca2) < 0) // same effect as string comparison s1 < s2
```

## Caller Is Responsible for Size of a Destination String

Concatenating or copying C-style strings is also very different from the same operations on library `strings`. For example, if we wanted to concatenate the two `strings` `s1` and `s2` defined above, we can do so directly:

```
// initialize largeStr as a concatenation of s1, a space, and s2
string largeStr = s1 + " " + s2;
```

Doing the same with our two arrays, `ca1` and `ca2`, would be an error. The expression `ca1 + ca2` tries to add two pointers, which is illegal and meaningless.

Instead we can use `strcat` and `strcpy`. However, to use these functions, we must pass an array to hold the resulting string. The array we pass *must* be large enough to hold the generated string, including the null character at the end. The code we show here, although a common usage pattern, is fraught with potential for serious error:

```
// because we're using arrays we can call size to figure out the size of the destination
// disastrous if we miscalculated the size of largeStr
strcpy(largeStr, ca1);      // copies ca1 into largeStr
strcat(largeStr, " ");      // adds a space at the end of largeStr
strcat(largeStr, ca2);      // concatenates ca2 onto largeStr
```

The problem is that we can easily miscalculate the size needed for `largeStr`. Moreover, any time we change the values that we want to store in `largeStr`, we have to remember to double-check that we calculated its size correctly. Unfortunately, programs similar to this code are widely distributed. Programs with such code are error-prone and often lead to serious security leaks.

**EXERCISES SECTION 3.5.4**

**Exercise 3.37:** What does the following program do?

```
const char ca[] = {'h', 'e', 'l', 'l', 'o'};
for (const char *cp = ca; *cp; ) {
    cout << *cp << endl;
    ++cp;
}
```

**Exercise 3.38:** In this section, we noted that it was not only illegal but meaningless to try to add two pointers. Why would adding two pointers be meaningless?

**Exercise 3.39:** Write a program to compare two `string`s. Now write a program to compare the values of two C-style character strings.

**Exercise 3.40:** Write a program to define two character arrays initialized from string literals. Now define a third character array to hold the concatenation of the two arrays. Use `strcpy` and `strcat` to copy the two arrays into the third.

### 3.5.5 Interfacing to Older Code

Many C++ programs predate the standard library and do not use the `string` and `vector` types. Moreover, many C++ programs interface to programs written in C or other languages that cannot use the C++ library. Hence, programs written in modern C++ may have to interface to code that uses arrays and/or C-style character strings. The C++ library offers facilities to make the interface easier to manage.

#### Mixing Library `strings` and C-Style Strings

In § 3.2.1 (p. 86) we saw that we can initialize a `string` from a string literal:

```
string s("Hello World");  // s holds Hello World
```

More generally, we can use a null-terminated character array anywhere that we can use a string literal:

- We can use a null-terminated character array to initialize or assign a `string`.
- We can use a null-terminated character array as one (but not both) operand to the `string` relational operators, the addition operator, or as the right-hand operand in the `string` compound assignment (`+=`) operator.

The reverse functionality is not provided: There is no direct way to use a library `string` when a C-style string is required. For example, there is no way to initialize a character pointer from a `string`. There is, however, a `string` member function named `c_str` that we can often use to accomplish what we want:

```
char *str = s; // error: can't initialize a char* from a string
const char *str = s.c_str(); // ok
```

The name `c_str` indicates that the function returns a C-style character string. That is, it returns a pointer to the beginning of a null-terminated character array that holds the same data as the characters in the `string`. The type of the pointer is `const char*`, which prevents us from changing the contents of the array.

The array returned by `c_str` is not guaranteed to be valid indefinitely. Any subsequent use of `s` that might change the size of `s` can invalidate this array.

WARNING

> If a program needs continued access to the contents of the array returned by `c_str()`, the program must copy the array returned by `c_str`.

## Watch Out for Nulls

When using a `char` array it is critical to know whether that array represents a C-style string or whether it is just an array. In a C-style string, the first null acts as a sentinel, marking the end of the string even if there are additional characters:

```
// ok: name[6] -- name[14] are value initialized to 0, i.e., they hold null characters
char name[15] = "Daniel";
cout << size(name) << endl;   // prints 15, i.e., the size of the array
cout << strlen(name) << endl; // prints 6, i.e., the size of the C-style string
name[7] = 'L';
cout << strlen(name) << endl; // still prints 6
```

Here we explicitly initialized the first six characters in `name` and implicitly value-initialized the remaining characters (§ 3.5.1, p. 120). Thus, the characters in `name` following `a[6]` are all nulls. The size of the array isn't changed, but when we use the array as a C-style string, it is as if the size has been truncated. Even if we later assign a non-null character to a subsequent position in `name`, if that character follows a null it will be ignored. So long as the null remains at `name[6]` that null effectively terminates the array for purposes of using `name` as a C-style string.

On the other hand, as far as the `string` library is concerned, a null just another character. When initializing a library `string` from a character array it is necessary to know whether that array represents a C-style string or whether it is just an array.

If it's a C-style string, then we should initialize the library `string` by passing a pointer to the first character in the array (§ 3.2.1, p. 86). When we pass a single pointer, only the character denoted by that pointer and those that follow up to—but not including—the first null are copied. If it's just an array, we should pass pointers to the first and one past the last character (§ 3.5.3, p. 126). When we pass pointers, every character—including the nulls—is copied into the `string`;

```
string s(name);   //  copies the C-style string, the null matters!
string s2(begin(name), end(name)); //  copy the array, nulls get copied too
cout << s.size() << " " << s2.size() << endl;   //  prints 6 15
```

When we pass a pair of pointers, we're saying we know how many characters we want to copy: We want every character in the range. When we pass a single pointer, there's no information about how many characters to copy. In this case, the C-style character string convention is followed: The pointer must point to a null-terminated array and the new `string` is a copy of every character from the one pointed to by `name` up to the one just before the null.

**WARNING** One perhaps surprising implication of the fact that library `strings` treat nulls as ordinary characters is that, although it happens only infrequently, a library `string` may contain null characters.

**ADVICE: USE LIBRARY TYPES INSTEAD OF ARRAYS**

**Pointers and arrays are surprisingly error-prone. Part of the problem is conceptual: Pointers are used for low-level manipulations and it is easy to make bookkeeping mistakes. Other problems arise because of the syntax, particularly the declaration syntax used with pointers.**

**Modern C++ programs should use `vectors` and iterators instead of built-in arrays and pointers, and use `strings` rather than C-style array-based character strings.**

**EXERCISES SECTION 3.5.5**

**Exercise 3.41:** Write a program to initialize a `vector` from an array of `ints`.

**Exercise 3.42:** Write a program to copy a `vector` of `ints` into an array of `ints`.

## 3.6 Multidimensional Arrays

Strictly speaking, there are no multidimensional arrays in C++. What are commonly referred to as multidimensional arrays are actually arrays of arrays. It can be helpful to keep this fact in mind when you use what appears to be a multidimensional array.

We define an array whose elements are arrays by providing two dimensions: the dimension of the array itself and the dimension of its elements:

```
int ia[3][4]; // array of size 3; each element is an array of ints of size 4
// array of size 10; each element is a 20-element array whose elements are arrays of 30 ints
int arr[10][20][30] = {0}; // initialize all elements to 0
```

As we saw in § 3.5.1 (p. 120), we can more easily understand these definitions by reading them from the inside out. We start with the name we're defining (`ia`) and see that `ia` is an array of size 3. Continuing to look to the right, we see that the elements of `ia` also have a dimension. Thus, the elements in `ia` are themselves arrays of size 4. Looking left, we see that the type of those elements is `int`. So, `ia` is an array of size 3, each of whose elements is an array of four `int`s.

We read the definition for `arr` in the same way. First we see that `arr` is an array of size 10. The elements of that array are themselves arrays of size 20. Each of those arrays has 30 elements that are of type `int`. There is no limit on how many subscripts are used. That is, we can have an array whose elements are arrays of elements that are arrays, and so on.

In a two-dimensional array, the first dimension is usually referred to as the row and the second as the column.

### Initializing the Elements of a Multidimensional Array

As with any array, we can list initialize the elements of a multidimensional array. Multidimensional arrays may be initialized by specifying bracketed values for each row:

```
int ia[3][4] = {      // three elements; each element is an array of size 4
    {0, 1, 2, 3},     // initializers for the row indexed by 0
    {4, 5, 6, 7},     // initializers for the row indexed by 1
    {8, 9, 10, 11}    // initializers for the row indexed by 2
};
```

The nested braces are optional. The following initialization is equivalent, although considerably less clear:

```
// equivalent initialization without the optional nested braces for each row
int ia[3][4] = {0,1,2,3,4,5,6,7,8,9,10,11};
```

As is the case for single-dimension arrays, elements may be left out of the initializer list. We can initialize only the first element of each row as follows:

```
// explicitly initialize only element 0 in each row
int ia[3][4] = {{ 0 }, { 4 }, { 8 }};
```

The remaining elements are value initialized in the same way as ordinary, single-dimension arrays (§ 3.5.1, p. 119). If the nested braces were omitted, the results would be very different. This code

```
// explicitly initialize row 0; the remaining elements are value initialized
int ix[3][4] = {0, 3, 6, 9};
```

initializes the elements of the first row. The remaining elements are initialized to 0.

## Subscripting a Multidimensional Array

As with any array, we can use a subscript to access the elements of a multidimensional array. To do so, we use a separate subscript for each dimension.

If an expression provides as many subscripts as there are dimensions, we get an element with the specified type. If we supply fewer subscripts than there are dimensions, then the result is the array at the specified index:

```
// assigns the first element of arr to the last element in the last row of ia
ia[2][3] = arr[0][0][0];
int (&row)[4] = ia[1]; // binds row to the second four-element array in ia
```

In the first example we supply indices for all the dimensions for both arrays. On the left-hand side, `ia[2]` returns the last row in `ia`. It does not fetch an element from that array but returns the array itself. We subscript that array, fetching element `[3]`, which is the last element in that array. Similarly, the right-hand operand has three dimensions. We first fetch the array at index `0` from the outermost array. The result of that operation is a (multidimensional) array of size 20. We take the first element from that 20-element array, yielding an array of size 30. We then fetch the first element from that array.

In the second example, we define `row` as a reference to an array of four `int`s. We bind that reference to the second row in `ia`.

As another example, it is common to use a pair of nested `for` loops to process the elements in a multidimensional array:

```
constexpr size_t rowCnt = 3, colCnt = 4;
int ia[rowCnt][colCnt];   // 12 uninitialized elements
// for each row
for (size_t i = 0; i != rowCnt; ++i) {
    // for each column within the row
    for (size_t j = 0; j != colCnt; ++j) {
        // assign the element's positional index as its value
        ia[i][j] = i * colCnt + j;
    }
}
```

The outer `for` loops through each of the array elements in `ia`. The inner `for` loops through the elements of those interior arrays. In this case, we set the value of each element as its index in the overall array.

## Using a Range `for` with Multidimensional Arrays

We can simplify the previous loop by using a range `for`:

```
size_t cnt = 0;
for (auto &row : ia) {        // for every element in the outer array
    for (auto &col : row) {   // for every element in the inner array
        col = cnt;            // give this element the next value
        ++cnt;                // increment cnt
    }
}
```

This loop gives the elements of `ia` the same values as the previous loop, but we let the compiler manage the indices. We want to change the value of the elements, so we declare our control variables, `row` and `col`, as references (§ 3.2.3, p. 96). The first `for` iterates through the elements in `ia`. Those elements are arrays of size 4. Thus, the type of `row` is a reference to an array of four `int`s. The second `for` iterates through one of those 4-element arrays. Hence, `col` is `int&`. On each iteration we assign the value of `cnt` to the next element in `ia` and increment `cnt`.

In the this example, we used references as our loop control variables because we wanted to change the elements in the array. However, there is a deeper reason for using references. As an example, consider the following loop:

```
for (const auto &row : ia)  // for every element in the outer array
    for (auto col : row)    // for every element in the inner array
        cout << col << endl;
```

This loop does not write to the elements, yet we still define the control variable of the outer loop as a reference. We do so in order to avoid the normal array to pointer conversion (§ 3.5.3, p. 123). Had we ommited the reference,

```
for (auto row : ia)
    for (auto col : row)
```

our program would not compile. As before, the first `for` iterates through `ia`, whose elements are arrays of size 4. Because `row` is not a reference, when the compiler initializes `row` it will convert each array element (like any other object of array type) to a pointer to that array's first element; in this loop `row` has type `int*`. The inner `for` loop is illegal. Despite our intentions, that loop attempts to iterate over an `int*`.

> Note: To use a multidimensional array in a range `for`, the loop control variable for all but the innermost array must be references.

### Pointers and Multidimensional Arrays

As with any array, when we use the name of a multidimensional array, it is automatically converted to a pointer to the first element in the array.

> Note: When you define a pointer to a multidimensional array, remember that a multidimensional array is really an array of arrays.

Because a multidimensional array is really an array of arrays, the pointer type to which the array converts is a pointer to the first inner array:

```
int ia[3][4];       // array of size 3; each element is an array of ints of size 4
int (*p)[4] = ia;  // p points to an array of four ints
p = &ia[2];         // p now points to the last element in ia
```

Using the strategy from § 3.5.1 (p. 120), we see that `(*p)` says `p` is a pointer. Looking right, we see that `p` points to an object with a dimension of size `4`, and looking left that the elements are `int`s. Thus, `p` is a pointer to an array of four `int`s.

Note

The parentheses in this declaration are essential:

```
int *ip[4];     // array of pointers to int
int (*ip)[4];   // pointer to an array of four ints
```

Fortunately, we can simplify our code by using `auto` or `decltype` (§ 2.5.2, p. 68):

```
auto p = ia;      //  p points to an array of four ints
auto p2 = ia[0]; //  p2 is int* and points to ia[0][0]
//  arr2 has the same type as ia
decltype(ia) arr2 = { {0,1,2,3}, {4,5,6,7}, {8,9,10,11} };
```

## Type Aliases Simplify Pointers to Multidimensional Arrays

A type alias (§ 2.5.1, p. 67) can make it easier to read, write, and understand pointers to multidimensional arrays. For example:

```
using int_array = int[4]; //  type alias declaration
//  print the value of each element in ia, with each inner array on its own line
for (int_array *p = ia; p != ia + 3; ++p) {
    for (int *q = *p; q != *p + 4; ++q)
         cout << *q << ' ';
    cout << endl;
}
```

Here we start by defining `int_array` as a name for the type "array of four `int`s." We use that type name to define our loop control variable in the outer `for` loop.

### EXERCISES SECTION 3.6

**Exercise 3.43:** Print the elements of `ia` using a range `for`, then using an ordinary `for` with subscripts, and again with pointers. Do not use a type alias, `auto`, or `decltype`.

**Exercise 3.44:** Redo the previous exercise first using a type alias for the type of the loop control variables and again using `auto`.

## Chapter Summary

Among the most important library types are `vector` and `string`. A `string` is a variable-length sequence of characters, and a `vector` is a container of objects of a single type.

Iterators allow indirect access to objects stored in a container. Iterators are used to access and navigate between the elements in `string`s and `vector`s.

Arrays and pointers to array elements provide low-level analogs to the `vector` and `string` libraries. In general, the library classes should be used in preference to low-level array and pointer alternatives built into the language.

## Defined Terms

**begin** Container member and ordinary library function that returns an iterator to the first element in a container.

**buffer overflow** Serious programming bug that results when we use an index that is out-of-range for a container.

**C-style strings** Null-terminated character array. String literals are C-style strings. C-style strings are inherently error-prone.

**class template** A blueprint from which specific class types can be created. To use a class template, we must specify additional information. For example, to define a `vector`, we specify the element type: `vector<int>` holds `int`s. Sometimes the compiler can infer this information from an initializer or other context.

**compiler extension** Feature that is added to the language by a particular compiler. Programs that rely on compiler extensions cannot be moved easily to other compilers.

**construct** Term used to describe the process of defining and initializing an object.

**container** Type, such as `vector`, that holds a collection of objects of a given type.

**copy initialization** Form of initialization that uses an `=`. The newly created object is a copy of the given initializer.

**difference_type** A `signed` integral type defined by `vector` and `string` that can hold the distance between two iterators that are associated with the same container.

**direct initialization** Form of initialization that does not include an `=`.

**empty** Container member and ordinary library function that returns `bool true` if `size` is zero, `false` otherwise.

**end** Container member and ordinary library function that returns an iterator one past the last element in a container.

**getline** Function that takes an `istream` and a `string` and reads the given stream up to the next newline, storing what it read into the given `string`. Returns the `istream`. The newline is read and discarded.

**index** Value used in a subscript operator to denote the element to retrieve.

**instantiation** Compiler process that generates a specific template class or function.

**iterator** A type used to access and navigate among the elements of a container.

**iterator arithmetic** Operations on `vector` or `string` iterators: Adding or subtracting an integral value and an iterator yields an iterator that many elements ahead of or behind the original iterator. Subtracting one iterator from another yields the distance between them. Iterators must refer to elements in, or off-the-end of the same container.

**null-terminated string** String whose last character is followed by the null character (`'\0'`).

**off-the-end iterator** The iterator returned by `end` that refers to a nonexistent element one past the end of a container.

**pointer arithmetic** The arithmetic operations that can be applied to pointers. Pointers to elements of arrays support the same operations as iterator arithmetic.

**ptrdiff_t** Machine-dependent signed integral type defined in the `cstddef` header that is large enough to hold the difference between two pointers into the largest possible array.

**push_back** Member of `vector`. Appends elements to the back of a `vector`.

**range for** Control statement that iterates through a specified collection of values.

**size** Container member and ordinary library function that returns the number of characters in a `string` or elements in a container. Ordinary function returns `size_t`, member returns the `size_type` for the type.

**size_t** Machine-dependent unsigned integral type defined in the `cstddef` header that is large enough to hold the size of any array.

**size_type** `unsigned` type defined by the `string` and `vector` classes that is large enough to hold the size of any `string` or `vector`, respectively.

**string** Library type that represents a sequence of characters.

**using declarations** Make a name from a namespace accessible directly.

```
using namespace::name;
```

makes *name* accessible without the *namespace*`::` prefix.

**value initialization** Initialization in which objects of built-in type are initialized to zero and class type objects are default-initialized. Objects of a class type can be value initialized only if the class has a default value.

**vector** Library type that holds a collection of elements of a specified type.

**`++` operator** The iterator types and pointers define the increment operator to "add one" by moving the iterator to refer to the next element.

**`[ ]` operator** Subscript operator. `obj[i]` yields the element at position `i` from the container object `obj`. Indices count from zero—the first element is element `0` and the last is the element indexed by `obj.size() - 1`. Subscript returns an object. If `p` is a pointer and `n` an integer, `p[n]` is a synonym for `*(p+n)`.

**`->` operator** Arrow operator. Combines the operations of dereference and dot operators: `a->b` is a synonym for `(*a).b`.

**`<<` operator** The `string` library type defines an output operator, which prints the characters in a `string`.

**`>>` operator** The `string` library type defines an input operator, which reads whitespace-delimited chunks of characters, storing what is read into the right-hand (`string`) operand.

**`!` operator** Logical NOT operator. Returns the inverse of the `bool` value of its operand. Result is `true` if operand is `false` (or `0`) and `false` otherwise.

**`&&` operator** Logical AND operator. Result is `true` if both operands are `true`. The right-hand operand is evaluated *only* if the left-hand operand is `true`.

**`||` operator** Logical OR operator. Yields `true` if either operand is `true`. The right-hand operand is evaluated *only* if the left-hand operand is `false`.

# CHAPTER 4

# EXPRESSIONS

## CONTENTS



C++ provides a rich set of operators and defines what these operators do when applied to operands of built-in types. Each operator specifies the type expected for its operand(s) and defines what type the operator yields. We can also define the meaning of most of the operators when applied to operands of class types. This chapter focuses on the operators as defined in the language and applied to operands of built-in type. We will also look at some of the operators defined by the library. Chapter 14 will show how we can define operators for our own types.

*An expression* is piece of code that when evaluated yields a **result**. The simplest form of an **expression** is a single literal or variable. The result of such an expression is the value of the variable or literal. More complex expressions are formed from **operators** and one or more **operands**, which are themselves expressions. The result of expressions involving operators depends on those operators.

## 4.1 Fundamentals

There are a few fundamental concepts that affect how expressions are evaluated. We start by briefly discussing the concepts that apply to most (if not all) expressions. Subsequent sections will cover these topics in more detail.

### 4.1.1 Basic Concepts

There are both ***unary operators*** and ***binary operators***. Unary operators, such as address-of (`&`) and dereference (`*`), act on one operand. Binary operators, such as equality (`==`) and multiplication (`*`), act on two operands. There is also one ternary operator that takes three operands, and one operator, function call, that takes an unlimited number of operands.

Some symbols, such as `*`, are used as both a unary (dereference) and a binary (multiplication) operator. The context in which a symbol is used determines whether the symbol represents a unary or binary operator. The uses of such symbols are independent; it can be helpful to think of them as two different symbols.

#### Grouping Operators and Operands

Understanding expressions with multiple operators usually requires understanding the ***precedence*** and ***associativity*** of the operators and may depend on the ***order of evaluation*** of the operands. For example, the result of the following expression depends on how the operands are grouped to the operators:

```
5 + 10 * 20/2;
```

The operands to the `*` operator could be `10` and `20`, or `10` and `20/2`, or `15` and `20`, or `15` and `20/2`. Understanding such expressions is the topic of the next section.

#### Operand Conversions

For each operator, C++ defines the type(s) of the operator's operand(s) and the type of its result. We can supply operands that differ from the expected type so long as the type that we supply can be converted (§ 2.1.2, p. 34) to the required type. For example, in general the two operands to a binary operator must have the same type. We can supply operands whose types differ from each other so long as those types can be converted to a common type.

Although the rules are somewhat complicated, conversions generally happen in unsurprising ways. For example, we can convert an integer to floating-point, and vice versa, but we cannot convert a pointer type to floating-point. We'll look in detail at conversions in § 4.11 (p. 169).

## Lvalues and Rvalues

Every expression in C++ is either an **rvalue** (pronounced "are-value") or an **lvalue** (pronounced "ell-value"). Just as an operator defines the type of its result and the type(s) of its operand(s), operators also define whether they return lvalues or rvalues and whether they require lvalue or rvalue operands.

The names lvalue and rvalue are inherited from C and originally had a simple mnemonic purpose: lvalues could appear on the left-hand side of an assignment whereas rvalues could not. In C++, the distinction isn't so neat. In particular, lvalue expressions whose result has a `const` type, may not be the left-hand operand of an assignment.

A better mnemonic in C++ is to realize that roughly speaking, an lvalue expression yields the identity (the location in memory) of an object; an rvalue expression yields the value of a computation or of an object (the object's contents). Operators require lvalue operands when they need to access a location in memory; they require rvalue operands when the operator needs only a value.

With one exception that we'll cover in § 13.5 (p. 584), we can use an lvalue when an rvalue is required. After all, an lvalue denotes a location in memory. When we use an lvalue as an rvalue, we use the value in that location. On the other hand, we (usually) cannot use an rvalue when an lvalue (i.e., a location) is required. However, when an rvalue has class or array type, there are some contexts in which the compiler will use the location of that class object or array rvalue.

We have already used several operators that involve lvalues.

- Assignment requires a modifiable (non`const`) lvalue as its left-hand operand and yields its left-hand operand as an lvalue.
- The address-of operator (§ 2.3.2, p. 52) requires an lvalue operand and returns a pointer to its operand as an rvalue.
- The dereference operator (§ 2.3.2, p. 53) requires an rvalue of pointer type and yields the object to which the pointer points as an lvalue. The lvalue is modifiable unless the pointer points to `const` (§ 2.4.2, p. 63).
- The dot operator (§ 1.5.2, p. 23) requires its left-hand operand to be of class type. If the left-hand operand is an lvalue, dot returns an lvalue; otherwise it returns an rvalue.
- Similarly, the subscript operator § 3.5.2, p. 121) requires a pointer or lvalue or rvalue operand of array type. If the operand was an rvalue array, then the operator returns an rvalue. Otherwise it returns an lvalue.
- The increment and decrement operators (§ 1.4.1, p. 12, and § 3.4.1, p. 112) require modifiable (non`const`) lvalue operands. The prefix versions (which are the ones we have used so far) yield lvalues.

As we present the operators, we will note whether an operand must be an lvalue and whether the operator returns an lvalue.

We have already seen that whether an expression is an lvalue or an rvalue affects the type deduced by `decltype` (§ 2.5.3, p. 70). When we apply `decltype` to an expression (other than a variable), the result is a reference type if the expression yields an lvalue. As an example, assume that `p` is an `int*`. Because dereference yields an lvalue, `decltype(*p)` is `int&`. On the other hand, because the address-of operator yields an rvalue, `decltype(&p)` is `int**` that is, a pointer to a pointer to type `int`.

### Overloaded Operators

The language defines what the operators mean when applied to built-in and compound types. We can also define what most operators mean when applied to class types. Because such definitions give an alternative meaning to an existing operator symbol, we refer to them as **overloaded operators**. We have used several overloaded operators already: the IO library >> and << operators and the operators we used with `string`s, `vector`s, and iterators are all overloaded operators.

When we use an overloaded operator, the meaning of the operator—including the type of its operand(s), the type of its result, and whether the result or operands are lvalues or rvalues—depend on the definition of that operator. However, the number of operands, the precedence, associativity, and (since C++17) order of operand evaluation of the operator cannot be changed.

New

Of the overloaded operators we've used so far, the IO operators (§ 1.2, p. 7 and § 1.2, p. 8), the iterator dereference operator (§ 3.4.1, p. 111), and the `string` and `vector` subscript operators (§ 3.2.3, p. 97 and § 3.3.4, p. 107) all yield lvalues.

### Exercises Section 4.1.1

**Exercise 4.1:** What are the types of `i` and `j` in the following:

```
int arr[] = {0,1,2,3,4,5,6,7,8,9};
auto i = arr[0];    decltype(arr[0]) j = i;
```

## 4.1.2 Precedence and Associativity

An expression with two or more operators is a **compound expression**. Evaluating a compound expression involves grouping the operands to the operators. Precedence and associativity determine how the operands are grouped. That is, they determine which parts of the expression are the operands for each of the operators in the expression. Programmers can override these rules by parenthesizing compound expressions to force a particular grouping.

In general, the value of an expression depends on how the subexpressions are grouped. Operands of operators with higher precedence group more tightly than operands of operators at lower precedence. Associativity determines how to group operands with the same precedence. For example, multiplication and division have the same precedence as each other, but they have higher precedence than addition. Therefore, operands to multiplication and division group before operands to addition and subtraction. The arithmetic operators are left associative, which means operators at the same precdence group left to right:

- Because of precedence, the expression `3+4*5` is `23`, not `35`.
- Because of associativity, the expression `20-15-3` is `2`, not `8`.

As a more complicated example, in C++ the following expression

```
6 + 3 * 4 / 2 + 2
```

yields 14 because it is equivalent to

```
// parentheses in this expression match default precedence and associativity
((6 + ((3 * 4) / 2)) + 2)
```

A left-to-right evaluation of this expression yields 20 and other imaginable results include 9, 14, and 36.

## Parentheses Override Precedence and Associativity

We can override the normal grouping with parentheses. Parenthesized expressions are evaluated by treating each parenthesized subexpression as a unit and otherwise applying the normal precedence rules. For example, we can parenthesize the expression above to force the result to be any of the four possible values:

```
// parentheses result in alternative groupings
cout << (6 + 3) * (4 / 2 + 2) << endl;     // prints 36
cout << ((6 + 3) * 4) / 2 + 2 << endl;     // prints 20
cout << 6 + 3 * 4 / (2 + 2) << endl;       // prints 9
```

## When Precedence and Associativity Matter

We have already seen examples where precedence affects the correctness of our programs. For example, consider the discussion in § 3.5.3 (p. 127) about dereference and pointer arithmetic:

```
int ia[] = {0,2,4,6,8}; // array with five elements of type int
int last = *(ia + 4);   // initializes last to 8, the value of ia[4]
last = *ia + 4;         // last = 4, equivalent to ia[0] + 4
```

If we want to access the element at the location `ia + 4`, then the parentheses around the addition are essential. Without parentheses, `*ia` is grouped first and `4` is added to the value in `*ia`.

The most common case that we've seen in which associativity matters is in input and output expressions. As we'll see in § 4.8 (p. 166), the operators used for IO are left associative. This associativity means we can combine several IO operations in a single expression:

```
cin >> v1 >> v2;  // read into v1 and then into v2
```

Table 4.12 (p. 176) lists all the operators organized into segments separated by double lines. Operators in each segment have the same precedence, and have higher precedence than operators in subsequent segments. For example, the prefix increment and dereference operators share the same precedence, which is higher than that of the arithmetic operators. The table includes a page reference to each operator's description. We have seen some of these operators already and will cover most of the rest in this chapter. However, there are a few operators that we will not cover until later.

**EXERCISES SECTION 4.1.2**

**Exercise 4.2:** What is the value returned by `5 + 10 * 20/2`?

**Exercise 4.3:** Using Table 4.12 (p. 176), parenthesize the following expressions to indicate the order in which the operands are grouped:

(a) `* vec.begin()` (b) `* vec.begin() + 1`

### 4.1.3 Order of Evaluation

Precedence and associativity specify how operands are grouped. They say nothing about the order in which the operands are evaluated. In many cases, the order is unspecified. For example, the arithmetic operators make no guarantees about when their operands are evaluated. In the following expression

```
int i = f1() - f2();
```

we know that `f1` and `f2` must be called before the subtraction can be done. After all, it is their results that are subtracted. However, we have no way of knowing whether `f1` will be called before `f2` or vice versa. So long as `f1` and `f2` are independent functions, the order in which they are called doesn't matter.

Now let's change our example and assume that we want to call a function named `readVal` that reads a value from an input stream. Further assume that our input stream holds two values, `42` and `0`. Thus, the first call to `readVal` returns `42` and the second returns `0`. The result of this seemingly similar expression:

```
i = readVal() - readVal(); // undefined
```

is unknowable. It should be clear that the result of this expression depends on whether the right-hand operand is evaluated before or after the left-hand. That is, this expression might be `42 - 0` or it might be `0 - 42`. Technically speaking, this code is undefined (§ 2.1.2, p. 36).

#### Operators that Define Order of Operand Evaluation

Historically, only four operators guaranteed the order in which operands were evaluated. We saw in § 3.2.3 (p. 98) that the logical AND operator guarantees that its left-hand operand is evaluated first. The other three operators that have always guaranteed the order in which operands are evaluated are the logical OR operator, the conditional operator, and the comma operator, all of which we cover later in this chapter.

Starting with C++17, several additional operators give guarantees about evaluation order. In this chapter we'll cover three of these: the assignment, subscript, and shift operators. New

## Order of Evaluation, Precedence, and Associativity

Order of operand evaluation is independent of precedence and associativity. In an expression such as `f() + g() * h() + j()`:

- Precedence guarantees that the results of `g()` and `h()` are multiplied.
- Associativity guarantees that the result of `f()` is added to the product of `g()` and `h()` and that the result of that addition is added to the value of `j()`.
- There are no guarantees as to the order in which these functions are called.

If `f`, `g`, `h`, and `j` are independent functions that do not affect the state of the same objects or perform IO, then the order in which the functions are called is irrelevant. If any of these functions do affect the same object, then the expression is in error and has undefined behavior.

### EXERCISES SECTION 4.1.3

**Exercise 4.4:** Order of evaluation for most of the binary operators is left undefined to give the compiler opportunities for optimization. This strategy presents a trade-off between efficient code generation and potential pitfalls in the use of the language by the programmer. Do you consider that an acceptable trade-off? Why or why not?

### ADVICE: MANAGING COMPOUND EXPRESSIONS

**When you write compound expressions, two rules of thumb can be helpful:**

1. **When in doubt—or when you think a reader of your code might be in doubt—parenthesize expressions to force the grouping that the logic of your program requires.**
2. **If you change the value of an operand, don't use that operand elsewhere in the same expresion.**

**An important exception to the second rule occurs when the subexpression that changes the operand is itself the operand of another subexpression. For example, in `*++iter`, the increment changes the value of `iter`. The (now changed) value of `iter` is the operand to the dereference operator. In this (and similar) expressions, order of evaluation isn't an issue. The increment (i.e., the subexpression that changes the operand) must be evaluated before the dereference can be evaluated. Such usage poses no problems and is quite common.**

## 4.2 Arithmetic Operators

**Table 4.1: Arithmetic Operators (Left Associative)**

| Operator | Function | Use |
|---|---|---|
| `+` | unary plus | `+ expr` |
| `-` | unary minus | `- expr` |
| `*` | multiplication | `expr * expr` |
| `/` | division | `expr / expr` |
| `%` | remainder | `expr % expr` |
| `+` | addition | `expr + expr` |
| `-` | subtraction | `expr - expr` |

Table 4.1, and the operator tables in subsequent sections, groups the operators by their precedence. The unary arithmetic operators have higher precedence than the multiplication and division operators, which in turn have higher precedence than the binary addition and subtraction operators. Operators of higher precedence group more tightly than do operators with lower precedence. These operators are all left associative, meaning that they group left to right when the precedence levels are the same. The order of evaluation of the operands to the binary arithmetic operators is unspecified.

Unless noted otherwise, the arithmetic operators may be applied to any of the arithmetic types (§ 2.1.1, p. 32) or to any type that can be converted to an arithmetic type. The operands and results of these operators are rvalues. As described in § 4.11 (p. 169), operands of small integral types (e.g., `bool`, `char`, `short`, etc.) are always **promoted** to a larger integral type, typically `int`. If after any promotions, the operands still have differing types, they are converted to a common type.

The unary plus operator and the addition and subtraction operators may also be applied to pointers. § 3.5.3 (p. 126) covered the use of binary `+` and `-` with pointer operands. When applied to a pointer or arithmetic value, unary plus returns a (possibly promoted) copy of the value of its operand.

The unary minus operator returns the result of negating a (possibly promoted) copy of the value of its operand:

```
int i = 1024;
int k = -i; //  k is -1024
bool b = true;
bool b2 = -b; //  b2 is true!
```

In § 2.1.1 (p. 35) we noted that `bool` values should not be used for computation. The result of `-b` is a good example of what we had in mind.

For most operators, operands of type `bool` are promoted to `int`. In this case, the value of `b` is `true`, which promotes to the `int` value `1` (§ 2.1.2, p. 35). That (promoted) value is negated, yielding `-1`. The value `-1` is converted back to `bool` and used to initialize `b2`. This initializer is a nonzero value, which when converted to `bool` is `true`. Thus, the value of `b2` is `true`!

### CAUTION: OVERFLOW AND OTHER ARITHMETIC EXCEPTIONS

**Some arithmetic expressions yield undefined results. Some of these undefined expressions are due to the nature of mathematics—for example, division by zero. Others are undefined due to the nature of computers—for example, due to overflow. Overflow happens when a value is computed that is outside the range of values that its type can represent.**

**Consider a machine on which `short`s are 16 bits. In that case, the maximum `short` is 32767. On such a machine, the following compound assignment overflows:**

```
short short_value = 32767; // max value if shorts are 16 bits
short_value += 1; // this calculation overflows
cout << "short_value: " << short_value << endl;
```

**The assignment to `short_value` is undefined. Representing a signed value of 32768 requires 17 bits, but only 16 are available. On many systems, there is *no* compile-time or run-time warning when an overflow occurs. As with any undefined behavior, what happens is unpredictable. On our system the program completes and writes**

```
short_value: -32768
```

**The value "wrapped around": The sign bit, which had been 0, was set to 1, resulting in a negative value. On another system, the result might be different, or the program might behave differently, including crashing entirely.**

When applied to objects of arithmetic types, the arithmetic operators, `+`, `-`, `*`, and `/`, have their obvious meanings: addition, subtraction, multiplication, and division. Division between integers returns an integer. If the quotient contains a fractional part, it is truncated toward zero:

```
int ival1 = 21/7;  // ival1 is 3; no remainder; result is an integral value
int ival2 = 21/6;  // ival2 is 3; result is truncated; remainder is discarded
int ival3 = 21/-6; // ival3 is -3; result is truncated towards zero
```

The `%` operator, known as the "remainder" or the "modulus" operator, computes the remainder that results from dividing the left-hand operand by the right-hand operand. The operands to `%` must have integral type:

```
int ival = 42;
double dval = 3.14;
ival % 12;   // ok: result is 6
ival % dval; // error: floating-point operand
```

The modulus operator is defined so that if `m` and `n` are integers and `n` is nonzero, then `(m/n)*n + m%n` is equal to `m`. By implication, if `m%n` is nonzero, it has the same sign as `m`. Except for the obscure case where `-m` overflows, `(-m)/n` and `m/(-n)` are always equal to `-(m/n)`, `m%(-n)` is equal to `m%n`, and `(-m)%n` is equal to `-(m%n)`. More concretely:

```
 21 % 6;   /* result is 3  */        21 / 6;   /* result is 3  */
 21 % 7;   /* result is 0  */        21 / 7;   /* result is 3  */
-21 % -8;  /* result is -5 */       -21 / -8;  /* result is 2  */
 21 % -5;  /* result is 1  */        21 / -5;  /* result is -4 */
```

**EXERCISES SECTION 4.2**

**Exercise 4.5:** Parenthesize the following expression to show how it is evaluated. Test your answer by compiling the expression (without parentheses) and printing its result.

```
12 / 3 * 4 + 5 * 15 + 24 % 4 / 2
```

**Exercise 4.6:** Determine the result of the following expressions.

(a) `-30 * 3 + 21 / 5`
(b) `-30 + 3 * 21 / 5`
(c) `30 / 3 * 21 % 5`
(d) `-30 / 3 * 21 % 4`

**Exercise 4.7:** Write an expression to determine whether an `int` value is even or odd.

**Exercise 4.8:** What does overflow mean? Show three expressions that will overflow.

## 4.3 Logical and Relational Operators

**Table 4.2: Logical and Relational Operators**

| Associativity | Operator | Function | Use |
|---|---|---|---|
| Right | `!` | logical NOT | `!expr` |
| Left | `<` | less than | `expr < expr` |
| Left | `<=` | less than or equal | `expr <= expr` |
| Left | `>` | greater than | `expr > expr` |
| Left | `>=` | greater than or equal | `expr >= expr` |
| Left | `==` | equality | `expr == expr` |
| Left | `!=` | inequality | `expr != expr` |
| Left | `&&` | logical AND | `expr && expr` |
| Left | `\|\|` | logical OR | `expr \|\| expr` |

The relational and logical operators return rvalues of type `bool`. The relational operators take rvalue operands of arithmetic or pointer type. the logical operators take operands of any type that can be converted to `bool`; arithmetic and pointer operand(s) with a value of zero convert to `false`; all other values convert to `true`.

### Logical AND and OR Operators

The logical AND operator yields `true` if and only if both operands (after converion to `bool`) evaluate as `true`. The logical OR (`||`) operator evaluates as `true` if and only if either or both of its operands (after converion to `bool`) evaluate as `true`.

The logical AND and OR operators always evaluate their left operand before the right. Moreover, the right operand is evaluated *if and only if* the left operand does not determine the result. This strategy is known as **short-circuit evaluation**:

- The right side of an `&&` is evaluated if and only if the left side is `true`.
- The right side of an `||` is evaluated if and only if the left side is `false`.

Several of the programs in Chapter 3 used the logical AND operator. Those programs used the left-hand operand to test whether it was safe to evaluate the right-hand operand. For example, the `for` condition on page 97:

```
index != s.size() && !isspace(s[index])
```

first checks that `index` has not reached the end of its associated `string`. We're guaranteed that the right operand won't be evaluated unless `index` is in range.

As an example that uses logical OR, imagine we have some text in a `vector` of `string`s. We want to print the `string`s, adding a newline after each empty `string` or after a `string` that ends with a period. We'll use a range-based `for` loop (§ 3.2.3, p. 93) to process each element:

```
// note s as a reference to const; the elements aren't copied and can't be changed
for (const auto &s : text) { // for each element in text
    cout << s;          // print the current element
    // blank lines and those that end with a period get a newline
    if (s.empty() || s[s.size() - 1] == '.')
        cout << endl;
    else
        cout << " ";   // otherwise just separate with a space
}
```

After we print the current element, we check to see if we need to print a newline. The condition in the `if` first checks whether `s` is an empty `string`. If so, we need to print a newline regardless of the value of the right-hand operand. Only if the `string` is not empty do we evaluate the second expression, which checks whether the `string` ends with a period. In this expression, we rely on short-circuit evaluation of `||` to ensure that we subscript `s` only if `s` is not empty.

It is worth noting that we declared `s` as a reference to `const` (§ 2.5.2, p. 69). The elements in `text` are `string`s, and might be large. By making `s` a reference, we avoid copying the elements. Because we don't need to write to the elements, we made `s` a reference to `const`.

## Logical NOT Operator

The logical NOT operator (`!`) returns the inverse of the truth value of its operand. We first used this operator in § 3.2.2 (p. 90). As another example, assuming `vec` is a `vector` of `int`s, we might use the logical NOT operator to see whether `vec` has elements by negating the value returned by `empty`:

```
// print the first element in vec if there is one
if (!vec.empty())
    cout << vec[0];
```

The subexpression

```
!vec.empty()
```

evaluates as `true` if the call to `empty` returns `false`.

## The Relational Operators

The relational operators (`<`, `<=`, `>`, `<=`, `==`, `!=`) take operands of arithmetic or pointer type, have their ordinary meanings, and return `bool` values. These operators are left associative; the order of evaluation of their operands is unspecified. The `==` and `!=` have lower precedence than the other four.

Because these operators return `bool`s, the result of chaining these operators together is likely to be surprising:

```
// oops! this condition compares k to the bool result of i < j
if (i < j < k) // true if k is greater than 1!
```

This condition groups `i` and `j` to the first `<` operator. The `bool` result of that expression is the left-hand operand of the second less-than operator. That is, `k` is compared to the `true`/`false` result of the first comparison! To accomplish the test we intended, we can rewrite the expression as follows:

```
// ok: condition is true if i is smaller than j and j is smaller than k
if (i < j && j < k) { /* ... */ }
```

## Equality Tests and the `bool` Literals

If we want to test the truth value of an arithmetic or pointer object, the most direct way is to use the value as a condition:

```
if (val)  { /* ... */ } // true if val is any nonzero value
if (!val) { /* ... */ } // true if val is zero
```

In both conditions, the compiler converts `val` to `bool`. The first condition succeeds so long as `val` is nonzero; the second succeeds if `val` is zero.

We might think we could rewrite a test of this kind as

```
if (val == true)  { /* ... */ } // true only if val is equal to 1!
```

There are two problems with this approach. First, it is longer and less direct than the previous code (although admittedly when first learning C++ this kind of abbreviation can be perplexing). Much more importantly, when `val` is not a `bool`, this comparison does not work as expected.

If `val` is not a `bool`, then `true` is converted to the type of `val` before the `==` operator is applied. That is, when `val` is not a `bool`, it is as if we had written

```
if (val == 1) { /* ... */ }
```

As we've seen, when a `bool` is converted to another arithmetic type, `false` converts to `0` and `true` converts to `1` (§ 2.1.2, p. 35). If we really cared whether `val` had the specific value `1`, we should write the condition to test that case directly.

**WARNING** It is usually a bad idea to use the boolean literals `true` and `false` as operands in a comparison. These literals should be used only to compare to an object of type `bool`.

### EXERCISES SECTION 4.3

**Exercise 4.9:** Explain when operands are evaluated in the logical AND, logical OR, and equality operators.

**Exercise 4.10:** Explain the behavior of the condition in the following `if`:

```
const char *cp = "Hello World";
if (cp && *cp)
```

**Exercise 4.11:** Write the condition for a `while` loop that would read `int`s from the standard input and stop when the value read is equal to `42`.

**Exercise 4.12:** Write an expression that tests four values, `a`, `b`, `c`, and `d`, and ensures that `a` is greater than `b`, which is greater than `c`, which is greater than `d`.

**Exercise 4.13:** Assuming `i`, `j`, and `k` are all `int`s, explain what `i != j < k` means. Also explain what `i < j == j < k` means.

## 4.4 Assignment Operators

The left-hand operand of an assignment operator must be a modifiable lvalue. For example, given

```
int i = 0, j = 0, k = 0; // initializations, not assignment
const int ci = i;        // initialization, not assignment
```

Each of these assignments is illegal:

```
1024 = k;      // error: literals are rvalues
i + j = k;     // error: arithmetic expressions are rvalues
ci = k;        // error: ci is a const (nonmodifiable) lvalue
```

The result of an assignment is its left-hand operand, which is an lvalue. The type of the result is the type of the left-hand operand. If the types of the left and right operands differ, the right-hand operand is converted to the type of the left:

```
i = 0;         // result: type int, value 0
i = 3.14159;   // result: type int, value 3
```

We can also use a braced initializer list (§ 2.2.1, p. 43) on the right-hand side. The initializer list may be empty. If it is empty, the compiler generates a value-initialized (§ 3.3.1, p. 101) value of the appropriate type and assigns that value to the left-hand operand. For the built-in types, the initializer list may contain at most one value and that value must not require a narrowing conversion (§ 2.2.1, p. 43):

```
i = {3.14};    // error: narrowing conversion
i = {1,2,3};   // error: too many initializers
i = {};        // ok: i is value-initialized to 0
```

### Assignment Is Right Associative

Unlike the other binary operators, assignment is right associative:

```
int ival, jval;
ival = jval = 0; // ok: each assigned 0
```

Because assignment is right associative, the right-most assignment, `jval = 0`, is the right-hand operand of the left-most assignment operator. Because assignment returns its left-hand operand, the result of the right-most assignment (i.e., `jval`) is assigned to `ival`.

Each object in a multiple assignment must have the same type as its right-hand neighbor or a type to which that neighbor can be converted (§ 4.11, p. 169):

```
int ival, *pval; // ival is an int; pval is a pointer to int
ival = pval = 0; // error: cannot assign the value of a pointer to an int
pval = 0;        // ok: 0 is a constant expression, okay to assign to a pointer
pval = ival = 0; // error: ival is not a constant expression
string s1, s2;
s1 = s2 = "OK";  // string literal "OK" converted to string
```

The first assignment is illegal because `ival` and `pval` have different types and there is no conversion from the type of `pval` (`int*`) to the type of `ival` (`int`). It is illegal even though zero is a value that can be assigned to either object.

On the other hand, the second assignment is fine. The string literal is converted to `string`, and that `string` is assigned to `s2`. The result of that assignment is `s2`, which has the same type as `s1`.

## The Right-Hand Operand is Evaluated before the Left

New Starting with C++17, we are guaranteed that the right-hand operand of an assignment is evaluated before the left. One might think that this guarantee doesn't matter—after all the right-hand operand has to be evaluated in order to calculate the value to store into the left-hand. When it matters is when both the left- and right-hand operands refer to the same object and one side changes that object:

```
// before C++17 this code was undefined
ia[i] = ia[++i];  // ok: i is incremented before ia[i] is evaluated
```

Here, the right-hand operand increments the value of `i` and we are guaranteed that the increment happens *before* the compiler uses `i` to fetch an element from `ia`. Thus, if `i` was `0` before this statement executes, the effect is to set `ia[1]` to `1`.

Prior to C++17 this code was undefined (§ 2.1.2, p. 36). In earlier releases of C++ assignment made no guarantees about when operands were evaluated. This code might have executed by first fetching `ia[i]` and then incrementing `i`. In that case, and again assuming `i` was `0`, the effect would be to set `ia[0]` to `1`.

## Assignment Has Low Precedence

Assignments often occur in conditions. Because assignment has relatively low precedence, we usually must parenthesize the assignment for the condition to work properly. To see why assignment in a condition is useful, consider the following loop. We want to call a function until it returns a desired value—say, 42:

```
// a verbose and therefore more error-prone way to write this loop
int i = get_value();  // get the first value
while (i != 42) {
    // do something that uses i . . .
    i = get_value();  // get remaining values
}
```

Here we start by calling `get_value` followed by a loop whose condition uses the value returned from that call. The last statement in this loop makes another call to `get_value`, and the loop repeats. We can write this code more directly as

```
int i;
// a better way to write our loop---what the condition does is now clearer
while ((i = get_value()) != 42) {
    // do something that uses i . . .
}
```

The condition now more clearly expresses our intent: We want to continue until `get_value` returns `42`. The condition executes by assigning the result returned by `get_value` to `i` and then comparing the result of that assignment with `42`.

Without the parentheses, the operands to `!=` would be the value returned from `get_value` and `42`. The `true` or `false` result of that test would be assigned to `i`—clearly not what we intended!

Note

Because assignment has lower precedence than the relational operators, parentheses are usually needed around assignments in conditions.

## Beware of Confusing Equality and Assignment Operators

The fact that we can use assignment in a condition can have surprising effects:

```
if (i = j)
```

The condition in this `if` assigns the value of `j` to `i` and then tests the result of the assignment. If `j` is nonzero, the condition will be `true`. The author of this code almost surely intended to test whether `i` and `j` have the same value:

```
if (i == j)
```

Bugs of this sort are notoriously difficult to find. Some, but not all, compilers are kind enough to warn about code such as this example.

## Compound Assignment Operators

We often apply an operator to an object and then assign the result to that same object. As an example, consider the sum program from § 1.4.2 (p. 13):

```
int sum = 0;
// sum values from 1 through 10 inclusive
for (int val = 1; val <= 10; ++val)
    sum += val;  // equivalent to sum = sum + val
```

This kind of operation is common not just for addition but for the other arithmetic operators and the bitwise operators, which we cover in § 4.8 (p. 162). There are compound assignments for each of these operators:

```
 +=    -=    *=    /=    %=       // arithmetic operators
<<=   >>=    &=    ^=    |=       // bitwise operators; see § 4.8 (p. 162)
```

Each compound operator is essentially equivalent to

```
a = a op b;
```

with the exception that, when we use the compound assignment, the left-hand operand is evaluated only once. If we use an ordinary assignment, that operand is evaluated twice: once in the expression on the right-hand side and again as the operand on the left hand. In many, perhaps most, contexts this difference is immaterial aside from possible performance consequences.

The compound assignment operators have the same precedence, associativity, and order of evaluation as the basic assignment operator. That is, these operators associate right-to-left and we are guaranteed that the right-hand operand is evaluated before the left-hand. Like basic assignment, the left-hand operand must be an lvalue and the result is that lvalue operand.

**EXERCISES SECTION 4.4**

**Exercise 4.14:** What are the values of `i` and `d` after each assignment?

```
int i;   double d;
(a) d = i = 3.5;     (b) i = d = 3.5;
```

**Exercise 4.15:** Explain what happens in each of the `if` tests:

```
if (42 = i)   // ...
if (i = 42)   // ...
```

**Exercise 4.16:** The following assignment is illegal. Why? How would you correct it?

```
double dval; int ival; int *pi;
dval = ival = pi = 0;
```

**Exercise 4.17:** Although the following are legal, they probably do not behave as the programmer expects. Why? Rewrite the expressions as you think they should be.

(a) `if (p = getPtr() != 0)`     (b) `if (i = 1024)`

## 4.5 Increment and Decrement Operators

The increment (`++`) and decrement (`--`) operators provide a convenient notational shorthand for adding or subtracting 1 from an object. The built-in versions of these operators take a single, lvalue, operand of a pointer type or any arithmetic type other than `bool`. The library also defines these operators for the iterators on library types such as `vector` and `string` (§ 3.4, p. 111). What we say in this section applies equally to the built-in operators used with built-in operands and the versions defined by library when applied to iterators.

There are two forms of these operators: prefix and postfix. So far, we have used only the prefix form. This form increments (or decrements) its operand and yields the *changed* object as its result. The postfix operators increment (or decrement) the operand but yield a copy of the original, *unchanged* value as its result:

```
int i = 0, j;
j = ++i; // j = 1, i = 1: prefix yields the incremented value
j = i++; // j = 1, i = 2: postfix yields the unincremented value
```

These operators require lvalue operands. The prefix operators return the object itself as an lvalue. The postfix operators return a copy of the object's original value as an rvalue.

**ADVICE: USE POSTFIX OPERATORS ONLY WHEN NECESSARY**

**Readers from a C background might be surprised that we use the prefix increment in the programs we've written. The reason is simple: The prefix version avoids unnecessary work. It increments the value and returns the incremented version. The postfix operator must store the original value so that it can return the unincremented value as its result. Unless we need the unincremented value, there's no need for the extra work done by the postfix operator.**

**For `ints` and pointers, the compiler can optimize away this extra work. For more complicated iterator types, this extra work potentially might be more costly. By habitually using the prefix versions, we do not have to worry about whether the performance difference matters. Moreover—and perhaps more importantly—we can express the intent of our programs more directly.**

## Combining Dereference and Increment in a Single Expression

The postfix versions of `++` and `--` are used when we want to use the current value of a variable and increment or decrement it in a single compound expression.

As one example, we can use postfix increment to write a loop to print the values in a `vector` up to but not including the first negative value:

```
auto pbeg = v.cbegin();
// print elements up to the first negative value
while (pbeg != v.cend() && *pbeg >= 0)
    cout << *pbeg++ << endl; // print the current value and advance pbeg
```

The expression `*pbeg++` is usually confusing to programmers new to both C++ and C. However, because this usage pattern is so common, C++ programmers must understand such expressions.

The precedence of postfix increment is higher than that of the dereference operator, so `*pbeg++` is equivalent to `*(pbeg++)`. The subexpression `pbeg++` increments `pbeg` and yields a copy of the previous value of `pbeg` as its result. Accordingly, the operand of `*` is the unincremented value of `pbeg`. Thus, the statement prints the element to which `pbeg` originally pointed and increments `pbeg`.

This usage relies on the fact that postfix increment returns a copy of its original, unincremented operand. If it returned the incremented value, we'd dereference the incremented value, with disastrous results: We'd skip the first element. Worse, if the sequence had no negative values, we would attempt to dereference one too many elements.

> **ADVICE: BREVITY CAN BE A VIRTUE**
>
> **Expressions such as `*pbeg++` can be bewildering—at first. However, it is a useful and widely used idiom. Once the notation is familiar, writing**
>
> ```
> cout << *iter++ << endl;
> ```
>
> **is easier and less error-prone than the more verbose equivalent**
>
> ```
> cout << *iter << endl;
> ++iter;
> ```
>
> **It is worthwhile to study examples of such code until their meanings are immediately clear. Most C++ programs use succinct expressions rather than more verbose equivalents. Therefore, C++ programmers must be comfortable with such usages. Moreover, once these expressions are familiar, you will find them less error-prone.**

### Using Increment/Decrement in Compound Expressions

It is essential to understand that increment and decrement change the value of the underlying object. Because the value changes, we must be particularly careful when writing compound expressions in which one subexpression increments or decrements an operand that is used in another subexpression.

To illustrate the problem, we'll rewrite the loop from § 3.4.1 (p. 113) that capitalizes the first word in the input. That example used a `for` loop:

```
for (auto it = s.begin(); it != s.end() && !isspace(*it); ++it)
    *it = toupper(*it); // capitalize the current character
```

which allowed us to separate the statement that dereferenced `beg` from the one that incremented it. Replacing the `for` with a seemingly equivalent `while`

```
// disaster -- this loop does the wrong thing and accesses a character off the end of s!
while (beg != s.end() && !isspace(*beg))
    *beg = toupper(*beg++);    // beg is incremented before the assignment
```

doesn't work. The problem is that in the revised version, the right-hand operand of the assignment changes `beg` *and* the left-hand operand uses `beg`. In an assignment, the right-hand operand is evaluated before the left (§ 4.4, p. 154). Therefore, it is as if we had written:

```
// assigns the uppercase version of *beg to the next character in the string
*(beg + 1) = toupper(*beg);  // beg incremented past s.end()
```

Not only are we changing the wrong character, but on its last iteration this loop will attempt to access a character past the end of `s`!

## Exercises Section 4.5

**Exercise 4.18:** Explain the difference between prefix and postfix increment. Give examples of when you might use each operator.

**Exercise 4.19:** What would happen if the `while` loop on page 157 that prints the elements from a `vector` used the prefix increment operator?

**Exercise 4.20:** Given that `ptr` points to an `int`, that `vec` is a `vector<int>`, and that `ival` is an `int`, explain the behavior of each of these expressions. Which, if any, are likely to be incorrect? Why? How might each be corrected?

(a) `ptr != 0 && *ptr++`  (b) `ival++ && ival`
(c) `vec[ival++] <= vec[ival]`

**Exercise 4.21:** Rewrite the `while` loop to capitalize the first word in a `string` so that it operates correctly.

# 4.6 Member and Element Access Operators

Like the call operator (§ 1.5.2, p. 23), the subscript (§ 3.5.2, p. 121), dot (§ 1.5.2, p. 23), and arrow (§ 3.4.1, p. 114) operators are "postfix" operators, so called because the operator follows the object to which it applies. The postfix operators group left to right. For example, assuming `items` is an array with at least one element:

```
items[0].isbn()
```

applies the subscript operator to obtain the first element in `items`. Next the dot operator fetches the `isbn` member of the element found by the subscript expression. Finally, the call operator calls the `isbn` member function. That is, this expression groups as if we had written:

```
((items[0]).isbn)()
```

## 4.6.1 Subscript Operator for Element Access

We've already used the built-in subscript operator (§ 3.5.2, p. 121), which gives access to elements in an array, and the subscript operators defined by the `string` (§ 3.2.3, p. 97) and `vector` (§ 3.3.4, p. 107) classes. As we've seen, the built-in subscript operator takes an array or a pointer and an index of integral type and returns the (lvalue) object at the given position:

```
int ia[] = {1,2,3};
auto i = ia[0]; //  initializes i to the value stored in the first element of ia, i.e., 1
```

Subscript is one of the operators that guarantees the order of evaluation for its operands: We are guaranteed that the left operand is evaluated before the index. As a somewhat contrived example, we might set the diagonal elements of a two-dimensional array to a specific value as follows: New

```
int md[10][10] = {}; // defualt initialize all elements to 0
int i = -1;
while (i != 9)
    md[++i][i] = 42; // set diagonal elements to 42
```

The expression in the while loop has two subscript operators. Because these operators group left to right, this expression is the same as (md[++i])[i]. That is, (md[++i]) is the array that we are subscripting with the index [i]. Because of order of evaluation, we know that md[++i] must be evaluated first. That expression increments i, setting it to 0 on the first iteration. Because we used prefix increment, the changed value of i is used to fetch the array at md[0]. We again use the value of i to set the 0'th element in that array to 42.

### 4.6.2 The Member Access Operators

The dot (§ 1.5.2, p. 23) and arrow (§ 3.4.1, p. 114) operators provide for member access. Like the subscript operator, these operators group left-to-right. The dot operator fetches a member from an object of class type; arrow is defined so that *ptr->mem* is a synonym for (**ptr*).*mem*:

New

```
string s1 = "a string", *p = &s1;
auto n = s1.size(); // run the size member of the string s1
n = (*p).size();    // run size on the object to which p points
n = p->size();      // equivalent to (*p).size()
```

Because dereference has a lower precedence than dot, we must parenthesize the dereference subexpression. If we omit the parentheses, this code means something quite different:

```
// run the size member of p, then dereference the result!
*p.size();    // error: p is a pointer and has no member named size
```

This expression attempts to fetch the size member of the object p. However, p is a pointer, which has no members; this code will not compile.

The arrow operator requires a pointer operand and yields an lvalue. The dot operator yields an lvalue if the object from which the member is fetched is an lvalue; otherwise the result is an rvalue.

**EXERCISES SECTION 4.6.2**

**Exercise 4.22:** Assuming that iter is a vector<string>::iterator, indicate which, if any, of the following expressions are legal. Explain the behavior of the legal expressions and why those that aren't legal are in error.

(a) *iter++;  (b) (*iter)++;  (c) *iter.empty()
(d) iter->empty();  (e) ++*iter;  (f) iter++->empty();

## 4.7 The Conditional Operator

The conditional operator (the `?:` **operator**) lets us embed simple if-else logic in an expression. The conditional operator has the following form:

```
cond ? expr1 : expr2
```

where *cond* is an expression that is used as a condition and *expr1* and *expr2* are expressions of the same type (or types that can be converted to a common type). The result of the conditional operator is an lvalue if both expressions are lvalues or if they convert to a common lvalue type. Otherwise the result is an rvalue.

The conditional operator first evaluates *cond*. If that expression evaluates as `true`, then *expr1* is evaluated; otherwise, *expr2* is evaluated. We are guaranteed that only one of *expr1* or *expr2* is evaluated.

As one example, assuming `grade` is a `string` and `g` is a student's grade, we can use a conditional operator to determine whether that grade is pass or fail:

```
grade = g < 60 ?  "fail" : "pass";
```

The condition checks whether `g` is less than `60`. If so, the result of the expression is `"fail"`; otherwise the result is `"pass"`.

### Nesting Conditional Operations

We can nest one conditional operator inside another by using a conditional expression as either *expr1* or *expr2*. For example, we can extend our expression to perform a three-way test to indicate whether a grade is a high pass, an ordinary pass, or fail:

```
grade = g > 90 ? "high pass"
               : g < 60 ? "fail" : "pass";
```

The first condition checks whether the grade is above `90`. If so, the expression after the `?` is evaluated, which yields `"high pass"`. If the condition fails, the `:` branch is executed, which is itself another conditional expression. This conditional asks whether the `grade` is less than `60`. If so, the `?` branch is evaluated and yields `"fail"`. If not, the `:` branch returns `"pass"`.

WARNING

Nested conditionals quickly become unreadable. It's a good idea to nest no more than two or three.

### Precedence and Associativity of the Conditional Operator

The conditional operator, like assignment (§ 4.4, p. 153), is right-associative and has has the same (fairly low) precedence as assignment. Consequently, all of the operands in our previous expression group right-to-left. Thus, the right-most conditional, which compares `g` to `60`, forms the `:` branch of the left-most conditional. That left-most conditional forms the right-hand operand of the assigment. In other words, this assignment, parenthesized to show the default precedence and associativity, groups as:

```
grade = ((g > 90) ? "high pass" : ((g < 60) ? "fail" : "pass"));
```

Because the conditional returns an lvalue if its operands are lvalues, we can use a conditional to select which variable to assign a give value. For example:

```
// assigns result of alloc to either tail or tail->next
(tail ? tail->next : tail) = alloc(args); // parentheses are necessary
```

Here we conditionally assign the result of a call to a function named `alloc` to either `tail` or `tail->next` depending on whether `tail` itself is nonzero. The parentheses in this expression are necessary. Without them,

```
// without parentheses, assigns to tail and discards result of conditional
tail ? tail->next : tail = alloc(args);   // wrong!
tail ? tail->next : (tail = alloc(args)); // equivalent expression
```

the assignment groups first. This expression unconditionally assigns the result of `alloc(args)` to `tail`. The result of that assignment, forms the `:` branch of the conditional. The result of the conditional itself is discarded.

Note: Conditional expressions embedded in a larger expression usually must be parenthesized.

### Exercises Section 4.7

**Exercise 4.23:** Write a program to use a conditional operator to find the elements in a `vector<int>` that have odd value and double the value of each such element.

**Exercise 4.24:** Extend the program that assigned high pass, pass, and fail grades to also assign low pass for grades between 60 and 75 inclusive. Write two versions: One version that uses only conditional operators; the other should use one or more `if` statements. Which version do you think is easier to understand and why?

**Exercise 4.25:** The following expression fails to compile due to operator precedence. Using Table 4.12 (p. 176), explain why it fails. How would you fix it?

```
string s = "word";
string pl = s + s[s.size() - 1] == 's' ? "" : "s" ;
```

**Exercise 4.26:** Our program that distinguished between high pass, pass, and fail depended on the fact that the conditional operator is right associative. Describe how that operator would be evaluated if the operator were left associative.

## 4.8 The Bitwise Operators

New

The bitwise operators take and return rvalues of integral type. These operators group left-to-right and guarantee that their operands are evaluated from left to right. These operators use their operands as a collection of bits—they test and set individual bits. As we'll see in § 17.3 (p. 790), the library `bitset` class, which represents a flexibly sized collection of bits, also defines these operators.

**Table 4.3: Bitwise Operators (Left Associative)**

| Operator | Function | Use |
|---|---|---|
| `~` | bitwise NOT | `~expr` |
| `<<` | left shift | `expr1 << expr2` |
| `>>` | right shift | `expr1 >> expr2` |
| `&` | bitwise AND | `expr1 & expr2` |
| `^` | bitwise XOR | `expr1 ^ expr2` |
| `\|` | bitwise OR | `expr1 \| expr2` |

As with the arithmetic operators, if an operand is a "small integer," its value is first promoted (§ 4.11.1, p. 170) to a larger integral type. The operand(s) can be either signed or unsigned. If the operand is signed and its value is negative, then the way that the "sign bit" is handled in a number of the bitwise operations is machine dependent. Moreover, doing a left shift that changes the value of the sign bit is undefined.

**WARNING** Because there are no guarantees for how the sign bit is handled, we strongly recommend using `unsigned` types with the bitwise operators.

## Bitwise Shift Operators

We have already used the overloaded versions of the `>>` and `<<` operators for input and output. The built-in meaning of these operators is that they perform a bitwise shift on their operands: They yield a value that is a copy of the (possibly promoted) left-hand operand with the bits shifted as directed by the right-hand operand. The right-hand operand must not be negative and must be a value that is strictly less than the number of bits in the result. Otherwise, the operation is undefined. The bits are shifted left (`<<`) or right (`>>`). Bits that are shifted off the end are discarded:

*These illustrations have the low-order bit on the right*
*These examples assume `char` has 8 bits, and `int` has 32*

```
    // 0233 is an octal literal (§ 2.1.3, p. 38)
    unsigned char bits = 0233;      |1 0 0 1 1 0 1 1|

bits << 8  // bits promoted to int and then shifted left by 8 bits
|0 0 0 0 0 0 0 0|0 0 0 0 0 0 0 0|1 0 0 1 1 0 1 1|0 0 0 0 0 0 0 0|

bits << 23  // ok: sign bit is unchanged
|0 1 0 0 1 1 0 1|1 0 0 0 0 0 0 0|0 0 0 0 0 0 0 0|0 0 0 0 0 0 0 0|

bits >> 3  // right shift 3 bits, 3 right-most bits discarded
|0 0 0 0 0 0 0 0|0 0 0 0 0 0 0 0|0 0 0 0 0 0 0 0|0 0 0 1 0 0 1 1|
```

The left-shift operator (the **`<<` operator**) inserts 0-valued bits on the right. When shifting `unsigned char` and `unsigned short` values, it is essential to remember that such values might promote to `signed int` (§ 4.11.1, p. 170). If the operand to the right-shift operator (the **>> operator**) is `unsigned` or has a nonnegative value, then the operator inserts 0-valued bits on the left; if it is a signed type and has a negative value, the result is implementation defined.

### Bitwise NOT Operator

The bitwise NOT operator (the **~operator**) generates a new value with the bits of its operand inverted. Each 1 bit is set to 0; each 0 bit is set to 1:

```
unsigned char bits = 0227;    |1 0 0 1 0 1 1 1|

~bits
|1 1 1 1 1 1 1 1|1 1 1 1 1 1 1 1|1 1 1 1 1 1 1 1|0 1 1 0 1 0 0 0|
```

Here, our `char` operand is first promoted to `int`. Promoting a `char` to `int` leaves the value unchanged but adds 0 bits to the high order positions. Thus, promoting `bits` to `int` adds 24 high order bits, all of which are 0-valued. The bits in the promoted value are inverted.

### Bitwise AND, OR, and XOR Operators

The AND (`&`), OR (`|`), and XOR (`^`) operators generate new values with the bit pattern composed from its two operands:

```
unsigned char b1 = 0145;                           |0 1 1 0 0 1 0 1|

unsigned char b2 = 0257;                           |1 0 1 0 1 1 1 1|

    b1 & b2     |24 high-order bits all 0|0 0 1 0 0 1 0 1|

    b1 | b2     |24 high-order bits all 0|1 1 1 0 1 1 1 1|

    b1 ^ b2     |24 high-order bits all 0|1 1 0 0 1 0 1 0|
```

For each bit position in the result of the bitwise AND operator (the **& operator**), the bit is 1 if both operands contain 1; otherwise, the result is 0. For the OR (inclusive or) operator (the **| operator**), the bit is 1 if either or both operands contain 1; otherwise, the result is 0. For the XOR (exclusive or) operator (the **^ operator**), the bit is 1 if either but not both operands contain 1; otherwise, the result is 0.

> **WARNING** It is a common error to confuse the bitwise and logical operators (§ 4.3, p. 150). For example to confuse the bitwise `&` with the logical `&&`, the bitwise `|` with the logical `||`, and the bitwise `~`and the logical `!`).

### Using Bitwise Operators

As an example of using the bitwise operators, let's assume a teacher has 30 students in a class. Each week the class is given a pass/fail quiz. We'll track the results of each quiz using one bit per student to represent the pass or fail grade on a given test. We might represent each quiz in an `unsigned` integral value:

```
unsigned long quiz1 = 0;   // we'll use this value as a collection of bits
```

We define `quiz1` as an `unsigned long`. Thus, `quiz1` will have at least 32 bits on any machine. We explicitly initialize `quiz1` to ensure that the bits start out with well-defined values.

The teacher must be able to set and test individual bits. For example, we'd like to be able to set the bit corresponding to student number 27 to indicate that this student passed the quiz. We can indicate that student number 27 passed by creating a value that has only bit 27 turned on. If we then bitwise OR that value with `quiz1`, all the bits except bit 27 will remain unchanged.

For the purpose of this example, we will count the bits of `quiz1` by assigning 0 to the low-order bit, 1 to the next bit, and so on.

We can obtain a value indicating that student 27 passed by using the left-shift operator and an `unsigned long` integer literal 1 (§ 2.1.3, p. 38):

```
1UL << 27   // generate a value with only bit number 27 set
```

`1UL` has a 1 in the low-order bit and (at least) 31 zero bits. We specified `unsigned long` because `int`s are only guaranteed to have 16 bits, and we need at least 27. This expression shifts the 1 bit left 27 positions inserting 0 bits behind it.

Next we OR this value with `quiz1`. Because we want to update the value of `quiz1`, we use a compound assignment (§ 4.4, p. 155):

```
quiz1 |= 1UL << 27;   // indicate student number 27 passed
```

The `|=` operator executes analogously to how `+=` does. It is equivalent to

```
quiz1 = quiz1 | 1UL << 27;   // equivalent to quiz1 |= 1UL << 27;
```

Imagine that the teacher reexamined the quiz and discovered that student 27 actually had failed the test. The teacher must now turn off bit 27. This time we need an integer that has bit 27 turned off and all the other bits turned on. We'll bitwise AND this value with `quiz1` to turn off just that bit:

```
quiz1 &= ~(1UL << 27);   // student number 27 failed
```

We obtain a value with all but bit 27 turned on by inverting our previous value. That value had 0 bits in all but bit 27, which was a 1. Applying the bitwise NOT to that value will turn off bit 27 and turn on all the others. When we bitwise AND this value with `quiz1`, all except bit 27 will remain unchanged.

Finally, we might want to know how the student at position 27 fared:

```
bool status = quiz1 & (1UL << 27);   // how did student number 27 do?
```

Here we AND a value that has bit 27 turned on with `quiz1`. The result is nonzero (i.e., true) if bit 27 of `quiz1` is also on; otherwise, it evaluates to zero.

## Shift Operators (aka IO Operators) Are Left Associative

Although many programmers never use the bitwise operators directly, most programmers do use overloaded versions of these operators for IO. An overloaded operator has the same precedence, associativity, and (after C++17) the same order of evaluation as the built-in version of that operator. Therefore, programmers need to understand the precedence, associativity, and order of evaluation of the shift operators even if they never use them with their built-in meaning.

As we've already seen (§ 1.2, p. 7), the shift operators are left associative, so

```
cout << "hi" << " there" << endl;
```

executes as

```
( (cout << "hi") << " there" ) << endl;
```

The operand `"hi"` is grouped with the first `<<` symbol. Its result, which is `cout`, is grouped with the second operand, and that result is grouped with the third.

The shift operators have midlevel precedence: lower than the arithmetic operators but higher than the relational, assignment, and conditional operators. These relative precedence levels mean we usually have to use parentheses to force the correct grouping of operators with lower precedence.

```
cout << 42 + 10;    // ok: + has higher precedence, so the sum is printed
cout << (10 < 42);  // ok: parentheses force intended grouping; prints 1
cout << 10 < 42;    // error: attempt to compare cout to 42!
```

The last `cout` is interpreted as

```
(cout << 10) < 42;
```

which says to "write `10` onto `cout` and then compare the result of that operation (i.e., `cout`) to `42`."

New Starting with C++17, we are guaranteed that the operands to the shift operators are evaluated from left to right. This ordering means that we can write a compound expression that uses variables that are changed by earlier subexpressions.

```
int a[] = {0,1,2,3,4,5,6,7,8,9};
auto sz = size(a);   // number of elements in a § 3.5.3 (p. 125)
int i = 0;
// print pairs of elements from a;
// legal in C++17, undefined in earlier versions of the language
while (i < sz)                    // this loop assumes sz is even
    // postfix increment, increments i but returns unincremented value
    cout << a[i++] << " " << a[i++] << endl;  // ok: in C++17
```

Because operands are evaluated left to right, we know that the left-most `a[i++]` is evaluated first, meaning that on the first iteration that operand is `a[0]`. As a side-effect, `i` is incremented so that when the right-most `a[i++]` is evaluated, `i` is `1`, meaning that the that operand evaluates as `a[1]`.

### EXERCISES SECTION 4.8

**Exercise 4.27:** What is the value of `~'q' << 6` on a machine with 32-bit `int`s and 8 bit `char`s, that uses Latin-1 character set in which `'q'` has the bit pattern `01110001`?

**Exercise 4.28:** In our grading example in this section, what would happen if we used `unsigned int` as the type for `quiz1`?

**Exercise 4.29:** What is the result of each of these expressions?

```
unsigned long ul1 = 3, ul2 = 7;
(a)  ul1 & ul2              (b)  ul1 | ul2
(c)  ul1 && ul2             (d)  ul1 || ul2
```

**Exercise 4.30:** Explain what each of these output statements do.

```
cout << (g < 60 ?  "fail" : "pass");
cout << g < 60 ?  "fail" : "pass";
cout << (g < 60) ?  "fail" : "pass";
```

Run this code to see whether you were correct. If you're surprised by the results, be sure you understand why the behavior differs from your expectations.

## 4.9 The sizeof Operator

The **`sizeof`** returns the size, in bytes, of the result of a given expression or of an object of the given type. The operator is right associative. The result of `sizeof` is a constant expression (§ 2.4.4, p. 66) of type `size_t` (§ 3.5.2, p. 121). The operator has two forms:

```
sizeof (type)
sizeof expr
```

In the first form, parentheses are required. In the second form, we can (as usual) surround *expr* with a pair of parentheses if we like.

Like `decltype`, `sizeof` does not evaluate its operand:

```
Sales_data data, *p;
sizeof(Sales_data); //  size required to hold an object of type Sales_data
sizeof data;    //  size of data's type, i.e., sizeof(Sales_data)
sizeof p;       //  size of a pointer
sizeof *p;      //  size of the type to which p points, i.e., sizeof(Sales_data)
sizeof data.revenue; //  size of the type of Sales_data's revenue member
sizeof Sales_data::revenue; //  alternative way to get the size of revenue
```

The most interesting of these examples is `sizeof *p`. First, because `sizeof` is right associative and has the same precedence as `*`, this expression groups right to left. That is, it is equivalent to `sizeof (*p)`. Second, because `sizeof` does not evaluate its operand, it doesn't matter that `p` is an invalid (i.e., uninitialized) pointer (§ 2.3.2, p. 53). Dereferencing an invalid pointer as the operand to `sizeof` is safe because the pointer is not actually used. `sizeof` doesn't need dereference the pointer to know what type it will return.

We can use the scope operator to ask for the size of a member of a class type. Ordinarily we can access the members of a class only through an object of that type. We don't need to supply an object, because `sizeof` does not need to fetch the member to know its size.

The result of applying `sizeof` depends in part on the type involved:

- `sizeof char` or an expression of type `char` is guaranteed to be 1.
- `sizeof` a reference type returns the size of an object of the referenced type.
- `sizeof` a pointer returns the size needed hold a pointer of that type.
- `sizeof` a dereferenced pointer returns the size of an object of the type to which the pointer points; the pointer need not be valid.
- `sizeof` an array is the size of the entire array. It is equivalent to taking the `sizeof` the element type times the number of elements in the array. Note that `sizeof` does not convert the array to a pointer.
- `sizeof` a `string` or a `vector` returns only the size of the fixed part of these types; it does not return the size used by the object's elements.

If your compiler does not yet support `size` (§ 3.5.3, p. 125), it is useful to know that we can use `sizeof` to determine the number of elements in an array:

```
// sizeof(ia)/sizeof(*ia) returns the number of elements in ia
constexpr size_t sz = sizeof(ia)/sizeof(*ia);
int arr2[sz];  // ok sizeof returns a constant expression § 2.4.4 (p. 66)
```

Because `sizeof` returns a constant expression, we can use the result of a `sizeof` expression to specify the dimension of an array.

## Exercises Section 4.9

**Exercise 4.31:** Write a program to print the size of each of the built-in types.

**Exercise 4.32:** Predict the output of the following code and explain your reasoning. Now run the program. Is the output what you expected? If not, figure out why.

```
int x[10];   int *p = x;
cout << sizeof(x)/sizeof(*x) << endl;
cout << sizeof(p)/sizeof(*p) << endl;
```

**Exercise 4.33:** Using Table 4.12 (p. 176), parenthesize the following expressions to match the default evaluation:

(a) `sizeof x + y`        (b) `sizeof p->mem[i]`
(c) `sizeof a < b`        (d) `sizeof f()`

## 4.10 Comma Operator

The **comma operator** takes two operands, which it evaluates from left to right. The left-hand expression is evaluated and its result is discarded. The result of a comma expression is the value of its right-hand expression. The result is an lvalue if the right-hand operand is an lvalue.

One common use for the comma operator is in a `for` loop:

```
vector<int>::size_type cnt = ivec.size();
// assign values from size . . . 1 to the elements in ivec
for(vector<int>::size_type ix = 0;
                ix != ivec.size(); ++ix, --cnt)
    ivec[ix] = cnt;
```

This loop increments `ix` and decrements `cnt` in the expression in the `for` header. Both `ix` and `cnt` are changed on each trip through the loop. As long as the test of `ix` succeeds, we reset the current element to the current value of `cnt`.

### EXERCISES SECTION 4.10

**Exercise 4.34:** The program in this section used the prefix increment and decrement operators. Explain why we used prefix and not postfix. What changes would have to be made to use the postfix versions? Rewrite the program using postfix operators.

**Exercise 4.35:** Explain the following loop.

```
constexpr int size = 5;
int ia[size] = {1,2,3,4,5};
for (int *ptr = ia, ix = 0;
     ix != size && ptr != ia+size;
     ++ix, ++ptr)   { /* ... */ }
```

**Exercise 4.36:** Using Table 4.12 (p. 176) explain what the following expression does:

```
someValue ? ++x, ++y : --x, --y
```

## 4.11 Type Conversions

In C++ some types are related to each other. When two types are related, we can use an object or value of one type where an operand of a related type is expected. Two types are related if there is a **conversion** between them. When an expression contains related types, the compiler converts the operands to a common type and then evaluates the resulting expression. These conversions are carried out automatically without programmer intervention—and sometimes without programmer knowledge. For that reason, they are referred to as **implicit conversions**.

The language defines a set of conversions among the arithmetic types. These conversions are defined to preserve precision, if possible. Most often, if an expression has both integral and floating-point operands, the integer is converted to floating-point. However, during initialization, the type of the object being initialized dominates; the initializer is converted to the object's type regardless of precision. As an example, two conversions take place in the following

```
int ival = 3.541 + 3; // the compiler might warn about loss of precision
```

First, the addition expression is evaluated. The compiler converts `3` to `double` and does floating-point addition. The result of that addition is a value of type `double`. Next, that `double` is converted to `int` and used to initialize `ival`. Converting a `double` to an `int` truncates the `double`'s value, discarding the decimal portion. In this expression, `ival` is initialized to `6`.

### When Implicit Conversions Occur

The compiler automatically converts operands in the following circumstances:

- In most expressions, values of integral types smaller than `int` are first promoted to an appropriate larger integral type.
- In conditions, non`bool` expressions are converted to `bool`.
- In initializations, the initializer is converted to the type of the variable; in assignments, the right-hand operand is converted to the type of the left-hand.
- In arithmetic and relational expressions with operands of mixed types, the types are converted to a common type.
- As we'll see in Chapter 6, conversions also happen during function calls.

## 4.11.1 The Arithmetic Conversions

The **arithmetic conversions**, which we introduced in § 2.1.2 (p. 34), convert one arithmetic type to another. The rules define a hierarchy of type conversions in which operands to an operator are converted to the widest type. For example, if one operand is of type `long double`, then the other operand is converted to type `long double` regardless of what the second type is. More generally, in expressions that mix floating-point and integral values, the integral value is converted to an appropriate floating-point type.

### Integral Promotions

The **integral promotions** convert the small integral types to a larger integral type. The types `bool`, `char`, `signed char`, `unsigned char`, `short`, and `unsigned short` are promoted to `int` if all possible values of that type fit in an `int`. Otherwise, the value is promoted to `unsigned int`. As we've seen many times, a `bool` that is `false` promotes to 0 and `true` to 1.

The larger `char` types (`wchar_t`, `char16_t`, and `char32_t`) are promoted to the smallest type of `int`, `unsigned int`, `long`, `unsigned long`, `long long`, or `unsigned long long` in which all possible values of that character type fit.

## Operands of Unsigned Type

If any operand has an `unsigned` type, the type to which the operands are converted depends on the relative sizes of the integral types on the machine.

As usual, integral promotions happen first. If both (possibly promoted) operands have the same signedness, but are not the same type, then the operand with the smaller type is converted to the larger type.

When the signedness differs, if the type of the unsigned operand is the same as or larger than that of the signed operand, the signed operand is converted to unsigned. For example, given an `unsigned int` and an `int`, the `int` is converted to `unsigned int`. It is worth noting that if the `int` has a negative value, the result will be converted as described in § 2.1.2 (p. 35), with the same results.

The remaining case is when the signed operand has a larger type than the unsigned operand. In this case, the result is machine dependent. If all values in the unsigned type fit in the larger type, then the unsigned operand is converted to the signed type. If the values don't fit, then the signed operand is converted to the unsigned type. For example, if the operands are `long` and `unsigned int`, and `int` and `long` have the same size, the `long` will be converted to `unsigned int`. If the `long` type has more bits, then the `unsigned int` will be converted to `long`.

## Understanding the Arithmetic Conversions

One way to understand the arithmetic conversions is to study lots of examples:

```
bool      flag;        char           cval;
short     sval;        unsigned short usval;
int       ival;        unsigned int   uival;
long      lval;        unsigned long  ulval;
float     fval;        double         dval;
3.14159L + 'a'; //  'a' promoted to int, then converted to long double
dval + ival;    //  ival converted to double
dval + fval;    //  fval converted to double
ival = dval;    //  dval converted (by truncation) to int
flag = dval;    //  if dval is 0, then flag is false, otherwise true
cval + fval;    //  cval promoted to int, then that int converted to float
sval + cval;    //  sval and cval promoted to int
cval + lval;    //  cval promoted to int, then converted to long
ival + ulval;   //  ival converted to unsigned long
usval + ival;   //  promotion depends on the size of unsigned short and int
uival + lval;   //  conversion depends on the size of unsigned int and long
```

In the first addition, the character constant lowercase `'a'` has type `char`, which is a numeric value (§ 2.1.1, p. 32). The specific value depends on the machine's character set. On our machine, `'a'` has the numeric value 97. When we add `'a'` to a `long double`, the `char` value is promoted to `int`, and then that `int` value is converted to a `long double`. The converted value is added to the literal. The other interesting cases are the last two expressions involving unsigned values. The type of the result in these expressions is machine dependent.

### EXERCISES SECTION 4.11.1

**Exercise 4.37:** Given the variable definitions in this section, explain what conversions take place in the following expressions:

(a) `if (fval)` (b) `dval = fval + ival;` (c) `dval + ival * cval;`

Remember that you may need to consider precedence and associativity.

**Exercise 4.38:** Given the following definitions,

```
char cval;      int ival;    unsigned int ui;
float fval;     double dval;
```

identify the implicit type conversions, if any, taking place:

(a) `cval = 'a' + 3;` (b) `fval = ui - ival * 1.0;`
(c) `dval = ui * fval;` (d) `cval = ival + fval + dval;`

## 4.11.2 Other Implicit Conversions

In addition to the arithmetic conversions, there are several additional kinds of implicit conversions. These include:

**Array to Pointer Conversions:** In most expressions, when we use an array, the array is automatically converted to a pointer to the first element in that array:

```
int ia[10];     // array of ten ints
int* ip = ia;   // convert ia to a pointer to the first element
```

This conversion is not performed when an array is used with `decltype` or as the operand of the address-of (`&`), `sizeof`, or `typeid` (which we'll cover in § 19.2.2 (p. 878)) operators. The conversion is also omitted when we initialize a reference to an array (§ 3.5.1, p. 120). As we'll see in § 6.7 (p. 261), a similar pointer conversion happens when we use a function type in an expression.

**Pointer Conversions:** There are several other pointer conversions: A constant integral value of 0 and the literal `nullptr` can be converted to any pointer type; a pointer to any non`const` type can be converted to `void*`, and a pointer to any type can be converted to a `const void*`. We'll see in § 15.2.2 (p. 651) that there is an additional pointer conversion that applies to types related by inheritance.

**Conversions to `bool`:** There is an automatic conversion from arithmetic or pointer types to `bool`. If the pointer or arithmetic value is zero, the conversion yields `false`; any other value yields `true`:

```
char *cp = get_string();
if (cp) /* ... */      // true if the pointer cp is not zero
while (*cp) /* ... */ // true if *cp is not the null character
```

**Conversion to pointer or reference to `const`:** We can convert a pointer (or reference) to a non`const` type to a pointer (or reference) to the corresponding `const` type. That is, if `T` is a type, we can convert a pointer or a reference to `T` into a pointer or reference to `const T`, respectively (§ 2.4.1, p. 62, and § 2.4.2, p. 63):

```
int i;
const int &j = i;  // convert a nonconst to a reference to const int
const int *p = &i; // convert address of a nonconst to the address of a const
int &r = j, *q = p; // error: conversion from const to nonconst not allowed
```

The reverse conversion—removing a low-level `const`—does not exist.

**Conversions Defined by Class Types:** Class types can define conversions that the compiler will apply automatically. The compiler will apply only one class-type conversion at a time. In § 7.5.4 (p. 309) we'll see an example of when multiple conversions might be required, and will be rejected.

Our programs have already used class-type conversions: We use a class-type conversion when we use a C-style character string where a library `string` is expected (§ 3.5.5, p. 131) and when we read from an `istream` in a condition:

```
for (int value; std::cin >> value; /* no expression */)
```

The condition (`cin >> val`) reads `cin` and yields `cin` as its result. Conditions expect a value of type `bool`, but this condition tests a value of type `istream`. The IO library defines a conversion from `istream` to `bool`. That conversion is used (automatically) to convert `cin` to `bool`. The resulting `bool` value depends on the state of the stream. If the last read succeeded, then the conversion yields `true`. If the last attempt failed, then the conversion to `bool` yields `false`.

## 4.11.3 Explicit Conversions

Sometimes we want to explicitly force an object to be converted to a different type. For example, we might want to use floating-point division in the following code:

```
int i, j;
double slope = i/j;
```

To do so, we'd need a way to explicitly convert `i` and/or `j` to `double`. We use a **cast** to request an explicit conversion.

WARNING

Although necessary at times, casts are inherently dangerous constructs.

### Named Casts

A named cast has the following form:

*cast-name*`<`*type*`>(`*expression*`);`

where *type* is the target type of the conversion, and *expression* is the value to be cast. If *type* is a reference, then the result is an lvalue. The *cast-name* may be one of **`static_cast`**, **`dynamic_cast`**, **`const_cast`**, and **`reinterpret_cast`**. We'll cover `dynamic_cast` in § 19.2 (p. 877). The *cast-name* determines what kind of conversion is performed.

### `static_cast`

Any legal type conversion, other than those involving low-level `const`, can be requested using a `static_cast`. For example, we can force an expression to use floating-point division by casting one of the operands to `double`:

```
// cast used to force floating-point division
double slope = static_cast<double>(j) / i;
```

A `static_cast` is often useful when a larger arithmetic type is assigned to a smaller type. The cast informs both the reader of the program and the compiler that we are aware of and are not concerned about the potential loss of precision. Compilers often generate a warning for assignments of a larger arithmetic type to a smaller type. When we do an explicit cast, the warning message is turned off.

A `static_cast` is also useful to perform a conversion that the compiler will not generate automatically. For example, we can use a `static_cast` to retrieve a pointer value that was stored in a `void*` pointer (§ 2.3.2, p. 56):

```
void* p = &d;    // ok: address of any nonconst object can be stored in a void*
// ok: converts void* back to the original pointer type
double *dp = static_cast<double*>(p);
```

When we store a pointer in a `void*` and then use a `static_cast` to cast the pointer back to its original type, we are guaranteed that the pointer value is preserved. That is, the result of the cast will be equal to the original address value. However, we must be certain that the type to which we cast the pointer is the actual type of that pointer; if the types do not match, the result is undefined.

### `const_cast`

A `const_cast` changes only a low-level (§ 2.4.3, p. 64) `const` in its operand:

```
const char *pc;
char *p = const_cast<char*>(pc); // ok: but writing through p is undefined
```

Conventionally we say that a cast that converts a `const` object to a non`const` type "casts away the `const`." Once we have cast away the `const` of an object, the compiler will no longer prevent us from writing to that object. If the object was originally not a `const`, using a cast to obtain write access is legal. However, using a `const_cast` in order to write to a `const` object is undefined.

Only a `const_cast` may be used to change the `const`ness of an expression. Trying to change whether an expression is `const` with any of the other forms of named cast is a compile-time error. Similarly, we cannot use a `const_cast` to change the type of an expression:

```
const char *cp = "hi Mom!";
// error: static_cast can't cast away const
char *q = static_cast<char*>(cp);
static_cast<string>(cp); // ok: converts C-style string to string
const_cast<string>(cp);  // error: const_cast only changes constness
```

A `const_cast` is most useful in the context of overloaded functions, which we'll describe in § 6.4.1 (p. 238).

### `reinterpret_cast`

A `reinterpret_cast` generally performs a low-level reinterpretation of the bit pattern of its operands. As an example, the following cast converts a pointer to `int` to pointer to `char`:

```
int *ip;
char *pc = reinterpret_cast<char*>(ip);
```

However, we must never forget that the actual object addressed by `pc` is an `int`, not a character. Any use of `pc` that assumes it's an ordinary character pointer is likely to fail at run time. For example:

```
string str(pc);
```

is likely to result in bizarre run-time behavior.

The use of `pc` to initialize `str` is a good example of why `reinterpret_cast` is dangerous. The problem is that types are changed, yet there are no warnings or errors from the compiler. When we initialized `pc` with the address of an `int`, there is no error or warning from the compiler because we explicitly said the conversion was okay. Any subsequent use of `pc` will assume that the value it holds is a `char*`. The compiler has no way of knowing that it actually holds a pointer to an `int`. Thus, the initialization of `str` with `pc` is absolutely correct—albeit in this case meaningless or worse! Tracking down the cause of this sort of problem can prove extremely difficult, especially if the cast of `ip` to `pc` occurs in a file separate from the one in which `pc` is used to initialize a `string`.

> **WARNING** A `reinterpret_cast` is inherently machine dependent. Safely using `reinterpret_cast` requires completely understanding the types involved as well as the details of how the compiler implements the cast.

#### ADVICE: AVOID CASTS

**Casts interfere with normal type checking (§ 2.2.2, p. 47). As a result, we strongly recommend that programmers avoid casts. This advice is particularly applicable to `reinterpret_casts`. Such casts are always hazardous. A `const_cast` can be useful in the context of overloaded functions, which we'll cover in § 6.4.1 (p. 238). Other uses of `const_cast` often indicate a design flaw. The other casts, `static_cast` and `dynamic_cast`, should be needed infrequently. Every time you write a cast, you should think hard about whether you can achieve the same result in a different way. If the cast is unavoidable, errors can be mitigated by limiting the scope in which the cast value is used and by documenting all assumptions about the types involved.**

## Old-Style Casts

In early versions of C++, an explicit cast took one of the following two forms:

```
type (expr);  // function-style cast notation
(type) expr;  // C-language-style cast notation
```

Depending on the types involved, an old-style cast has the same behavior as a `const_cast`, a `static_cast`, or a `reinterpret_cast`. When we use an old-style cast where a `static_cast` or a `const_cast` would be legal, the old-style cast does the same conversion as the respective named cast. If neither cast is legal, then an old-style cast performs a `reinterpret_cast`. For example:

```
char *pc = (char*) ip;   // ip is a pointer to int
```

has the same effect as using a `reinterpret_cast`.

**WARNING** Old-style casts are less visible than are named casts. Because they are easily overlooked, it is more difficult to track down a rogue cast.

### EXERCISES SECTION 4.11.3

**Exercise 4.39:** Assuming `i` is an `int` and `d` is a `double` write the expression `i *= d` so that it does integral, rather than floating-point, multiplication.

**Exercise 4.40:** Rewrite each of the following old-style casts to use a named cast:

```
int i;  double d;  const string *ps;  char *pc;  void *pv;
(a) pv = (void*)ps;    (b) i = int(*pc);
(c) pv = &d;           (d) pc = (char*) pv;
```

**Exercise 4.41:** Explain the following expression:

```
double slope = static_cast<double>(j/i);
```

## 4.12 Operator Precedence Table

| Associativity | and Operator | Function | Use | See Page |
|---|---|---|---|---|
| L | `::` | global scope | `::`name | 300 |
| L | `::` | class scope | class`::`name | 90 |
| L | `::` | namespace scope | namespace`::`name | 84 |
| L | `.` | member selectors | object`.`member | 23 |
| L | `->` | member selectors | pointer`->`member | 114 |
| L | `[]` | subscript | expr`[` expr `]` | 121 |
| L | `()` | function call | name`(`expr_list`)` | 23 |
| L | `()` | type construction | type`(`expr_list`)` | 175 |
| L | `++` | postfix increment | lvalue`++` | 156 |
| L | `--` | postfix decrement | lvalue`--` | 156 |
| L | `typeid` | type ID | `typeid`(type) | 878 |
| L | `typeid` | run-time type ID | `typeid`(expr) | 878 |
| L | explicit cast | type conversion | *cast_name*<type>(expr) | 173 |

*Continued on next page*

**Table 4.4: Operator Precedence (continued)**

| Associativity | and Operator | Function | Use | See Page |
|---|---|---|---|---|
| R | `++` | prefix increment | `++`lvalue | 156 |
| R | `--` | prefix decrement | `--`lvalue | 156 |
| R | `~` | bitwise NOT | `~`expr | 162 |
| R | `!` | logical NOT | `!`expr | 150 |
| R | `-` | unary minus | `-`expr | 148 |
| R | `+` | unary plus | `+`expr | 148 |
| R | `*` | dereference | `*`expr | 53 |
| R | `&` | address-of | `&`lvalue | 52 |
| R | `()` | type conversion | (type) expr | 175 |
| R | `sizeof` | size of object | `sizeof` expr | 167 |
| R | `sizeof` | size of type | `sizeof`( type ) | 167 |
| R | `sizeof...` | size of parameter pack | `sizeof...`( name ) | 757 |
| R | `new` | allocate object | `new` type | 536 |
| R | `new[]` | allocate array | `new` type[size] | 536 |
| R | `delete` | deallocate object | `delete` expr | 538 |
| R | `delete[]` | deallocate array | `delete[]` expr | 538 |
| R | `noexcept` | can expr throw | `noexcept` ( expr ) | 832 |
| L | `->*` | ptr to member select | ptr`->*`ptr_to_member | 889 |
| L | `.*` | ptr to member select | obj`.*`ptr_to_member | 889 |
| L | `*` | multiply | expr `*` expr | 148 |
| L | `/` | divide | expr `/` expr | 148 |
| L | `%` | modulo (remainder) | expr `%` expr | 148 |
| L | `+` | add | expr `+` expr | 148 |
| L | `-` | subtract | expr `-` expr | 148 |
| L | `<<` | bitwise shift left | expr `<<` expr | 162 |
| L | `>>` | bitwise shift right | expr `>>` expr | 162 |
| L | `<` | less than | expr `<` expr | 150 |
| L | `<=` | less than or equal | expr `<=` expr | 150 |
| L | `>` | greater than | expr `>` expr | 150 |
| L | `>=` | greater than or equal | expr `>=` expr | 150 |
| L | `==` | equality | expr `==` expr | 150 |
| L | `!=` | inequality | expr `!=` expr | 150 |
| L | `&` | bitwise AND | expr `&` expr | 162 |
| L | `^` | bitwise XOR | expr `^` expr | 162 |
| L | `\|` | bitwise OR | expr `\|` expr | 162 |
| L | `&&` | logical AND | expr `&&` expr | 150 |
| L | `\|\|` | logical OR | expr `\|\|` expr | 150 |
| R | `?:` | conditional | expr `?` expr `:` expr | 161 |
| R | `=` | assignment | lvalue `=` expr | 153 |
| R | `*=`, `/=`, `%=`, | compound assign | lvalue `+=` expr, etc. | 153 |
| R | `+=`, `-=`, | | | 153 |
| R | `<<=`, `>>=`, | | | 153 |
| R | `&=`, `\|=`, `^=` | | | 153 |
| R | `throw` | throw exception | throw expr | 206 |
| L | `,` | comma | expr `,` expr | 169 |

## Chapter Summary

C++ provides a rich set of operators and defines their meanings when applied to values of the built-in types. Additionally, the language supports operator overloading, which allows us to define the meaning of the operators for class types.

Each operator has a precedence level and associativity. Precedence determines how operators are grouped in a compound expression. Associativity determines how operators at the same precedence level are grouped.

Most operators do not specify the order in which operands are evaluated: Order of evaluation matters in expressions that refer to the same object more than once and in which one of the operators *changes* that object.

Finally, operands are often converted automatically from their initial type to another related type. For example, small integral types are promoted to a larger integral type in every expression. Conversions exist for both built-in and class types. Conversions can also be done explicitly through a cast.

## Defined Terms

**arithmetic conversion** Conversion from one arithmetic type to another. These conversions try to preserve precision by converting smaller types to larger types (e.g., integral types are converted to floating point).

**associativity** Defines the grouping of operands to operators with the same precedence. Right associative operators group from right to left; left associative operators group from left to right.

**binary operator** Operator with two operands.

**cast** An explicit conversion.

**compound expression** An expression involving more than one operator.

**const_cast** A cast that converts a low-level `const` object to the corresponding non`const` type or vice versa.

**conversion** Process whereby a value of one type is transformed into a value of another type. The language defines conversions among the built-in types. Conversions to and from class types are also possible.

**dynamic_cast** Explicit conversion used in the context of inheritance.

**expression** The lowest level of computation in a C++ program. Expressions generally apply an operator to one or more operands and yield a result.

**implicit conversion** Conversion that is automatically generated by the compiler.

**integral promotions** Conversion of a small integral type to the most closely related larger integral type. Operands of small integral types are always promoted.

**lvalue** An expression that yields an object or function.

**operands** Values on which an expression operates.

**operator** Symbol that determines the action of an expression. The language defines what the operators do when applied to values of built-in types. Operators may be overloaded and applied to values of class types.

**order of evaluation** Order, if any, in which the operands to an operator are evaluated.

**overloaded operator** Version of an operator that is defined for use with a class type.

**precedence** Defines the grouping of operands in compound expressions. Operators with higher precedence group more tightly than those with lower precedence.

**promoted** See integral promotions.

**reinterpret_cast** Converts an operand to an unrelated type. Inherently machine dependent and dangerous.

**result** Value or object obtained by evaluating an expression.

**rvalue** Expression that yields a value but not the associated location of that value.

**short-circuit evaluation** Form of evaluation used by logical AND and logical OR operators. The second operand is evaluated only if the first operand cannot by itself determine the result.

**sizeof** Operator that returns the size, in bytes, of a given expression or an object of a given type.

**static_cast** Explicit conversion to a related type.

**unary operator** Operator with one operand.

**, operator** Comma operator. Binary operator that is evaluated left to right. The result is the value of the right-hand operand and is an lvalue if that operand is an lvalue, rvalue otherwise.

**?: operator** Conditional operator takes a condition and two expressions. If the condition is true, then first expression is evaluated. Otherwise, the second one is evaluated. The expressions must have the same type or be convertible to a common type.

**&& operator** Logical AND operator. Result is `true` if both operands are `true`. The right-hand operand is evaluated *only* if the left-hand operand is `true`.

**& operator** Bitwise AND operator. Generates a new integral value in which each bit position is 1 if both operands have a 1 in that position; otherwise the bit is 0.

**^ operator** Bitwise exclusive or operator. Generates a new integral value in which each bit position is 1 if either but not both operands contain a 1 in that bit position; otherwise, the bit is 0.

**|| operator** Logical OR operator. Yields `true` if either operand is `true`. The right-hand operand is evaluated *only* if the left-hand operand is `false`.

**| operator** Bitwise OR operator. Generates a new integral value in which each bit position is 1 if either operand has a 1 in that position; otherwise the bit is 0.

**++ operator** Increment operator, which has two forms, prefix and postfix. Both add 1 to an lvalue operand. Prefix returns the changed value of that operand as an lvalue. Postfix returns an rvalue that is a copy of the original, unchanged value.

**-- operator** Decrement operator, which has two forms, prefix and postfix. Both subtract 1 to an lvalue operand. Prefix returns the changed value of that operand as an lvalue. Postfix returns an rvalue that is a copy of the original, unchanged value of the operand.

**<< operator** Left-shift operator. Shifts bits in a copy of the left-hand operand to the left. Shifts as many bits as indicated by the right-hand operand. The right-hand operand must be `>= 0` and less than the number of bits in the result. Undefined if the left-hand operand is `signed` and a different bit is shifted into the sign bit.

**>> operator** Right-shift operator. Like the left-shift operator except that bits are shifted to the right. Implementation-defined whether bits shifted in a `signed` operand are 0 or copies of the sign bit.

**~ operator** Bitwise NOT operator. Generates a new integral value in which each bit is an inverted copy of the corresponding bit in the (possibly promoted) operand.

**! operator** Logical NOT operator. Returns the inverse of the `bool` value of its operand. Result is `true` if operand is `false` and vice versa.

# CHAPTER 5

# STATEMENTS

## CONTENTS



Like most languages, C++ provides statements for conditional execution, loops that repeatedly execute the same body of code, and jump statements that interrupt the flow of control. This chapter looks in detail at the statements supported by C++.

*Statements* define the actions that our programs take. In C++, there are declaration statements that declare and define variables, functions, and classes; expression statements that evaluate an expression; and various ***flow-of-control*** statements that control the execution path of the program.

## 5.1 Simple Statements

Most statements in C++ end with a semicolon. An expression, such as `ival + 5`, becomes an **expression statement** when it is followed by a semicolon. Expression statements cause the expression to be evaluated and its result discarded:

```
ival + 5;       // rather useless expression statement
cout << ival;   // useful expression statement
```

The first statement is pretty useless: The addition is done but the result is not used. More commonly, an expression statement contains an expression that has a side effect—such as assigning a new value to a variable, or printing a result—when it is evaluated.

### Null Statements

The simplest statement is the empty statement, also known as a **null statement**. A null statement is a single semicolon:

```
;   // null statement
```

A null statement is useful where the language requires a statement but the program's logic does not. Such usage is most common when a loop's work can be done within its condition. For example, we might want to read an input stream, ignoring everything we read until we encounter a particular value:

```
// read until we hit end-of-file or find an input equal to sought
while (cin >> s && s != sought)
    ; // null statement
```

This condition reads a value from the standard input and implicitly tests `cin` to see whether the read was successful. Assuming the read succeeded, the second part of the condition tests whether the value we read is equal to the value in `sought`. If we found the value we want, the `while` loop is exited. Otherwise, the condition is evaluated again, which reads another value from `cin`.

Best Practices

Null statements should be commented. That way anyone reading the code can see that the statement was omitted intentionally.

### Beware of Missing or Extraneous Semicolons

Because a null statement is a statement, it is legal anywhere a statement is expected. For this reason, semicolons that might appear illegal are often nothing more than null statements. The following fragment contains two statements—the expression statement and the null statement:

```
ival = v1 + v2;; // ok: second semicolon is a superfluous null statement
```

Although an unnecessary null statement is often harmless, an extra semicolon following the condition in a `while` or `if` can drastically alter the programmer's intent. For example, the following code will loop indefinitely:

```
// disaster: extra semicolon: loop body is this null statement
while (iter != svec.end()) ; // the while body is the empty statement
    ++iter;       // increment is not part of the loop
```

Contrary to the indentation, the increment is not part of the loop. The loop body is the null statement formed by the semicolon that follows the condition.

WARNING

Extraneous null statements are not always harmless.

## Compound Statements (Blocks)

A **compound statement**, usually referred to as a **block**, is a (possibly empty) sequence of statements and declarations surrounded by a pair of curly braces. A block is a scope (§ 2.2.4, p. 48). Names introduced inside a block are accessible only in that block and in blocks nested inside that block. Names are visible from where they are defined until the end of the (immediately) enclosing block.

Compound statements are used when the language requires a single statement but the logic of our program needs more than one. For example, the body of a `while` or `for` loop must be a single statement, yet we often need to execute more than one statement. We do so by enclosing the statements in curly braces, thus turning the sequence of statements into a block.

As one example, recall the `while` loop in the program in § 1.4.1 (p. 11):

```
while (val <= 10) {
    sum += val;  // assigns sum + val to sum
    ++val;       // add 1 to val
}
```

The logic of our program needed two statements but a `while` loop may contain only one statement. By enclosing these statements in curly braces, we made them into a single (compound) statement.

Note

A block is *not* terminated by a semicolon.

We also can define an empty block by writing a pair of curlies with no statements. An empty block is equivalent to a null statement:

```
while (cin >> s && s != sought)
    { } // empty block
```

**EXERCISES SECTION 5.1**

**Exercise 5.1:** What is a null statement? When might you use a null statement?

**Exercise 5.2:** What is a block? When might you might use a block?

**Exercise 5.3:** Use the comma operator (§ 4.10, p. 169) to rewrite the `while` loop from § 1.4.1 (p. 11) so that it no longer requires a block.

**Exercise 5.4:** Do you think rewriting the `while` in the previous exercise improved or diminished the readability of this code. Explain your reasoning.

## 5.2 Statement Scope

Several statements allow us to define variable(s) inside their control structures. Depending on the statement, a variable can be defined as the condition that controls the execution path or inside an initialization statement that is part of the control structure.

### Variables Defined in the Expression that Controls Execution

Some statements use a value to control their action. Such statements generally can define and initialize a single variable whose value will be used control execution. Because the variable determines the flow of control, it must be initialized. A variable defined in a `switch` must be convertible to an integral type; in all other statements, the variable is used as a condition and must be convertible to `bool`.

For example, we might have a function `readVal` that returns an integer whose value we want to test. If we only need to test the value we could do so directly:

```
if (readVal())
    // do whatever
```

The `int` returned by `readVal` is converted to `bool` and tested for its truth value. After the condition, that value is discarded. It is much more likely that we want not only to test, but also to use the value returned by `readVal`. In that case, we need a variable to hold the return value:

```
if (auto i = readVal())
    // processing that uses i
```

Here the condition defines and initializes `i` and then converts the value in `i` to `bool`. The variable `i` remains in scope throughout the `if` statement.

### Statements with Separate Initialization Statements

A problem arises when we have a value that we want to initialize but that is not convertible to `bool`. For example, assume that instead of reading an `int`, we wanted to read a `string` and determine what to do based on whether that `string` was empty. Starting with C++17, we can use a seperate initialization statement inside an `if` or a `switch`:

New

```
// note the semicolon that terminates the init-statement
if (auto s = readVals(); s.empty())
    // do whatever processing is appropriate if s is empty
else
    // process s
```

The initialization statement can be either a declaration statement or an expression statement.

Although we can define names either in the condition that controls a statement's execution or in a separate initialization statement, there are some crucial differences between these two forms. We can define only one variable in a condition and that variable must be initialized. An initialization statement is an ordinary declaration statement. As in any other declaration, we can define multiple names and initializers are (generally) optional. However, there can be only one initialization statement in a condition, which means that all the variables must have the same base type (§ 2.3, p. 50):

```
// define 3 variables in the init-statement and then test the value of r as the condition
if (int i = readVal(), *p, &r = i; r)
    // do whatever
```

In this example, we've defined an `int`, a pointer to `int` and a reference to an `int`. As usual, we must initialize the reference, but initializers for nonreference variables are optional.

### Scope of Names Defined in a Control Structure

Variables defined in the control structure are visible only within that statement and are out of scope after the statement ends:

```
while (int i = get_num()) // i is created and initialized on each iteration
    cout << i << endl;
i = 0;  // error: i is not accessible outside the loop
```

If we need access to the control variable, after the statment completes then that variable must be defined outside the statement:

```
// find the first negative element
auto beg = v.cbegin();
while (beg != v.cend() && *beg >= 0)
    ++beg;
if (beg == v.cend())
    // we know that all elements in v are greater than or equal to zero
```

Note

A variable defined in a control structure is in scope *only* inside the body of the statement.

## 5.3 Selection Statements

C++ provides two statements that allow for conditional execution. The `if` statement determines the flow of control based on a condition. The `switch` statement

**Exercises Section 5.2**

**Exercise 5.5:** Explain each of the following examples, and correct any problems you detect.

(a)
```
while (string::iterator iter != s.end()) { /* ... */ }
```
(b)
```
while (bool status = find(word)) { /* ... */ }
   if (!status) { /* ... */ }
```

evaluates an integral expression and chooses one of several execution paths based on the expression's value.

### 5.3.1 The `if` Statement

An **`if` statement** conditionally executes another statement based on whether a specified condition is true. There are two forms of the `if`: one with an `else` branch and one without. The syntactic form of the simple `if` is

```
if constexpr_opt (init-statement_opt condition)
      statement1
```

An **`if else` statement** has the form

```
if constexpr_opt (init-statement_opt condition)
      statement1
else
      statement2
```

In both versions, the *condition* can be an expression or an initialized variable declaration (§ 5.2, p. 184). The expression or the variable must have a type that is convertible (§ 4.11, p. 169) to `bool`. C++17 added two new facilities to `if` statements: the optional use of `constexpr`, which we'll cover on page 189, and the ability to define a variable of any type in an optional *init-statement* (§ 5.2, p. 184).

If *condition* is `true`, then *statement1* is executed. After *statement1* completes, execution continues with the statement following the `if`.

If *condition* is `false`, *statement1* is skipped. In a simple `if`, execution continues with the statement following the `if`. In an `if else`, *statement2* is executed.

As usual, either or both *statement1* and *statement2* can be a block.

#### Using an `if else` Statement

To illustrate an `if` statement, we'll calculate a letter grade from a numeric grade. We'll assume that the numeric grades range from zero to 100 inclusive. A grade of 100 gets an "A++," grades below 60 get an "F," and the others range in clumps of ten: grades from 60 to 69 inclusive get a "D," 70 to 79 a "C," and so on. We'll use a `vector` to hold the possible letter grades:

```
const vector<string> scores = {"F", "D", "C", "B", "A", "A++"};
```

To solve this problem, we can use an `if else` statement to execute different actions for failing and passing grades:

```
// if grade is less than 60 it's an F, otherwise compute a subscript
string lettergrade;
if (grade < 60)
    lettergrade = scores[0];
else
    lettergrade = scores[(grade - 50)/10];
```

Depending on the value of `grade`, we execute the statement after the `if` or the one after the `else`. In the `else`, we compute a subscript from a grade by reducing the grade to account for the larger range of failing grades. Then we use integer division (§ 4.2, p. 149), which truncates the remainder, to calculate the appropriate `scores` index.

## Nested `if` Statements

To make our program more interesting, we'll add a plus or minus to passing grades. We'll give a plus to grades ending in 8 or 9, and a minus to those ending in 0, 1, or 2:

```
if (grade % 10 > 7)
    lettergrade += '+';      // grades ending in 8 or 9 get a +
else if (grade % 10 < 3)
    lettergrade += '-';      // those ending in 0, 1, or 2 get a -
```

Here we use the modulus operator (§ 4.2, p. 149) to get the remainder and decide based on the remainder whether to add plus or minus.

Next we'll incorporate the code that adds a plus or minus to the code that fetches the letter grade from scores:

```
// if failing grade, no need to check for a plus or minus
if (grade < 60)
    lettergrade = scores[0];
else {
    lettergrade = scores[(grade - 50)/10];  // fetch the letter grade
    if (grade != 100) {     // add plus or minus only if not already an A++
        if (grade % 10 > 7)
            lettergrade += '+';    // grades ending in 8 or 9 get a +
        else if (grade % 10 < 3)
            lettergrade += '-';    // grades ending in 0, 1, or 2 get a -
    }
}
```

Note that we use a block to enclose the two statements that follow the first `else`. If the `grade` is `60` or more, we have two actions that we need to do: Fetch the letter grade from `scores`, and conditionally set the plus or minus. The braces after the second `if` (that surround the innermost `if`) are not strictly necessary. We include them as a matter of style to make it clear how we intend this code to be interpreted.

### Watch Your Braces

It is a common mistake to forget the curly braces when multiple statements must be executed as a block. In the following example, contrary to the indentation, the code to add a plus or minus happens unconditionally:

```
if (grade < 60)
    lettergrade = scores[0];
else  // WRONG: missing curly
    lettergrade = scores[(grade - 50)/10];
    // despite appearances, without the curly brace, this code is always executed
    // failing grades will incorrectly get a - or a +
    if (grade != 100)
        if (grade % 10 > 7)
            lettergrade += '+';    // grades ending in 8 or 9 get a +
        else if (grade % 10 < 3)
            lettergrade += '-';    // grades ending in 0, 1, or 2 get a -
```

Uncovering this error may be very difficult because the program looks correct.

To avoid such problems, some coding styles recommend always using braces after an `if` or an `else` (and also around the bodies of `while` and `for` statements). Doing so avoids any possible confusion. It also means that the braces are already in place if later modifications of the code require adding statements.

Best Practices

Many editors and development environments have tools to automatically indent source code to match its structure. It is a good idea to use such tools if they are available.

### Dangling `else`

When we nest an `if` inside another `if`, it is possible that there will be more `if` branches than `else` branches. Indeed, our grading program has four `if`s and two `else`s. The question arises: How do we know to which `if` a given `else` belongs?

This problem, usually referred to as a **dangling `else`**, is common to many programming languages that have both `if` and `if else` statements. Different languages solve this problem in different ways. C++ resolves the ambiguity by specifying that each `else` is matched with the closest preceding unmatched `if`.

Programmers sometimes get into trouble when they write code that contains more `if` than `else` branches. To illustrate the problem, we'll rewrite the innermost `if else` that adds a plus or minus using a different set of conditions:

```
// WRONG: execution does NOT match indentation; the else goes with the inner if
if (grade % 10 >= 3)
    if (grade % 10 > 7)
        lettergrade += '+';  // grades ending in 8 or 9 get a +
else
    lettergrade += '-'; // grades ending in 3, 4, 5, 6, or 7 get a minus!
```

The indentation in our code indicates that we intend the `else` to go with the outer `if`—we intend for the `else` branch to be executed when the `grade` ends in a digit less than 3. However, despite our intentions, and contrary to the indentation, the else branch is part of the inner if. This code adds a '-' to grades ending in 3 to 7 inclusive! Properly indented to match the actual execution, what we wrote is:

```
// indentation matches the execution path, not the programmer's intent
if (grade % 10 >= 3)
    if (grade % 10 > 7)
        lettergrade += '+';  // grades ending in 8 or 9 get a +
    else
        lettergrade += '-';  // grades ending in 3, 4, 5, 6, or 7 get a minus!
```

## Controlling the Execution Path with Braces

We can make the else part of the outer if by enclosing the inner if in a block:

```
// add a plus for grades that end in 8 or 9 and a minus for those ending in 0, 1, or 2
if (grade % 10 >= 3) {
    if (grade % 10 > 7)
        lettergrade += '+';  // grades ending in 8 or 9 get a +
} else                       // curlies force the else to go with the outer if
    lettergrade += '-'; // grades ending in 0, 1, or 2 will get a minus
```

Statements do not span block boundaries, so the inner if ends at the close curly before the else. The else cannot be part of the inner if. Now, the nearest unmatched if is the outer if, which is what we intended all along.

## `constexpr if`

New

C++17 introduced a new form of if, the **constexpr if** statement. A constexpr if differs from an ordinary if in that the condition must be a constant expression (§ 2.4.4, p. 66), which the compiler evaluates at compile time. The compiler generates code for the if body only if the condition evaluates as true; otherwise the code in the if body is discarded. Similarly, code in the else branch is compiled only if the condition evaluates as false.

As an example, imagine we're writing code that should be able to run on multiple different machines using different compilers. On some machines ints and longs might be the same size; on others longs might be bigger. In this case, we might want to check for overflow (§ 4.2, p. 149) based on the relative sizes of int and long. We could do so as follows:

```
long sum = 0;
int i = readVal(), j = readVal();
// sizeof see § 4.9 (p. 167)
if constexpr (sizeof(long) >= 2* sizeof(int))
    sum = i + j; // ok: we know we're safe on such machines
else
    sum = checkover(i, j);
}
```

This condition checks whether the number of bytes in a long is at least twice the number in an int. If so, it's safe to add two ints storing their result in sum.

If longs aren't large enough to hold any sum, we call the checkover function, which we assume will do the addition and manage any overflow.

If the condition is true, the call to checkover will be discarded. If this call to checkover was the only one in the program, the checkover function would not even have to have been defined. However, even if no call is generated, a declaration for checkover must be in scope.

**EXERCISES SECTION 5.3.1**

**Exercise 5.6:** Using an if–else statement, write your own version of the program to generate the letter grade from a numeric grade.

**Exercise 5.7:** Rewrite your grading program to use the conditional operator (§ 4.7, p. 161) in place of the if–else statement.

**Exercise 5.8:** Correct the errors in each of the following code fragments:

(a)
```
if (ival1 != ival2)
     ival1 = ival2
else ival1 = ival2 = 0;
```
(b)
```
if (ival < minval)
     minval = ival;
     occurs = 1;
```
(c)
```
if (int ival = get_value())
     cout << "ival = " << ival << endl;
if (!ival)
     cout << "ival = 0\n";
```
(d)
```
if (ival = 0)
     ival = get_value();
```

**Exercise 5.9:** What is a "dangling else"? How are else clauses resolved in C++?

## 5.3.2 The switch Statement

A **switch statement** provides a convenient way of selecting among a (possibly large) number of fixed alternatives. Its form is:

switch (*init-statement*$_{opt}$ *condition*)

New

As we saw in § 5.2 (p. 184), the optional *init-statement* must be a declaration statement or an expression statement. As such it must end with a semicolon. The variable(s) introduced in the *init-statement* are in scope throughout the switch statement and are inaccessible after it.

### Using a switch Statement

As one example, suppose that we want to count how often each of the five vowels appears in some segment of text. Our program logic is as follows:

- Read every character in the input.
- Compare each character to the set of vowels.
- If the character matches one of the vowels, add 1 to that vowel's count.
- Display the results.

For example, when we run the program on the text of this chapter, the output is

```
Number of vowel a:  3557
Number of vowel e:  6816
Number of vowel i:  3478
Number of vowel o:  3590
Number of vowel u:  1116
```

We can solve our problem most directly using a `switch` statement:

```
// initialize counters for each vowel
unsigned aCnt = 0, eCnt = 0, iCnt = 0, oCnt = 0, uCnt = 0;
for (char ch; cin >> ch; ) {
    // if ch is a vowel, increment the appropriate counter
    switch (ch) {
        case 'a':
            ++aCnt;
            break;
        case 'e':
            ++eCnt;
            break;
        case 'i':
            ++iCnt;
            break;
        case 'o':
            ++oCnt;
            break;
        case 'u':
            ++uCnt;
            break;
    }
}
// print results
cout << "Number of vowel a: \t" << aCnt << '\n'
     << "Number of vowel e: \t" << eCnt << '\n'
     << "Number of vowel i: \t" << iCnt << '\n'
     << "Number of vowel o: \t" << oCnt << '\n'
     << "Number of vowel u: \t" << uCnt << endl;
```

A `switch` statement executes by evaluating the parenthesized expression that follows the keyword `switch`. That expression may be an initialized variable declaration (§ 5.2, p. 184). The expression is converted to integral type. The result of the expression is compared with the value associated with each **case label**. A case label is the keyword `case` followed by an integral value.

If the expression matches the value of a `case` label, execution begins with the first statement following that label. Execution continues normally from that statement through the end of the `switch` or until a `break` statement.

We'll look at `break` statements in detail in § 5.5.1 (p. 202), but, briefly, a `break` interrupts the current control flow. In this case, the `break` transfers control out of the `switch`. In this program, the `switch` appears inside the body of a `while`. Breaking out of this `switch` returns control to that enclosing `while`. Because there are no other statements in this `while`, execution continues at the condition in the `while`.

If no matching case label is found, execution falls through to the first statement following the `switch`. As we already know, in this example, exiting the `switch` returns control to the condition in the `while`.

The value of a case label must be an integral constant expression (§ 2.4.4, p. 66):

```
char ch = getVal();
int ival = 42;
switch(ch) {
    case 3.14: // error: noninteger as case label
    case ival: // error: nonconstant as case label
    // ...
```

It is an error for any two `case` labels to have the same value. There is also a special-case label, `default`, which we cover on page 193.

## Control Flow within a `switch`

It is important to understand that execution flows across `case` labels. After a `case` label is matched, execution starts at that label and continues across all the remaining `case`s or until the program explicitly interrupts it. To avoid executing code for subsequent `case`s, we must explicitly tell the compiler to stop execution. Under most conditions, the last statement before the next `case` label is `break`.

However, there are situations where the default `switch` behavior is exactly what is needed. Each `case` label can have only a single value, but sometimes we have two or more values that share a common set of actions. In such instances, we omit a `break` statement, allowing the program to *fall through* multiple `case` labels.

For example, we might want to count only the total number of vowels:

```
unsigned vowelCnt = 0;
// ...
switch (ch)
{
    // any occurrence of a, e, i, o, or u increments vowelCnt
    case 'a':
    case 'e':
    case 'i':
    case 'o':
    case 'u':
       ++vowelCnt;
       break;
}
```

Here we stacked several `case` labels together with no intervening `break`. The same code will be executed whenever `ch` is a vowel.

Because C++ programs are free-form, `case` labels need not appear on a new line. We can emphasize that the `case`s represent a range of values by listing them all on a single line:

```
switch (ch)
{
    // alternative legal syntax
    case 'a': case 'e': case 'i': case 'o': case 'u':
       ++vowelCnt;
       break;
}
```

Best Practices

Omitting a `break` at the end of a `case` happens rarely. If you do omit a `break`, include a comment explaining the logic.

## Forgetting a `break` Is a Common Source of Bugs

It is a common misconception to think that only the statements associated with the matched `case` label are executed. For example, here is an *incorrect* implementation of our vowel-counting `switch` statement:

```
// warning: deliberately incorrect!
switch (ch) {
    case 'a':
        ++aCnt;    // oops: should have a break statement
    case 'e':
        ++eCnt;    // oops: should have a break statement
    case 'i':
        ++iCnt;    // oops: should have a break statement
    case 'o':
        ++oCnt;    // oops: should have a break statement
    case 'u':
        ++uCnt;
}
```

To understand what happens, assume that the value of `ch` is `'e'`. Execution jumps to the code following the `case 'e'` label, which increments `eCnt`. Execution *continues* across the `case` labels, incrementing `iCnt`, `oCnt`, and `uCnt` as well.

Best Practices

Although it is not necessary to include a `break` after the last label of a `switch`, the safest course is to provide one. That way, if an additional `case` is added later, the `break` is already in place.

## The `default` Label

The statements following the **`default` label** are executed when no `case` label matches the value of the `switch` expression. For example, we might add a counter to track how many nonvowels we read. We'll increment this counter, which we'll name `otherCnt`, in the `default` case:

```
        // if ch is a vowel, increment the appropriate counter
        switch (ch) {
            case 'a': case 'e': case 'i': case 'o': case 'u':
                ++vowelCnt;
                break;
            default:
                ++otherCnt;
                break;
        }
    }
```

In this version, if `ch` is not a vowel, execution will start at the `default` label and we'll increment `otherCnt`.

Best Practices

It can be useful to define a `default` label even if there is no work for the `default` case. Defining an empty `default` section indicates to subsequent readers that the case was considered.

A label may not stand alone; it must precede a statement or another `case` label. If a `switch` ends with a `default` case that has no work to do, then the `default` label must be followed by a null statement or an empty block.

## Variable Definitions inside the Body of a `switch`

As we've seen, execution in a `switch` can jump across `case` labels. When execution jumps to a particular `case`, any code that occurred inside the `switch` before that label is ignored. The fact that code is bypassed raises an interesting question: What happens if the code that is skipped includes a variable definition?

The answer is that it is illegal to jump from a place where a variable with an initializer is out of scope to a place where that variable is in scope:

```
case true:
    // this switch statement is illegal because these initializations might be bypassed
    string file_name; // error: control bypasses an implicitly initialized variable
    int ival = 0;     // error: control bypasses an explicitly initialized variable
    int jval;         // ok: because jval is not initialized
    break;
case false:
    // ok: jval is in scope but is uninitialized
    jval = next_num(); // ok: assign a value to jval
```

If this code were legal, then any time control jumped to the `false` case, it would bypass the initialization of `file_name` and `ival`. Those variables would be in scope. Code following `false` could use those variables. However, these variables would not have been initialized. As a result, the language does not allow us to jump over an initialization if the initialized variable is in scope at the point to which control transfers.

If we need to define and initialize a variable for a particular `case`, we can do so by defining the variable inside a block, thereby ensuring that the variable is out of scope at the point of any subsequent label.

```
case true:
    {
       // ok: declaration statement within a statement block
       string file_name = get_file_name();
       // ...
    }
    break;
case false:
       if (file_name.empty())  // error: file_name is not in scope
```

### EXERCISES SECTION 5.3.2

**Exercise 5.10:** Write a program using a series of `if` statements to count the number of vowels in text read from `cin`.

**Exercise 5.11:** There is one problem with our vowel-counting program as we've implemented it: It doesn't count capital letters as vowels. Write a program that counts both lower- and uppercase letters as the appropriate vowel—that is, your program should count both `'a'` and `'A'` as part of `aCnt`, and so forth.

**Exercise 5.12:** Modify our vowel-counting program so that it also counts the number of blank spaces, tabs, and newlines read.

**Exercise 5.13:** Modify our vowel-counting program so that it counts the number of occurrences of the following two-character sequences: `ff`, `fl`, and `fi`.

**Exercise 5.14:** Each of the programs in the highlighted text on page 196 contains a common programming error. Identify and correct each error.

## 5.4 Iterative Statements

Iterative statements, commonly called loops, provide for repeated execution until a condition is true. The `while` and `for` statements test the condition before executing the body. The `do while` executes the body and then tests its condition.

### 5.4.1 The `while` Statement

A **`while` statement** repeatedly executes a target statement as long as a condition is true. Its syntactic form is

```
while (condition)
      statement
```

**Code for Exercise 5.14**

(a)
```
unsigned aCnt = 0, eCnt = 0, iouCnt = 0;
char ch = next_text();
switch (ch) {
    case 'a': aCnt++;
    case 'e': eCnt++;
    default: iouCnt++;
}
```

(b)
```
unsigned index = some_value();
switch (index) {
    case 1:
        int ix = get_value();
        ivec[ ix ] = index;
        break;
    default:
        ix = ivec.size()-1;
        ivec[ ix ] = index;
}
```

(c)
```
unsigned evenCnt = 0, oddCnt = 0;
int digit = get_num() % 10;
switch (digit) {
    case 1, 3, 5, 7, 9:
        oddcnt++;
        break;
    case 2, 4, 6, 8, 10:
        evencnt++;
        break;
}
```

(d)
```
unsigned ival=512, jval=1024, kval=4096;
unsigned bufsize;
unsigned swt = get_bufCnt();
switch(swt) {
    case ival:
        bufsize = ival * sizeof(int);
        break;
    case jval:
        bufsize = jval * sizeof(int);
        break;
    case kval:
        bufsize = kval * sizeof(int);
        break;
}
```

where *statement* (which is often a block) is executed as long as *condition* evaluates as `true`. *condition* may not be empty. If the first evaluation of *condition* yields `false`, *statement* is not executed.

The *cond* can be an expression or an initialized variable declaration (§ 5.2, p. 185). Ordinarily, *cond* itself or the loop body must do something to change the value of *cond*. Otherwise, the loop might never terminate.

> **Note** Variables defined in a `while` condition or `while` body are created and destroyed on each iteration.

### Using a `while` Loop

Both `while` and `for` loops begin by testing a condition and depending on the value of the condition execute the loop body or exit. We can rewrite any particular `while` as a `for` and vice versa. As a result, whether to use a `while` or a `for` is a matter of style or habit.

We like to use a `while` when we need access to the loop control variable after the loop finishes:

```
// find the first negative element
auto beg = v.cbegin();
while (beg != v.cend() && *beg >= 0)
    ++beg;
if (beg == v.cend())
    // we know that all elements in v are greater than or equal to zero
```

This loop continues until we find a negative value. When the loop terminates, `beg` is either equal to `v.end()`, or it denotes an element in `v` whose value is less than zero. We then use the state of `beg` outside the `while` to determine further processing.

#### EXERCISES SECTION 5.4.1

**Exercise 5.15:** Write a program to read `strings` from standard input looking for duplicated words. The program should find places in the input where one word is followed immediately by itself. Keep track of the largest number of times a single repetition occurs and which word is repeated. Print the maximum number of duplicates, or else print a message saying that no word was repeated. For example, if the input is

```
how now now now brown cow cow
```

the output should indicate that the word `now` occurred three times.

## 5.4.2 Traditional `for` Statement

The syntactic form of the **`for` statement** is:

```
for (init-statement condition; expression)
    statement
```

The `for` and the part inside the parentheses is often referred to as the `for` header.

*init-statement* must be a declaration statement, an expression statement, or a null statement. Each of these statements ends with a semicolon, so the syntactic form can also be thought of as

```
for (initializer; condition; expression)
      statement
```

In general, *init-statement* is used to initialize or assign a starting value that is modified over the course of the loop. *condition* serves as the loop control. As long as *condition* evaluates as `true`, *statement* is executed. If the first evaluation of *condition* yields `false`, *statement* is not executed. *expression* usually modifies the variable(s) initialized in *init-statement* and tested in *condition*. *expression* is evaluated after each iteration of the loop. As usual, *statement* can be a block.

## Execution Flow in a Traditional `for` Loop

Given the following `for` loop from § 3.2.3 (p. 97):

```
// process characters in s until we run out of characters or we hit a whitespace
for (decltype(s.size()) index = 0;
     index != s.size() && !isspace(s[index]); ++index)
        s[index] = toupper(s[index]); // capitalize the current character
```

the order of evaluation is as follows:

1. *init-statement* is executed once at the start of the loop. In this example, `index` is defined and initialized to zero.
2. Next, *condition* is evaluated. If `index` is not equal to `s.size()` and the character at `s[index]` is not whitespace, the `for` body is executed. Otherwise, the loop terminates. If the condition is `false` on the first iteration, then the `for` body is not executed at all.
3. If the condition is `true`, the `for` body executes. In this case, the `for` body makes the character at `s[index]` uppercase.
4. Finally, *expression* is evaluated. In this example, `index` is incremented by 1.

These four steps represent the first iteration of the `for` loop. Step 1 is executed only once on entry to the loop. Steps 2, 3, and 4 are repeated until the condition evaluates as `false`—that is, when we encounter a whitespace character in `s`, or `index` is greater than `s.size()`.

> Note
> It is worth remembering that the visibility of any object defined within the `for` header is limited to the body of the `for` loop. Thus, in this example, `index` is inaccessible after the `for` completes.

### Multiple Definitions in the `for` Header

As in any other declaration, *init-statement* can define several objects. However, *init-statement* may be only a single statement. Therefore, all the variables must have the same base type (§ 2.3, p. 50). As one example, we might write a loop to duplicate the elements of a `vector` on the end as follows:

```
// remember the size of v and stop when we get to the original last element
for (decltype(v.size()) i = 0, sz = v.size(); i != sz; ++i)
    v.push_back(v[i]);
```

In this loop we define both the index, `i`, and the loop control, `sz`, in *init-statement*.

### Omitting Parts of the `for` Header

A `for` header can omit any (or all) of *init-statement*, *condition*, or *expression*.

We can use a null statement for *init-statement* when an initialization is unnecessary. For example, we might rewrite the loop that looked for the first negative number in a `vector` so that it uses a `for`:

```
auto beg = v.cbegin();
for ( /* null */; beg != v.cend() && *beg >= 0; ++beg)
    ; // no work to do
```

Note that the semicolon is necessary to indicate the absence of *init-statement*—more precisely, the semicolon represents a null *init-statement*. In this loop, the `for` body is also empty because all the work of the loop is done inside the `for` condition and expression. The condition decides when it's time to stop looking and the expression increments the iterator.

Omitting *condition* is equivalent to writing `true` as the condition. Because the condition always evaluates as `true`, the `for` body must contain a statement that exits the loop. Otherwise the loop will execute indefinitely:

```
for (int i = 0; /* no condition */ ; ++i) {
    // process i; code inside the loop must stop the iteration!
}
```

Typically a `for` body in a loop without a condition includes at least one `break` statement (§ 5.3.2, p. 190), which we will cover in more detail in § 5.5.1 (p. 202).

We've already seen several programs that omit the *expression*. In such loops, either the condition or the body must do something to advance the iteration. For example, in the `for` loop in our program from § 1.4.3 (p. 15):

```
for (int value; std::cin >> value; /* no expression */)
    sum += value; // equivalent to sum = sum + value
```

the condition reads a new value on each iteration, thus advancing the iteration.

## 5.4.3 Range `for` Statement

The range `for` statement iterates through the elements of a container or other sequence. The syntactic form of the **range `for` statement** is:

### Exercises Section 5.4.2

**Exercise 5.16:** Explain each of the following loops. Correct any problems you detect.

(a)
```
for (int ix = 0; ix != sz; ++ix)  { /* ... */ }
if (ix != sz)
     // ...
```
(b)
```
int ix;
for (ix != sz; ++ix) { /* ... */ }
```
(c)
```
for (int ix = 0; ix != sz; ++ix, ++ sz)  { /* ... */ }
```

**Exercise 5.17:** The `while` loop is particularly good at executing while some condition holds; for example, when we need to read values until end-of-file. The `for` loop is generally thought of as a step loop: An index steps through a range of values in a collection. Write an idiomatic use of each loop and then rewrite each using the other loop construct. If you could use only one loop, which would you choose? Why?

**Exercise 5.18:** Given two `vector`s of `int`s, write a program to determine whether one `vector` is a prefix of the other. For `vector`s of unequal length, compare the number of elements of the smaller `vector`. For example, given the `vector`s containing `0`, `1`, `1`, and `2` and `0`, `1`, `1`, `2`, `3`, `5`, `8`, respectively your program should return `true`.

```
for (declaration : expression)
    statement
```

where *expression* must represent a sequence, such as a braced initializer list (§ 3.3.1, p. 102), an array (§ 3.5, p. 119), or an object of a type such as `vector` or `string` that has `begin` and `end` members that return iterators (§ 3.4, p. 111).

*declaration* defines a variable. It must be possible to convert each element of the sequence to the variable's type (§ 4.11, p. 169). The easiest way to ensure that the types match is to use the `auto` type specifier (§ 2.5.2, p. 68). That way the compiler will deduce the type for us. If we want to write to the elements in the sequence, the loop variable must be a reference type.

On each iteration, the control variable is defined and initialized by the next value in the sequence, after which *statement* is executed. As usual, *statement* can be a block. Execution ends once all the elements have been processed.

We have already seen several such loops, but for completeness, here is one that doubles the value of each element in a `vector`:

```
vector<int> v = {0,1,2,3,4,5,6,7,8,9};
// range variable must be a reference so we can write to the elements
for (auto &r : v)     // for each element in v
    r *= 2;           // double the value of each element in v
```

The `for` header declares the loop control variable, `r`, and associates it with `v`. We use `auto` to let the compiler infer the correct type for `r`. Because we want to change the value of the elements in `v`, we declare `r` as a reference. When we assign to `r` inside the loop, that assignment changes the value of the element to which `r` is bound.

A range `for` can be understood in terms of the equivalent traditional `for`:

```
{
    auto beg = v.begin();
    auto end = v.end();
    for ( ; beg != end; ++beg) {
        auto &r = *beg; // r must be a reference so we can change the element
        r *= 2;         // double the value of each element in v
    }
}
```

Now that we know how a range `for` works, we can understand why we said in § 3.3.3 (p. 106) that we cannot use a range `for` to add elements to a `vector` (or other container). In a range `for`, the value of `end()` is cached. If we add elements to (or remove them from) the sequence, the value of `end` might be invalidated (§ 3.4.1, p. 116). We'll have more to say about these matters in § 9.3.6 (p. 389).

### 5.4.4 The `do while` Statement

A **`do while` statement** is like a `while` but the condition is tested after the statement body completes. Regardless of the value of the condition, we execute the loop at least once. The syntactic form is as follows:

```
do
        statement
while (condition);
```

Note: A `do while` ends with a semicolon after the parenthesized condition.

In a `do`, *statement* is executed before *condition* is evaluated. *condition* cannot be empty. If *condition* evaluates as `false`, then the loop terminates; otherwise, the loop is repeated. Variables used in *condition* must be defined outside the body of the `do while` statement.

We can write a program that (indefinitely) does sums using a `do while`:

```
// repeatedly ask the user for a pair of numbers to sum
string rsp;  // used in the condition; can't be defined inside the do
do {
    cout << "please enter two values: ";
    int val1 = 0, val2 = 0;
    cin  >> val1 >> val2;
    cout << "The sum of " << val1 << " and " << val2
         << " = " << val1 + val2 << "\n\n"
         << "More? Enter yes or no: ";
    cin  >> rsp;
} while (!rsp.empty() && rsp[0] != 'n');
```

The loop starts by prompting the user for two numbers. It then prints their sum and asks whether the user wishes to do another sum. The condition checks that the user gave a response. If not, or if the input starts with an `n`, the loop is exited. Otherwise the loop is repeated.

Because the condition is not evaluated until after the statement or block is executed, the `do while` loop does not allow variable definitions inside the condition:

```
do {
    // ...
    mumble(foo);
} while (int foo = get_foo()); //  error: declaration in a do condition
```

If we could define variables in the condition, then any use of the variable would happen *before* the variable was defined!

**EXERCISES SECTION 5.4.4**

**Exercise 5.19:** Explain each of the following loops. Correct any problems you detect.

(a)
```
do
     int v1, v2;
     cout << "Please enter two numbers to sum:" ;
     if (cin >> v1 >> v2)
         cout << "Sum is: " << v1 + v2 << endl;
while (cin);
```
(b)
```
do {
     // ...
} while (int ival = get_response());
```
(c)
```
do {
     int ival = get_response();
 } while (ival);
```

**Exercise 5.20:** Write a program that uses a `do while` loop to repetitively request two `string`s from the user and report which `string` is less than the other.

## 5.5 Jump Statements

Jump statements interrupt the flow of execution. C++ offers four jumps: `break`, `continue`, and `goto`, which we cover in this chapter, and the `return` statement, which we'll describe in § 6.3 (p. 227).

### 5.5.1 The `break` Statement

A **`break` statement** terminates the nearest enclosing `while`, `do while`, `for`, or `switch` statement. Execution resumes at the statement immediately following the terminated statement.

A `break` can appear only within an iteration statement or `switch` statement (including inside statements or blocks nested inside such loops). A `break` affects only the nearest enclosing loop or `switch`:

```
for (string buf; cin >> buf && !buf.empty(); ) {
    switch(buf[0]) {
    case '-':
        // process up to the first blank
        for (auto it = buf.begin()+1; it != buf.end(); ++it) {
              if (*it == ' ')
                   break; // #1, leaves the for loop
              // . . .
        }
        // break #1 transfers control here
        // remaining '-' processing:
        break; // #2, leaves the switch statement
    case '+':
        // . . .
    } // end switch
   // end of switch: break #2 transfers control here
} // end while
```

The `break` labeled #1 terminates the `for` loop that follows the hyphen `case` label. It does not terminate the enclosing `switch` statement and in fact does not even terminate the processing for the current case. Processing continues with the first statement following the `for`, which might be additional code to handle a hyphen or the `break` that completes that section.

The `break` labeled #2 terminates the `switch` but does not terminate the enclosing `while` loop. Processing continues after that `break` by executing the condition in the `while`.

**EXERCISES SECTION 5.5.1**

**Exercise 5.21:** Write a program to read a sequence of `string`s from the standard input until either the same word occurs twice in succession or all the words have been read. Use a `while` loop to read the text one word at a time. Use the `break` statement to terminate the loop if a word occurs twice in succession. Print the word if it occurs twice in succession, or else print a message saying that no word was repeated.

**Exercise 5.22:** Rewrite the program from § 3.2.3 (p. 97) that capitalized the first word in a `string` using a range `for` statement.

### 5.5.2 The `continue` Statement

A **`continue` statement** terminates the current iteration of the nearest enclosing loop and immediately begins the next iteration. A `continue` can appear only inside a `for`, `while`, or `do while` loop, including inside statements or blocks nested inside such loops. Like the `break` statement, a `continue` inside a nested loop affects only the nearest enclosing loop. Unlike a `break`, a `continue` may appear inside a `switch` only if that `switch` is embedded inside an iterative statement.

A `continue` interrupts the current iteration; execution stays inside the loop. In the case of a `while` or a `do while`, execution continues by evaluating the condition. In a traditional `for` loop, execution continues at the *expression* inside the `for` header. In a range `for`, execution continues by initializing the control variable from the next element in the sequence.

For example, the following loop reads the standard input one word at a time. Only words that begin with an underscore will be processed. For any other value, we terminate the current iteration and get the next input:

```
for (string buf; cin >> buf && !buf.empty(); ) {
    if (buf[0] != '_')
        continue; // get another input
    // still here? the input starts with an underscore; process buf . . .
}
```

**EXERCISES SECTION 5.5.2**

**Exercise 5.23:** Revise the program from the exercise in § 5.5.1 (p. 203) so that it looks only for duplicated words that start with an uppercase letter.

### 5.5.3 The `goto` Statement

A **`goto` statement** provides an unconditional jump from the `goto` to a another statement in the same function.

Best Practices: Programs should not use `goto`s. `goto`s make programs hard to understand and hard to modify.

The syntactic form of a `goto` statement is

```
goto label;
```

where *label* is an identifier that identifies a statement. A **labeled statement** is any statement that is preceded by an identifier followed by a colon:

```
end: ret = 0;   // labeled statement; may be the target of a goto
```

Label identifiers are independent of names used for variables and other identifiers. Hence, a label may have the same identifier as another entity in the program without interfering with the other uses of that identifier. The `goto` and the labeled statement to which it transfers control must be in the same function.

Like a `switch` statement, a `goto` cannot transfer control from a point where an initialized variable is out of scope to a point where that variable is in scope:

```
    // ...
    goto end;
    int ix = 10; // error: goto bypasses an initialized variable definition
```

```
end:
     // error: code here could use ix but the goto bypassed its declaration
     ix = 42;
```

A jump backward over an already executed definition is okay. Jumping back to a point before a variable is defined destroys the variable and constructs it again:

```
// backward jump over an initialized variable definition is okay
  begin:
    int sz = get_size();
    if (sz <= 0) {
          goto begin;
    }
```

Here `sz` is destroyed when the `goto` executes. It is defined and initialized anew when control passes back through its definition after the jump back to `begin`.

**EXERCISES SECTION 5.5.3**

**Exercise 5.24:** The last example in this section that jumped back to `begin` could be better written using a loop. Rewrite the code to eliminate the `goto`.

## 5.6 `try` Blocks and Exception Handling

Exceptions are run-time anomalies—such as losing a database connection or encountering unexpected input—that exist outside the normal functioning of a program. Dealing with anomalous behavior can be one of the most difficult parts of designing any system.

Exception handling is generally used when one part of a program detects a problem that it cannot resolve and the problem is such that the detecting part of the program cannot continue. In such cases, the detecting part needs a way to signal that something happened and that this part of the program cannot continue. Moreover, the detecting part needs a way to signal the problem without knowing what part of the program will deal with the exceptional condition. Having signaled what happened, the detecting part stops processing.

A program that contains code that might raise an exception (usually) has another part to handle whatever happened. For example, if the problem is invalid input, the handling part might ask the user to provide correct input. If the database was lost, the handling part might alert an operator.

Exception handling supports this cooperation between the detecting and handling parts of a program. In C++, exception handling involves

- **`throw` expressions**, which the detecting part uses to indicate that it encountered something it can't handle. We say that a `throw` **raises** an exception.
- **`try` blocks**, which the handling part uses to deal with an exception. A `try` block starts with the keyword `try` and ends with one or more **`catch` clauses**.

Exceptions thrown from code executed inside a `try` block are usually handled by one of the `catch` clauses. Because they "handle" the exception, `catch` clauses are also known as **exception handlers** or just **handlers**.

- A set of **`exception` classes** that are used to pass information about what happened between a `throw` and an associated `catch`.

In the remainder of this section, we'll introduce these three components of exception handling. We'll also have more to say about exceptions in § 18.1 (p. 824).

### 5.6.1 A `throw` Expression

The detecting part of a program uses a `throw` expression to raise an exception. A `throw` consists of the keyword `throw` followed by an expression. The type of the expression determines what kind of exception is thrown. A `throw` expression is usually followed by a semicolon, making it into an expression statement.

As a simple example, recall the program in § 1.5.2 (p. 23) that added two objects of type `Sales_item`. That program checked whether the records it read referred to the same book. If not, it printed a message and exited.

```
Sales_item item1, item2;
cin >> item1 >> item2;
// first check that item1 and item2 represent the same book
if (item1.isbn() == item2.isbn()) {
    cout << item1 + item2 << endl;
    return 0;   // indicate success
} else {
    cerr << "Data must refer to same ISBN"
              << endl;
    return -1;  // indicate failure
}
```

In a more realistic program, the part that adds the objects might be separated from the part that manages the interaction with a user. For example, the code that checks ISBNs and does the addition might itself be inside a separate function. In this case, we might rewrite the test to throw an exception rather than returning an error indicator:

```
// if the transactions aren't for the same isbn, throw a runtime_error exception
if (item1.isbn() != item2.isbn())
   throw runtime_error("Data must refer to same ISBN");
// if we're still here, the ISBNs are the same
return item1 + item2;
```

In this code, if the ISBNs differ, we throw an expression that is an object of type `runtime_error`. Throwing an exception terminates the current function and transfers control to a handler that will know how to handle this error.

The type `runtime_error` is one of the standard library exception types and is defined in the `stdexcept` header. We'll have more to say about these types in

§ 5.6.2 (p. 209). We must initialize a `runtime_error` by giving it a `string` or a C-style character string (§ 3.5.4, p. 128). That string provides additional information about the problem.

## 5.6.2 The `try` Block

The general form of a `try` block is

```
try {
    program-statements
} catch (exception-declaration) {
    handler-statements
} catch (exception-declaration) {
    handler-statements
} // ...
```

A `try` block begins with the keyword `try` followed by a block, which, as usual, is a sequence of statements enclosed in curly braces.

Following the `try` block is a list of one or more `catch` clauses. A `catch` consists of three parts: the keyword `catch`, the declaration of a (possibly unnamed) object within parentheses (referred to as an **exception declaration**), and a block. When a `catch` is selected to handle an exception, the associated block is executed. Once the `catch` finishes, execution continues with the statement immediately following the last `catch` clause of the `try` block.

The *program-statements* inside the `try` constitute the normal logic of the program and can contain any C++ statement, including declarations. As in any other block, variables declared inside a `try` block are inaccessible outside the block—in particular, they are not accessible to the `catch` clauses.

### Writing a Handler

With this knowledge, we can now write the part of our addition program that handles interaction with the user. Assuming that the code to add two `Sales_items` is in a function named `add_items` we might handle a possible exception as follows:

```
while (cin >> item1 >> item2) {
    try {
        cout << add_items(item1, item2) << endl;
    } catch (runtime_error err) {
        // remind the user that the ISBNs must match and prompt for another pair
        cout << err.what()
             << "\nTry Again?  Enter y or n" << endl;
        char c;
        cin >> c;
        if (!cin || c == 'n')
            break;      // break out of the while loop
    }
}
```

The ordinary logic of the program appears inside the `try` block. In this case, we call `add_items` to add `item1` and `item2`. This part of the program is wrapped inside a `try` because it might throw an exception of type `runtime_error`.

This `try` block has a single `catch` clause, which handles exceptions of type `runtime_error`. The statements in the block following the `catch` are executed if code inside the `try` block throws a `runtime_error`. Our `catch` handles the error by printing a message and asking the user to indicate whether to continue. If the user enters 'n', then the `break` is executed and we exit the `while`. Otherwise, execution falls through to the closing brace of the `while`, which transfers control back to the `while` condition for the next iteration.

The prompt to the user prints the return from `err.what()`. We know that `err` has type `runtime_error`, so we can infer that `what` is a member function (§ 1.5.2, p. 23) of the `runtime_error` class. Each of the library exception classes defines a member function named `what`. These functions take no arguments and return a C-style character string (i.e., a `const char*`). The `what` member of `runtime_error` returns a copy of the `string` used to initialize the particular object. If the code described in the previous section threw an exception, then this `catch` would print

```
Data must refer to same ISBN
Try Again?  Enter y or n
```

**CAUTION: WRITING EXCEPTION SAFE CODE IS *Hard***

**It is important to realize that exceptions interrupt the normal flow of a program. At the point where the exception occurs, some of the computations that the caller requested may have been done, while others remain undone. In general, bypassing part of the program might mean that an object is left in an invalid or incomplete state, or that a resource is not freed, and so on. Programs that properly "clean up" during exception handling are said to be *exception safe*. Writing exception safe code is surprisingly hard, and (largely) beyond the scope of this language Primer.**

**Some programs use exceptions simply to terminate the program when an exceptional condition occurs. Such programs generally don't worry about exception safety.**

**Programs that do handle exceptions and continue processing generally must be constantly aware of whether an exception might occur and what the program must do to ensure that objects are valid, that resources don't leak, and that the program is restored to an appropriate state.**

**We will occasionally point out particularly common techniques used to promote exception safety. However, readers whose programs require robust exception handling should be aware that the techniques we cover are insufficient by themselves to achieve exception safety.**

### Functions Are Exited during the Search for a Handler

In complicated systems, the execution path of a program may pass through multiple `try` blocks before encountering code that throws an exception. For example, a `try` block might call a function that contains a `try`, which calls another function with its own `try`, and so on.

The search for a handler reverses the call chain. When an exception is thrown, the function that threw the exception is searched first. If no matching `catch` is found, that function terminates. The function that called the one that threw is searched next. If no handler is found, that function also exits. That function's caller is searched next, and so on back up the execution path until a `catch` of an appropriate type is found.

If no appropriate `catch` is found, execution is transferred to a library function named **`terminate`**. The behavior of that function is system dependent but is guaranteed to stop further execution of the program.

Exceptions that occur in programs that do not define any `try` blocks are handled in the same manner: After all, if there are no `try` blocks, there can be no handlers. If a program has no `try` blocks and an exception occurs, then `terminate` is called and the program is exited.

## Standard Exceptions

The C++ library defines several classes that it uses to report problems encountered in the functions in the standard library. These exception classes are also intended to be used in the programs we write. These classes are defined in four headers:

- The `exception` header defines the most general kind of exception class named `exception`. It communicates only that an exception occurred but provides no additional information.
- The `stdexcept` header defines several general-purpose exception classes, which are listed in Table 5.1.
- The `new` header defines the `bad_alloc` exception type, which we cover in § 12.3 (p. 536).
- The `type_info` header defines the `bad_cast` exception type, which we cover in § 19.2 (p. 877).

**Table 5.1: Standard Exception Classes Defined in `<stdexcept>`**

| | |
|---|---|
| `exception` | The most general kind of problem. |
| `runtime_error` | Problem that can be detected only at run time. |
| `range_error` | Run-time error: result generated outside the range of values that are meaningful. |
| `overflow_error` | Run-time error: computation that overflowed. |
| `underflow_error` | Run-time error: computation that underflowed. |
| `logic_error` | Error in the logic of the program. |
| `domain_error` | Logic error: argument for which no result exists. |
| `invalid_argument` | Logic error: inappropriate argument. |
| `length_error` | Logic error: attempt to create an object larger than the maximum size for that type. |
| `out_of_range` | Logic error: used a value outside the valid range. |

The library exception classes have only a few operations. We can create, copy, and assign objects of any of the exception types.

We can only default initialize (§ 2.2.1, p. 44) `exception`, `bad_alloc`, and `bad_cast` objects; it is not possible to provide an initializer for objects of these exception types.

The other exception types have the opposite behavior: We can initialize those objects from either a `string` or a C-style string, but we *cannot* default initialize them. When we create objects of any of these other exception types, we must supply an initializer. That initializer is used to provide additional information about the error that occurred.

The exception types define an operation named `what`. That function takes no arguments and returns a `const char*` that points to a C-style character string (§ 3.5.4, p. 128). The purpose of this C-style character string is to provide some sort of textual description of the exception thrown.

The contents of the C-style string that `what` returns depends on the type of the exception object. For the types that take a string initializer, the `what` function returns that string. For the other types, the value of the string that `what` returns varies by compiler.

## EXERCISES SECTION 5.6.2

**Exercise 5.25:** Write a program that reads two integers from the standard input and prints the result of dividing the first number by the second.

**Exercise 5.26:** Revise your program to throw an exception if the second number is zero. Test your program with a zero input to see what happens on your system if you don't `catch` an exception.

**Exercise 5.27:** Revise your program from the previous exercise to use a `try` block to `catch` the exception. The `catch` clause should print a message to the user and ask them to supply a new number and repeat the code inside the `try`.

## Chapter Summary

C++ provides a limited number of statements. Most of these affect the flow of control within a program:

- `while`, `for`, and `do while` statements, which provide iterative execution.
- `if` and `switch`, which provide conditional execution.
- `constexpr if`, which provides for conditional compilation.
- `break`, `continue`, and `goto` which interrupt the flow of control.
- `try` and `catch`, which define a `try` block enclosing a sequence of statements that might throw an exception. The `catch` clause(s) are intended to handle the exception(s) that the enclosed code might throw.
- `throw` expression statements, which exit a block of code, transferring control to an associated `catch` clause.
- `return`, which stops execution of a function. (We'll cover `return` statements in Chapter 6.)

In addition, there are expression statements and declaration statements. An expression statement causes the subject expression to be evaluated. Declarations and definitions of variables were described in Chapter 2.

## Defined Terms

**block** Sequence of zero or more statements enclosed in curly braces. A block is a statement, so it can appear anywhere a statement is expected.

**break statement** Terminates the nearest enclosing loop or `switch` statement. Execution transfers to the first statement following the terminated loop or `switch`.

**case label** Keyword `case` followed by a constant expression (§ 2.4.4, p. 66) and a colon in a `switch` statement. No two `case` labels in the same `switch` statement may have the same value.

**catch clause** The `catch` keyword, an exception declaration in parentheses, and a block of statements. The code inside a `catch` clause does whatever is necessary to handle an exception of the type defined in its exception declaration.

**compound statement** Synonym for block.

**constexpr if statement** An `if` statement in which the condition is checked at compile time. Depending on the value of the condition, the `if` body or the `else` body is discarded and not compiled.

**continue statement** Terminates the current iteration of the nearest enclosing loop. Execution transfers to the loop condition in a `while` or `do`, to the next iteration in a range `for`, or to the expression in the header of a traditional `for` loop.

**dangling else** Colloquial term used to refer to the problem of how to process nested `if` statements in which there are more `if`s than `else`s. In C++, an `else` is always paired with the closest preceding unmatched `if`. Note that curly braces can be used to effectively hide an inner `if` so that the programmer can control which `if` a given `else` should match.

**default label** `case` label that matches any otherwise unmatched value computed in the `switch` expression.

**do while statement** Like a `while`, except that the condition is tested at the end of the loop, not the beginning. The statement inside the `do` is executed at least once.

**exception classes** Set of classes defined by the standard library to be used to represent errors. Table 5.1 (p. 209) lists the general-purpose exception classes.

**exception declaration** The declaration in a `catch` clause. This declaration specifies the type of exceptions the `catch` can handle.

**exception handler** Code that deals with an exception raised in another part of the program. Synonym for `catch` clause.

**exception safe** Term used to describe programs that behave correctly when exceptions are thrown.

**expression statement** An expression followed by a semicolon. Evaluates the expression and discards the result.

**flow of control** Execution path through a program.

**for statement** Iteration statement that provides iterative execution. Ordinarily used to step through a container or to repeat a calculation a given number of times.

**goto statement** Statement that causes an unconditional transfer of control to a specified labeled statement elsewhere in the same function. `goto`s obfuscate the flow of control within a program and should be avoided.

**handler** Synonym for exception handler.

**if else statement** Conditional execution of code following the `if` or the `else`, depending on the truth value of the condition.

**if statement** Conditional execution based on the value of the specified condition. If the condition is `true`, then the `if` body is executed. If not, control flows to the statement following the `if`.

**labeled statement** Statement preceded by a label. A label is an identifier followed by a colon. Label identifiers are independent of other uses of the same identifier.

**null statement** An empty statement. Indicated by a single semicolon.

**raise** Often used as a synonym for throw. C++ programmers speak of "throwing" or "raising" an exception interchangeably.

**range for statement** Statement that iterates through a sequence.

**switch statement** A conditional statement that starts by evaluating the expression that follows the `switch` keyword. Control passes to the labeled statement with a `case` label that matches the value of the expression. If there is no matching label, execution either continues at the `default` label, if there is one, or falls out of the `switch` if there is no `default` label.

**terminate** Library function that is called if an exception is not caught. `terminate` aborts the program.

**throw expression** Expression that interrupts the current execution path. Each `throw` throws an object and transfers control to the nearest enclosing `catch` clause that can handle the type of exception that is thrown.

**try block** Block enclosed by the keyword `try` and one or more `catch` clauses. If the code inside a `try` block raises an exception and one of the `catch` clauses matches the type of the exception, then the exception is handled by that `catch`. Otherwise, the exception is handled by an enclosing `try` block or the program terminates.

**while statement** Iteration statement that executes its target statement as long as a specified condition is `true`. The statement is executed zero or more times, depending on the truth value of the condition.

# CHAPTER 6

## FUNCTIONS

### CONTENTS



This chapter describes how to define and declare functions. We'll cover how arguments are passed to and values returned from functions. In C++, functions can be overloaded, which means that we can use the same name for several different functions. We'll cover both how to overload functions and how the compiler selects the matching version for a particular call from several overloaded functions.

*A function* is a block of code with a name. We execute the code by calling the function. A function may take zero or more arguments and (usually) yields a result. Functions can be overloaded, meaning that the same name may refer to several different functions.

## 6.1 Function Basics

A ***function*** definition typically consists of a ***return type***, a name, a list of zero or more ***parameters***, and a body. The parameters are specified in a comma-separated list enclosed in parentheses. The actions that the function performs are specified in a block (§ 5.1, p. 183), referred to as the ***function body***.

We execute a function through the **call operator**, which is a pair of parentheses. The call operator takes an expression that is a function or points to a function. Inside the parentheses is a comma-separated list of ***arguments***, which are used to initialize the function's parameters. The type of a call expression is the return type of the function.

### Writing a Function

As an example, we'll write a function to determine the factorial of a given number. The factorial of a number `n` is the product of the numbers from 1 through `n`. The factorial of 5, for example, is 120.

```
1 * 2 * 3 * 4 * 5 = 120
```

We might define this function as follows:

```
// factorial of val is val * (val - 1) * (val - 2) . . . * ((val - (val - 1)) * 1)
int fact(int val)
{
    int ret = 1; // local variable to hold the result as we calculate it
    while (val > 1)
        ret *= val--;  // assign ret * val to ret and decrement val
    return ret;        // return the result
}
```

Our function is named `fact`. It takes one `int` parameter and returns an `int` value. Inside the `while` loop, we compute the factorial using the postfix decrement operator (§ 4.5, p. 156) to reduce the value of `val` by 1 on each iteration. The `return` statement ends execution of `fact` and returns the value of `ret`.

### Calling a Function

To call `fact`, we must supply an `int` value. The result of the call is also an `int`:

```
int main()
{
    int j = fact(5);  // j equals 120, i.e., the result of fact(5)
    cout << "5! is " << j << endl;
    return 0;
}
```

A function call does two things: It initializes the function's parameters from the corresponding arguments, and it transfers control to that function. Execution of the *calling* function is suspended and execution of the *called* function begins.

Execution of a function begins with the (implicit) definition and initialization of its parameters. Thus, when we call `fact`, the first thing that happens is that an `int` variable named `val` is created. This variable is initialized by the argument in the call to `fact`, which in this case is `5`.

Execution of a function ends when a `return` statement is encountered. Like a function call, the `return` statement does two things: It returns the value (if any) in the `return`, and it transfers control out of the *called* function back to the *calling* function. The value in the `return` statement is the result of the call expression. Execution continues with whatever remains of the expression in which the call appeared. Thus, our call to `fact` is equivalent to the following:

```
int val = 5;       // initialize val from the argument 5
int ret = 1;       // code from the body of fact
while (val > 1)
    ret *= val--;
int j = ret;       // use ret to initialize j
```

## Parameters and Arguments

Arguments are the initializers for a function's parameters. The first argument initializes the first parameter, the second argument initializes the second parameter, and so on. Although we know which argument initializes which parameter, we have no guarantees about the order in which arguments are evaluated (§ 4.1.3, p. 146): The compiler is free to evaluate the arguments in whatever order it prefers.

The type of each argument must match the corresponding parameter in the same way that the type of any initializer must match the type of the object it initializes. We must pass exactly the same number of arguments as the function has parameters. Because every call is guaranteed to pass as many arguments as the function has parameters, parameters are always initialized.

Our `fact` function has one parameter of type `int`; every time we call it we must supply a single argument that can be converted (§ 4.11, p. 169) to `int`:

```
fact("hello");        // error: wrong argument type
fact();               // error: too few arguments
fact(42, 10, 0);      // error: too many arguments
fact(3.14);           // ok: argument is converted to int
```

The first call fails because there is no conversion from `const char*` to `int`. The second and third calls pass the wrong number of arguments. The `fact` function must be called with one argument; it is an error to call it with any other number. The last call is legal because there is a conversion from `double` to `int`. In this call, the argument is implicitly converted to `int` (through truncation). After the conversion, this call is equivalent to

```
fact(3);
```

### Function Parameter List

A function's parameter list can be empty but cannot be omitted. Typically we define a function with no parameters by writing an empty parameter list. For compatibility with C, we also can use the keyword `void` to indicate that there are no parameters:

```
void f1(){ /* ... */ }      // implicit void parameter list
void f2(void){ /* ... */ } // explicit void parameter list
```

A parameter list typically consists of a comma-separated list of parameters, each of which looks like a declaration with a single declarator. Even when the types of two parameters are the same, the type must be repeated:

```
int f3(int v1, v2) { /* ... */ }      // error
int f4(int v1, int v2) { /* ... */ } // ok
```

No two parameters can have the same name. Moreover, local variables at the outermost scope of the function may not use the same name as any parameter.

Parameter names are optional. However, there is no way to use an unnamed parameter. Therefore, parameters ordinarily have names. Occasionally a function has a parameter that is not used. Such parameters are often left unnamed, to indicate that they aren't used. Leaving a parameter unnamed doesn't change the number of arguments that a call must supply: A call must supply an argument for every parameter, even if that parameter isn't used.

### Function Return Type

Most types, including references, can be used as the return type of a function. In particular, the return type can be `auto`, which we'll cover in § 6.3.3 (p. 233). A function can also return type `void`, which means that the function does not return a value. However, the return type may not be an array type (§ 3.5, p. 119) or a function type. However, a function may return a pointer to an array or a function. We'll see how to define functions that return pointers (or references) to arrays in § 6.6.3 (p. 259) and how to return pointers to functions in § 6.7 (p. 261).

## 6.1.1 Local Objects

In C++, names have scope (§ 2.2.4, p. 48), and objects have **lifetimes**. It is important to understand both of these concepts.

- Scope is *the part of the program's text* in which that name is visible.
- Lifetime is *the time during the program's execution* that the object exists.

As we've seen, the body of a function is a statement block. As usual, the block forms a new scope in which we can define variables. Parameters and variables defined inside a function body are referred to as **local variables**. They are "local" to that function and **hide** declarations of the same name made in an outer scope.

## EXERCISES SECTION 6.1

**Exercise 6.1:** What is the difference between a parameter and an argument?

**Exercise 6.2:** Indicate which of the following functions are in error and why. Suggest how you might correct the problems.

(a) `int f() { string s; // ... return s; }`
(b) `f2(int i) { /* ... */ }`
(c) `int calc(int v1, int v1) /* ... */ }`
(d) `double square(double x) return x * x;`

**Exercise 6.3:** Write and test your own version of `fact`.

**Exercise 6.4:** Write a function that interacts with the user, asking for a number and generating the factorial of that number. Call this function from `main`.

**Exercise 6.5:** Write a function to return the absolute value of its argument.

Objects defined outside any function are in scope once their declaration is seen. Such objects are known as **global objects** or more colloquially, globals.

Global objects exist throughout the program's execution: They are created when the program starts and are destroyed when the program ends. In contrast, the lifetime of a local variable depends on how it is defined.

### Automatic Objects

Objects defined as ordinary local variables are created when the execution path passes through the variable's definition and are destroyed when control passes through the end of the block in which the variable is defined. Objects that exist only while a block is executing are known as **automatic objects**. After execution exits a block, those objects no longer exist.

Parameters are automatic objects. Storage for the parameters is allocated when the function begins. Parameters are defined in the scope of the function body. Hence they are destroyed when the function terminates.

Automatic objects corresponding to the function's parameters are initialized by the arguments passed to the function. Automatic objects corresponding to local variables are initialized if their definition contains an initializer. Otherwise, they are default initialized (§ 2.2.1, p. 44), which means that uninitialized local variables of built-in type have undefined values.

### Local `static` Objects

It can be useful to have a local variable whose lifetime continues across calls to the function. We obtain such objects by defining a local variable as `static`. Each **local `static` object** is initialized before the first time execution passes through the object's definition. Local `static`s are not destroyed when a function ends; they are destroyed when the program terminates.

As a trivial example, here is a function that counts how many times it is called:

```
size_t count_calls()
{
    static size_t ctr = 0;  // value will persist across calls
    return ++ctr;
}
int main()
{
    for (size_t i = 0; i != 10; ++i)
        cout << count_calls() << endl;
    return 0;
}
```

This program will print the numbers from 1 through 10 inclusive.

Before control flows through the definition of `ctr` for the first time, `ctr` is created and given an initial value of `0`. Each call increments `ctr` and returns its new value. Whenever `count_calls` is executed, the variable `ctr` already exists and has whatever value was in that variable the last time the function exited. Thus, on the second invocation, the value of `ctr` is `1`, on the third it is `2`, and so on.

If a local `static` has no explicit initializer, it is value initialized (§ 3.3.1, p. 101), meaning that local `static`s of built-in type are initialized to zero.

**EXERCISES SECTION 6.1.1**

**Exercise 6.6:** Explain the differences between a parameter, a local variable, and a local `static` variable. Give an example of a function in which each might be useful.

**Exercise 6.7:** Write a function that returns 0 when it is first called and then generates numbers in sequence each time it is called again.

### 6.1.2 Function Declarations

Like any other name, the name of a function must be declared before we can use it. As with variables (§ 2.2.2, p. 45), a function may be defined only once but may be declared multiple times. With one exception that we'll cover in § 15.3 (p. 658), we can declare a function that is not defined so long as we never use that function.

A function declaration is just like a function definition except that it has no function body. In a declaration, a semicolon follows the parameter list.

Because a function declaration has no body, there is no need for parameter names. Hence, parameter names are often omitted in a declaration. Although parameter names are not required, they can be used to help users of the function understand what the function does:

```
// parameter names chosen to indicate that the iterators denote a range of values to print
void print(vector<int>::const_iterator beg,
           vector<int>::const_iterator end);
```

These three elements—the return type, function name, and parameter types—describe the function's interface. They specify all the information we need to call the function. Function declarations are also known as the **function prototype**.

### Function Declarations Go in Header Files

Recall that variables are declared in header files (§ 2.6.3, p. 78) and defined in source files. For the same reasons, functions should be declared in header files and defined in source files.

It may be tempting—and would be legal—to put a function declaration directly in each source file that uses the function. However, doing so is tedious and error-prone. When we use header files for our function declarations, we can ensure that all the declarations for a given function agree. Moreover, if the interface to the function changes, only one declaration has to be changed.

The source file that defines a function should include the header that contains that function's declaration. That way the compiler will verify that the definition and declaration are consistent.

Best Practices: The header that *declares* a function should be included in the source file that *defines* that function.

#### EXERCISES SECTION 6.1.2

**Exercise 6.8:** Write a header file named `Chapter6.h` that contains declarations for the functions you wrote for the exercises in § 6.1 (p. 217).

## 6.1.3 Separate Compilation

As our programs get more complicated, we'll want to store the various parts of the program in separate files. For example, we might store the functions we wrote for the exercises in § 6.1 (p. 217) in one file and store code that uses these functions in other source files.

To allow programs to be written in logical parts, C++ supports what is commonly known as ***separate compilation***. Separate compilation lets us split our programs into several files, each of which can be compiled independently.

### Compiling and Linking Multiple Source Files

As an example, assume that the definition of our `fact` function is in a file named `fact.cc` and its declaration is in a header file named `Chapter6.h`. Our `fact.cc` file, like any file that uses these functions, will include the `Chapter6.h` header. We'll store a `main` function that calls `fact` in a second file named `factMain.cc`. To produce an ***executable file***, we must tell the compiler where to find all of the code we use. We might compile these files as follows:

```
$ CC factMain.cc fact.cc   # generates factMain.exe or a.out
$ CC factMain.cc fact.cc -o main # generates main or main.exe
```

Here `CC` is the name of our compiler, `$` is our system prompt, and `#` begins a command-line comment. We can now run the executable file, which will run our `main` function.

If we have changed only one of our source files, we'd like to recompile only the file that actually changed. Most compilers provide a way to separately compile each file. This process usually yields a file with the `.obj` (Windows) or `.o` (UNIX) file extension, indicating that the file contains ***object code***.

The compiler lets us ***link*** object files together to form an executable. On the system we use, we would separately compile our program as follows:

```
$ CC -c factMain.cc     # generates factMain.o
$ CC -c fact.cc         # generates fact.o
$ CC factMain.o fact.o  # generates factMain.exe or a.out
$ CC factMain.o fact.o -o main # generates main or main.exe
```

You'll need to check with your compiler's user's guide to understand how to compile and execute programs made up of multiple source files.

**EXERCISES SECTION 6.1.3**

**Exercise 6.9:** Write your own versions of the `fact.cc` and `factMain.cc` files. These files should include your `Chapter6.h` from the exercises in the previous section. Use these files to understand how your compiler supports separate compilation.

## 6.2 Argument Passing

As we've seen, each time we call a function, its parameters are created and initialized by the arguments passed in the call.

Note: Parameter initialization works the same way as variable initialization.

As with any other variable, the type of a parameter determines the interaction between the parameter and its argument. If the parameter is a reference (§ 2.3.1, p. 50), then the parameter is bound to its argument. Parameters of a non-reference type are copy-initialized (§ 3.2.1, p. 86) by the argument's value.

When a parameter is a reference, we say that its corresponding argument is **"passed by reference"** or that the function is **"called by reference."** Like any other reference, a reference parameter is an alias for the object to which it is bound; that is, the parameter is just another name for its corresponding argument.

When the argument value is copied, the parameter and argument are independent objects. We say such arguments are **"passed by value"** or alternatively that the function is **"called by value."**

### 6.2.1 Passing Arguments by Value

When we initialize a nonreference type variable, the value of the initializer is copied. Changes made to the variable have no effect on the initializer:

```
int n = 0;             // ordinary variable of type int
int i = n;             // i is a copy of the value in n
i = 42;                // value in i is changed; n is unchanged
```

Passing an argument by value works exactly the same way; nothing the function does to the parameter can affect the argument. For example, inside `fact` (§ 6.1, p. 214) the parameter `val` is decremented:

```
ret *= val--;  // decrements the value of val
```

Although `fact` changes the value of `val`, that change has no effect on the argument passed to `fact`. Calling `fact(i)` does not change the value of `i`.

For the same reasons that `const` doesn't matter when we initialize an object of a plain (nonreference) type (§ 2.4, p. 60), `const` also has no effect on what arguments can be bound to a by-value parameter (aka a plain, nonreference type parameter):

```
void f1(const int);    // parameter has top-level const § 2.4.3 (p. 64)
void f2(int);

int i = 0;
const int ci = i;      // ci has top-level const

f1(i);  // ok: when copying, top-level consts are ignored
f2(ci); // ok: when copying, top-level consts are ignored
```

`const` on a parameter or an argument matters only when that `const` is low-level (§ 2.4.3, p. 64). That is, it matters when a parameter or an argument is a pointer (or reference) to a `const` type:

```
void f3(int * const);         // parameter has top-level const
void f4(const int * const);   // parameter has low-level and top-level const
int *p = &i;
const int *cp = p;     // ok: can convert nonconst int* to const int*
f3(p);  // ok: when copying, top-level consts are ignored
f3(cp); // error: can't convert const int* to plain int*
f4(cp); // ok: low-level consts agree ant top-level const is ignored
```

### 6.2.2 Passing Arguments by Reference

Recall that operations on a reference are actually operations on the object to which the reference refers (§ 2.3.1, p. 50):

```
int n = 0, &r = n; // r is bound to n (i.e., r is another name for n)
r = 42; // changes to the reference change the referred to object; n is now 42
n = 0;  // changes to the referred to object change the reference; r is now 0
```

Reference parameters exploit this behavior. We use references to allow a function to change the value of one or more of its arguments.

As one example, we'll define a function named `pad` that will take three parameters: a reference to a `string`, a size, and a character value. If the `string` is larger than the given size, our function will leave its `string` parameter unchanged. Otherwise, it will pad the `string` with the given character up to the given size:

```
// s is just another name for the object passed to pad
void pad(string &s, string::size_type sz, char c)
{
    auto i = s.size();
    if (i < sz)
        // changes the value of the object to which i refers
        s += string(sz - i, c); // append the sz - i copies of c to s
}
```

Like any other reference, a reference parameter is bound to the object from which it is initialized. When we call `pad`, `s` will be bound to whatever `string` object we pass. Changes `pad` makes to `s` will actually change the object to which `s` is bound. Thus, when we assign to `s` inside `pad`, we're assigning a new value to the object to which `s` is bound:

```
string str("hi");
pad(str, 10, '!');   // str is passed by reference; the value in str is changed
cout << "str = " << str  << endl;  // prints hi!!!!!!!!
```

In this call, `s` is bound to `str`. When `pad` appends characters to `s`, it's actually appending those characters to `str`.

### Using References to Avoid Copies

Another common use of reference parameters is to avoid the overhead involved in copying an argument. In particular, it can be inefficient to copy objects of large class types or large containers. Moreover, some class types (including the IO types) cannot be copied. Functions must use reference parameters to operate on objects of a type that cannot be copied.

As an example, we'll write a function to compare the length of two `string`s. Because `string`s can be long, we'd like to avoid copying them, so we'll make our parameters references. Because comparing two `string`s does not involve changing the `string`s, we'll make the parameters references to `const` (§ 2.4.1, p. 61):

```
// compare the length of two strings
bool isShorter(const string &s1, const string &s2)
{
    return s1.size() < s2.size();
}
```

**Best Practices**

> Reference parameters that are not changed inside a function should be references to `const`.

## Arguments to Reference Parameters

When we call a function that has reference parameters, we must remember the general initialization rules from § 2.4.1 (p. 62): We can initialize a reference to `const` from any expression that can be converted to the base type of the reference but we can initialize a plain reference *only* from an object of the same type. Exactly the same initialization rules apply to parameter passing.

Using the terminology from § 4.1.1 (p. 143) we can now more succinctly note that we can bind a plain reference only to a modifiable lvalue.

```
string s = "hi";
pad(s, 10, '!');        // ok: s is a string lvalue
const string& cs = s; // ok: can bind a const string& to a plain string
pad(cs, 10, '!');       // error: cs is not modifiable
pad("hi", 10, '!');     // error: literals aren't modifiable and isn't type string
pad(s + s, 10, '!');   // error: string addition yields an rvalue
```

Thus, we cannot pass a `const string` object, a literal, an expression that yields an rvalue, or an object that requires conversion (such as a C-style character string (§ 3.5.4, p. 128)) to `pad`. On the other hand, we can pass any of these rvalues to `isShorter` because its parameters are references to `const`:

```
isShorter("this string", "or that one");
```

We can bind a reference to `const` to any expression that can be converted to the base type of the reference itself.

It can be important—and admittedly tricky—to understand exactly what happens in a call such as this one. In particular, although `isShorter` takes its arguments by reference, the arguments in this call are actually copied. The copy happens when the literals are converted to `const string&`. Converting a literal (or other rvalue expression) to a reference to `const` creates a temporary object (§ 2.4.1, p. 62). A reference to that temporary object is passed to the function. For example, in this call to `isShorter`, two `string`s are created to hold the characters of the given literals. The reference parameters in `isShorter` are then bound to those temporary `string`s.

For many, perhaps even most, programs the performance impact of creating a temporary to bind to a reference is of no consequence. However, in the particular case of `string`s, the library provides a companion class—that we'll cover in § 9.5.1 (p. 396)—that can be used to avoid copies in functions such as `isShorter`.

## If You're Not Changing the Argument, Use Reference to `const`

It is a somewhat common mistake to define parameters that a function does not change as (plain) references. Doing so gives the function's caller the misleading impression that the function might change its argument's value. Moreover, using a reference instead of a reference to `const` unduly limits the type of arguments that can be used with the function. As we've just seen, we cannot pass a `const` object, or a literal, or an object that requires conversion to a plain reference parameter.

The effect of this mistake can be surprisingly pervasive. As an example, consider our `isShorter` function from § 6.3.2 (p. 229). That function (correctly) made its `string` parameters references to `const`. Had we defined that parameter as a plain `string&`:

```
// oops! parameters should be const
bool isShorter(string &s1, string &s2)
{
    return s1.size() < s2.size();
}
```

we could call `isShorter` only on `string` objects. A call such as

```
isShorter("this string", "or that one");
```

would fail at compile time.

More subtly, we could not use this version of `isShorter` from other functions that (correctly) define their parameters as references to `const`. For example, we might want to use `isShorter` inside a function that returns the length of the shorter of two given `string`s:

```
string::size_type
shorterLen(const string &s1, const string &s2)
{
    // conditional operator (? :) see § 4.7 (p. 161)
    return isShorter(s1, s2) ?  s1.size() : s2.size();
}
```

If `isShorter` takes plain `string&`, then this call to `isShorter` would be a compile-time error. The problem is that the `string`s in `shorterLen` are references to `const string`, but `isShorter` was (incorrectly) defined to take plain references.

**WARNING**

Using a non`const` reference parameter when an argument is not changed means that the function cannot be called from other functions that correctly made their reference parameters reference to `const`.

### 6.2.3 Default Arguments

Some functions have parameters that are passed a particular value in many, but not all, calls. In such cases, we can declare that common value as a **default argument** for the function. Functions with default arguments can be called with or without that argument.

For example, we might use a `string` to represent the contents of a window. By default, we might want the window to have a particular height, width, and background character. However, we might also want to allow users to pass values other than the defaults. To accommodate both default and specified values we would declare our function to define the window as follows:

```
using sz = string::size_type sz; // type alias see § 2.5.1 (p. 67)
string screen(sz ht = 24, sz wid = 80, char backgrnd = ' ');
```

## Exercises Section 6.2.2

**Exercise 6.10:** Write and test your own versions of `pad` and `isShorter`. Experiment with passing differing expressions as arguments to the two functions.

**Exercise 6.11:** Assuming `T` is the name of a type, explain the difference among functions declared as `void f(T)`, `void f(T&)`, and `void f(const T&)`.

**Exercise 6.12:** The following function, although legal, is less useful than it might be. Identify and correct the limitation on this function:

```
bool is_empty(string& s) { return s.empty(); }
```

**Exercise 6.13:** Write a function to determine whether a `string` contains any capital letters. Write a function to change a `string` to all lowercase. Do the parameters you used in these functions have the same type? If so, why? If not, why not?

**Exercise 6.14:** Give an example of when a parameter should be a reference type. Give an example of when a parameter should not be a reference.

**Exercise 6.15:** When should reference parameters be references to `const`? What happens if we make a parameter a plain reference when it could be a reference to `const`?

Here we've provided a default for each parameter. A default argument is specified as an initializer for a parameter in the parameter list. We may define defaults for one or more parameters. However, if a parameter has a default argument, all the parameters that follow it must also have default arguments.

### Calling Functions with Default Arguments

If we want to use the default argument, we omit that argument when we call the function. Because `screen` provides defaults for all of its parameters, we can call `screen` with zero, one, two, or three arguments:

```
string window;
window = screen();  // equivalent to screen(24,80,' ')
window = screen(66);// equivalent to screen(66,80,' ')
window = screen(66, 256);      // screen(66,256,' ')
window = screen(66, 256, '#'); // screen(66,256,'#')
```

Arguments in the call are resolved by position. The default arguments are used for the trailing (right-most) arguments of a call. For example, to override the default for `backgrnd`, we must also supply arguments for `ht` and `wid`:

```
window = screen(, , '?'); // error: can omit only trailing arguments
window = screen('?');     // calls screen('?',80,' ')
```

Note that the second call, which passes a single character value, is legal. Although legal, it is unlikely to be what was intended. The call is legal because `'?'` is a `char`, and a `char` can be converted (§ 4.11.1, p. 170) to the type of the left-most parameter. That parameter is `string::size_type`, which is an `unsigned` integral type. Our `char` argument is implicitly converted to `string::size_type`, and is passed as the argument to `height`. On our machine, `'?'` has the hexadecimal value `0x3F`, which is decimal `63`. Thus, this call passes `63` to the `height` parameter.

Part of the work of designing a function with default arguments is ordering the parameters so that those least likely to use a default value appear first and those most likely to use a default appear last.

### Default Argument Declarations

Although it is normal practice to declare a function once inside a header, it is legal to redeclare a function multiple times. However, each parameter can have its default specified only once in a given scope. Thus, any subsequent declaration can add a default only for a parameter that has not previously had a default specified. As usual, defaults can be specified only if all parameters to the right already have defaults. For example, given

```
// no default for the height or width parameters
string screen(sz, sz, char = ' ');
```

we cannot change an already declared default value:

```
string screen(sz, sz, char = '*'); // error: redeclaration
```

but we can add a default argument as follows:

```
string screen(sz = 24, sz = 80, char);  // ok: adds default arguments
```

Best Practices

Default arguments ordinarily should be specified with the function declaration in an appropriate header.

### Default Argument Initializers

Local variables may not be used as a default argument. Excepting that restriction, a default argument can be any expression that has a type that is convertible to the type of the parameter:

```
// the declarations of wd, def, and ht must appear outside a function
sz wd = 80;
char def = ' ';
sz ht();
string screen(sz = ht(), sz = wd, char = def);
string window = screen(); // calls screen(ht(), 80, ' ')
```

Names used as default arguments are resolved in the scope of the function declaration. The value that those names represent is evaluated at the time of the call:

```
void f2()
{
    def = '*';    // changes the value of a default argument
    sz wd = 100; // hides the outer definition of wd but does not change the default
    window = screen(); // calls screen(ht(), 80, '*')
}
```

Inside `f2`, we change the value of the global variable `def`. Our function also declares a local variable named `wd`. That variable hides the outer `wd`. Our local variable named `wd` is unrelated to, and has no effect on, the default argument passed to `screen`. Thus, the call to `screen` passes the updated value of `def` and the unchanged value of `wd`.

**EXERCISES SECTION 6.2.3**

**Exercise 6.16:** Which, if either, of the following declarations are errors? Why?

(a) `int ff(int a, int b = 0, int c = 0);`
(b) `char *init(int ht = 24, int wd, char bckgrnd);`

**Exercise 6.17:** Given the following declaration for `init`, which if any, of the following calls are illegal? Why?

```
char *init(int ht, int wd = 80, char bckgrnd = ' ');
```

(a) `init();`   (b) `init(24,10);`   (c) `init(14, '*');`

**Exercise 6.18:** For the legal calls in the previous exercise, which, if any, are unlikely to match the programmer's intent? Why?

**Exercise 6.19:** Write a version of `pad` (§ 6.2.2, p. 222) that takes a space as the default for its `char` parameter.

## 6.3 Return Types and the `return` Statement

A `return` statement terminates the function that is currently executing and returns control to the point from which the function was called. There are two forms of `return` statements:

```
return;
return expression;
```

### 6.3.1 Functions with No Return Value

A `return` with no value is used in functions that have a return type of `void`. Functions that return `void` are not required to contain a `return` statement at all. In a `void` function, an implicit `return` takes place after the function's last statement.

Typically, `void` functions use a `return` to exit the function at an intermediate point. This use of `return` is analogous to the use of a `break` statement (§ 5.5.1, p. 202) to exit a loop. As an example, we'll write a function that takes two `strings` and conditionally appends the second `string` to the first, separated by a comma. If the first `string` is empty, we don't want to append, but instead want to replace the contents of the first `string` by the second:

```
void
listify(string &s1, const string &s2)
{
    if (s1.empty()) {
        s1 = s2;
        return;
    }
    // if we're here we may need to append s2 to s1
    if (!s2.empty())
        s1 += ", " + s2;
    // no explicit return needed
}
```

This function checks whether its first parameter is `empty`. If so, it replaces that empty string by the `string` in its second parameter and exits. If the first `string` isn't `empty`, the function appends a comma, a space, and the second `string`. An implicit return occurs after the second `if` statement.

A function with a `void` return type may use the second form of the `return` statement only to return the result of calling another function that returns `void`. Returning any other expression from a `void` function is a compile-time error.

### 6.3.2 Functions That Return a Value

The second form of the `return` statement provides the function's result. Every return in a function with a return type other than `void` must return a value. The value returned must have the same type as the function's return type, or it must have a type that can be implicitly converted (§ 4.11, p. 169) to that type.

Although C++ cannot guarantee the correctness of a result, it can guarantee that every `return` includes a result of the appropriate type. The compiler also attempts to ensure that functions that return a value are exited only through a valid `return` statement. However, the compiler cannot detect all such errors:

```
// incorrect return values, this code will not compile
bool str_subrange(const string &str1, const string &str2)
{
    // same sizes: return normal equality test
    if (str1.size() == str2.size())
        return str1 == str2;   // ok: == returns bool
    // find the size of the smaller string; conditional operator, see § 4.7 (p. 161)
    auto size = (str1.size() < str2.size())
                ? str1.size() : str2.size();
    // look at each element up to the size of the smaller string
    for (decltype(size) i = 0; i != size; ++i) {
        if (str1[i] != str2[i])
            return; // error #1: no return value; compiler should detect this error
    }
    // error #2: control might flow off the end of the function without a return
    // the compiler might not detect this error
}
```

The `return` from within the `for` loop is an error because it fails to return a value. The compiler should detect this error.

The second error occurs because the function fails to provide a `return` after the loop. If we call this function with one `string` that is a subset of the other, execution would fall out of the `for`. There should be a return to handle this case. The compiler may or may not detect this error. If it does not detect the error, what happens at run time is undefined.

WARNING

Failing to provide a `return` after a loop that contains a `return` is an error. However, many compilers will not detect such errors.

## How Values Are Returned

Values are returned in exactly the same way as variables and parameters are initialized: The value in the `return` initializes the result of the function call operator. As with parameter types, if the return type is not a reference type, the result is copy-initialized by the returned value; if the return type is a reference the result is bound to the object that is returned.

Functions often return local variables. As an example, we might write a function that, given a counter, a word, and an ending, gives us back the plural version of the word if the counter is greater than 1:

```
// return the plural version of word if ctr is greater than 1
string make_plural(size_t ctr, const string &word,
                               const string &ending)
{
    return (ctr > 1) ? word + ending : word;
}
```

The return type of this function is `string`, which means the result of the call will be a copy of the value that is returned. In this case, the result will be either a copy of the local object `word`, or of the result of adding `word` and `ending`.

As with any other reference, when a function returns a reference, that reference is just another name for the object to which it refers. As an example, consider a function that returns a reference to the shorter of its two `string` parameters:

```
// return a reference to the shorter of two strings
const string &shorterString(const string &s1, const string &s2)
{
    return s1.size() <= s2.size() ? s1 : s2;
}
```

The parameters and return type are references to `const string`. The `strings` are not copied when the function is called or when the result is returned.

## Never Return a Reference or Pointer to a Local Object

When a function completes, its storage is freed (§ 6.1.1, p. 216). After a function terminates, references to local objects refer to memory that is no longer valid:

```
// disaster: this function returns a reference to a local object
const string &manip()
{
    string ret;
    // transform ret in some way
    if (!ret.empty())
        return ret;      // WRONG: returning a reference to a local object!
    else
        return "Empty"; // WRONG: "Empty" is a local temporary string
}
```

Both of these `return` statements return an undefined value—what happens if we try to use the value returned from `manip` is undefined. In the first `return`, it should be obvious that the function returns a reference to a local object. In the second case, the string literal is converted to a local temporary `string` object. That object, like the `string` named `ret`, is local to `manip`. The storage in which the temporary resides is freed when the function ends. Both `return`s refer to memory that is no longer available.

**Tip:** One good way to ensure that the return is safe is to ask: To what *preexisting* object is the reference referring?

For the same reasons that it is wrong to return a reference to a local object, it is also wrong to return a pointer to a local object. Once the function completes, the local objects are freed. The pointer would point to a nonexistent object.

## Functions That Return Class Types and the Call Operator

Like any other operator, the call operator has associativity, precedence, and order of evaluation (§ 4.1.2, p. 144). The call operator has the same precedence as the dot and arrow operators (§ 4.6.2, p. 160). Like those operators, the call operator is left associative. When a function returns a pointer, reference, or object of class type, we can use the result of a function call to call a member of the resulting object.

For example, we can determine the size of the shorter `string` as follows:

```
// call the size member of the string returned by shorterString
auto sz = shorterString(s1, s2).size();
auto sz = (shorterString(s1, s2)).size(); // equivalent declaration
```

Because these operators are left associative, the result of `shorterString` is the left-hand operand of the dot operator. That operator fetches the `size` member of that `string`. That member is the left-hand operand of the second call operator.

## Reference Returns Are Lvalues

Whether a function call is an lvalue (§ 4.1.1, p. 143) depends on the return type of the function. Calls to functions that return references are lvalues; other return types yield rvalues. A call to a function that returns a reference can be used in the same ways as any other lvalue. In particular, we can assign to the result of a function that returns a reference to non`const`:

```
char &get_val(string &str, string::size_type ix)
{
    return str[ix]; // get_val assumes the given index is valid
}
int main()
{
    string s("a value");
    cout << s << endl;   // prints a value
    get_val(s, 0) = 'A'; // changes s[0] to A
    cout << s << endl;   // prints A value
    return 0;
}
```

It may be surprising to see a function call on the left-hand side of an assignment. However, nothing special is involved: The return value is a reference, so the call is an lvalue. Like any other lvalue, it may appear as the left-hand operand of the assignment operator.

If the return type is a reference to `const`, then (as usual) we may not assign to the result of the call:

```
shorterString("hi", "bye") = "X"; // error: return value is const
```

## List Initializing the Return Value

A function can return a braced list of values. As in any other return, the list is used to initialize the result of the function call. If the list is empty, the result is value initialized (§ 3.3.1, p. 101). Otherwise, the value of the return depends on the function's return type.

As an example, we might write a function that returns a `vector` of `strings` that describes the results of a particular process:

```
vector<string> process()
{
    // . . .
    // expected and actual are strings
    if (expected.empty())
        return {};  // return an empty vector
    else if (expected == actual)
        return {"functionX", "okay"}; // return list-initialized vector
    else
        return {"functionX", expected, actual};
}
```

In the first return statement, we return an empty list. In this case, the `vector` that `process` returns will be empty. Otherwise, we return a `vector` initialized with two or three elements depending on whether `expected` and `actual` are equal.

In a function that returns a built-in type, a braced list may contain at most one value, and that value must not require a narrowing conversion (§ 2.2.1, p. 43). If the function returns a class type, then the class itself defines how the intiailizers are used (§ 3.3.1, p. 102).

## Recursion

A function that calls itself, either directly or indirectly, is a ***recursive function***. As an example, we can rewrite our factorial function to use recursion:

```
int factorial(int val)
{
    if (val > 1)
        return factorial(val-1) * val;
    return 1;
}
```

In this implementation, we recursively call `factorial` to compute the factorial of the numbers counting down from the original value in `val`. Once we have reduced `val` to `1`, we stop the recursion by returning `1`.

There must always be a path through a recursive function that does not involve a recursive call; otherwise, the function will recurse "forever," meaning that the function will continue to call itself until the program stack is exhausted. Such functions are sometimes described as containing a **recursion loop**. In the case of `factorial`, the stopping condition occurs when `val` is `1`.

The following table traces the execution of `factorial(5)`.

**Trace of `factorial(5)`**

| Call | Returns | Value |
|---|---|---|
| `factorial(5)` | `factorial(4) * 5` | `120` |
| `factorial(4)` | `factorial(3) * 4` | `24` |
| `factorial(3)` | `factorial(2) * 3` | `6` |
| `factorial(2)` | `factorial(1) * 2` | `2` |
| `factorial(1)` | `1` | `1` |

Note: The `main` function may *not* call itself.

## Return from `main`

There is one exception to the rule that a function with a non`void` return type must return a value: `main` is allowed to terminate without a return. If control reaches the end of `main`, the compiler implicitly inserts a return of 0.

`main`'s return is used as a status indicator (§ 1.1, p. 2): Zero indicates success; other values indicate failure. Nonzero returns have machine-dependent meaning. To make return values machine independent, the `cstdlib` header defines two preprocessor variables (§ 2.3.2, p. 54) that we can use to indicate success or failure:

```
int main()
{
    if (some_failure)
        return EXIT_FAILURE;  // defined in cstdlib
```

```
    else
        return EXIT_SUCCESS;   // defined in cstdlib
}
```

Because these are preprocessor variables, we must not precede them with `std::`, nor may we mention them in `using` declarations.

**EXERCISES SECTION 6.3.2**

**Exercise 6.20:** Compile the version of `str_subrange` as presented on page 228 to see what your compiler does with the indicated errors.

**Exercise 6.21:** Write and test your own version of `make_plural`, but use a default argument for the second parameter.

**Exercise 6.22:** When is it valid to return a reference? A reference to `const`?

**Exercise 6.23:** Write and test your own version of `get_val` from page 231. What would happen if you declare the first parameter ss `const string&`? Try it and see if your prediction was correct.

**Exercise 6.24:** Write a recursive function to print the contents of a `vector`.

**Exercise 6.25:** What would happen if the stopping condition in `factorial` were

```
if (val != 0)
```

**Exercise 6.26:** In the call to `factorial`, why did we pass `val - 1` rather than `val--`?

## 6.3.3 `auto` Return Type

There are two different ways to use `auto` as the return type for a function: Since C++11, we can use `auto` to signal that the return type appears after the parameter list. Since C++14 we can also `auto` to let the compiler infer the return type based on the value(s) the function returns.

Using `auto` in either of these ways is of limited utility for functions with simple return types, such as those we have written so far. However, as we'll see in § 6.6.3 (p. 260), trailing returns can simplify writing functions with return types such as pointers (or references) to arrays. Moreover, as we'll see in § 16.2.3 (p. 739) and § 16.2.2 (p. 737) `auto` returns are quite useful in the context of templates.

### Trailing Return Type

A **trailing return type** uses `auto` to signal that return type follows the parameter list. Functions with trailing returns have the following form:

```
auto f(parameters) -> return-type
```

which declares `f` as a function with the given *parameters* that returns *return-type*. Unlike other uses of `auto`, this use of `auto` does not involve type deduction. Instead, `auto` signals that the return type *follows* the parameter list rather than preceding it.

This form of declaration is particularly useful when the return type depends in some way on the parameter type(s). As a trivial example:

```
auto scale(int i, double d) -> decltype(i * d);
```

This declaration says that `scale` returns the same type as is returned by multiplying an `int` and a `double`.

### Return Type Deduction

New Starting with C++14 we can use `auto` as an ordinary return type specifier. That is, we can use `auto` without a trailing return. As when we use `auto` to define a variable, the compiler will deduce the return type. When we define a variable, the compiler uses the type of the initializer to deduce the type of the variable. For a function, the value(s) in the `return` statement(s) determine the return type:

```
auto find_char(const string& s, char c)
{
    for (string::size_type i = 0; i != s.size(); ++i)
        if (s[i] == c)
            return i;
    return s.size();
}
```

This function returns the index of the first appearance of a given character in a given `string`. We know the return type, `string::size_type` but have used `auto` to avoid having to write the type out.

Every `return` in a function that uses type deduction must return exactly the same type. For example, this version of `find_char` will not compile:

```
auto find_char(const string& s, char c)
{
    // sloppy practice using int when should use string::size_type
    for (int i = 0; i != s.size(); ++i)
        if (s[i] == c)
            return i; // returns an int
    return s.size();   // error: return type differs from previously seen return
}
```

The problem is that this version used `int` as the index type in the `for` loop. Thus, the first `return` statement deduces `int` as the return type for `find_char`. The second `return` returns `s.size()`, which is a `string::size_type`. We don't know the exact type of `string::size_type`, but we do know it is an unsigned type, which means it is not `int`. The types deduced for these two return statements differ; the function is in error. Even though there is a conversion between `int` and `string::size_type`, the compiler will still reject this program.

Unlike ordinary non-void functions, an `auto` function may use a `return` that does not return a value. In this case, the function is deduced to have `void` as its return type. Similarly, like a `void` function, an `auto` function may omit a `return` statement altogether. In this case, the compiler will also deduce `void` as the function's return type:

```
auto f1() { /* empty body */ } // equivalent to void() f1()
auto f2() { return; }          // equivalent to void() f2()
```

### Separate Compilation and `auto` Deduced Functions

With ordinary functions, we typically put declarations in header files and definitions in a source file (§ 6.1.3, p. 219). This approach does not work for `auto`-deduced functions. In order to deduce the type returned by a function, the compiler must be able to see the definition of the function, not just its declaration.

In practice, this restriction rarely matters because `auto` deduction for functions is most useful in the context of templates. As we'll see in § 16.1.1 (p. 712), template compilation requires that the compiler have access a template's definition, not just its declaration. Therefore, the requirement that an auto-deduced function's definition be available at compile time is no additional burden.

Moreover, we'll see in § 6.5.1 (p. 246) that it is possible to make the definition of an ordinary (non-template) function available in multiple source files.

### Recursion and Deduced Return Type

We can use `auto` with a recursive function so long as a return statement with an actual type precedes the first recursive call. For example, our `factorial` function as originally written could not use `auto`

```
auto factorial(int val)
{
    if (val > 1)
        return factorial(val-1) * val; // error: can't deduce return type
    return 1;
}
```

However, we can easily rewrite `factorial` by swapping the `return` statements:

```
auto factorial(int val)
{
    if (val == 1)
        return 1;  // ok: compiler can deduce return type as int
    else
        return factorial(val-1) * val;
}
```

We still stop the recursion when `val` is `1`, but now the `return` that handles this case comes first. That `return` establishes the type of `factorial` so that when the compiler sees the recursive call it knows what type `factorial` returns.

## 6.4 Overloaded Functions

Functions that have the same name but different parameter lists and that appear in the same scope are **overloaded**. Typically we use overloading to define a set of

**EXERCISES SECTION 6.3.3**

**Exercise 6.27:** Rewrite the functions from the exercises in § 6.1 (p. 217) using `auto` for the return type.

**Exercise 6.28:** What if any changes are needed in `Chapter6.h` that you wrote for exercise 6.8 in § 6.1.2 (p. 219)? Make any changes you think needed and recompile your test code for these functions.

functions that perform the same general action but apply to different parameter types. When we call these functions, the compiler deduces which function we want based on the argument type we pass.

Function overloading eliminates the need to invent—and remember—names that exist only to help the compiler figure out which function to call.

Note: The `main` function may *not* be overloaded.

## 6.4.1 Defining Overloaded Functions

A database application might have several functions to find a record based on different criteria, such as name, or phone number, or account number, and so on. Function overloading lets us define a collection of functions, each named `lookup`, that differ in terms of how they do the search. We can call `lookup` passing a value of any of several types:

```
Record lookup(const Account&);  // find by Account
Record lookup(const Phone&);    // find by Phone
Record lookup(const Name&);     // find by Name
Account acct;
Phone phone;
Record r1 = lookup(acct);  // call version that takes an Account
Record r2 = lookup(phone); // call version that takes a Phone
```

Here, all three functions share the same name, yet they are three distinct functions. The compiler uses the argument type(s) to figure out which function to call.

Overloaded functions must differ in the number or the type(s) of their parameters. Each of the functions above takes a single parameter, but the parameters have different types.

It is an error for two functions to differ only in terms of their return types. If the parameter lists of two functions match but the return types differ, then the second declaration is an error:

```
Record lookup(const Account&);
bool lookup(const Account&);    // error: only the return type is different
```

## Determining Whether Two Parameter Types Differ

Two parameter lists can be identical, even if they don't look the same:

```
// each pair declares the same function
Record lookup(const Account &acct);
Record lookup(const Account&); // parameter names are ignored

using Telno = Phone;
Record lookup(const Phone&);   // Telno is a synonym for Phone
Record lookup(const Telno&);   // Telno and Phone are the same type
```

In the first pair, the first declaration names its parameter. Parameter names are only a documentation aid. They do not change the parameter list.

In the second pair, it looks like the types are different, but `Telno` is not a new type; it is a synonym for `Phone`. A type alias (§ 2.5.1, p. 68) provides an alternative name for an existing type; it does not create a new type. Therefore, two parameters that differ only in that one uses an alias and the other uses the type to which the alias corresponds are not different.

### ADVICE: WHEN NOT TO OVERLOAD A FUNCTION NAME

**Although overloading lets us avoid having to invent (and remember) names for common operations, we should only overload operations that actually do similar things. There are some cases where providing different function names adds information that makes the program easier to understand. Consider a set of functions that move the cursor on a `Screen`.**

```
Screen& moveHome();
Screen& moveAbs(int, int);
Screen& moveRel(int, int, string direction);
```

**It might at first seem better to overload this set of functions under the name `move`:**

```
Screen& move();
Screen& move(int, int);
Screen& move(int, int, string direction);
```

**However, by overloading these functions, we've lost information that was inherent in the function names. Even though each of these functions moves the cursor, the specific nature of that movement is unique to each function. Whether to overload these functions depends on which of these two calls is easier to understand:**

```
// which is easier to understand?
myScreen.moveHome(); // we think this one!
myScreen.move();
```

## Overloading and `const` Parameters

As we saw in § 6.2.1 (p. 221), when a parameter has a plain, nonreference type, top-level `const` (§ 2.4.3, p. 64) has no effect on the types of arguments that can be passed to that parameter. Thus, a parameter that has a top-level `const` is indistinguishable from one without a top-level `const`:

```
Record lookup(Phone);
Record lookup(const Phone);   // redeclares Record lookup(Phone)
Record lookup(Phone*);
Record lookup(Phone* const); // redeclares Record lookup(Phone*)
```

In these declarations, the second declaration declares the same function as the first.

On the other hand, we can overload based on whether the parameter is a reference (or pointer) to the `const` or non`const` version of a given type; such `const`s are low-level:

```
// functions taking const and nonconst references or pointers have different parameters
// declarations for four independent, overloaded functions
Record lookup(Account&);       // function that takes a reference to Account
Record lookup(const Account&); // new function that takes a const reference
Record lookup(Account*);       // new function, takes a pointer to Account
Record lookup(const Account*); // new function, takes a pointer to const
```

In these cases, the compiler can use the `const`ness of the argument to distinguish which function to call. Because there is no conversion (§ 4.11.2, p. 172) *from* `const`, we can pass a `const` object (or a pointer to `const`) only to the version with a `const` parameter. Because there is a conversion *to* `const`, we can call either function on a non`const` object or a pointer to non`const`. However, as we'll see in § 6.4.4 (p. 245), the compiler will prefer the non`const` versions when we pass a non`const` object or pointer to non`const`.

## `const_cast` and Overloading

In § 4.11.3 (p. 174) we noted that `const_cast`s are most useful in the context of overloaded functions. As one example, recall our `shorterString` function from § 6.3.2 (p. 229):

```
// return a reference to the shorter of two strings
const string &shorterString(const string &s1, const string &s2)
{
    return s1.size() <= s2.size() ? s1 : s2;
}
```

This function takes and returns references to `const string`. We can call the function on a pair of non`const` `string` arguments, but we'll get a reference to a `const string` as the result.

We might want to have a second version of `shorterString` that, when given non`const` arguments, yields a plain reference. We can write that version as:

```
string &shorterString(string &s1, string &s2)
{
    // we must cast at least one of arguments to force the
    // compiler to call the const version of shorterString
    auto &r = shorterString(const_cast<const string &> (s1),
                            const_cast<const string &> (s2));
    // r has type const string&, but we know it is bound to a nonconst object
    return const_cast<string&>(r); // ok: r refers to a nonconst object
}
```

This version calls the `const` version of `shorterString` by casting its arguments to references to `const`. That function returns a reference to a `const string`, which we know is bound to one of our original, non`const` arguments. Therefore, we know it is safe to cast that `string` back to a plain `string&` in the return.

### Calling an Overloaded Function

Once we have defined a set of overloaded functions, we need to be able to call them with appropriate arguments. **Function matching** (also known as **overload resolution**) is the process by which a particular function call is associated with a specific function from a set of overloaded functions. The compiler determines which function to call by comparing the arguments in the call with the parameters offered by each function in the overload set.

In many—probably most—cases, it is straightforward for a programmer to determine whether a particular call is legal and, if so, which function will be called. Often the functions in the overload set differ in terms of the number of arguments, or the types of the arguments are unrelated. In such cases, it is easy to determine which function is called. Determining which function is called when the overloaded functions have the same number of parameters and those parameters are related by conversions (§ 4.11, p. 169) can be less obvious. We'll look at how the compiler resolves calls involving conversions in § 6.4.3 (p. 241).

For now, what's important to realize is that for any given call to an overloaded function, there are three possible outcomes:

- The compiler finds exactly one function that is a **best match** for the actual arguments and generates code to call that function.
- There is no function with parameters that match the arguments in the call, in which case the compiler issues an error message that there was **no match**.
- There is more than one function that matches and none of the matches is clearly best. This case is also an error; it is an **ambiguous call**.

## 6.4.2 Overloading and Scope

**WARNING**

Ordinarily, it is a bad idea to declare a function locally. However, to explain how scope interacts with overloading, we will violate this practice and use local function declarations.

Programmers new to C++ are often confused about the interaction between scope and overloading. However, overloading has no special properties with respect to scope: When we declare a name in an inner scope, that name *hides* uses of that name declared in an outer scope. Names do not overload across scopes:

```
string read();
void print(const string &);
void print(double);   // overloads the print function
```

## Exercises Section 6.4.1

**Exercise 6.29:** Explain why the second version of `shorterString` on page 238 that has `string&` parameters needs to cast its arguments to `const string&`. What would happen if those casts were omitted? Run the code without the casts to see whether yor were correct.

**Exercise 6.30:** Explain the effect of the second declaration in each one of the following sets of declarations. Indicate which, if any, are illegal.

(a) `int calc(int, int);`
    `int calc(const int, const int);`
(b) `int get();`
    `double get();`
(c) `int *reset(int *);`
    `double *reset(double *);`

```
void fooBar(int ival)
{
    bool read = false; //  new scope: hides the outer declaration of read
    string s = read(); //  error: read is a bool variable, not a function
    //  bad practice: usually it's a bad idea to declare functions at local scope
    void print(int);   //  new scope: hides previous instances of print
    print("Value: "); //  error: print(const string &) is hidden
    print(ival);       //  ok: print(int) is visible
    print(3.14);       //  ok: calls print(int); print(double) is hidden
}
```

Most readers will not be surprised that the call to `read` is in error. When the compiler sees the call to `read`, it finds the local definition of `read`. That name is a `bool` variable, and we cannot call a `bool`. Hence, the call is illegal.

Exactly the same process is used to resolve the calls to `print`. The declaration of `print(int)` in `fooBar` hides the earlier declarations of `print`. It is as if there is only one `print` function available: the one that takes a single `int` parameter.

When we call `print`, the compiler first looks for a declaration of that name. It finds the local declaration for `print` that takes an `int`. Once a name is found, the compiler ignores uses of that name in any outer scope. Instead, the compiler assumes that the declaration it found is the one for the name we are using. What remains is to see if the use of the name is valid.

> Note: In C++, name lookup happens before type checking.

The first call to `print` passes a string literal, but the only declaration for `print` that is in scope has a parameter that is an `int`. A string literal cannot be converted to an `int`, so this call is an error. The `print(const string&)` function, which would have matched this call, is hidden and is not considered.

When we call `print` passing a `double`, the process is repeated. The compiler finds the local definition of `print(int)`. The `double` argument can be converted to an `int`, so the call is legal.

Had we declared `print(int)` in the same scope as the other `print` functions, then it would be another overloaded version of `print`. In that case, these calls would be resolved differently, because the compiler will see all three functions:

```
void print(const string &);
void print(double); // overloads the print function
void print(int);    // another overloaded instance
void fooBar2(int ival)
{
    print("Value: "); // calls print(const string &)
    print(ival);      // calls print(int)
    print(3.14);      // calls print(double)
}
```

### 6.4.3 Function Matching

In many (if not most) cases, it is easy to figure out which overloaded function matches a given call. However, it is not so simple when the overloaded functions have the same number of parameters and when one or more of the parameters have types that are related by conversions. As an example, consider the following set of functions and function call:

```
void f();
void f(int);
void f(int, int);
void f(double, double = 3.14);
f(5.6);  // calls void f(double, double)
```

#### Determining the Candidate and Viable Functions

The first step of function matching identifies the set of overloaded functions considered for the call. These functions are referred to as the **candidate functions** for the call. A candidate function is a function with the same name as the called function and for which a declaration is visible at the point of the call. In this example, there are four candidate functions named `f`.

The second step selects from the set of candidate functions those functions that can be called with the arguments in the given call. The selected functions are the **viable functions**. To be viable, a function must have the same number of parameters as there are arguments in the call, and the type of each argument must match—or be convertible to—the type of its corresponding parameter.

We can eliminate two of our candidate functions based on the number of arguments. The function that has no parameters and the one that has two `int` parameters are not viable for this call. Our call has only one argument, and these functions have zero and two parameters, respectively.

The function that takes a single `int` and the function that takes two `doubles` are viable. Either of these functions can be called with a single argument. The function taking two `double`s has a default argument, which means it can be called with a single argument.

> Note: When a function has default arguments (§ 6.2.3, p. 224), a call may appear to have fewer arguments than it actually does.

Having used the number of arguments to winnow the candidate functions, we next look at whether the argument types match those of the parameters. As with any call, an argument might match its parameter either because the types match exactly or because there is a conversion from the argument type to the type of the parameter. In this example, both of our remaining functions are viable:

- `f(int)` is viable because a conversion exists that can convert the argument of type `double` to the parameter of type `int`.
- `f(double, double)` is viable because a default argument is provided for the function's second parameter and its first parameter is of type `double`, which exactly matches the type of the argument in the call.

> Note: If there are no viable functions, the compiler will complain that there is no matching function.

### Finding the Best Match, If Any

The third step of function matching determines which viable function provides the best match for the call. This process looks at each argument in the call and selects the viable function (or functions) for which the corresponding parameter best matches the argument. We'll explain the details of "best" in the next section, but the idea is that the closer the types of the argument and parameter are to each other, the better the match.

In our case, there is only one (explicit) argument in the call. That argument has type `double`. To call `f(int)`, the argument would have to be converted from `double` to `int`. The other viable function, `f(double, double)`, is an exact match for this argument. An exact match is better than a match that requires a conversion. Therefore, the compiler will resolve the call `f(5.6)` as a call to the function that has two `double` parameters. The compiler will add the default argument for the second, missing argument.

### Function Matching with Multiple Parameters

Function matching is more complicated if there are two or more arguments. Given the same functions named `f`, let's analyze the following call:

```
f(42, 2.56);
```

The set of viable functions is selected in the same way as when there was only one argument. The compiler selects those functions that have the required number of parameters and for which the argument types match the parameter types. In this case, the viable functions are `f(int, int)` and `f(double, double)`. The compiler then determines, argument by argument, which function is (or functions are) the best match. There is an overall best match if there is one and only one function for which

- The match for each argument is no worse than the match required by any other viable function
- There is at least one argument for which the match is better than the match provided by any other viable function

If after looking at each argument, there is no single function that is preferable, then the call is in error. The compiler will complain that the call is ambiguous.

In this call, when we look only at the first argument, we find that the function `f(int, int)` is an exact match. To match the second function, the `int` argument `42` must be converted to `double`. A match through a built-in conversion is "less good" than one that is exact. Considering only the first argument, `f(int, int)` is a better match than `f(double, double)`.

When we look at the second argument, `f(double, double)` is an exact match to the argument `2.56`. Calling `f(int, int)` would require that `2.56` be converted from `double` to `int`. When we consider only the second parameter, the function `f(double, double)` is a better match.

The compiler will reject this call because it is ambiguous: Each viable function is a better match than the other on one of the arguments to the call. It might be tempting to force a match by explicitly casting (§ 4.11.3, p. 173) one of our arguments. However, in well-designed systems, argument casts should not be necessary.

Best Practices

Casts should not be needed to call an overloaded function. The need for a cast suggests that the parameter sets are designed poorly.

**EXERCISES SECTION 6.4.3**

**Exercise 6.31:** What is a candidate function? What is a viable function?

**Exercise 6.32:** Given the declarations for `f` from page 241, list the viable functions, if any for each of the following calls. Indicate which function is the best match, or if the call is illegal whether there is no match or why the call is ambiguous.

(a) `f(2.56, 42)` (b) `f(42)` (c) `f(42, 0)` (d) `f(2.56, 3.14)`

**Exercise 6.33:** Write all four versions of `f`. Each function should print a distinguishing message. Check your answers for the previous exercise. If your answers were incorrect, study this section until you understand why your answers were wrong.

### 6.4.4 Argument Type Conversions

In order to determine the best match, the compiler ranks the conversions that could be used to convert each argument to the type of its corresponding parameter. Conversions are ranked as follows:

1. An exact match. An exact match happens when:
   - The argument and parameter types are identical.
   - The argument is converted from an array or function type to the corresponding pointer type. (§ 6.7 (p. 261) covers function pointers.)
   - A top-level `const` is added to or discarded from the argument.
2. Match through a `const` conversion (§ 4.11.2, p. 172).
3. Match through a promotion (§ 4.11.1, p. 170).
4. Match through an arithmetic (§ 4.11.1, p. 170) or pointer conversion (§ 4.11.2, p. 172).
5. Match through a class-type conversion. (§ 14.9 (p. 634) covers these conversions.)

### Matches Requiring Promotion or Arithmetic Conversion

**WARNING** Promotions and conversions among the built-in types can yield surprising results in the context of function matching. Fortunately, well-designed systems rarely need to use functions with parameters as closely related as those in the following examples.

In order to analyze a call, it is important to remember that the small integral types always promote to `int` or to a larger integral type. Given two functions, one of which takes an `int` and the other a `short`, the `short` version will be called only on values of type `short`. Even though the smaller integral values might appear to be a closer match, those values are promoted to `int`, whereas calling the `short` version would require a conversion:

```
void ff(int);
void ff(short);
ff('a');    // char promotes to int; calls f(int)
```

All the arithmetic conversions are treated as equivalent to each other. The conversion from `int` to `unsigned int`, for example, does not take precedence over the conversion from `int` to `double`. As a concrete example, consider

```
void manip(long);
void manip(float);
manip(42); // error: ambiguous call
```

The literal `42` is an `int` and an `int` can be converted to either `long` or `float`. Because there are two possible arithmetic conversions, the call is ambiguous.

## Function Matching and const Arguments

When we call an overloaded function that differs on whether a reference or pointer parameter refers or points to const, the compiler uses the constness of the argument to decide which function to call:

```
Record lookup(Account&);       // function that takes a reference to Account
Record lookup(const Account&); // new function that takes a const reference
const Account a;
Account b;
lookup(a);   // calls lookup(const Account&)
lookup(b);   // calls lookup(Account&)
```

In the first call, we pass the const object a. We cannot bind a plain reference to a const object. In this case, the only viable function is the version that takes a reference to const. Moreover, that call is an exact match to the argument a.

In the second call, we pass the nonconst object b. For this call, both functions are viable. We can use b to initialize a reference to either const or nonconst type. However, initializing a reference to const from a nonconst object requires a conversion. The version that takes a nonconst parameter is an exact match for b. Hence, the nonconst version is preferred.

Pointer parameters work in a similar way. If two functions differ only as to whether a pointer parameter points to const or nonconst, the compiler can distinguish which function to call based on the constness of the argument: If the argument is a pointer to const, the call will match the function that takes a const*; otherwise, if the argument is a pointer to nonconst, the function taking a plain pointer is called.

### Exercises Section 6.4.4

**Exercise 6.34:** Given the following declarations,

```
void manip(int, int);
double dobj;
```

what is the rank (§ 6.4.4, p. 243) of each conversion in the following calls?

(a) `manip('a', 'z');` (b) `manip(55.4, dobj);`

**Exercise 6.35:** Explain the effect of the second declaration in each one of the following sets of declarations. Indicate which, if any, are illegal.

```
(a) int calc(int&, int&);
    int calc(const int&, const int&);
(b) int calc(char*, char*);
    int calc(const char*, const char*);
(c) int calc(char*, char*);
    int calc(char* const, char* const);
```

## 6.5 Features for Specialized Uses

In this section, we'll cover three function-related features that are useful in many, but not all, programs: inline and `constexpr` functions, functions with a varying number of parameters, and some facilities that can be useful during debugging.

### 6.5.1 Inline and `constexpr` Functions

In § 6.3.2 (p. 229), we wrote a small function that returned a reference to the shorter of its two `string` parameters. The benefits of defining a function for such a small operation include the following:

- It is easier to read and understand a call to `shorterString` than it would be to read and understand the equivalent conditional expression.
- Using a function ensures uniform behavior: Each test is guaranteed to be done the same way.
- If we need to change the computation, it is easier to change the function than to find and change every occurrence of the equivalent expression.
- The function can be reused rather than rewritten for other applications.

There is, however, one potential drawback to making `shorterString` a function: Calling a function is apt to be slower than evaluating the equivalent expression. On most machines, a function call does a lot of work: Registers are saved before the call and restored after the return; arguments may be copied; and the program branches to a new location.

#### `inline` Functions Avoid Function Call Overhead

A function specified as **`inline`** (usually) is expanded "in line" at each call. If `shorterString` were defined as `inline`, then this call

```
cout << shorterString(s1, s2) << endl;
```

(probably) would be expanded during compilation into something like

```
cout << (s1.size() < s2.size() ? s1 : s2) << endl;
```

The run-time overhead of making `shorterString` a function is thus removed.

We can define `shorterString` as an inline function by putting the keyword `inline` before the function's return type:

```
// inline version: find the shorter of two strings
inline const string &
shorterString(const string &s1, const string &s2)
{
        return s1.size() <= s2.size() ? s1 : s2;
}
```

> **Note:** The `inline` specification is only a *request* to the compiler. The compiler may choose to ignore this request.

In general, the `inline` mechanism is meant to optimize small, straight-line functions that are called frequently. Many compilers will not inline a recursive function. A 75-line function will almost surely not be expanded inline.

## `constexpr` Functions

A function specified as **`constexpr`** can be used in a constant expression (§ 2.4.4, p. 66). We define or declare a `constexpr` function by putting the `constexpr` before the function's return type. We must include `constexpr` on the function's definition and on all its declarations.

`constexpr` functions must meet certain restrictions, among which are that the parameters and return type must be literal types (§ 2.4.4, p. 67) and there must be at least one argument value for which the function will return a constant expression:

```
// triple(arg) is a constant expression if cnt is a constant expression
constexpr size_t triple(size_t cnt) { return 3 * cnt; }
```

Here we defined `triple` as a `constexpr` function that takes one argument of type `int` and returns that value tripled. Our `triple` function will return a constant expression if its argument is a constant expression:

```
int arr[triple(2)];   // ok: triple(2) is a constant expression
int i = 2;            // i is not a constant expression
int a2[triple(i)];    // error: triple(i) is not a constant expression
```

When we pass a constant expression—such as the literal `2`—`triple` returns a constant expression; we may use that value where a constant expression is required such as to specify an array dimension. If we call `triple` with an expression that is not a constant expression—such as with the `int` object `i`—then the return is not a constant expression. If we use this return where a constant expression is required, the compiler will produce an error message.

> **Note:** A `constexpr` function is not required to return a constant expression.

When a `constexpr` function returns a constant expression, the compiler will replace the call with the resulting value. In order to be able to expand the function immediately, `constexpr` functions are implicitly `inline`.

Prior to C++14, `constexpr` functions were limited as to the actions they could perform. They could contain various kinds of declarations, but the only executable statement they could contain was a single `return` statement. New

Such functions now have very few restrictions: `constexpr` functions may contain variables of a non-literal type but such variables must be initialized. They may contain most statements, excepting `goto`s or statement labels (§ 5.5.3, p. 204) or `try` blocks (§ 5.6, p. 205).

For example, prior to C++14, we could have written the recursive version of our `factorial` function (§ 6.3.2, p. 232) as a `constexpr`:

```
// legal in both C++11 and C++14 and after
constexpr int factorial(int val)
{
    if (val > 1)
        return factorial(val-1) * val;
    return 1;
}
```

but could not have written the iterative one (§ 6.1, p. 214) as a `constexpr`:

```
// illegal in C++11, legal in C++14 and after
int fact(int val)
{
    int ret = 1; // local variable to hold the result as we calculate it
    while (val > 1)
        ret *= val--;   // assign ret * val to ret and decrement val
    return ret;         // return the result
}
```

### Put `inline` and `constexpr` Functions in Header Files

Unlike other functions, `inline` and `constexpr` functions may be defined multiple times in the program. After all, the compiler needs the definition, not just the declaration, in order to expand the code. However, all of the definitions of a given `inline` or `constexpr` must match exactly. As a result, like `inline` and `constexpr` variables (§ 2.6.3, p. 78), `inline` and `constexpr` functions usually are defined in headers.

**EXERCISES SECTION 6.5.1**

**Exercise 6.36:** Which one of the following declarations and definitions would you put in a header? In a source file? Explain why.

(a) `inline bool eq(const BigInt&, const BigInt&) {...}`
(b) `void putValues(int *arr, int size);`

**Exercise 6.37:** Rewrite the `isShorter` function from § 6.2.2 (p. 222) to be `inline`.

**Exercise 6.38:** Review the programs you've written for the earlier exercises and decide whether they should be defined as `inline`. If so, do so. If not, explain why they should not be `inline`.

**Exercise 6.39:** Would it be possible to define `isShorter` as a `constexpr`? If so, do so. If not, explain why not.

## 6.5.2 Functions with Varying Number of Parameters

Sometimes we do not know in advance how many arguments we need to pass to a function. C++ provides two primary ways to write a function that takes a varying number of arguments: If all the arguments have the same type, we can pass a library type named `initializer_list`. If the argument types vary, we can write a special kind of function, known as a variadic template, which we'll cover in § 16.4 (p. 756).

C++ also has a special parameter type, ellipsis, that can be used to pass a varying number of arguments. We'll look briefly at ellipsis parameters in this section. However, it is worth noting that this facility ordinarily should be used only in programs that need to interface to C functions.

## Library `initializer_list` Type

We can write a function that takes an unknown number of arguments of a single type by using an **`initializer_list`** parameter. An `initializer_list` is a library type that represents an array (§ 3.5, p. 119) of values of a specified type. This type is defined in the `initializer_list` header. The operations that `initializer_list` provides are listed in Table 6.1. Unlike `vector` or `string` and other library types that we'll cover in Part II, `initializer_list` is a specialized type. It is primarily intended to be used as a parameter type for functions that take a varying number of arguments of a particular type.

**Table 6.1: Operations on `initializer_lists`**

| | |
|---|---|
| `initializer_list<T> lst;` | Default initialization; an empty list of elements of type `T`. |
| `initializer_list<T> lst{a,b,c...};` | `lst` has as many elements as there are initializers; elements are copies of the corresponding initializers. Elements in the list are `const`. |
| `lst2(lst)`<br>`lst2 = lst` | Copying or assigning an `initializer_list` does not copy the elements in the list. After the copy, the original and the copy share the elements. |
| `lst.size()` | Number of elements in the list. `size()` is `constexpr`. |
| `lst.begin()`<br>`lst.end()` | Returns a pointer to the first and one past the last element in `lst`. These functions are `constexpr`. |

Like `vector`, `initializer_list` is a template type (§ 3.3, p. 100). When we define an `initializer_list`, we must specify the element type:

```
initializer_list<string> ls; // initializer_list of strings
initializer_list<int> li;    // initializer_list of ints
```

Aside from the fact they both are templates, in most other respects, `vectors` and `initializer_lists` are quite different from one another. In particular, the elements in an `initializer_list` are always `const` values; there is no way to change the value of an element in an `initializer_list`. Moreover, the standard specifies that the elements in an `initializer_list` are stored in an array. As a result, `initializer_lists` have a fixed size.

## What Copying an `initializer_list` Means

Another significant difference between `initialier_list` and most other library types is what it means to copy an `initializer_list`. When we copy a `vector`, the elements themselves are copied—after the copy each `vector` has elements that are independent of one another.

When we copy an `initializer_list`, the elements are not copied. Instead, after the copy, both the original and the copy refer to the same underlying array. Typically, copying an `initializer_list` copies a pointer to the first element of the array and a length or it copies pointers to the first and one past the last element in the array.

Because there is no way to change the number of or the value of the elements in an `initializer_list`, this behavior generally makes sense. In particular, the fact that copying an `initializer_list` does not copy the elements means that we can efficiently pass one as an argument to a function. However, this efficiency comes at a cost: we are responsible for coordinating the lifetime of the original list and any copies.

## `initializer_list` Parameters

When we pass an `initializer_list` to a function, lifetime is not an issue—we know that the list will exist while the called function is executing. As an example, we'll write a function to produce error messages from a varying number of arguments as follows:

```
void error_msg(initializer_list<string> il)
{
    for (auto beg = il.begin(); beg != il.end(); ++beg)
        cout << *beg << " " ;
    cout << endl;
}
```

The `begin` and `end` operations on `initializer_list` objects are analogous to the corresponding `vector` members (§ 3.4.1, p. 111). The `begin()` member gives us a pointer to the first element in the list, and `end()` is an off-the-end pointer one past the last element. Our function initializes `beg` to denote the first element and iterates through each element in the `initializer_list`. In the body of the loop we dereference `beg` in order to access the current element and print its value.

When we pass a sequence of values to an `initializer_list` parameter, we must enclose the sequence in curly braces:

```
// expected, actual are strings
if (expected != actual)
    error_msg({"functionX", expected, actual});
else
    error_msg({"functionX", "okay"});
```

Here we're calling the same function, `error_msg`, passing three values in the first call and two values in the second.

A function with an `initializer_list` parameter can have other parameters as well. For example, our debugging system might have a class, named `ErrCode`, that represents various kinds of errors. We can revise our program to take an `ErrCode` in addition to an `initializer_list` as follows:

```
void error_msg(ErrCode e, initializer_list<string> il)
{
    cout << e.msg() << ": ";
    for (const auto &elem : il)  // for each string in il
        cout << elem << " " ;    // write that string on cout
    cout << endl;
}
```

Because `initializer_list` has `begin` and `end` members, we can use a range `for` (§ 5.4.3, p. 199) to process the elements. This program, like our previous version, iterates an element at a time through the braced list of values passed to the `il` parameter.

To call this version, we need to revise our calls to pass an `ErrCode` argument:

```
if (expected != actual)
    error_msg(ErrCode(42), {"functionX", expected, actual});
else
    error_msg(ErrCode(0), {"functionX", "okay"});
```

### Functions Generally Should Not Return `initializer_list`

Returning an `initializer_list` from a function is almost surely an error (although one the compiler may not catch). For example:

```
initializer_list<int> foo()
{
    // local_list points to elements in an array that exists only while foo is running
    initializer_list<int> local_list = {1,2,3,4};
    if (some_condition)
        return local_list; // error: returning pointer to local variable
    else
        return {1,2,3,4};  // error: returning pointer to local temporary object
    // when foo completes, the memory in which both arrays were stored is freed
}
```

The problem is that when the compiler creates an `initializer_list` object, it creates an array in the current execution block. As with any other local variable, that (unnamed) array is deleted when the block in which it was defined is exited (§ 6.1.1, p. 217). When `foo` returns `local_list`, it is effectively returning a pointer to an array that exists only while `local_list` exists. That variable is local to `foo` and so goes out of existence when `foo` ends. Similarly, the `else` branch returns a pointer to the array that holds the elements of the unnamed temporary `initializer_list` object that is created in the `return`. Like the array associated with `local_obj`, this array no longer exists once `foo` terminates.

WARNING

For the same reasons that we don't return pointers to local variables (§ 6.3.2, p. 230), it is almost surely an error for a function to return an `initializer_list`.

### Ellipsis Parameters

Ellipsis parameters are in C++ to allow programs to interface to C code that uses a C library facility named `varargs`. Generally an ellipsis parameter should not be used for other purposes. Your C compiler documentation will describe how to use `varargs`.

> **WARNING** Ellipsis parameters should be used only for types that are common to both C and C++. In particular, objects of most class types are not copied properly when passed to an ellipsis parameter.

An ellipsis parameter may appear only as the last element in a parameter list and may take either of two forms:

```
void foo(parm_list, ...);
void foo(...);
```

The first form specifies the type(s) for some of `foo`'s parameters. Arguments that correspond to the specified parameters are type checked as usual. No type checking is done for the arguments that correspond to the ellipsis parameter. In this first form, the comma following the parameter declarations is optional.

**EXERCISES SECTION 6.5.2**

**Exercise 6.40:** Write a function that takes an `initializer_list<int>` and produces the sum of the elements in the list.

**Exercise 6.41:** In the second version of `error_msg` that has an `ErrCode` parameter, what is the type of `elem` in the `for` loop?

**Exercise 6.42:** When you use an `initializer_list` in a range `for` would you ever use a reference as the loop control variable? If so, why? If not, why not?

## 6.5.3 Aids for Debugging

C++ programmers sometimes use a technique similar to header guards (§ 2.6.3, p. 79) to conditionally execute debugging code. The idea is that the program will contain debugging code that is executed only while the program is being developed. When the application is completed and ready to ship, the debugging code is turned off. This approach uses two preprocessor facilities: `assert` and `NDEBUG`.

### The `assert` Preprocessor Macro

**`assert`** is a **preprocessor macro**. A preprocessor macro is a preprocessor variable that acts somewhat like an inline function. The `assert` macro takes a single expression, which it uses as a condition:

```
assert(expr);
```

evaluates *expr* and if the expression is false (i.e., zero), then `assert` writes a message and terminates the program. If the expression is true (i.e., is nonzero), then `assert` does nothing.

The `assert` macro is defined in the `cassert` header. As we've seen, preprocessor names are managed by the preprocessor, not the compiler (§ 2.3.2, p. 54). As a result, we use preprocessor names directly and do not provide a `using` declaration for them. That is, we refer to `assert`, not `std::assert`, and provide no `using` declaration for `assert`.

Like preprocessor variables, macro names must be unique within the program. Programs that include the `cassert` header may not define a variable, function, or other entity named `assert`. In practice, it is a good idea to avoid using the name `assert` for our own purposes even if we don't include `cassert`. Many headers include the `cassert` header, which means that even if you don't directly include that file, your programs are likely to have it included anyway.

The `assert` macro is often used to check for conditions that "cannot happen." For example, a program might know that every word in its input is always longer than a threshold. That program might contain a statement such as

```
assert(word.size() > threshold);
```

## The `NDEBUG` Preprocessor Variable

The behavior of `assert` depends on the status of a preprocessor variable named `NDEBUG`. If `NDEBUG` is defined, `assert` does nothing. By default, `NDEBUG` is *not* defined, so, by default, `assert` performs a run-time check.

We can "turn off" debugging by providing a `#define` to define `NDEBUG`. Alternatively, most compilers provide a command-line option that lets us define preprocessor variables:

```
$ CC -D NDEBUG main.C  # use /D with the Microsoft compiler
```

has the same effect as writing `#define NDEBUG` at the beginning of `main.C`.

If `NDEBUG` is defined, we avoid the potential run-time overhead involved in checking various conditions. Of course, there is also no run-time check. Therefore, `assert` should be used only to verify things that truly should not be possible. It can be useful as an aid in getting a program debugged but should not be used to substitute for run-time logic checks or error checking that the program should do.

In addition to using `assert`, we can write our own conditional debugging code using `NDEBUG`. If `NDEBUG` is *not* defined, the code between the `#ifndef` and the `#endif` is executed. If `NDEBUG` *is* defined, that code is ignored:

```
void print(const int ia[], size_t size)
{
#ifndef NDEBUG
// __func__ is a local static defined by the compiler that holds the function's name
cerr << __func__ << ": array size is " << size << endl;
#endif
// ...
```

Here we use a variable named `__func__` to print the name of the function we are debugging. The compiler defines `__func__` in every function. It is a local `static` array of `const char` that holds the name of the function.

In addition to `__func__`, which the C++ compiler defines, the preprocessor defines four other names that can be useful in debugging:

`__FILE__` string literal containing the name of the file

`__LINE__` integer literal containing the current line number

`__TIME__` string literal containing the time the file was compiled

`__DATE__` string literal containing the date the file was compiled

We might use these constants to report additional information in error messages:

```
if (word.size() < threshold)
    cerr << "Error: " << __FILE__
         << " : in function " << __func__
         << " at line " << __LINE__ << endl
         << "       Compiled on " << __DATE__
         << " at " << __TIME__ << endl
         << "       Word read was \"" << word
         << "\":  Length too short" << endl;
```

If we give this program a `string` that is shorter than the `threshold`, then the following error message will be generated:

```
Error: wdebug.cc : in function main at line 27
       Compiled on Jul 19 2018 at 13:17:57
       Word read was "foo":  Length too short
```

**EXERCISES SECTION 6.5.3**

**Exercise 6.43:** Revise the program you wrote in the exercises in § 6.3.2 (p. 233) that used recursion to print the contents of a `vector` to conditionally print information about its execution. For example, you might print the size of the `vector` on each call. Compile and run the program with debugging turned on and again with it turned off.

**Exercise 6.44:** Explain what this loop does and whether it is a good use of `assert`:

```
string s;
while (cin >> s && s != sought) { }   // empty body
assert(cin);
```

## 6.6 Functions that Operate on Arrays

Arrays have two special properties that affect how we define and use functions that operate on arrays: We cannot copy an array (§ 3.5.1, p. 120), and when we use an array it is (usually) converted to a pointer (§ 3.5.3, p. 123). Because we cannot copy an array, we cannot pass an array by value. Because arrays are converted to pointers, when we pass an array to (or return an array from) a function, we are actually passing (or returning) a pointer to the array's first element.

## 6.6.1 Array Parameters

Although we cannot pass an array by value, we can write a parameter with a type that looks like an array type:

```
// despite appearances, these three declarations of print are equivalent
// each function has a single parameter of type const int*
void print(const int*);
void print(const int[]);    // shows the intent that the function takes an array
void print(const int[10]); // dimension for documentation purposes (at best)
```

Regardless of appearances, these declarations are equivalent: Each declares a function with a single parameter of type `const int*`. When the compiler checks a call to `print`, it checks only that the argument has type `const int*`:

```
int i = 0, j[2] = {0, 1};
print(&i); // ok: &i is int*
print(j);  // ok: j is converted to an int* that points to j[0]
```

If we pass an array to `print`, that argument is automatically converted to a pointer to the first element in the array; the size of the array is irrelevant.

> **WARNING** As with any code that uses arrays, functions that take array parameters must ensure that all uses of the array stay within the array bounds.

Because arrays are passed as pointers, functions ordinarily don't know the size of the array they are given. They must rely on additional information provided by the caller. There are three common techniques used to manage array parameters.

### Using a Marker to Specify the Extent of an Array

The first approach to managing array arguments requires the array itself to contain an end marker, often called a sentinel. C-style character strings (§ 3.5.4, p. 128) are an example of this approach. C-style strings are stored in character arrays in which the last character of the string is followed by a null character. Functions that deal with C-style strings stop processing the array when they see a null character:

```
void print(const char *cp)
{
    if (cp)          // if cp is not a null pointer
        while (*cp)  // so long as the character it points to is not a null character
            cout << *cp++;   // print the character and advance the pointer
}
```

This convention works well for data where there is an obvious end-marker value (such as the null character) that does not appear in ordinary data. It works less well with data, such as `int`s, where every value in the range is a legitimate value.

### Using the Standard Library Conventions

A second technique used to manage array arguments is to pass pointers to the first and one past the last element in the array. This approach is inspired by techniques used in the standard library. We'll learn more about this style of programming in Part II. Using this approach, we'll print the elements in an array as follows:

```
void print(const int *beg, const int *end)
{
    // print every element starting at beg up to but not including end
    while (beg != end)
        cout << *beg++ << endl; // print the current element
                                // and advance the pointer
}
```

The `while` uses the dereference and postfix increment operators (§ 4.5, p. 157) to print the current element and advance `beg` one element at a time through the array. The loop stops when `beg` is equal to `end`.

To call this function, we pass two pointers—one to the first element we want to print and one just past the last element:

```
int j[2] = {0, 1};
// j is converted to a pointer to the first element in j
// the second argument is a pointer to one past the end of j
print(begin(j), end(j));  // begin and end functions, see § 3.5.3 (p. 124)
```

This function is safe, as long as the caller correctly calculates the pointers. Here we let the library `begin` and `end` functions (§ 3.5.3, p. 124) provide those pointers.

### Explicitly Passing a Size Parameter

A third approach for array arguments, which is common in C programs and older C++ programs, is to define a second parameter that indicates the size of the array. Using this approach, we'll rewrite `print` as follows:

```
// const int ia[] is equivalent to const int* ia
// size is passed explicitly and used to control access to elements of ia
void print(const int ia[], size_t size)
{
    for (size_t i = 0; i != size; ++i) {
        cout << ia[i] << endl;
    }
}
```

This version uses the `size` parameter to determine how many elements there are to print. When we call `print`, we must pass this additional parameter:

```
int j[] = { 0, 1 };  // int array of size 2
print(j, size(j));   // size on array see § 3.5.3 (p. 125)
```

The function executes safely as long as the size passed is no greater than the actual size of the array.

## Array Parameters and `const`

Note that all three versions of our `print` function defined their array parameters as pointers to `const`. The discussion in § 6.2.2 (p. 223) applies equally to pointers as to references. When a function does not need write access to the array elements, the array parameter should be a pointer to `const` (§ 2.4.2, p. 63). A parameter should be a plain pointer to a non`const` type only if the function needs to change element values.

## Array Reference Parameters

Just as we can define a variable that is a reference to an array (§ 3.5.1, p. 120), we can define a parameter that is a reference to an array. As usual, the reference parameter is bound to the corresponding argument, which in this case is an array:

```
// ok: parameter is a reference to an array; the dimension is part of the type
void print(int (&arr)[10])
{
    for (auto elem : arr)
        cout << elem << endl;
}
```

> **Note:** The parentheses around `&arr` are necessary (§ 3.5.1, p. 120):
>
> ```
> f(int &arr[10])    // error: declares arr as an array of references
> f(int (&arr)[10])  // ok: arr is a reference to an array of ten ints
> ```

Because the size of an array is part of its type, it is safe to rely on the dimension in the body of the function. However, the fact that the size is part of the type limits the usefulness of this version of `print`. We may call this function only for an array of exactly ten `int`s:

```
int i = 0, j[2] = {0, 1};
int k[10] = {0,1,2,3,4,5,6,7,8,9};
print(&i);    // error: argument is not an array of ten ints
print(j);     // error: argument is not an array of ten ints
print(k);     // ok: argument is an array of ten ints
```

We'll see in § 16.1.1 (p. 710) how we might write this function in a way that would allow us to pass a reference parameter to an array of any size.

## Passing a Multidimensional Array

Recall that there are no multidimensional arrays in C++ (§ 3.6, p. 133). Instead, what appears to be a multidimensional array is an array of arrays.

Like any other array, a multidimensional array is passed as a pointer to its first element (§ 3.6, p. 136). Because we are dealing with an array of arrays, that element is an array, so the pointer is a pointer to an array. The size of the second (and any subsequent) dimension is part of the element type and must be specified:

```
// matrix points to the first element in an array whose elements are arrays of ten ints
void print(int (*matrix)[10], int rowSize) { /* ... */ }
```

declares `matrix` as a pointer to an array of ten `int`s.

> Note: Again, the parentheses around `*matrix` are necessary:
>
> ```
> int *matrix[10];    // array of ten pointers
> int (*matrix)[10];  // pointer to an array of ten ints
> ```

We can also define our function using array syntax. As usual, the compiler ignores the first dimension, so it is best not to include it:

```
// equivalent definition
void print(int matrix[][10], int rowSize) { /* ... */ }
```

declares `matrix` to be what looks like a two-dimensional array. In fact, the parameter is a pointer to an array of ten `int`s.

### Exercises Section 6.6.1

**Exercise 6.45:** Write a function that takes an `int` and a pointer to an `int` and returns the larger of the `int` value or the value to which the pointer points.

**Exercise 6.46:** Write a function to swap two `int` pointers.

**Exercise 6.47:** Write your own versions of each of the `print` functions presented in this section. Call each of these functions to print `i` and `j` defined as follows:

```
int i = 0, j[2] = {0, 1};
```

**Exercise 6.48:** Explain the behavior of the following function. If there are problems in the code, explain what they are and how you might fix them.

```
void print(const int ia[10])
{
    for (size_t i = 0; i != 10; ++i)
        cout << ia[i] << endl;
}
```

## 6.6.2 `main`: Handling Command-Line Options

It turns out that `main` is a good example of how C++ programs pass arrays to functions. Up to now, we have defined `main` with an empty parameter list:

```
int main() { ... }
```

However, sometimes we have values we want the operating system to pass to `main`. The most common use of arguments to `main` is to let the user specify a set of options to guide the operation of the program. For example, assuming our `main` program is in an executable file named `prog`, we might pass options to the program as follows:

```
prog -d -o ofile data0
```

Such command-line options are passed to `main` in two (optional) parameters:

```
int main(int argc, char *argv[]) { ... }
```

The second parameter, `argv`, is a pointer to the first element in an array of pointers to C-style character strings. The first parameter, `argc`, passes the number of strings in that array. Because the second parameter is an array, we might alternatively indicate that `argv` points to a `char*` by defining `main` as

```
int main(int argc, char **argv) { ... }
```

When arguments are passed to `main`, the first element in `argv` points to a C-style character string holding the name of the program. If the program's name is unavailable for some reason, the string is null. Subsequent elements pass the arguments provided on the command line. The element just past the last pointer is guaranteed to be 0.

Given the previous command line, `argc` would be 5, and `argv` would hold the following C-style character strings:

```
argv[0] = "prog";   // or argv[0] might point to an empty string
argv[1] = "-d";
argv[2] = "-o";
argv[3] = "ofile";
argv[4] = "data0";
argv[5] = 0;
```

**WARNING**

The optional arguments to `main` begin in `argv[1]`; `argv[0]` contains the program's name, not user input.

### Exercises Section 6.6.2

**Exercise 6.49:** Write a `main` function that takes two arguments. Concatenate the supplied arguments and print the resulting `string`.

**Exercise 6.50:** Write a program that accepts the options presented in this section. Print the values of the arguments passed to `main`.

## 6.6.3 Returning a Pointer to an Array

Because we cannot copy an array, a function cannot return an array. However, a function can return a pointer or a reference to an array (§ 3.5.1, p. 120). When writing such functions, it is essential to remember that it is an error to return a reference or pointer to a local object (§ 6.1.1, p. 216). If a function returns a pointer or reference to an array, that array must exist outside the context of the function.

### Simplifying Functions that Return Pointers to Arrays

The syntax used to define functions that return pointers or references to arrays can be intimidating. For example, the following declaration (which we shall explain shortly) declares that `func` is a pointer to a function with one `int` parameter that returns an array of ten `int`s:

```
int (*func(int))[10];
```

Fortunately, there are ways to simplify such declarations. The most straightforward way to declare such functions is to use a trailing return (§ 6.3.3, p. 233):

```
// func takes an int argument and returns a pointer to an array of ten ints
auto func(int i) -> int(*)[10];
```

Because the return type comes after the parameter list, it is easier to see that `func` returns a pointer and that that pointer points to an array of ten `int`s.

Another alternative is to use a type alias (§ 2.5.1, p. 68):

```
using arrT = int[10]; // arrT is a synonym for the type array of ten ints
arrT* func(int i);    // func returns a pointer to an array of ten ints
```

Here `arrT` is a synonym for the type array of ten `int`s. We use `arrT` to specify the return type for `func`, which is a pointer to an array of ten `int`s.

### Declaring a Function That Returns a Pointer to an Array

To declare `func` without using a type alias, we must remember that the dimension of an array follows the name being defined:

```
int arr[10];          // arr is an array of ten ints
int *p1[10];          // p1 is an array of ten pointers
int (*p2)[10] = &arr; // p2 points to an array of ten ints
```

As with these declarations, if we want to define a function that returns a pointer to an array, the dimension must follow the function's name. However, a function includes a parameter list, which also follows the name. The parameter list precedes the dimension. Hence, the form of a function that returns a pointer to an array is:

*Type* `(*`*function*`(`*parameter_list*`))[`*dimension*`]`

As in any other array declaration, *Type* is the type of the elements and *dimension* is the size of the array. The parentheses around `(*`*function*`(`*parameter_list*`))` are necessary for the same reason that they were required when we defined `p2`. Without them, we would be defining a function that returns an array of pointers.

As a concrete example, the following declares `func` without using a type alias:

```
int (*func(int))[10];
```

To understand this declaration, it can be helpful to think about it as follows:

- `func(int)` says that we can call `func` with an `int` argument.
- `(*func(int))` says we can dereference the result of that call.

- `(*func(int))[10]` says that dereferencing the result of a call to `func` yields an array of size ten.
- `int (*func(int))[10]` says the element type in that array is `int`.

**EXERCISES SECTION 6.6.3**

**Exercise 6.51:** Write the declaration for a function that returns a reference to an array of ten `string`s, without using either a trailing return, `decltype`, or a type alias.

**Exercise 6.52:** Rewrite the declaration from the function in the previous exercise using a type alias, a trailing return, and `decltype`. Which form do you prefer and why?

**Exercise 6.53:** Revise the `arrPtr` function on to return a reference to the array.

## 6.7 Pointers to Functions

A function pointer is just that—a pointer that denotes a function rather than an object. Like any other pointer, a function pointer points to a particular type. A function's type is determined by its return type and the types of its parameters. The function's name is not part of its type. For example:

```
// compares lengths of two strings
bool lengthCompare(const string &, const string &);
```

has type `bool(const string&, const string&)`. To declare a pointer that can point at this function, we declare a pointer in place of the function name:

```
// pf points to a function returning bool that takes two const string references
bool (*pf)(const string &, const string &);  // uninitialized
```

Starting from the name we are declaring, we see that `pf` is preceded by a `*`, so `pf` is a pointer. To the right is a parameter list, which means that `pf` points to a function. Looking left, we find that the type the function returns is `bool`. Thus, `pf` points to a function that has two `const string&` parameters and returns `bool`.

Note

The parentheses around `*pf` are necessary. If we omit the parentheses, then we declare `pf` as a function that returns a pointer to `bool`:

```
// declares a function named pf that returns a bool*
bool *pf(const string &, const string &);
```

### Using Function Pointers

When we use the name of a function as a value, the function is automatically converted to a pointer. For example, we can assign the address of `lengthCompare` to `pf` as follows:

```
pf = lengthCompare;  // pf now points to the function named lengthCompare
pf = &lengthCompare; // equivalent assignment: address-of operator is optional
```

Moreover, we can use a pointer to a function to call the function to which the pointer points. We can do so directly—there is no need to dereference the pointer:

```
bool b1 = pf("hello", "goodbye");    // calls lengthCompare
bool b2 = (*pf)("hello", "goodbye"); // equivalent call
bool b3 = lengthCompare("hello", "goodbye");  // equivalent call
```

There is no conversion between pointers to one function type and pointers to another function type. However, as usual, we can assign `nullptr` (§ 2.3.2, p. 54) or a zero-valued integer constant expression to a function pointer to indicate that the pointer does not point to any function:

```
string::size_type sumLength(const string&, const string&);
bool cstringCompare(const char*, const char*);
pf = 0;              // ok: pf points to no function
pf = sumLength;      // error: return type differs
pf = cstringCompare; // error: parameter types differ
pf = lengthCompare;  // ok: function and pointer types match exactly
```

### Pointers to Overloaded Functions

As usual, when we use an overloaded function, the context must make it clear which version is being used. When we use the name of an overloaded function to initialize a pointer

```
void ff(int*);
void ff(unsigned int);
void (*pf1)(unsigned int) = ff;  // pf1 points to ff(unsigned)
```

the compiler uses the type of the pointer to determine which overloaded function to use. The type of the pointer must match one of the overloaded functions exactly:

```
void (*pf2)(int) = ff;    // error: no ff with a matching parameter list
double (*pf3)(int*) = ff; // error: return type of ff and pf3 don't match
```

### Function Pointer Parameters

Just as with arrays (§ 6.6.1, p. 255), we cannot define parameters of function type but we can have a parameter that is a pointer to function. As with arrays, we can write a parameter that looks like a function type, but it will be treated as a pointer:

```
// third parameter is a function type and is automatically treated as a pointer to function
void useBigger(const string &s1, const string &s2,
               bool pf(const string &, const string &));
// equivalent declaration: explicitly define the parameter as a pointer to function
void useBigger(const string &s1, const string &s2,
               bool (*pf)(const string &, const string &));
```

When we pass a function as an argument, we can do so directly. It will be automatically converted to a pointer:

```
// automatically converts the function lengthCompare to a pointer to function
useBigger(s1, s2, lengthCompare);
```

As we've just seen in the declaration of `useBigger`, writing function pointer types quickly gets tedious. Type aliases (§ 2.5.1, p. 68), along with `decltype` (§ 2.5.3, p. 70), let us simplify code that uses function pointers:

```
// Func and Func2 have function type
using Func =  bool (const string&, const string&);
using Func2 = decltype(lengthCompare); // equivalent type
// FuncP and FuncP2 have pointer to function type
using FuncP = bool(*)(const string&, const string&);
using FuncP2 = decltype(lengthCompare)*;  // equivalent type
```

Here we've used an alias declaration to define our types. Both `Func` and `Func2` are function types, and `FuncP` and `FuncP2` are pointer types. It is important to note that `decltype` returns the function type; the automatic conversion to pointer is not done. Because `decltype` returns a function type, if we want a pointer we must add the `*` ourselves. We can redeclare `useBigger` using any of these types:

```
// equivalent declarations of useBigger using type aliases
void useBigger(const string&, const string&, Func);
void useBigger(const string&, const string&, FuncP2);
```

Both declarations declare the same function. In the first case, the compiler will automatically convert the function type represented by `Func` to a pointer.

## Returning a Pointer to Function

As with arrays (§ 6.6.3, p. 259), we can't return a function type but can return a pointer to a function type. Similarly, we must write the return type as a pointer type; the compiler will not automatically treat a function return type as the corresponding pointer type. Also as with array returns, by far the easiest way to declare a function that returns a pointer to function is by using a type alias:

```
using F = int(int*, int);       // F is a function type, not a pointer
using PF = int(*)(int*, int); // PF is a pointer type
```

Unlike parameters of function type, a return type is not automatically converted to a pointer type. We must explicitly specify that the return is a pointer:

```
PF f1(int);   // ok: PF is a pointer to function; f1 returns a pointer to function
F f1(int);    // error: F is a function type; f1 can't return a function
F *f1(int);   // ok: explicitly specify that the return type is a pointer to function
```

Of course, we can also declare `f1` directly, which we'd do as

```
int (*f1(int))(int*, int);
```

Reading this declaration from the inside out, we see that `f1` has a parameter list, so `f1` is a function. `f1` is preceded by a `*` so `f1` returns a pointer. The type of that pointer itself has a parameter list, so the pointer points to a function. That function returns an `int`.

For completeness, it's worth noting that we can simplify declarations of functions that return pointers to function by using a trailing return (§ 6.3.3, p. 233):

```
auto f1(int) -> int (*)(int*, int);
```

### Using `decltype` for Function Pointer Types

If we know which function(s) we want to return, we can use `decltype` to simplify writing a function pointer return type. For example, assume we have two functions, both of which return a `string::size_type` and have two `const string&` parameters. We can write a third function that takes a `string` parameter and returns a pointer to one of these two functions as follows:

```
string::size_type sumLength(const string&, const string&);
string::size_type largerLength(const string&, const string&);
// depending on the value of its string parameter,
// getFcn returns a pointer to sumLength or to largerLength
decltype(sumLength) *getFcn(const string &);
```

Of course, when we use `decltype`, we get a function type. We must add a `*` to indicate that we are returning a pointer, not a function.

#### Exercises Section 6.7

**Exercise 6.54:** Write a declaration for a function that takes two `int` parameters and returns an `int`, and declare a `vector` whose elements have this function pointer type.

**Exercise 6.55:** Write functions that add, subtract, multiply, and divide two `int` values. Store pointers to these functions in your `vector` from the previous exercise. Call each element in the `vector` and print the result.

## Chapter Summary

Functions are named units of computation and are essential to structuring even modest programs. Every function has a return type, a name, a (possibly empty) list of parameters, and a function body. The function body is a block that is executed when the function is called. When a function is called, the arguments passed to the function must be compatible with the types of the corresponding parameters.

In C++, functions may be overloaded: The same name may be used to define different functions as long as the number or types of the parameters in the functions differ. The compiler automatically figures out which function to call based on the arguments in a call. The process of selecting the right function from a set of overloaded functions is referred to as function matching.

## Defined Terms

**ambiguous call** Compile-time error that results during function matching when two or more functions provide an equally good match for a call.

**argument** Value supplied in a function call. Arguments initialize the function's parameters.

**assert** Preprocessor macro that takes a single expression. If the preprocessor variable `NDEBUG` is defined, `assert` does nothing. Otherwise, it evaluates the expression as a condition; if the condition is false, `assert` prints a message and terminates the program.

**automatic object** Object that exists only during the execution of a block. They are created when control passes through their definition and are destroyed at the end of the block in which they are defined.

**best match** Function selected from a set of overloaded functions for a call. If a best match exists, the selected function is a better match than all the other viable candidates for at least one argument in the call and is no worse on the rest of the arguments.

**call by reference** See pass by reference.

**call by value** See pass by value.

**candidate functions** Set of functions that are considered when resolving a function call. The candidate functions are all the functions with the name used in the call for which a declaration is in scope at the point of the call.

**constexpr** Function that may return a constant expression. A `constexpr` function is implicitly `inline`.

**default argument** Value specified to be used when an argument is omitted in a call to the function.

**executable file** File, which the operating system executes, that contains code corresponding to our program.

**function** Callable unit of computation.

**function body** Block that defines the actions of a function.

**function matching** Compiler process by which a call to an overloaded function is resolved. Arguments used in the call are compared to the parameter list of each overloaded function.

**function prototype** Function declaration, consisting of the function's name, return type, and its parameter types.

**global object** Object defined outside any function. Global objects are created during program startup and destroyed when the program ends.

**hidden name** Names declared inside a scope hide previously declared entities with the same names declared outside that scope.

**initializer_list** Library class that represents a comma-separated list of objects of a single type enclosed inside curly braces.

**inline function** Request to the compiler to expand a function at the point of call, if possible. Inline functions avoid the normal function-calling overhead.

**link** Compilation step in which multiple object files are put together to form an executable program.

**local static object** Local object whose value persists across calls to the function. Local `static` objects are created and initialized before control reaches their first use and are destroyed when the program ends.

**local variable** Variable defined inside a block.

**no match** Compile-time error that results during function matching when there is no function with parameters that match the arguments in a given call.

**object code** Format into which the compiler transforms our source code.

**object file** File holding object code generated by the compiler from a given source file. An executable file is generated from one or more object files after the files are linked together.

**object lifetime** Every object has an associated lifetime. Non`static` objects that are defined inside a block exist from when their definition is encountered until the end of the block in which they are defined. Global objects are created during program startup. Local `static` objects are created before the first time execution passes through the object's definition. Global objects and local `static` objects are destroyed when the `main` function ends.

**overload resolution** See function matching.

**overloaded function** Function that has the same name as at least one other function. Overloaded functions must differ in the number or type of their parameters.

**parameter** Local variable declared inside the function parameter list. Parameters are initialized by the arguments provided in each function call.

**pass by reference** How arguments are passed to parameters of reference type. Reference parameters work the same way as any other use of references; the parameter is bound to its corresponding argument.

**pass by value** How arguments are passed to parameters of a nonreference type. A nonreference parameter is copy-initialized by its corresponding argument.

**preprocessor macro** Preprocessor facility that behaves like an inline function. Aside from `assert`, modern C++ programs make very little use of preprocessor macros.

**recursion loop** What happens when a recursive function omits a stopping condition and repeatedly calls itself until exhasuting the program stack.

**recursive function** Function that calls itself directly or indirectly.

**return type** Part of a function declaration that specifies the type of the value that the function returns.

**separate compilation** Ability to split a program into multiple separate source files.

**trailing return type** Return type specified after the parameter list.

**viable functions** Subset of the candidate functions that could match a given call. Viable functions have the same number of parameters as arguments to the call, and each argument type can be converted to the corresponding parameter type.

**`()` operator** Call operator. Executes a function. The name of a function or a function pointer precedes the parentheses, which enclose a (possibly empty) comma-separated list of arguments.

# CHAPTER 7

# CLASSES

## CONTENTS



In C++ we use classes to define our own data types. By defining types that mirror concepts in the problems we are trying to solve, we can make our programs easier to write, debug, and modify.

This chapter continues the coverage of classes begun in Chapter 2. Here we will focus on the importance of data abstraction, which lets us separate the implementation of an object from the operations that that object can perform. Subsequent chapters will add to the material introduced here: Chapter 13 covers how we can control what happens when class objects are copied, moved, assigned, or destroyed. Chapter 14 will show how to define our own operators. Chapter 15 and Chapter 16 cover classes related by inheritance and template classes, respectively. Finally, Chapter 18 and Chapter 19 cover various other class-related features of a more specialized nature.

*The fundamental ideas* behind **classes** are **data abstraction** and **encapsulation**. Data abstraction is a programming (and design) technique that relies on the separation of **interface** and **implementation**. The interface of a class consists of the operations that users of the class can execute. The implementation includes the class' data members, the bodies of the functions that constitute the interface, and any functions needed to define the class that are not intended for general use.

Encapsulation enforces the separation of a class' interface and implementation. A class that is encapsulated hides its implementation—users of the class can use the interface but have no access to the implementation.

A class that uses data abstraction and encapsulation defines an **abstract data type**. In an abstract data type, the class designer worries about how the class is implemented. Programmers who use the class need not know how the type works. They can instead think *abstractly* about what the type does.

## 7.1 Defining Abstract Data Types

The `Sales_item` class that we used in Chapter 1 is an abstract data type: We use a `Sales_item` object by using its interface (i.e., the operations described in § 1.5.1 (p. 20)). We have no access to the data members stored in a `Sales_item` object. Indeed, we don't even know what data members that class has.

Our `Sales_data` class (§ 2.6.1, p. 73) is not an abstract data type: It lets users of the class access its data members and forces users to write their own operations. To make `Sales_data` an abstract type, we need to define operations for users of `Sales_data` to use. Once `Sales_data` defines its own operations, we can encapsulate (that is, hide) its data members.

### 7.1.1 Designing the `Sales_data` Class

Ultimately, we want `Sales_data` to support the same set of operations as the `Sales_item` class. The `Sales_item` class had one **member function** (§ 1.5.2, p. 23), named `isbn`, and supported the `+`, `=`, `+=`, `<<`, and `>>` operators.

We'll learn how to define our own operators in Chapter 14. For now, we'll define ordinary (named) functions for these operations. For reasons that we will explain in § 14.1 (p. 607), the functions that do addition and IO will not be members of `Sales_data`. Instead, we'll define those functions as ordinary functions. The function that handles compound assignment will be a member, and for reasons we'll explain in § 7.1.5 (p. 280), our class doesn't need to define assignment.

Thus, the interface to `Sales_data` consists of the following operations:

- An `isbn` member function to return the object's ISBN
- A `combine` member function to add one `Sales_data` object into another
- A function named `add` to add two `Sales_data` objects
- A `read` function to read data from an `istream` into a `Sales_data` object
- A `print` function to print the value of a `Sales_data` object on an `ostream`

**KEY CONCEPT: DIFFERENT KINDS OF PROGRAMMING ROLES**

**Programmers tend to think about the people who will run their applications as *users*. Similarly, a class designer designs and implements a class for *users* of that class. In this case, the user is a programmer, not the ultimate user of the application.**

**When we refer to a *user*, the context makes it clear which kind of user is meant. If we speak of *user code* or the *user* of the `Sales_data` class, we mean a programmer who is using that class. If we speak of the *user* of the bookstore application, we mean the manager of the store who is running the application. When we need to distinguish users who are programmers from users of the application, we tend to refer to the latter as "end users."**

Note

**C++ programmers tend to speak of *users* interchangeably as users of the application or users of a class.**

**In simple applications, the user of a class and the designer of the class might be the same person. Even in such cases, it is useful to keep the roles distinct. When we design the interface of a class, we should think about how easy it will be to use the class. When we use the class, we shouldn't think about how the class works.**

**Authors of successful applications do a good job of understanding and implementing the needs of the application's users. Similarly, good class designers pay close attention to the needs of the programmers who will use the class. A well-designed class has an interface that is intuitive and easy to use and has an implementation that is efficient enough for its intended use.**

## Using the Revised `Sales_data` Class

Before we think about how to implement our class, let's look at how we can use our interface functions. As one example, we can use these functions to write a version of the bookstore program from § 1.6 (p. 24) that works with `Sales_data` objects rather than `Sales_items`:

```
// total will hold the running sum (definition inside if see § 5.2 (p. 184))
if (Sales_data total; read(cin, total)) { // read the first transaction
    // read and process the remaining transactions
    for (Sales_data trans; read(cin, trans); ) {
        if (total.isbn() == trans.isbn()) // check the isbns
            total.combine(trans);  // update the running total
        else {
            print(cout, total) << endl;    // print the results
            total = trans;                 // process the next book
        }
    }
    print(cout, total) << endl;            // print the last transaction
}
```

We start by defining a `Sales_data` object to hold the running total. Inside the `if` condition, we call `read` to read the first transaction into `total`. This condition works like other loops we've written that used the `>>` operator. Like the `>>` operator, our `read` function will return its stream parameter, which the condition checks (§ 4.11.2, p. 173). If the `read` fails, there were no records to process and we fall through to the close curly just before the end of the function.

The `for` loop defines `trans`, which we'll use to hold each transaction. The condition in the `for` reads the next transaction and checks the stream returned by `read`. So long as the input operations in `read` succeed, the condition succeeds and we have another transaction to process.

In the body of the `for`, we call the `isbn` members of `total` and `trans` to fetch their respective ISBNs. If `total` and `trans` refer to the same book, we call `combine` to add the components of `trans` into the running total in `total`. If `trans` represents a new book, we call `print` to print the total for the previous book. Because `print` returns a reference to its stream parameter, we can use the result of `print` as the left-hand operand of the `<<`. We do so to print a newline following the output generated by `print`. We next assign `trans` to `total`, thus setting up to process the records for the next book in the file.

After we have exhausted the input, we have to remember to print the data for the last transaction, which we do in the call to `print` following the `for` loop.

### Exercises Section 7.1.1

**Exercise 7.1:** Write a version of the transaction-processing program from § 1.6 (p. 24) using the `Sales_data` class you defined for the exercises in § 2.6.1 (p. 73).

## 7.1.2 Defining the Revised `Sales_data` Class

Our revised class will have the same data members as the version we defined in § 2.6.1 (p. 73): `bookNo`, a `string` representing the ISBN; `units_sold`, an `unsigned` that says how many copies of the book were sold; and `revenue`, a `double` representing the total revenue for those sales.

As we've seen, our class will also have two member functions, `combine` and `isbn`. In addition, we'll give `Sales_data` another member function to return the average price at which the books were sold. This function, which we'll name `avg_price`, is part of `Sales_data`'s implementation, not part of the interface.

We define (§ 6.1, p. 214) and declare (§ 6.1.2, p. 218) member functions similarly to ordinary functions. Like data members, member functions *must* be declared inside the class. Member functions *may* be defined inside or outside the class body. Nonmember functions that are part of the interface, such as `add`, `read`, and `print`, are declared and defined outside the class.

With this knowledge, we're ready to write our revised version of `Sales_data`:

```
struct Sales_data {
    // new members: operations on Sales_data objects
    std::string isbn() const { return bookNo; }
    Sales_data& combine(const Sales_data&);
    double avg_price() const;
    // data members are unchanged from § 2.6.1 (p. 73)
```

```
    std::string bookNo;
    unsigned units_sold = 0;
    double revenue = 0.0;
};
// nonmember Sales_data interface functions
Sales_data add(const Sales_data&, const Sales_data&);
std::ostream &print(std::ostream&, const Sales_data&);
std::istream &read(std::istream&, Sales_data&);
```

> Note: Functions defined in the class are implicitly `inline` (§ 6.5.1, p. 246).

## Defining Member Functions

Although every member must be declared inside its class, we can define a member function's body either inside or outside of the class body. In `Sales_data`, `isbn` is defined inside the class; `combine` and `avg_price` will be defined elsewhere.

We'll start by explaining the `isbn` function, which returns a `string` and has an empty parameter list:

```
std::string isbn() const { return bookNo; }
```

As with any function, the body of a member function is a block. In this case, the block contains a single `return` statement that returns the `bookNo` data member of a `Sales_data` object. The interesting thing about this function is how it gets the object from which to fetch the `bookNo` member.

## Introducing `this`

Let's look again at a call to the `isbn` member function:

```
total.isbn()
```

Here we use the dot operator (§ 4.6.2, p. 160) to fetch the `isbn` member of the object named `total`, which we then call.

With one exception that we'll cover in § 7.7 (p. 315), when we call a member function we do so on behalf of an object. When `isbn` refers to members of `Sales_data` (e.g., `bookNo`), it is referring implicitly to the members of the object on which the function was called. In this call, when `isbn` returns `bookNo`, it is implicitly returning `total.bookNo`.

Member functions access the object on which they were called through an extra, implicit parameter named **`this`**. When we call a member function, `this` is initialized with the address of the object on which the function was invoked. For example, when we call

```
total.isbn()
```

the compiler passes the address of `total` to the implicit `this` parameter in `isbn`. It is as if the compiler rewrites this call as

```
// pseudo-code illustration of how a call to a member function is translated
Sales_data::isbn(&total)
```

which calls the `isbn` member of `Sales_data` passing the address of `total`.

Inside a member function, we can refer directly to the members of the object on which the function was called. We do not have to explicitly access them through the `this` pointer. Any direct use of a member of the class is assumed to be an implicit reference through `this`. That is, when `isbn` uses `bookNo`, it is as if we had written `this->bookNo`.

The `this` parameter is defined for us implicitly. Indeed, it is illegal for us to define a parameter or variable named `this`. Inside the body of a member function, we can use `this`. It would be legal, although unnecessary, to define `isbn` as

```
std::string isbn() const { return this->bookNo; }
```

Because `this` is intended to always refer to "this" object, `this` is a `const` pointer (§ 2.4.2, p. 63). We cannot change the address that `this` holds.

## Introducing `const` Member Functions

The other important aspect of the `isbn` function is the keyword `const` that follows the parameter list. The purpose of that `const` is to promise that the member function does not change its object. The compiler enforces this promise by modifying the type of the implicit `this` pointer.

By default, the type of `this` is a `const` pointer to the non`const` version of the class type. For example, by default, the type of `this` in a `Sales_data` member function is `Sales_data *const`. Although `this` is implicit, it follows the normal initialization rules, which means that (by default) we cannot bind `this` to a `const` object (§ 2.4.2, p. 63). This fact, in turn, means that we cannot call an ordinary member function on a `const` object.

If `isbn` were an ordinary function and if `this` were an ordinary pointer parameter, we would declare `this` as `const Sales_data *const`. After all, the body of `isbn` doesn't change the object to which `this` points, so our function would be more flexible if `this` were a pointer to `const` (§ 6.2.2, p. 223).

However, `this` is implicit and does not appear in the parameter list, so there is no place to indicate that `this` should be a pointer to `const`. The language resolves this problem by letting us put `const` after the parameter list of a member function. The `this` is a pointer to `const` in member functions that use `const` in this way. Such functions are known as **`const` member functions**.

We can think of the body of `isbn` as if it were written as

```
// pseudo-code illustration of how the implicit this pointer is used
// this code is illegal: we may not explicitly define the this pointer ourselves
// note that this is a pointer to const because isbn is a const member
std::string Sales_data::isbn(const Sales_data *const this)
{ return this->isbn; }
```

The fact that `this` is a pointer to `const` means that `const` member functions cannot change the object on which they are called. Thus, `isbn` may read but not write the data members of the objects on which it is called.

> **Note:** Objects that are `const`, and references or pointers to `const` objects, may call only `const` member functions.

## Class Scope and Member Functions

Recall that a class is itself a scope (§ 2.6.1, p. 73). The definitions of the member functions of a class are nested inside the scope of the class itself. Hence, `isbn`'s use of the name `bookNo` is resolved as the data member defined inside `Sales_data`.

It is worth noting that `isbn` can use `bookNo` even though `bookNo` is defined *after* `isbn`. As we'll see in § 7.4.1 (p. 297), the compiler processes classes in two steps—the member declarations are compiled first, after which the member function bodies, if any, are processed. Thus, member function bodies may use other members of their class regardless of where in the class those members appear.

## Defining a Member Function outside the Class

As with any other function, when we define a member function outside the class body, the member's definition must match its declaration. That is, the return type, parameter list, and name must match the declaration in the class body. If the member was declared as a `const` member function, then the definition must also specify `const` after the parameter list. The name of a member defined outside the class must include the name of the class of which it is a member:

```
double Sales_data::avg_price() const {
    if (units_sold)
        return revenue/units_sold;
    else
        return 0;
}
```

The function name, `Sales_data::avg_price`, uses the scope operator (§ 1.2, p. 8) to say that we are defining the function named `avg_price` that is declared in the scope of the `Sales_data` class. Once the compiler sees the function name, the rest of the code is interpreted as being inside the scope of the class. Thus, when `avg_price` refers to `revenue` and `units_sold`, it is implicitly referring to the members of `Sales_data`.

## Defining a Function to Return "This" Object

The `combine` function is intended to act like the compound assignment operator, `+=`. The object on which this function is called represents the left-hand operand of the assignment. The right-hand operand is passed as an explicit argument:

```
Sales_data& Sales_data::combine(const Sales_data &rhs)
{
    units_sold += rhs.units_sold; // add the members of rhs into
    revenue += rhs.revenue;       // the members of ''this'' object
    return *this; // return the object on which the function was called
}
```

When our transaction-processing program calls

```
total.combine(trans);   // update the running total
```

the address of `total` is bound to the implicit `this` parameter and `rhs` is bound to `trans`. Thus, when `combine` executes

```
units_sold += rhs.units_sold; // add the members of rhs into
```

the effect is to add `total.units_sold` and `trans.units_sold`, storing the result back into `total.units_sold`.

The interesting part about this function is its return type and the `return` statement. Ordinarily, when we define a function that operates like a built-in operator, our function should mimic the behavior of that operator. The built-in assignment operators return their left-hand operand as an lvalue (§ 4.4, p. 153). To return an lvalue, our `combine` function must return a reference (§ 6.3.2, p. 230). Because the left-hand operand is a `Sales_data` object, the return type is `Sales_data&`.

As we've seen, we do not need to use the implicit `this` pointer to access the members of the object on which a member function is executing. However, we do need to use `this` to access the object as a whole:

```
return *this; // return the object on which the function was called
```

Here the `return` statement dereferences `this` to obtain the object on which the function is executing. That is, for the call above, we return a reference to `total`.

**EXERCISES SECTION 7.1.2**

**Exercise 7.2:** Add the `combine` and `isbn` members to the `Sales_data` class you wrote for the exercises in § 2.6.2 (p. 78).

**Exercise 7.3:** Revise your transaction-processing program from § 7.1.1 (p. 270) to use these members.

**Exercise 7.4:** Write a class named `Person` that represents the name and address of a person. Use a `string` to hold each of these elements. Subsequent exercises will incrementally add features to this class.

**Exercise 7.5:** Provide operations in your `Person` class to return the name and address. Should these functions be `const`? Explain your choice.

### 7.1.3 Defining Nonmember Class-Related Functions

Class authors often define auxiliary functions, such as our `add`, `read`, and `print` functions. Although such functions define operations that are conceptually part of the interface of the class, they are not part of the class itself.

We define nonmember functions as we would any other function. As with any other function, we normally separate the declaration of the function from its definition (§ 6.1.2, p. 218). Functions that are conceptually part of a class, but not defined inside the class, are typically declared (but not defined) in the same header as the class itself. That way users need to include only one file to use any part of the interface.

Note: Ordinarily, nonmember functions that are part of the interface of a class should be declared in the same header as the class itself.

## Defining the `read` and `print` Functions

The `read` and `print` functions do the same job as the code in § 2.6.2 (p. 76) and not surprisingly, the bodies of our functions look a lot like the code presented there:

```
// input transactions contain ISBN, number of copies sold, and sales price
istream &read(istream &is, Sales_data &item)
{
    double price = 0;
    is >> item.bookNo >> item.units_sold >> price;
    item.revenue = price * item.units_sold;
    return is;
}
ostream &print(ostream &os, const Sales_data &item)
{
    os << item.isbn() << " " << item.units_sold << " "
       << item.revenue << " " << item.avg_price();
    return os;
}
```

The `read` function reads data from the given stream into the given object. The `print` function prints the contents of the given object on the given stream.

However, there are two points worth noting about these functions. First, both `read` and `print` take a reference to their respective IO class types. The IO classes are types that cannot be copied, so we may only pass them by reference (§ 6.2.2, p. 221). Moreover, reading or writing to a stream changes that stream, so both functions take ordinary references, not references to `const`.

The second thing to note is that `print` does not print a newline. Ordinarily, functions that do output should do minimal formatting. That way user code can decide whether the newline is needed.

## Defining the `add` Function

The `add` function takes two `Sales_data` objects and returns a new `Sales_data` representing their sum:

```
Sales_data add(const Sales_data &lhs, const Sales_data &rhs)
{
    Sales_data sum = lhs;  // copy data members from lhs into sum
    sum.combine(rhs);      // add data members from rhs into sum
    return sum;
}
```

In the body of the function, we define a new `Sales_data` object named `sum` to hold the sum of our two transactions. We initialize `sum` as a copy of `lhs`. By default, copying a class object copies that object's members. After the copy, the `bookNo`, `units_sold`, and `revenue` members of `sum` will have the same values as those in `lhs`. Next we call `combine` to add the `units_sold` and `revenue` members of `rhs` into `sum`. When we're done, we return a copy of `sum`.

### EXERCISES SECTION 7.1.3

**Exercise 7.6:** Define your own versions of the `add`, `read`, and `print` functions.

**Exercise 7.7:** Rewrite the transaction-processing program you wrote for the exercises in § 7.1.2 (p. 274) to use these new functions.

**Exercise 7.8:** Why does `read` define its `Sales_data` parameter as a plain reference and `print` define its parameter as a reference to `const`?

**Exercise 7.9:** Add operations to read and print `Person` objects to the code you wrote for the exercises in § 7.1.2 (p. 274).

**Exercise 7.10:** What does the condition in the following `if` statement do?

```
if (read(read(cin, data1), data2))
```

## 7.1.4 Constructors

Each class defines how objects of its type can be initialized. Classes control object initialization by defining one or more special member functions known as **constructors**. A constructor runs as part of creating every object of class type; its job is to initialize the object's data members.

In this section, we'll introduce the basics of how to define a constructor. Constructors are a surprisingly complex topic. Indeed, we'll have more to say about constructors in § 7.5 (p. 301), § 15.7 (p. 677), and § 18.1.3 (p. 829), and in Chapter 13.

Constructors have the same name as the class. Unlike other functions, constructors have no return type. Like other functions, constructors have a (possibly empty) parameter list and a (possibly empty) function body, which can be defined inside or outside of the class definition. A class can have multiple constructors. Like any other overloaded function (§ 6.4, p. 235), the constructors must differ from each other in the number or types of their parameters.

Unlike other member functions, constructors may not be declared as `const` (§ 7.1.2, p. 272). When we create a `const` object of a class type, the object does not assume its "`const`ness" until after the constructor completes the object's initialization. Thus, constructors can write to `const` objects during their construction.

### The Synthesized Default Constructor

Our `Sales_data` class does not define any constructors, yet the programs we've written that use `Sales_data` objects compile and run correctly. As an example, the program on page 269 defined two `sales_data` objects, `total` and `trans`. The question naturally arises: How are these objects initialized?

We did not supply an initializer for either object, so we know that they are default initialized (§ 2.2.1, p. 44). Classes control default initialization by defining a special constructor, known as the **default constructor**. The default constructor is one that takes no arguments.

As we'll, see the default constructor is special in various ways, one of which is that if our class does not *explicitly* define any constructors, the compiler will *implicitly* define the default constructor for us

The compiler-generated constructor is known as the **synthesized default constructor**. For most classes, this synthesized constructor initializes each data member of the class as follows:

- If there is an member initializer (§ 2.6.1, p. 74), use it to initialize the member.
- Otherwise, default-initialize (§ 2.2.1, p. 44) the member.

Because `Sales_data` provides initializers for `units_sold` and `revenue`, the synthesized default constructor uses those values to initialize those members. It default initializes `bookNo` to the empty string.

## Some Classes Cannot Rely on the Synthesized Default Constructor

Only fairly simple classes—such as the current version of `Sales_data`—can rely on the synthesized default constructor. The most common reason that a class must define its own default constructor is that the compiler generates the default for us *only if we do not define any other constructors for the class*. If we define any constructors at all, the class will not have a default constructor unless we define that constructor ourselves. The basis for this rule is that if a class requires control to initialize an object in one case, then the class is likely to require control in all cases.

> Note: The compiler generates a default constructor automatically only if a class declares *no* constructors at all.

A second reason to define the default constructor is that for some classes, the synthesized default constructor does the wrong thing. Remember that objects of built-in or compound type (such as arrays and pointers) that are defined inside a block have undefined value when they are default initialized (§ 2.2.1, p. 44). The same rule applies to members of built-in type that are default initialized. Therefore, classes that have members of built-in or compound type should ordinarily either initialize those members inside the class or define their own version of the default constructor. Otherwise, users could create objects with members that have undefined value.

> WARNING: Classes that have members of built-in or compound type usually should rely on the synthesized default constructor *only* if all such members have member initializers.

A third reason that some classes must define their own default constructor is that sometimes the compiler is unable to synthesize one. For example, if class `C` has a member, mem, that has a class type M, and M doesn't have a default constructor, then the compiler can't initialize mem. Therefore, the compiler cannot synthesize a default constructor for class C. C will only have a default constructor if it explicitly defines one itself. We'll see in § 13.1.6 (p. 564) additional circumstances that prevent the compiler from generating an appropriate default constructor.

Best Practices

For reasons we'll cover in § 7.5.3 (p. 306), classes generally should provide a default constructor.

### Defining the `Sales_data` Constructors

For Sales_data, we'll define four constructors with the following parameters:

- An istream& from which to read a transaction.
- A const string& representing an ISBN, an unsigned representing the count of how many books were sold, and a double representing the price at which the books sold.
- A const string& representing an ISBN. This constructor will use default values for the other members.
- An empty parameter list (i.e., the default constructor) which, as we've just seen, we must define because we have defined other constructors.

Adding these members to our class, we now have

```
struct Sales_data {
    // constructors added
    Sales_data() = default;
    Sales_data(const std::string &s): bookNo(s) { }
    Sales_data(const std::string &s, unsigned n, double p):
               bookNo(s), units_sold(n), revenue(p*n) { }
    Sales_data(std::istream &);
    // other members as before
    std::string isbn() const { return bookNo; }
    Sales_data& combine(const Sales_data&);
    double avg_price() const;
    std::string bookNo;
    unsigned units_sold = 0;
    double revenue = 0.0;
};
```

### What `= default` Means

We'll start by explaining the default constructor:

```
Sales_data() = default;
```

First, note that this constructor defines the default constructor because it takes no arguments. We are defining this constructor *only* because we want to provide other constructors as well as the default constructor. We want this constructor to do exactly the same work as the synthesized version we had been using.

We ask the compiler to generate a constructor with the default behavior by writing **= default** after the parameter list. The = default can appear inside the class body or on the definition outside the class body. Specifying = default is a defintion; as with any other function, if the = default appears inside the class body, the constructor will be inlined.

WARNING

The synthesized constructor works for Sales_data only because the class has initializers for the members with built-in type.

## Constructor Initializer List

Next we'll look at the other two constructors that were defined inside the class:

```
Sales_data(const std::string &s): bookNo(s) { }
Sales_data(const std::string &s, unsigned n, double p):
          bookNo(s), units_sold(n), revenue(p*n) { }
```

The new parts in these definitions are the colon and the code between it and the curly braces that define the (empty) function bodies. This new part is a **constructor initializer list**, which specifies initial values for one or more data members of the object being created. The constructor initializer is a list of member names, each of which is followed by that member's initial value in parentheses (or inside curly braces). Multiple member initializations are separated by commas.

The constructor that has three parameters uses its first two parameters to initialize the bookNo and units_sold members. The initializer for revenue is calculated by multiplying the number of books sold by the price per book.

The constructor that has a single string parameter uses that string to initialize bookNo but does not explicitly initialize the units_sold and revenue members. When a member is omitted from the constructor initializer list, it is implicitly initialized using the same process as is used by the synthesized default constructor. In this case, those members are initialized by their member initializers. Thus, the constructor that takes a string is equivalent to

```
// has the same behavior as the original constructor defined above
Sales_data(const std::string &s):
          bookNo(s), units_sold(0), revenue(0){ }
```

It is usually best for a constructor to use a member initializer if one exists that gives the member the correct value.

Best Practices

Constructors should initialize members of built-in type that do not have in-class initializers. Members with in-class initializers should be initialized explicitly only if a different initial value is required.

It is worth noting that both constructors have empty function bodies. The only work these constructors need to do is give the data members their values. If there is no further work, then the function body is empty.

### Defining a Constructor outside the Class Body

Unlike our other constructors, the constructor that takes an `istream` does have work to do. Inside its function body, this constructor calls `read` to give the data members new values:

```
Sales_data::Sales_data(std::istream &is)
{
    read(is, *this); // read will read a transaction from is into this object
}
```

Constructors have no return type, so this definition starts with the name of the function that we are defining. As with any other member function, when we define a constructor outside of the class body, we must specify the class of which the constructor is a member. Thus, `Sales_data::Sales_data` says that we're defining the `Sales_data` member named `Sales_data`. This member is a constructor because it has the same name as its class.

In this constructor the constructor initializer list is empty. Even though the constructor initializer list is empty, the members of this object are still initialized before the constructor body is executed.

Members that do not appear in the constructor initializer list are initialized by the corresponding member initializer (if there is one) or are default initialized. For `Sales_data` that means that when the function body starts executing, `bookNo` will be the empty `string`, and `units_sold` and `revenue` will both be `0`.

To understand the call to `read`, remember that `read`'s second parameter is a reference to a `Sales_data` object. In § 7.1.2 (p. 273), we noted that we use `this` to access the object as a whole, rather than a member of the object. In this case, we use `*this` to pass "this" object as an argument to the `read` function.

#### EXERCISES SECTION 7.1.4

**Exercise 7.11:** Add constructors to your `Sales_data` class and write a program to use each of the constructors.

**Exercise 7.12:** Move the definition of the `Sales_data` constructor that takes an `istream` into the body of the `Sales_data` class.

**Exercise 7.13:** Rewrite the program from page 269 to use the `istream` constructor.

**Exercise 7.14:** Write a version of the default constructor that explicitly initializes the members to the values we have provided as member initializers.

**Exercise 7.15:** Add appropriate constructors to your `Person` class.

### 7.1.5 Copy, Assignment, and Destruction

In addition to defining how objects of the class type are initialized, classes also control what happens when we copy, assign, or destroy objects of the class type.

Objects are copied in several contexts, such as when we copy-initialize a variable or when we pass or return an object by value (§ 6.2.1, p. 221, and § 6.3.2, p. 229). Objects are assigned when we use the assignment operator (§ 4.4, p. 153). Objects are destroyed when they cease to exist, such as when a local object is destroyed on exit from the block in which it was created (§ 6.1.1, p. 216). Objects stored in a `vector` (or an array) are destroyed when that `vector` (or array) is destroyed.

If we do not define these operations, the compiler will synthesize them for us. Ordinarily, the versions that the compiler generates for us execute by copying, assigning, or destroying each member of the object. For example, in our bookstore program in § 7.1.1 (p. 269), when the compiler executes this assignment

```
total = trans;                // process the next book
```

it executes as if we had written

```
// default assignment for Sales_data is equivalent to:
total.bookNo = trans.bookNo;
total.units_sold = trans.units_sold;
total.revenue = trans.revenue;
```

We'll show how we can define our own versions of these operations in Chapter 13.

### Some Classes Cannot Rely on the Synthesized Versions

Although the compiler will synthesize the copy, assignment, and destruction operations for us, it is important to understand that for some classes the default versions do not behave appropriately. In particular, the synthesized versions are unlikely to work correctly for classes that allocate and manage resources that reside outside the class objects themselves.

It is worth noting that the synthesized versions for copy, assignment, and destruction work correctly for classes that have `vector` or `string` members. When we copy or assign an object that has a `vector` member, the `vector` class takes care of copying or assigning the elements in that member. When the object is destroyed, the `vector` member is destroyed, which in turn destroys the elements in the `vector`. Similarly for `string`s.

**WARNING** Until you know how to define the operations covered in Chapter 13, the resources that your classes allocate should be stored directly as data members of the class.

## 7.2 Access Control and Encapsulation

At this point, we have defined an interface for our class; but nothing forces users to use that interface. Our class is not yet encapsulated—users can reach inside a `Sales_data` object and meddle with its implementation. In C++ we use **access specifiers** to enforce encapsulation:

- Members defined after a **`public`** specifier are accessible to all parts of the program. The `public` members define the interface to the class.

- Members defined after a **`private`** specifier are accessible to the member functions of the class but are not accessible to code that uses the class. The `private` sections encapsulate (i.e., hide) the implementation.

Redefining `Sales_data` once again, we now have

```
class Sales_data {      // note use of class instead of struct
public:                 //  access specifier added
    Sales_data() = default;
    Sales_data(const std::string &s, unsigned n, double p):
               bookNo(s), units_sold(n), revenue(p*n) { }
    Sales_data(const std::string &s): bookNo(s) { }
    Sales_data(std::istream&);
    std::string isbn() const { return bookNo; }
    Sales_data &combine(const Sales_data&);
private:                //  access specifier added
    double avg_price() const
        { return units_sold ? revenue/units_sold : 0; }
    std::string bookNo;
    unsigned units_sold = 0;
    double revenue = 0.0;
};
```

The constructors and member functions that are part of the interface (e.g., `isbn` and `combine`) follow the `public` specifier; the data members and the functions that are part of the implementation follow the `private` specifier.

A class may contain zero or more access specifiers, and there are no restrictions on how often an access specifier may appear. Each access specifier specifies the access level of the succeeding members. The specified access level remains in effect until the next access specifier or the end of the class body.

## Using the `class` or `struct` Keyword

We also made another, more subtle, change: We used the **`class` keyword** rather than **`struct`** to open the class definition. This change is strictly stylistic; we can define a class type using either keyword. The only difference between `struct` and `class` is the default access level.

A class may define members before the first access specifier. Access to such members depends on how the class is defined. If we use the `struct` keyword, the members defined before the first access specifier are `public`; if we use `class`, then the members are `private`.

As a matter of programming style, when we define a class intending for all of its members to be `public`, we use `struct`. If we intend to have `private` members, then we use `class`.

Note

The *only* difference between using `class` and using `struct` to define a class is the default access level.

**EXERCISES SECTION 7.2**

**Exercise 7.16:** What, if any, are the constraints on where and how often an access specifier may appear inside a class definition? What kinds of members should be defined after a `public` specifier? What kinds should be `private`?

**Exercise 7.17:** What, if any, are the differences between using `class` or `struct`?

**Exercise 7.18:** What is encapsulation? Why is it useful?

**Exercise 7.19:** Indicate which members of your `Person` class you would declare as `public` and which you would declare as `private`. Explain your choice.

### 7.2.1 Friends

Now that the data members of `Sales_data` are `private`, our `read`, `print`, and `add` functions will no longer compile. The problem is that although these functions are part of the `Sales_data` interface, they are not members of the class.

A class can allow another class or function to access its non`public` members by making that class or function a **friend**. A class makes a function its friend by including a declaration for that function preceded by the keyword `friend`:

```
class Sales_data {
// friend declarations for nonmember Sales_data operations added
friend Sales_data add(const Sales_data&, const Sales_data&);
friend std::istream &read(std::istream&, Sales_data&);
friend std::ostream &print(std::ostream&, const Sales_data&);
// other members and access specifiers as before
public:
    Sales_data() = default;
    Sales_data(const std::string &s, unsigned n, double p):
               bookNo(s), units_sold(n), revenue(p*n) { }
    Sales_data(const std::string &s): bookNo(s) { }
    Sales_data(std::istream&);
    std::string isbn() const { return bookNo; }
    Sales_data &combine(const Sales_data&);
private:
    std::string bookNo;
    unsigned units_sold = 0;
    double revenue = 0.0;
};
// declarations for nonmember parts of the Sales_data interface
Sales_data add(const Sales_data&, const Sales_data&);
std::istream &read(std::istream&, Sales_data&);
std::ostream &print(std::ostream&, const Sales_data&);
```

Friend declarations may appear only inside a class definition; they may appear anywhere in the class. Friends are not members of the class and are not affected by the access control of the section in which they are declared. We'll have more to say about friendship in § 7.3.4 (p. 293).

Ordinarily it is a good idea to group friend declarations together at the beginning or end of the class definition.

**KEY CONCEPT: BENEFITS OF ENCAPSULATION**

**Encapsulation provides two important advantages:**

- **User code cannot inadvertently corrupt the state of an encapsulated object.**
- **The implementation of an encapsulated class can change over time without requiring changes in user code.**

**By defining data members as `private`, the class author is free to make changes in the data. If the implementation changes, only the class code needs to be examined to see what effect the change may have. User code needs to change only when the interface changes. If the data are `public`, then any code that used the old data members would have to be found and rewritten.**

**Another advantage of making data members `private` is that the data are protected from mistakes that users might introduce. If there is a bug that corrupts an object's state, the places to look for the bug are localized: Only code that is part of the implementation could be responsible for the error. The search for the mistake is limited, greatly easing the problems of maintenance and program correctness.**

**Although user code need not change when a class definition changes, the source files that use a class must be recompiled any time the class changes.**

### Declarations for Friends

A friend declaration only specifies access. It is not a general declaration of the function. If we want users of the class to be able to call a friend function, then we must also declare the function separately from the friend declaration.

To make a friend visible to users of the class, we usually declare each friend (outside the class) in the same header as the class itself. Thus, our `Sales_data` header should provide separate declarations (aside from the friend declarations inside the class body) for `read`, `print`, and `add`.

Many compilers do not enforce the rule that friend functions must be declared *outside* the class before they can be used.

Some compilers allow calls to a `friend` function when there is no ordinary declaration for that function. Even if your compiler allows such calls, it is a good idea to provide separate declarations for `friends`. That way you won't have to change your code if you use a compiler that enforces this rule.

## 7.3 Additional Class Features

The `Sales_data` class is pretty simple, yet it allowed us to explore quite a bit of the language support for classes. In this section, we'll cover some additional class-

**EXERCISES SECTION 7.2.1**

**Exercise 7.20:** When are friends useful? Discuss the pros and cons of using friends.

**Exercise 7.21:** Update your `Sales_data` class to hide its implementation. The programs you've written to use `Sales_data` operations should still continue to work. Recompile those programs with your new class definition to verify that they still work.

**Exercise 7.22:** Update your `Person` class to hide its implementation.

related features that `Sales_data` doesn't need to use. These features include type members, member initializers for members of class type, `mutable` data members, `inline` member functions, returning `*this` from a member function, more about how we define and use class types, and class friendship.

## 7.3.1 Class Members Revisited

To explore several of these additional features, we'll define a pair of cooperating classes named `Screen` and `Window_mgr`.

### Defining a Type Member

A `Screen` represents a window on a display. Each `Screen` has a `string` member that holds the `Screen`'s contents, and three `string::size_type` members that represent the position of the cursor, and the height and width of the screen.

In addition to defining data and function members, a class can define its own local names for types. Type names defined by a class are subject to the same access controls as any other member and may be either `public` or `private`:

```
class Screen {
public:
    using pos = std::string::size_type;
private:
    pos cursor = 0;
    pos height = 0, width = 0;
    std::string contents;
};
```

We defined `pos` in the `public` part of `Screen` because we want users to use that name. Users of `Screen` shouldn't know that `Screen` uses a `string` to hold its data. By defining `pos` as a `public` member, we can hide this detail of how `Screen` is implemented.

For reasons we'll explain in § 7.4.1 (p. 299), unlike ordinary members, members that define types must appear before they are used. As a result, type members usually appear at the beginning of the class.

## Member Functions of class `Screen`

To make our class more useful, we'll add a constructor that will let users define the size and contents of the screen, along with members to move the cursor and to get the character at a given location:

```
class Screen {
public:
    using pos = std::string::size_type;
    Screen() = default;   // needed because Screen has another constructor
    //  cursor initialized to 0 by its in-class initializer
    Screen(pos ht, pos wd, char c): height(ht), width(wd),
                                    contents(ht * wd, c) { }
    char get() const              // get the character at the cursor
        { return contents[cursor]; }       // implicitly inline
    inline char get(pos ht, pos wd) const; // explicitly inline
    Screen &move(pos r, pos c);       // can be made inline later
private:
    pos cursor = 0;
    pos height = 0, width = 0;
    std::string contents;
};
```

Because we have provided a constructor, the compiler will not automatically generate a default constructor for us. If our class is to have a default constructor, we must say so explicitly. In this case, we use `= default` to ask the compiler to synthesize the default constructor's definition for us (§ 7.1.4, p. 278).

It's also worth noting that our second constructor (that takes three arguments) implicitly uses the member initializer for the `cursor` member (§ 7.1.4, p. 280). If our class did not have an member initializer for `cursor`, we would have explicitly initialized `cursor` along with the other members.

## Making Members `inline`

Classes often have small functions that can benefit from being inlined. As we've seen, member functions defined inside the class are automatically `inline` (§ 6.5.1, p. 246). Thus, `Screen`'s constructors and the version of `get` that returns the character denoted by the cursor are `inline` by default.

We can explicitly declare a member function as `inline` as part of its declaration inside the class body. Alternatively, we can specify `inline` on the function definition that appears outside the class body. Our `Screen` class uses both forms: it defines the `move` function as `inline` outside the class, and declares the two-parameter version of `get` as `inline` in the class:

```
inline                        // we can specify inline on the definition
Screen &Screen::move(pos r, pos c)
{
    pos row = r * width; // compute the row location
    cursor = row + c;    // move cursor to the column within that row
    return *this;        // return this object as an lvalue
}
```

```
char Screen::get(pos r, pos c) const // declared as inline in the class
{
    pos row = r * width;        // compute row location
    return contents[row + c]; // return character at the given column
}
```

Although we are not required to do so, it is legal to specify `inline` on both the declaration and the definition. However, specifying `inline` only on the definition outside the class can make the class easier to read.

> **Note:** For the same reasons that we define `inline` functions in headers (§ 6.5.1, p. 248), `inline` member functions should be defined in the same header as the corresponding class definition.

## Overloading Member Functions

Like nonmember functions, member functions may be overloaded (§ 6.4, p. 235) so long as the functions differ by the number and/or types of parameters. The same function-matching (§ 6.4.1, p. 239) process is used for calls to member functions as for nonmember functions.

For example, our `Screen` class defined two versions of `get`. One version returns the character currently denoted by the cursor; the other returns the character at a given position specified by its row and column. The compiler uses the number of arguments to determine which version to run:

```
Screen myscreen;
char ch = myscreen.get();// calls Screen::get()
ch = myscreen.get(0,0);  // calls Screen::get(pos, pos)
```

## Initializers for Data Members of Class Type

In addition to defining the `Screen` class, we'll define a window manager class that represents a collection of `Screens` on a given display. This class will have a `vector` of `Screens` in which each element represents a particular `Screen`. By default, we'd like our `Window_mgr` class to start up with a single, default-initialized `Screen`. The best way to specify this default value is as an member initializer (§ 2.6.1, p. 74):

```
class Window_mgr {
private:
    // Screens this Window_mgr is tracking
    // by default, a Window_mgr has one standard sized blank Screen
    std::vector<Screen> screens{Screen(24, 80, ' ')};
};
```

When we initialize a member of class type, we are supplying arguments to a constructor of that member's type. In this case, we list initialize our `vector` member (§ 3.3.1, p. 102) with a single element initializer. That initializer contains a `Screen` value that is passed to the `vector<Screen>` constructor to create a one-element vector. That value is created by the Screen constructor that takes two size parameters and a character to create a blank screen of the given size.

As we've seen, member initializers must use either the = form of initialization (which we used when we initialized the the data members of Screen) or the direct form of initialization using curly braces (as we do for screens).

Note: When we provide an member initializer, we must do so following an = sign or inside braces.

### Exercises Section 7.3.1

**Exercise 7.23:** Write your own version of the Screen class including the move and get functions.

**Exercise 7.24:** Give your Screen class three constructors: a default constructor; a constructor that takes values for height and width and initializes the contents to hold the given number of blanks; and a constructor that takes values for height, width, and a character to use as the contents of the screen.

**Exercise 7.25:** Can Screen safely rely on the default versions of copy and assignment? If so, why? If not, why not?

**Exercise 7.26:** Define Sales_data::avg_price as an inline function.

**Exercise 7.27:** It is legal but redundant to refer to members through the this pointer. Discuss the pros and cons of explicitly using the this pointer to access members.

## 7.3.2 Functions That Return *this

Next we'll add functions to set the character at the cursor or at a given location:

```
class Screen {
public:
    Screen &set(char);
    Screen &set(pos, pos, char);
    // other members as before
};
inline Screen &Screen::set(char c)
{
    contents[cursor] = c; // set the new value at the current cursor location
    return *this;         // return this object as an lvalue
}
inline Screen &Screen::set(pos r, pos col, char ch)
{
    contents[r*width + col] = ch;  // set specified location to given value
    return *this;                  // return this object as an lvalue
}
```

Like the `move` operation, our `set` members return a reference to the object on which they are called (§ 7.1.2, p. 273). Functions that return a reference are lvalues (§ 6.3.2, p. 230), which means that they return the object itself, not a copy of the object. If we concatenate a sequence of these actions into a single expression:

```
// move the cursor to a given position, and set that character
myScreen.move(4,0).set('#');
```

these operations will execute on the same object. In this expression, we first `move` the `cursor` inside `myScreen` and then `set` a character in `myScreen`'s `contents` member. That is, this statement is equivalent to

```
myScreen.move(4,0);
myScreen.set('#');
```

Had we defined `move` and `set` to return `Screen`, rather than `Screen&`, this statement would execute quite differently. In this case it would be equivalent to:

```
// if move returns Screen not Screen&
Screen temp = myScreen.move(4,0);   // the return value would be copied
temp.set('#'); // the contents inside myScreen would be unchanged
```

If `move` had a nonreference return type, then the return value of `move` would be a copy of `*this` (§ 6.3.2, p. 229). The call to `set` would change the temporary copy, not `myScreen`.

## Returning `*this` from a `const` Member Function

Next, we'll add an operation, which we'll name `display`, to print the contents of the `Screen`. We'd like to be able to include this operation in a sequence of `set` and `move` operations. Therefore, like `set` and `move`, our `display` function will return a reference to the object on which it executes.

Logically, displaying a `Screen` doesn't change the object, so we should make `display` a `const` member. Moreover, `display` must be a `const` member if we want to be able to print the contents of a `const Screen`. However, if `display` is a `const` member, then `this` is a pointer to `const` and `*this` is a `const` object. Hence, the return type of `display` must be `const Screen&`. However, if `display` returns a reference to `const`, we won't be able to embed `display` into a series of actions:

```
Screen myScreen;
// if display returns a const reference, the call to set is an error
myScreen.display(cout).set('*');
```

Even though `myScreen` is a nonconst object, the call to `set` won't compile. The problem is that the `const` version of `display` returns a reference to `const` and we cannot call `set` on a `const` object.

> **Note** A `const` member function that returns `*this` as a reference should have a return type that is a reference to `const`.

## Overloading Based on `const`

We can overload a member function based on whether it is `const` for the same reasons that we can overload a function based on whether a pointer parameter points to `const` (§ 6.4.4, p. 245). When we call the function on a `const` object, only the `const` version will be viable. We could call either the `const` or non`const` version on a non`const` object, but the non`const` version will be a better match.

In this example, we'll define a `private` member named `do_display` to do the actual work of printing the `Screen`. Each of the `display` operations will call this function and then return the object on which it is executing:

```
class Screen {
public:
     // display overloaded on whether the object is const or not
     Screen &display(std::ostream &os)
                   { do_display(os); return *this; }
     const Screen &display(std::ostream &os) const
                   { do_display(os); return *this; }
private:
      // function to do the work of displaying a Screen
      void do_display(std::ostream &os) const {os << contents;}
     // other members as before
};
```

As in any other context, when one member calls another the `this` pointer is passed implicitly. Thus, when `display` calls `do_display`, its own `this` pointer is implicitly passed to `do_display`. When the non`const` version of `display` calls `do_display`, its `this` pointer is implicitly converted from a pointer to non`const` to a pointer to `const` (§ 4.11.2, p. 172).

When `do_display` completes, the `display` functions each return the object on which they execute by dereferencing `this`. In the non`const` version, `this` points to a non`const` object, so that version of `display` returns an ordinary (non`const`) reference; the `const` member returns a reference to `const`.

When we call `display` on an object, whether that object is `const` determines which version of `display` is called:

```
Screen myScreen(5,3);
const Screen blank(5, 3);
myScreen.set('#').display(cout);   // calls nonconst version
blank.display(cout);               // calls const version
```

## `mutable` Data Members

It sometimes (but not very often) happens that a class has a data member that we want to be able to modify, even inside a `const` member function. We indicate such members by including the `mutable` keyword in their declaration. A **`mutable` data member** is never `const`, even when it is a member of a `const` object. Accordingly, a `const` member function may change a `mutable` member.

As an example, it might be logical for our class to allow the cursor position to be moved even if the underlying `Screen` is `const`. After all, moving the cur-

**ADVICE: USE PRIVATE UTILITY FUNCTIONS FOR COMMON CODE**

**Some readers might be surprised that we bothered to define a separate `do_display` operation. After all, the calls to `do_display` aren't much simpler than the action done inside `do_display`. Why bother? We do so for several reasons:**

- **A general desire to avoid writing the same code in more than one place.**
- **We expect that the `display` operation will become more complicated as our class evolves. As the actions involved become more complicated, it makes more obvious sense to write those actions in one place, not two.**
- **It is likely that we might want to add debugging information to `do_display` during development that would be eliminated in the final product version of the code. It will be easier to do so if only one definition of `do_display` needs to be changed to add or remove the debugging code.**
- **There needn't be any overhead involved in this extra function call. We defined `do_display` inside the class body, so it is implicitly `inline`. Thus, there likely be no run-time overhead associating with calling `do_display`.**

**In practice, well-designed C++ programs tend to have lots of small functions such as `do_display` that are called to do the "real" work of some other set of functions.**

sor doesn't change the contents of a screen, it just gives us access to a particular character on the display. We might accommodate this new design as follows:

```
class Screen {
public:
    // other members as before
    Screen &move(pos r, pos c);
    const Screen &move(pos r, pos c) const;  // new const version
private:
    mutable pos cursor = 0;                  // mutable added
    pos height = 0, width = 0;
    std::string contents;
};
```

Because `cursor` is `mutable`, we can write a `const` version of `move` (which we leave as an exercise) that does the same work as our original `move` (§ 7.3.1, p. 286) but, which can be run on `const` objects:

```
const Screen cs (myScreen); // copy myScreen into cs, cs is const
// move changes the value of cursor, but cursor is mutable
cs.move(1,2); // ok: calls const version of move
cs.set('#');  // error: cs is const
```

## 7.3.3 Class Types

Every class defines a unique type. Two different classes define two different types even if they define the same members. For example:

### EXERCISES SECTION 7.3.2

**Exercise 7.28:** Add the `set`, and `display` operations to your version of `Screen`. Test your class by executing the following code:

```
Screen myScreen(5, 5, 'X');
myScreen.move(4,0).set('#').display(cout); cout << "\n";
myScreen.display(cout); cout << "\n";
```

**Exercise 7.29:** What would happen in the previous exercise if the return type of `move`, `set`, and `display` were `Screen` rather than `Screen&`?

**Exercise 7.30:** Revise your `Screen` class so that `move`, `set`, and `display` functions return `Screen` and check your prediction from the previous exercise.

**Exercise 7.31:** Revise your class so that `cursor` is `mutable` and `move` is `const`. Remember, you'll need to define `const` and nonconst versions of `move`.

**Exercise 7.32:** What does the following code do:

```
const Screen cs (myScreen); // copy myScreen into cs, cs is const
```

```
struct First {
    int memi;
    int getMem();
};
struct Second {
    int memi;
    int getMem();
};
First obj1;
Second obj2 = obj1; // error: obj1 and obj2 have different types
```

*Note*

Even if two classes have exactly the same member list, they are different types. The members of each class are distinct from the members of any other class (or any other scope).

We can refer to a class type directly, by using the class name as a type name. Alternatively, we can use the class name following the keyword `class` or `struct`:

```
Sales_data item1;       // default-initialized object of type Sales_data
class Sales_data item1; // equivalent declaration
```

Both methods of referring to a class type are equivalent. The second method is inherited from C and is also valid in C++.

## Class Declarations

Just as we can declare a function apart from its definition (§ 6.1.2, p. 218), we can also declare a class without defining it:

```
class Screen; // declaration of the Screen class
```

This declaration, sometimes referred to as a **forward declaration**, introduces the name `Screen` into the program and indicates that `Screen` refers to a class type. After a declaration and before a definition is seen, the type `Screen` is an **incomplete type**—it's known that `Screen` is a class type but not known what members that type contains.

We can use an incomplete type in only limited ways: We can define pointers or references to such types, and we can declare (but not define) functions that use an incomplete type as a parameter or return type.

A class must be defined—not just declared—before we can write code that creates objects of that type. Otherwise, the compiler does not know how much storage such objects need. Similarly, the class must be defined before a reference or pointer is used to access a member of the type. After all, if the class has not been defined, the compiler can't know what members the class has.

With one exception that we'll describe in § 7.7 (p. 315), data members can be specified to be of a class type only if the class has been defined. The type must be complete because the compiler needs to know how much storage the data member requires. Because a class is not defined until its class body is complete, a class cannot have data members of its own type. However, a class is considered declared (but not yet defined) as soon as its class name has been seen; as with any other incomplete type, a class can have data members that are pointers or references to its own type:

```
class Link_screen {
    Screen window;
    Link_screen *next;
    Link_screen *prev;
};
```

**EXERCISES SECTION 7.3.3**

**Exercise 7.33:** Define a pair of classes `X` and `Y`, in which `X` has a pointer to `Y`, and `Y` has an object of type `X`.

### 7.3.4 Friendship Revisited

Our `Sales_data` class defined three ordinary nonmember functions (`add`, `read`, and `print`) as friends (§ 7.2.1, p. 283). A class can also make another class its friend or it can declare specific member functions of another (previously defined) class as friends. In addition, a friend function can be defined inside the class body. Such functions are implicitly `inline`.

#### Friendship between Classes

As an example of class friendship, our `Window_mgr` class (§ 7.3.1, p. 287) will have members that will need access to the internal data of the `Screen` objects it manages. For example, let's assume that we want to add a member, named `clear` to `Window_mgr` that will reset the contents of a particular `Screen` to all blanks. To do this job, `clear` needs to access the `private` data members of `Screen`. To allow this access, `Screen` can designate `Window_mgr` as its friend:

```
class Screen {
    // Window_mgr members can access the private parts of class Screen
    friend class Window_mgr;
    // ... rest of the Screen class
};
```

The member functions of a friend class can access all the members, including the non`public` members, of the class granting friendship. Now that `Window_mgr` is a friend of `Screen`, we can write the `clear` member of `Window_mgr` as follows:

```
class Window_mgr {
public:
    // location ID for each screen on the window
    using ScreenIndex = std::vector<Screen>::size_type;
    // reset the Screen at the given position to all blanks
    void clear(ScreenIndex);
private:
    std::vector<Screen> screens{Screen(24, 80, ' ')};
};
void Window_mgr::clear(ScreenIndex i)
{
    // s is a reference to the Screen we want to clear
    Screen &s = screens[i];
    // reset the contents of that Screen to all blanks
    s.contents = string(s.height * s.width, ' ');
}
```

We start by defining `s` as a reference to the `Screen` at position `i` in the `screens` `vector`. We then use the `height` and `width` members of that `Screen` to compute a new `string` that has the appropriate number of blank characters. We assign that string of blanks to the `contents` member.

If `clear` were not a friend of `Screen`, this code would not compile. The `clear` function would not be allowed to use the `height`, `width`, or `contents` members of `Screen`. Because `Screen` grants friendship to `Window_mgr`, all the members of `Screen` are accessible to the functions in `Window_mgr`.

It is important to understand that friendship is not transitive. That is, if class `Window_mgr` has its own friends, those friends have no special access to `Screen`.

> Note: Each class controls which classes or functions are its friends.

### Making a Member Function a Friend

Rather than making the entire `Window_mgr` class a friend, `Screen` could restrict its friendship to just the `clear` member. When we declare a member function to be a friend, we must specify the class of which that function is a member:

```
class Screen {
    // Window_mgr::clear must have been declared before class Screen
    friend void Window_mgr::clear(ScreenIndex);
    // ... rest of the Screen class
};
```

Making a member function a friend requires careful structuring of our programs to accommodate interdependencies among the declarations and definitions. In this example, we must order our program as follows:

- Define the `Window_mgr` class, which declares, but cannot define, `clear`. `Screen` must be declared before `clear` can use the members of `Screen`.
- Then, define class `Screen`, including a friend declaration for `clear`.
- Finally, define `clear`, which can now refer to the members in `Screen`.

## Overloaded Functions and Friendship

Although overloaded functions share a common name, they are still different functions. Therefore, a class must declare as a friend each function in a set of overloaded functions that it wishes to make a friend:

```
// overloaded storeOn functions
std::ostream& storeOn(std::ostream &, Screen &);
std::string& storeOn(std::string &, Screen &);
class Screen {
    // ostream version of storeOn may access the private parts of Screen objects
    friend std::ostream& storeOn(std::ostream &, Screen &);
    // ...
};
```

Class `Screen` makes the version of `storeOn` that takes an `ostream&` its friend. The version that takes a `string&` has no special access to `Screen`.

## Friend Declarations and Scope

Classes and nonmember functions need not have been declared before they are used in a friend declaration. When a name first appears in a friend declaration, that name is implicitly *assumed* to be part of the surrounding scope. However, the friend itself is not actually declared in that scope (§ 7.2.1, p. 284). That is, a `friend` declaration specifies access; it is not a declaration in an ordinary sense. Even if we define the function inside the class, we must still provide a declaration outside of the class itself to make that function visible. A declaration must exist even if we only call the friend from members of the friendship granting class:

```
struct X {
    friend void f() { /* friend function can be defined in the class body */ }
    X() { f(); } // error: no declaration for f
    void g();
    void h();
};
```

```
void X::g() { return f(); } // error: f hasn't been declared
void f();                   // declares the function defined inside X
void X::h() { return f(); } // ok: declaration for f is now in scope
```

Note

Remember, some compilers do not enforce the lookup rules for friends (§ 7.2.1, p. 284).

**EXERCISES SECTION 7.3.4**

**Exercise 7.34:** Define your own version of `Window_mgr` and update your `Screen` class as necessary.

## 7.4 Class Scope

Every class defines its own new scope. Outside the class scope, ordinary data and function members may be accessed only through an object, a reference, or a pointer using a member access operator (§ 4.6.2, p. 160). Type members may be accessed using the scope operator. In either case, the name that follows the operator must be a member of the associated class.

```
Screen::pos ht = 24, wd = 80; // use the pos type defined by Screen
Screen scr(ht, wd, ' ');
Screen *p = &scr;
char c = scr.get(); // fetches the get member from the object scr
c = p->get();       // fetches the get member from the object to which p points
```

### Scope and Members Defined outside the Class

The fact that a class is a scope explains why we must provide the class name as well as the function name when we define a member function outside its class (§ 7.1.2, p. 273). Outside of the class, the names of the members are hidden.

Once the class name is seen, the remainder of the definition—including the parameter list and the function body—is in the scope of the class. As a result, we can refer to other class members without qualification.

For example, recall the `clear` member of class `Window_mgr` (§ 7.3.4, p. 294). That function's parameter uses a type that is defined by `Window_mgr`:

```
void Window_mgr::clear(ScreenIndex i)
{
    Screen &s = screens[i];
    s.contents = string(s.height * s.width, ' ');
}
```

Because the compiler sees the parameter list after noting that we are in the scope of class `Window_mgr`, there is no need to specify that we want the `ScreenIndex` that is defined by `Window_mgr`. For the same reason, the use of `screens` in the function body refers to name declared inside class `Window_mgr`.

On the other hand, the return type of a function normally appears before the function's name. When a member function is defined outside the class body, names used in the return type are outside the class scope.

As an example, we might give `Window_mgr` a function, named `addScreen`, to add a screen to the display. This member will return a `ScreenIndex` value that the user can subsequently use to locate this `Screen`:

```
class Window_mgr {
public:
    // add a Screen to the window and returns its index
    ScreenIndex addScreen(const Screen&);
    // other members as before
};
// return type is seen after we're in the scope of Window_mgr
auto Window_mgr::addScreen(const Screen &s) -> ScreenIndex
{
    screens.push_back(s);
    return screens.size() - 1;
}
```

In modern C++, the easiest way to use a member type as a return type is to use a trailing return (§ 6.3.3, p. 233). In this definition, `SceenIndex` appears after the name of the function. The compiler—and readers of our code—know that we are in the scope of `WindowMgr`.

To write our function without a trailing return, we'd have to specify the class in which that type is defined:

```
Window_mgr::ScreenIndex Window_mgr::addScreen(const Screen &s)
```

**EXERCISES SECTION 7.4**

**Exercise 7.35:** What would happen if we gave `Screen` a `size` member defined as follows? Fix any problems you identify.

```
pos Screen::size() const
{
    return height * width;
}
```

### 7.4.1 Name Lookup and Class Scope

In the programs we've written so far, **name lookup** (the process of finding which declarations match the use of a name) has been relatively straightforward:

- First, look for a declaration of the name in the block in which the name was used. Only names declared before the use are considered.

- If the name isn't found, look in the enclosing scope(s).
- If no declaration is found, then the program is in error.

The way names are resolved inside member functions defined inside the class may seem to behave differently than these lookup rules. However, in this case, appearances are deceiving. Class definitions are processed in two phases:

- First, the member declarations are compiled.
- Function bodies are compiled only after the entire class has been seen.

> Note: Member function definitions are processed *after* the compiler processes all of the declarations in the class.

Classes are processed in this two-phase way to make it easier to organize class code. Because member function bodies are not processed until the entire class is seen, they can use any name defined inside the class. If function definitions were processed at the same time as the member declarations, then we would have to order the member functions so that they referred only to names already seen.

### Name Lookup for Class Member Declarations

This two-step process applies only to names used in the body of a member function. Names used in declarations, including names used for the return type and types in the parameter list, must be seen before they are used. If a member declaration uses a name that has not yet been seen inside the class, the compiler will look for that name in the scope(s) in which the class is defined. For example:

```
using Money = double;
string bal;
class Account {
public:
    Money balance() { return bal; }
private:
    Money bal;
    // ...
};
```

When the compiler sees the declaration of the `balance` function, it will look for a declaration of `Money` in the `Account` class. The compiler considers only declarations inside `Account` that appear before the use of `Money`. Because no matching member is found, the compiler then looks for a declaration in the enclosing scope(s). In this example, the compiler will find the type alias `Money`. That type will be used for the return type of the function `balance` and as the type for the data member `bal`. On the other hand, the function body of `balance` is processed only after the entire class is seen. Thus, the `return` inside that function returns the member named `bal`, not the `string` from the outer scope.

## Type Names Are Special

Ordinarily, an inner scope can redefine a name from an outer scope even if that name has already been used in the inner scope. However, in a class, if a member uses a name from an outer scope and that name is a type, then the class may not subsequently redefine that name:

```
using Money = double;
class Account {
public:
    Money balance() { return bal; }  // uses Money from the outer scope
private:
    using Money = double; // error: cannot redefine Money
    Money bal;
    // ...
};
```

Even though the definition of `Money` inside `Account` uses the same type as the definition in the outer scope, this code is still in error. However, compilers are not required to diagnose this error and some compilers will quietly accept such code.

> **Tip:** Definitions of type names usually should appear at the beginning of a class. That way any member that uses that type will be seen after the type name has already been defined.

## Normal Block-Scope Name Lookup inside Member Definitions

A name used in the body of a member function is resolved as follows:

- First, look inside the member function. As usual, only declarations in the function body that precede the use of the name are considered.
- If the declaration is not found inside the member function, look for a declaration inside the class. All the members of the class are considered.
- If a declaration for the name is not found in the class, look for a declaration that is in scope before the member function definition.

Ordinarily, it is a bad idea to use the name of another member as the name for a parameter in a member function. However, in order to show how names are resolved, we'll violate that normal practice:

```
// note: this code is for illustration purposes only and reflects bad practice
// it is generally a bad idea to use the same name for a parameter and a member
int height;    // defines a name subsequently used inside Screen
class Screen {
public:
    using pos = std::string::size_type;
    pos dummy1() {
        return width * height; // which height? the member
    }
```

```
    pos dummy2(pos height) {
        return width * height; //   which height? the parameter
    }
private:
    pos cursor = 0;
    pos height = 0, width = 0;
};
```

When the compiler processes the multiplication expression inside `dummy1`, it first looks for the names used in that expression in the scope of that function. Finding no local variable named `height` or `width`, it next looks for a data member with the matching names. In this case, both names are data members and the multiplication uses those data members as operands. In `dummy2`, the same process is repeated. This time, the compiler finds the parameter named `height`. A function's parameters are in the function's scope. Thus, the name `height`, used in the body of `dummy2`, refers to the parameter and `width` to the data member.

In this case, the `height` parameter hides the member named `height`. If we wanted to override the normal lookup rules, we can do so:

```
// bad practice: names local to member functions shouldn't hide member names
void Screen::dummy_fcn(pos height) {
    cursor = width * this->height;    // member height
    // alternative way to indicate the member
    cursor = width * Screen::height; // member height
}
```

> Note
>
> Even though the class member is hidden, it is still possible to use that member by qualifying the member's name with the name of its class or by using the `this` pointer explicitly.

A much better way to ensure that we get the member named `height` would be to give the parameter a different name:

```
// good practice: don't use a member name for a parameter or other local variable
void Screen::dummy_fcn(pos ht) {
    cursor = width * height;  // member height
}
```

In this case, when the compiler looks for the name `height`, it won't be found inside `dummy_fcn`. The compiler next looks at all the declarations in `Screen`. Even though the declaration of `height` appears after its use inside `dummy_fcn`, the compiler resolves this use to the data member named `height`.

## After Class Scope, Look in the Surrounding Scope

If the compiler doesn't find the name in function or class scope, it looks for the name in the surrounding scope. In our example, the name `height` is defined in the outer scope before the definition of `Screen`. However, the object in the outer scope is hidden by our member named `height`. If we want the name from the outer scope, we can ask for it explicitly using the scope operator:

```
// bad practice: don't hide names that are needed from surrounding scopes
void Screen::dummy_fcn(pos height) {
    cursor = width * ::height;// which height? the global one
}
```

Note: Even though the outer object is hidden, it is still possible to access that object by using the scope operator.

### Names Are Resolved Where They Appear within a File

When a member is defined outside its class, the third step of name lookup includes names declared in the scope of the member definition as well as those that appear in the scope of the class definition. For example:

```
int height;   // defines a name subsequently used inside Screen
class Screen {
public:
    using pos = std::string::size_type;
    void setHeight(pos);
    pos height = 0;  // hides the declaration of height in the outer scope
};
Screen::pos verify(Screen::pos);
void Screen::setHeight(pos var) {
    // var: refers to the parameter
    // height: refers to the class member
    // verify: refers to the global function
    height = verify(var);
}
```

Notice that the declaration of the global function `verify` is not visible before the definition of the class `Screen`. However, the third step of name lookup includes the scope in which the member definition appears. In this example, the declaration for `verify` appears before `setHeight` is defined and may, therefore, be used.

# 7.5 Constructors Revisited

We covered the basics of constructors in § 7.1.4 (p. 276). In this section we'll cover some additional capabilities of constructors, and deepen our coverage of the material introduced earlier.

## 7.5.1 Constructor Initializer List

When we define variables, we typically initialize them immediately rather than defining them and then assigning to them:

```
string foo = "Hello World!"; // define and initialize
string bar;                  // default initialized to the empty string
bar = "Hello World!";        // assign a new value to bar
```

## Exercises Section 7.4.1

**Exercise 7.36:** What would happen if we defined `pos` as the last line in the `Screen` class page 299?

**Exercise 7.37:** Explain the following code, indicating which definition of `Type` or `initVal` is used for each use of those names. Say how you would fix any errors.

```
using Type = string;
Type initVal();
class Exercise {
public:
    using Type = double;
    Type setVal(Type);
    Type initVal();
private:
    int val;
};
Type Exercise::setVal(Type parm) {
    val = parm + initVal();
    return val;
}
```

Exactly the same distinction between initialization and assignment applies to the data members of objects. If we do not explicitly initialize a member with a member initializer or in the constructor initializer list, that member is default initialized before the constructor body starts executing. For example:

```
// legal but sloppier way to write the Sales_data constructor: no constructor initializers
Sales_data::Sales_data(const string &s,
                       unsigned cnt, double price)
{
    bookNo = s;
    units_sold = cnt;
    revenue = cnt * price;
}
```

This version and our original definition on page 278 have the same effect: When the constructor finishes, the data members will hold the same values. The difference is that the original version *initializes* its data members, whereas this version *assigns* values to the data members. How significant this distinction is depends on the type of the data member.

### Constructor Initializers Are Sometimes Required

We can often, *but not always*, ignore the distinction between whether a member is initialized or assigned. Members that are `const` or references must be initialized, because there is no way of assigning to them later. Similarly, members that are of a class type that does not define a default constructor also must be initialized. For example:

```
class ConstRef {
public:
    ConstRef(int ii);
private:
    int i;
    const int ci;
    int &ri;
};
```

Like any other `const` object or reference, the members `ci` and `ri` must be initialized. As a result, omitting a constructor initializer for these members is an error:

```
// error: ci and ri must be initialized
ConstRef::ConstRef(int ii)
{              // assignments:
     i = ii;   // ok
     ci = ii;  // error: cannot assign to a const
     ri = i;   // error: ri was never initialized
}
```

By the time the body of the constructor begins executing, initialization is complete. Our only chance to initialize `const` or reference data members is in the constructor initializer. The correct way to write this constructor is

```
// ok: explicitly initialize reference and const members
ConstRef::ConstRef(int ii): i(ii), ci(ii), ri(i) {  }
```

Note

We *must* use the constructor initializer list to provide values for members that are `const`, reference, or of a class type that does not have a default constructor.

**ADVICE: USE CONSTRUCTOR INITIALIZERS**

**In many classes, the distinction between initialization and assignment is strictly a matter of low-level efficiency: A data member is initialized and then assigned when it could have been initialized directly.**

**More important than the efficiency issue is the fact that some data members must be initialized. By routinely using constructor initializers, you can avoid being surprised by compile-time errors when you have a class with a member that requires a constructor initializer.**

## Order of Member Initialization

Not surprisingly, each member may be named only once in the constructor initializer. After all, what might it mean to give a member two initial values?

What may be more surprising is that the constructor initializer list specifies only the values used to initialize the members, not the order in which those initializations are performed.

Members are initialized in the order in which they appear in the class definition: The first member is initialized first, then the next, and so on. The order in which initializers appear in the constructor initializer list does not change the order of initialization.

The order of initialization often doesn't matter. However, if one member is initialized in terms of another, then the order in which members are initialized is crucially important.

As an example, consider the following class:

```
class X {
    int i;
    int j;
public:
    // undefined: i is initialized before j
    X(int val): j(val), i(j) { }
};
```

In this case, the constructor initializer makes it *appear* as if `j` is initialized with `val` and then `j` is used to initialize `i`. However, `i` is initialized first. The effect of this initializer is to initialize `i` with the undefined value of `j`!

Some compilers are kind enough to generate a warning if the data members are listed in the constructor initializer in a different order from the order in which the members are declared.

Best Practices

It is a good idea to write constructor initializers in the same order as the members are declared. Moreover, when possible, avoid using members to initialize other members.

If possible, it is a good idea write member initializers to use the constructor's parameters rather than another data member from the same object. That way we don't even have to think about the order of member initialization. For example, it would be better to write the constructor for `X` as

```
X(int val): i(val), j(val) { }
```

In this version, the order in which `i` and `j` are initialized doesn't matter.

## Default Arguments and Constructors

The actions of the `Sales_data` default constructor are similar to those of the `Sales_data` constructor that takes a single `string` argument (§ 7.1.4, p. 279). The only difference is that the constructor that takes a `string` argument uses that argument to initialize `bookNo`. The default constructor (implicitly) uses the `string` default constructor to initialize `bookNo`.

We can rewrite these constructors as a single constructor with a default argument (§ 6.2.3, p. 224):

```
class Sales_data {
public:
    // defines the default constructor as well as one that takes a string argument
    Sales_data(std::string s = ""): bookNo(s) { }
```

```
    // remaining constructors unchanged
    Sales_data(std::string s, unsigned cnt, double rev):
          bookNo(s), units_sold(cnt), revenue(rev*cnt) { }
    Sales_data(std::istream &is) { read(is, *this); }
    // remaining members as before
};
```

This version of our class provides the same interface as our original on page 278. Both versions create the same object when given no arguments or when given a single `string` argument. Because we can call this constructor with no arguments, this constructor defines a default constructor for our class.

> Note: A constructor that supplies default arguments for all its parameters also defines the default constructor.

It is worth noting that we probably should not use default arguments with the `Sales_data` constructor that takes three arguments. If a user supplies a nonzero count for the number of books sold, we want to ensure that the user also supplies the price at which those books were sold.

## EXERCISES SECTION 7.5.1

**Exercise 7.38:** The following initializer is in error. Identify and fix the problem.

```
struct X {
    X (int i, int j): base(i), rem(base % j) { }
    int rem, base;
};
```

**Exercise 7.39:** Using the version of `Sales_data` from this section, determine which constructor is used to initialize each of the following variables and list the values of the data members in each object:

```
Sales_data first_item(cin);

int main() {
    Sales_data next;
    Sales_data last("9-999-99999-9");
}
```

**Exercise 7.40:** Write a version of the `Sales_data` constructor that takes an `istream&` that uses `cin` as a default argument.

**Exercise 7.41:** Would it be legal for both the constructor that takes a `string` and the one that takes an `istream&` to have default arguments? If not, why not?

**Exercise 7.42:** Choose one of the following abstractions (or an abstraction of your own choosing). Determine what data are needed in the class. Provide an appropriate set of constructors. Explain your decisions.

(a) `Book` (b) `Date` (c) `Employee`
(d) `Vehicle` (e) `Object` (f) `Tree`

### 7.5.2 Delegating Constructors

A **delegating constructor** uses another constructor from its own class to perform its initialization. It is said to "delegate" some (or all) of its work to this other constructor.

Like any other constructor, a delegating constructor has a member initializer list and a function body. In a delegating constructor, the member initializer list has a single entry that is the name of the class itself. As with other member initializers, the name of the class is followed by a parenthesized list of zero or more arguments. The argument list must match another constructor in the class.

As an example, we'll rewrite the `Sales_data` class to use delegating constructors as follows:

```
class Sales_data {
public:
    // nondelegating constructor initializes members from corresponding arguments
    Sales_data(std::string s, unsigned cnt, double price):
            bookNo(s), units_sold(cnt), revenue(cnt*price) { }
    // remaining constructors all delegate to another constructor
    Sales_data(): Sales_data("", 0, 0) {}
    Sales_data(std::string s): Sales_data(s, 0,0) {}
    Sales_data(std::istream &is): Sales_data()
                                    { read(is, *this); }
    // other members as before
};
```

In this version of `Sales_data`, all but one of the constructors delegate their work. The first constructor takes three arguments, uses those arguments to initialize the data members, and does no further work. In this version of the class, we define the default constructor to use the three-argument constructor to do its initialization. It too has no additional work, as indicated by the empty constructor body. The constructor that takes a `string` also delegates to the three-argument version.

The constructor that takes an `istream&` also delegates. It delegates to the default constructor, which in turn delegates to the three-argument constructor. Once those constructors complete their work, the body of the `istream&` constructor is run. Its constructor body calls `read` to read the given `istream`.

When a constructor delegates to another constructor, the constructor initializer list and function body of the delegated-to constructor are both executed. In `Sales_data`, the function bodies of the delegated-to constructors happen to be empty. Had the function bodies contained code, that code would be run before control returned to the function body of the delegating constructor.

### 7.5.3 The Role of the Default Constructor

The default constructor is used automatically whenever an object is default- or value-initialized. Default initialization happens

- When we define non`static` variables (§ 2.2.1, p. 44) or arrays (§ 3.5.1, p. 119) at block scope without initializers

## Exercises Section 7.5.2

**Exercise 7.43:** Rewrite your own version of the `Sales_data` class to use delegating constructors. Add a statement to the body of each of the constructors that prints a message whenever it is executed. Write declarations to construct a `Sales_data` object in every way possible. Study the output until you are certain you understand the order of execution among delegating constructors.

**Exercise 7.44:** For the class you wrote for exercise 7.42 in § 7.5.1 (p. 305), decide whether any of the constructors might use delegation. If so, write the delegating constructor(s) for your class. If not, look at the list of abstractions and choose one that you think would use a delegating constructor. Write the class definition for that abstraction.

- When a class that itself has members of class type uses the synthesized default constructor (§ 7.1.4, p. 276)
- When members of class type are not explicitly initialized in a constructor initializer list (§ 7.1.4, p. 279)

Value initialization happens

- During array initialization when we provide fewer initializers than the size of the array (§ 3.5.1, p. 119)
- When we define a local static object without an initializer (§ 6.1.1, p. 217)
- When we explicitly request value initialization by writing an expressions of the form `T()` where `T` is the name of a type (The `vector` constructor that takes a single argument to specify the `vector`'s size (§ 3.3.1, p. 101) uses an argument of this kind to value initialize its element initializer.)

Classes must have a default constructor in order to be used in these contexts. Most of these contexts should be fairly obvious.

What may be less obvious is the impact on classes that have data members that do not have a default constructor:

```
class NoDefault {
public:
    NoDefault(const std::string&);
    // additional members follow, but no other constructors
};
struct A {   // my_mem is public by default; see § 7.2 (p. 282)
    NoDefault my_mem;
};
A a;         // error: cannot synthesize a constructor for A
struct B {
    B() {}   // error: no initializer for b_member
    NoDefault b_member;
};
```

**Best Practices** In practice, it is almost always right to provide a default constructor if other constructors are being defined.

### Using the Default Constructor

The following declaration of `obj` compiles without complaint. However, when we try to use `obj`

```
Sales_data obj();     // ok: but defines a function, not an object
if (obj.isbn() == Primer_5th_ed.isbn())  // error: obj is a function
```

the compiler complains that we cannot apply member access notation to a function. The problem is that, although we intended to declare a default-initialized object, `obj` actually declares a function taking no parameters and returning an object of type `Sales_data`.

The correct way to define an object that uses the default constructor for initialization is to leave off the trailing, empty parentheses:

```
// ok: obj is a default-initialized object
Sales_data obj;
```

**WARNING** It is a common mistake among programmers new to C++ to try to declare an object initialized with the default constructor as follows:

```
Sales_data obj(); // oops! declares a function, not an object
Sales_data obj2;  // ok: obj2 is an object, not a function
```

#### EXERCISES SECTION 7.5.3

**Exercise 7.45:** Assume we have a class named `NoDefault` that has a constructor that takes an `int`, but has no default constructor. Define a class `C` that has a member of type `NoDefault`. Define the default constructor for `C`.

**Exercise 7.46:** Is the following declaration legal? If not, why not?

```
vector<NoDefault> vec(10);
```

**Exercise 7.47:** What if we defined the `vector` in the previous execercise to hold objects of type `C`?

**Exercise 7.48:** Which, if any, of the following statements are untrue? Why?

(a) A class must provide at least one constructor.
(b) A default constructor is a constructor with an empty parameter list.
(c) If there are no meaningful default values for a class, the class should not provide a default constructor.
(d) If a class does not define a default constructor, the compiler generates one that initializes each data member to the default value of its associated type.

### 7.5.4 Implicit Class-Type Conversions

As we saw in § 4.11 (p. 169), the language defines several automatic conversions among the built-in types. We also noted that classes can define implicit conversions as well. Ordinarily, constructors that can be called with a single argument define implicit conversions *to* a class type. Such constructors are sometimes referred to as **converting constructors**. We'll see in § 14.9 (p. 634) how to define conversions *from* a class type to another type.

Note: An ordinary constructor that can be called with a single argument defines an implicit conversion from that parameter type to the class type.

The `Sales_data` constructors that take a `string` and that take an `istream` both define implicit conversions from those types to `Sales_data`. That is, we can use a `string` or an `istream` where an object of type `Sales_data` is expected:

```
string null_book = "9-999-99999-9";
// uses Sales_data(const string&) to construct a temporary Sales_data object
// and initializes the const Sales_data& parameter in combine to refer to that object
item.combine(null_book);
```

Here we call the `Sales_data` `combine` member function with a `string` argument. This call is perfectly legal; the compiler automatically creates a `Sales_data` object from the given `string`. That newly generated (temporary) `Sales_data` is passed to `combine`. Because `combine`'s parameter is a reference to `const`, we can bind that parameter to a temporary.

#### Only One Class-Type Conversion Is Allowed

In § 4.11.2 (p. 173) we noted that the compiler will automatically apply only one class-type conversion. For example, the following code is in error because it needs two conversions:

```
// error: requires two user-defined conversions:
//     (1) convert "9-999-99999-9" to string
//     (2) convert that (temporary) string to Sales_data
item.combine("9-999-99999-9");
```

If we wanted to make this call, we can do so by explicitly converting the character string to either a `string` or a `Sales_data` object:

```
// ok: explicit conversion to string, implicit conversion to Sales_data
item.combine(string("9-999-99999-9"));
// ok: implicit conversion to string, explicit conversion to Sales_data
item.combine(Sales_data("9-999-99999-9"));
```

#### Class-Type Conversions Are Not Always Useful

Whether the conversion of a `string` to `Sales_data` is desired depends on how we think our users will use the conversion. In this case, there is no obviously right or wrong choice. The conversion from `istream` to `Sales_data` is more questionable:

```
// uses Sales_data(istream&) to construct a temporary Sales_data object
// and initializes the const Sales_data& parameter in combine to refer to that object
item.combine(cin);
```

This code implicitly converts `cin` to `Sales_data`. This conversion executes the `Sales_data` constructor that takes an `istream`. That constructor creates a (temporary) `Sales_data` object by reading the standard input. That object is then passed to `combine`.

This `Sales_data` object is a temporary (§ 2.4.1, p. 62). We have no access to it once `combine` finishes. Effectively, we have read an object from the input that is discarded after we add its value into `item`.

## Suppressing Implicit Conversions Defined by Constructors

We can prevent the use of a constructor in a context that requires an implicit conversion by declaring the constructor as **`explicit`**:

```
class Sales_data {
public:
    Sales_data() = default;
    Sales_data(const std::string &s, unsigned n, double p):
              bookNo(s), units_sold(n), revenue(p*n) { }
    explicit Sales_data(const std::string &s): bookNo(s) { }
    explicit Sales_data(std::istream&);
    // remaining members as before
};
```

Now, neither constructor can be used to implicitly create a `Sales_data` object. Neither of our previous uses will compile:

```
item.combine(null_book);  // error: string constructor is explicit
item.combine(cin);        // error: istream constructor is explicit
```

The `explicit` keyword is used on the constructor declaration inside the class. It is not repeated on a definition made outside the class body:

```
// error: explicit allowed only on a constructor declaration in a class header
explicit Sales_data::Sales_data(istream& is)
{
    read(is, *this);
}
```

## `explicit` Constructors Can Be Used Only for Direct Initialization

Recall that C++ has both direct- and copy-initialization (§ 7.1.4, p. 279). `explicit` constructors are considered only in a direct-initialization context. In addition variable initializations, direct-initialization also happens in other contexts, including casts (§ 4.11.3, p. 173) and in constructor initializers (§ 7.1.4, p. 279).

`explicit` constructors are ignored when we use the copy form of initialization (with an =):

```
Sales_data item1(null_book);  // ok: direct initialization
// error: cannot use the copy form of initialization with an explicit constructor
Sales_data item2 = null_book;
```

They are also ignored in other copy-initialization contexts, which include when we pass or return an object by value from a function (§ 6.2.1, p. 221 and § 6.3.2, p. 229); when a data member is initialized from an in-class member initializer (§ 2.6.1, p. 74), and when we brace initialize elements in an array or members of an aggregate class (§ 7.6.1, p. 312).

Note

`explicit` constructors are not converting constructors; the compiler will *not* use an `explicit` constructor in an automatic conversion.

### Explicitly Using Constructors for Conversions

Although the compiler will not use an `explicit` constructor for an implicit conversion, we can use such constructors explicitly to force a conversion:

```
// ok: the argument is an explicitly constructed Sales_data object
item.combine(Sales_data(null_book));
// ok: static_cast can use an explicit constructor
item.combine(static_cast<Sales_data>(cin));
```

In the first call, we use the `Sales_data` constructor directly. This call constructs a temporary `Sales_data` object using the `Sales_data` constructor that takes a `string`. In the second call, we use a `static_cast` (§ 4.11.3, p. 174) to perform an explicit, rather than an implicit, conversion. In this call, the `static_cast` uses the `istream` constructor to construct a temporary `Sales_data` object.

### Library Classes with `explicit` Constructors

Some of the library classes that we've used have single-parameter constructors:

- The `string` constructor that takes a single parameter of type `const char*` (§ 3.2.1, p. 86) is not `explicit`.
- The `vector` constructor that takes a size (§ 3.3.1, p. 101) is `explicit`.

## 7.6 Aggregate Classes

An **aggregate class** lets users directly access its members. A class is an aggregate if

- All of its data members are `public`
- It does not define (or inherit see § 15.7.4 (p. 683)) any constructors
- It has no `virtual` functions, and no `private`, `protected`, or `virtual` base classes, which are features that we cover in Chapter 15 and § 18.3 (p. 854).

### 7.6.1 Uses and Limitations of Aggregate Classes

Aggregate classes are defined the same way as any other class:

## Exercises Section 7.5.4

**Exercise 7.49:** Explain whether the `Sales_data` constructor that takes a `string` should be `explicit`. What are the benefits of making the constructor `explicit`? What are the drawbacks?

**Exercise 7.50:** Assuming the `Sales_data` constructors are not `explicit`, what operations happen during the following definitions

```
string null_isbn("9-999-99999-9");
Sales_data item1(null_isbn);
Sales_data item2("9-999-99999-9");
```

What happens if the `Sales_data` constructors are `explicit`?

**Exercise 7.51:** For each of the three following declarations of `combine`, explain what happens if we call `i.combine(s)`, where `i` is a `Sales_data` and `s` is a `string`:

(a) `Sales_data &combine(Sales_data);`
(b) `Sales_data &combine(Sales_data&);`
(c) `Sales_data &combine(const Sales_data&) const;`

**Exercise 7.52:** Determine whether any of your `Person` class constructors should be `explicit`.

**Exercise 7.53:** Why do you think `vector` defines its single-argument constructor as `explicit`, but `string` does not?

```
struct Data {
    string s;
    int ival = 42;   // legal in C++14; illegal in C++11
};
```

Whether a class is an aggregate depends solely on whether it meets the constraints outlined above. However, prior to C++14, aggregate classes could not define member initializers. That is, in earlier releases of the language, a class that had member initializers would not be treated as an aggregate.

New

### Aggregate Initialization

We can initialize the data members of an aggregate similarly to how we initialize an array (§ 3.5.1, p. 119), by providing a braced list of member initializers:

```
Data val1 = { "Anna" , 1024 };
```

The initializers must appear in the same order as the the data members are declared. That is, the initializer for the first member is first, for the second is next, and so on. The following, for example, is an error:

```
Data val2 = { 0, "Anna" };
```

The list of initializers must not contain more elements than the class has members. However, the list of initializers may have fewer elements and the list can use an empty pair of curlies in place of an initializer for one or more members. In both cases, an otherwise uninitialized member is initialized by its member initializer (§ 2.6.1, p. 74), if there is one, and is value initialized (§ 3.5.1, p. 119) otherwise:

```
// default initializes Data::s and initializes Data::ival from its member initializer
Data val1 = {};         // val1.s is the empty string and val1.ival is 42
// default initializes Data::s and initializes Data::ival to 0
Data val2 = {{}, 0}; // val2.s is the empty string and val2.ival is 0
```

## Structured Bindings and Aggregate Classes

Just as we can use a structured binding (§ 2.6.2, p. 77) to give names to the elements in an array, we can also use a structured binding to access each member of an aggregate: New

```
Data val1 = { "Anna" , 1024 };
auto [name, age] = val1; // name is a string, age is an int
```

which is equivalent to:

```
// equivalent declarations
auto name = val1.s;
auto age = val1.ival;
```

## Drawbacks of Aggregate Classes

It is worth noting that there are three significant drawbacks to explicitly initializing the members of an object of class type:

- It requires that all the data members of the class be `public`.
- It puts the burden on the user of the class (rather than on the class author) to correctly initialize every member of every object. Such initialization is tedious and error-prone because it is easy to forget an initializer or to supply an inappropriate initializer.
- If a member is added or removed, all initializations have to be updated.

In practice, it is usually a good idea to limit aggregate classes to classes with only a few members.

### EXERCISES SECTION 7.6.1

**Exercise 7.54:** Using our first version of `Sales_data` from § 2.6.1 (p. 73), explain the following initialization. Identify and fix any problems.

```
Sales_data item = {"978-0590353403", 25, 15.99};
```

## 7.6.2 Literal Classes

In § 6.5.1 (p. 247) we noted that the parameters and return type of a `constexpr` function must be literal types. In addition to the arithmetic types, references, and pointers, certain classes are also literal types. An aggregate class (§ 7.6, p. 311) whose data members are all of literal type is a literal class. A nonaggregate class, that meets the following restrictions, is also a literal class:

- All of the data members must have literal type.
- If a data member has a member initializer, the initializer must be a constant expression (§ 2.4.4, p. 66).
- The class must have at least one `constexpr` constructor (described below).
- The class must use the default definition for its destructor, which is the member that destroys objects of the class type (which we cover in § 7.1.5 (p. 280)).

As an example, we'll define a class to hold various debug flags, each of which is of type `bool`. Our class will have members to test these flags as well as constructors to initialize them:

```
class Debug {
public:
    constexpr Debug() = default;
    constexpr Debug(bool b) : hw(b), io(b), other(b) { }
    constexpr Debug(bool h, bool i, bool o):
                                    hw(h), io(i), other(o) { }
    //  bitwise operators see § 4.8 (p. 162)
    constexpr bool any() const { return hw || io || other; }
    constexpr bool hardware() const { return hw || io; }
private:
    bool hw = true;     //  hardware errors other than IO errors
    bool io = true;     //  IO errors
    bool other = true; //  other errors
};
```

### `constexpr` Member Functions

Literal classes often have function members that are `constexpr`. Such members must meet the requirements of a `constexpr` function (§ 6.5.1, p. 247):

```
//  bitwise operators see § 4.8 (p. 162)
constexpr bool any() const { return hw || io || other; }
constexpr bool hardware() const { return hw || io; }
```

WARNING

In C++11, `constexpr` member functions were implicitly `const` (§ 7.1.2, p. 272). In C++14 and later, they are not. If a `constexpr` member function should be a `const` member, it must be explicitly specified as such.

### `constexpr` Constructors

Although constructors can't be `const` (§ 7.1.4, p. 276), constructors in a literal class can be `constexpr` functions. Indeed, a literal class must provide at least one `constexpr` constructor.

A `constexpr` constructor can be defined as `= default` (§ 7.1.4, p. 278) (or as a deleted function, which we cover in § 13.1.6 (p. 563)). A `constexpr` constructor must meet the usual requirements of a constructor (§ 7.1.4, p. 276) and of a `constexpr` function (§ 6.5.1, p. 247). We define a `constexpr` constructor by preceding its declaration with the keyword `constexpr`:

```
constexpr Debug() = default;
constexpr Debug(bool b) : hw(b), io(b), other(b) { }
constexpr Debug(bool h, bool i, bool o):
                                hw(h), io(i), other(o) { }
```

A `constexpr` constructor must initialize every data member. Each initializer must either use a `constexpr` constructor or be a constant expression.

A `constexpr` constructor is used to generate objects that are `constexpr` and for parameters or return types in `constexpr` functions:

```
constexpr Debug io_sub(false, true, false);  // debugging IO
if (io_sub.any())  // equivalent to if(true)
    cerr << "print appropriate error messages" << endl;
constexpr Debug prod(false); // no debugging during production
if (prod.any())    // equivalent to if(false)
    cerr << "print an error message" << endl;
```

#### EXERCISES SECTION 7.6.2

**Exercise 7.55:** Define your own version of `Debug`.

**Exercise 7.56:** Add members to `Debug` to set each flag. Should these members be declared as `constexpr`? If not, why not?

**Exercise 7.57:** Is the `Data` class from § 7.6 (p. 311) a literal class? If not, why not? If so, explain why it is literal.

## 7.7 `static` Class Members

Classes sometimes need members that are associated with the class, rather than with individual objects of the class type. For example, a bank account class might need a data member to represent the current prime interest rate. In this case, we'd want to associate the rate with the class, not with each individual object. From an efficiency standpoint, there'd be no reason for each object to store the rate. Much more importantly, if the rate changes, we'd want each object to use the new rate.

## Declaring `static` Members

We say that a member is associated with the class by adding the keyword `static` to its declaration. Like any other member, `static` members can be `public` or `private`. The type of a `static` data member can be `const`, reference, array, class type, and so forth. Unlike ordinary data members, `static` data members can be `inline` (§ 2.2.2, p. 46) or `constexpr` (§ 2.4.4, p. 66)

As an example, we'll define a class to represent an account record at a bank:

```
class Account {
public:
    // constructors initialize only non-static data members
    Account() : amount(0.0), daily_tbl{0} { }
    void calculate() { amount += amount * interestRate; }
    static double rate() { return interestRate; }
    static void newRate(double);
private:
    static double initRate();

    static double interestRate;
    inline static const std::string accountType = "Savings";
    static constexpr int period = 30;

    std::string owner;
    double amount;
    double daily_tbl[period];
};
```

The `static` members of a class exist outside any object. There is one object associated with each `static` data member. That object is allocated and initialized as part of program start-up.

Objects of a class type contain only the non`static` data members of that class. Thus, each `Account` object will contain three data members—`owner`, `amount`, and `daily_tbl`. Because the `static` data members are not part of `Account` objects, the `Account` constructor does not—indeed cannot—initialize the `static` data members.

Similarly, `static` member functions are not bound to any object; they do not have a `this` pointer. As a result, `static` member functions may not be declared as `const`, and we may not refer to `this` in the body of a `static` member. This restriction applies both to explicit uses of `this` and to implicit uses of `this` by calling a non`static` member.

## Using a Class `static` Member

We can access a `static` member directly through the scope operator:

```
double r;
r = Account::rate(); // access a static member using the scope operator
```

Even though `static` members are not part of the objects of its class, we can use an object, reference, or pointer of the class type to access a `static` member:

```
Account ac1;
Account *ac2 = &ac1;
// equivalent ways to call the static member rate function
r = ac1.rate();        // through an Account object or reference
r = ac2->rate();       // through a pointer to an Account object
```

Member functions can use `static` members directly, without the scope operator:

```
void calculate() { amount += amount * interestRate; }
```

Here the (ordinary) member function `calculate` refers to both the `static` member `interestRate` and the ordinary data member `amount` without any qualification.

## Defining `static` Members

As with any other member function, we can define a `static` member function inside or outside of the class body. When we define a `static` member outside the class, we do not repeat the `static` keyword:

```
void Account::newRate(double r)
{
    interestRate = r;
}
```

> **Note:** As with any class member, when we refer to a class `static` member outside the class body, we must specify the class in which the member is defined. The `static` keyword, however, is used *only* on the declaration inside the class body.

Like global objects (§ 6.1.1, p. 216), `static` data members are defined outside any function. They are created at program start-up and destroyed when the program terminates. A `static` data member gets defined and initialized in one of two ways:

- `inline` and `constexpr` `static` data members are defined as part of the class definition and are initialized from a member initializer (§ 2.6.1, p. 74) if there is one. As usual, the intializer for a `constexpr` member must be a constant expression. If we do not initialize an these members, they are value-initialized.
- With one exception (covered below), all other `static` data members must be defined and initialized outside the class body. They may not be initialized inside the class.

We define a `static` data member similarly to how we define class member functions outside the class. We name the object's type, followed by the name of the class, the scope operator, and the member's own name:

```
// define and initialize a static class member
double Account::interestRate = initRate();
```

This statement defines the object named `interestRate` that is a `static` member of class `Account` and has type `double`. Once the class name is seen, the remainder of the definition is in the scope of the class. As a result, we can use `initRate` without qualification as the initializer for `interestrate`. Note also that although `initRate` is `private`, we can use it to initialize `interestRate`. As with any other member definition, a static data member definition may access the `private` members of its class.

Tip: The best way to ensure that the object is defined exactly once is to put the definition of `static` data members in the same file that contains the definitions of the class noninline member functions.

New: For compatability with C++11 we can redefine `constexpr static` data members outside the class, but the definition must not contain an initializer.

```
// C++14 required separate definition for constexpr static members
// C++17 redefinition of a constexpr static member is allowed, but deprecated
constexpr int Account::period; // initializer provided in the class definition
```

Such definitions are **deprecated**. A deprecated feature is one that future releases of C++ may not support.

For historical reasons, `const static` data members of integral type may be initialized inside the class so long as the initializer is a constant expression. If such a member is only used in contexts in which the compiler can substitute the member's value, then the member does not have to be separately defined. Otherwise, there must be a definition for that member and thet definition must not contain an initializer.

Best Practices: Modern C++ programs should use `constexpr` to define `static` data members that represent constant expressions.

## `static` Members Can Be Used in Ways Ordinary Members Can't

As we've seen, `static` members exist independently of any other object. As a result, they can be used in ways that would be illegal for non`static` data members. As one example, a `static` data member can have incomplete type (§ 7.3.3, p. 292). In particular, a `static` data member can have the same type as the class type of which it is a member. A non`static` data member is restricted to being declared as a pointer or a reference to an object of its class:

```
class Bar {
private:
    static Bar mem1; // ok: static member can have incomplete type
    Bar *mem2;       // ok: pointer member can have incomplete type
    Bar mem3;        // error: data members must have complete type
    // ...
};
```

Another difference between `static` and ordinary members is that we can use a `static` member as a default argument (§ 6.2.3, p. 224):

```
class Screen {
public:
    // bkground refers to the static member declared later in the class definition
    Screen& clear(char = bkground);
private:
    static const char bkground;
};
```

A nonstatic data member may not be used as a default argument because its value is part of the object of which it is a member. Using a nonstatic data member as a default argument provides no object from which to obtain the member's value and so is an error.

## EXERCISES SECTION 7.7

**Exercise 7.58:** What is a `static` class member? How do they differ from ordinary members?

**Exercise 7.59:** Write your own version of the `Account` class.

**Exercise 7.60:** Which, if any, of the following `static` data member declarations and definitions are errors? Explain why.

```
// example.h
class Example {
public:
    static double rate = 6.5;
    static const int vecSize = 20;
    static vector<double> vec(vecSize);
};
// example.C
#include "example.h"
double Example::rate;
vector<double> Example::vec;
```

## Chapter Summary

Classes are the most fundamental feature in C++. Classes let us define new types for our applications, making our programs shorter and easier to modify.

Data abstraction—the ability to define both data and function members—and encapsulation—the ability to protect class members from general access—are fundamental to classes. We encapsulate a class by defining its implementation members as `private`. Classes may grant access to their non`public` member by designating another class or function as a `friend`.

Classes may define constructors, which are special member functions that control how objects are initialized. Constructors may be overloaded. Constructors should use a constructor initializer list to initialize all the data members.

Classes may also define `mutable` or `static` members. A `mutable` member is a data member that is never `const`; its value may be changed inside a `const` member function. A `static` member can be either function or data; `static` members exist independently of the objects of the class type.

## Defined Terms

**abstract data type** Data structure that encapsulates (hides) its implementation.

**access specifier** Keywords `public` and `private`. Used to define whether members are accessible to users of the class or only to friends and members of the class. Specifiers may appear multiple times within a class. Each specifier sets the access of the following members up to the next specifier.

**aggregate class** Class with only `public` data members and no constructors. Members of an aggregate can be initialized by a brace-enclosed list of initializers.

**class** C++ mechanism for defining our own abstract data types. Classes may have data, function, or type members. A class defines a new type and a new scope.

**class declaration** The keyword `class` (or `struct`) followed by the class name followed by a semicolon. If a class is declared but not defined, it is an incomplete type.

**class keyword** Keyword used to define a class; by default members are `private`.

**class scope** Each class defines a scope. Class scopes are more complicated than other scopes—member functions defined within the class body may use names that appear even after the definition.

**const member function** A member function that may not change an object's ordinary (i.e., neither `static` nor `mutable`) data members. The `this` pointer in a `const` member is a pointer to `const`. A member function may be overloaded based on whether the function is `const`.

**constructor** A special member function used to initialize objects. Each constructor should give each data member a well-defined initial value.

**constructor initializer list** Specifies initial values of the data members of a class. The members are initialized to the values specified in the initializer list before the body of the constructor executes. Class members that are not initialized in the initializer list are default initialized.

**converting constructor** A non`explicit` constructor that can be called with one argument. Such constructors implicitly convert from the argument's type to the class type.

**data abstraction** Programming technique that focuses on the interface to a type. Data abstraction lets programmers ignore the details of how a type is represented and focus instead on the operations the type performs. Data abstraction is fundamental to both object-oriented and generic programming.

**default constructor** Constructor that is used if no initializer is supplied.

**delegating constructor** Constructor with a constructor-initializer list that has one entry that designates another constructor of the same class to do the initialization.

**deprecated** A feature provided for backward compatability and that may be omitted in future releases.

**encapsulation** Separation of implementation from interface; encapsulation hides the implementation details of a type. In C++, encapsulation is enforced by putting the implementation in the `private` part of a class.

**explicit constructor** Constructor that can be called with a single argument but cannot be used in an implicit conversion. A constructor is made explicit by prepending the keyword `explicit` to its declaration.

**forward declaration** Declaration of an as yet undefined name. Most often used to declare the name of a class whose definition appears later. See incomplete type.

**friend** Mechanism by which a class grants access to its non`public` members to another class or function. Friends have the same access rights as members.

**implementation** The (usually `private`) members of a class that define the data and any operations that are not intended for use by code that uses the type.

**incomplete type** Type that is declared but not defined. It is not possible to use an incomplete type to define a variable or class member. It is legal to define references or pointers to incomplete types.

**interface** The (`public`) operations supported by a type. Ordinarily, the interface does not include data members.

**member function** Class member that is a function. Ordinary member functions are bound to an object through the implicit `this` pointer. `static` member functions are not bound to an object and have no `this` pointer. Member functions may be overloaded.

**mutable data member** Data member that is never `const`, even when it is a member of a `const` object. A `mutable` member can be changed inside a `const` function.

**name lookup** Process by which the use of a name is matched to its declaration.

**private members** Members that are accessible only to the friends and other class members. Data members and utility functions used by the class that are not part of the type's interface are usually declared `private`.

**public members** Members that are accessible to any user of the class. Ordinarily, only the functions that define the interface to the class should be defined in the `public` sections.

**struct keyword** Keyword used to define a class; by default members are `public`.

**synthesized default constructor** The default constructor created (synthesized) by the compiler for classes that do not explicitly define any constructors. This constructor initializes the data members from their member initializers, if present; otherwise it default initializes the data members.

**this pointer** Implicit parameter in every non`static` member function. `this` points to the object on which the function is invoked. The implicit `this` pointer participates in the function matching.

**= default** Syntax used after the parameter list of the default constructor to signal to the compiler that it should generate the constructor. Useful when a class wants the synthesized default constructor but has other constructors.